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Measurement of the branching fraction of $B^+ \rightarrow \tau^+ \nu_\tau$ decays with the semileptonic tagging method at Belle II

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Abstract

In this note we present the measurement of the branching fraction of the decay $B^+ \rightarrow \tau^+ \nu_\tau$ using the semileptonic tagging method at Belle II. This measurement uses a data sample of 387 million $B\bar{B}$ meson pairs recorded by the Belle II detector at the SuperKEKB electron-positron collider between 2019 and 2022.

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1 **CHANGELOG**

2 **Version 7.0**

3 Unblinding procedure done, BR measured.

- 4 • Unblinding and Final result sections updated (see Sections 9 and 10).
- 5 • Added additional post-fit checks in Appendix G.
- 6 • Updated the number of NPs used in the fit with the new gathering of eigen-
7 vectors in Section 8.
- 8 • Added D_{tag}^0 corrections for μ mode in Section 5.1.
- 9 • Updated Asimov fit results in Section 7.1.

10 **Version 6.0**

- 11 • Found a bug in $\ell\ell XX$ normalization (assumed 1444fb^{-1} instead of 360fb^{-1}),
12 updated tables and fit. No major changes in the results.
- 13 • Modified and added more details on the unblinding procedure in Section 9.
- 14 • Add $E_{\text{extra}}^{\text{ROE}}$ checks in low and high BDT sidebands in Appendix D.b.

15 **Version 5.0**

16 Further RC comments addressed.

- 17 • Updated embedded sample efficiencies using new embedded MC samples.
18 Added a 11% systematic uncertainty to the signal efficiency (see Section 6.1).
- 19 • Use official rare MC samples, updated Section 8.7.2 and fitting results. Num-
20 bers are slightly lower than before, but same conclusion applies.
- 21 • Added more details for SCF in Appendix E.b.
- 22 • After checking control samples, added p_{CMS} correction + systematic to signal
23 sample, updated fit.

- Re-ran the ROE mask optimization, taking into account data/MC discrepancy and the uncertainty of the ROE corrections. No improvement was found (see Appendix B).

Version 4.0

Further RC comments addressed.

- ROE correction transferability checks. For signal weights, now using the control samples ($B^+ \rightarrow \bar{D}^{*0}\ell^+\nu_\ell$ and double tagged) instead of the extra track sample. For $B\bar{B}$ background, now using BDT sideband instead of the extra track sample. Updated all related sections and plots (see Sections 5.2, 6.3.4 and 6.3.5 and appendix D.c).
- Increase SCF shape syst to a conservative 10% value, update the fit results.
- Added impact of each NPs in Appendix F.d.

Ongoing changes for next version:

- Use official rare MC samples (we found out only exp7 was skimmed. The experts are looking to fix the issue in the workflow and reprocess the files).

Version 3.0

Latest RC comments addressed.

- Added Appendix E with extra track and SCF studies.
- Updated $X_u\ell\nu$ and other BR in the systematic uncertainties (see Section 8.7.1).
- Added SCF shape systematic.
- Update fit results in Section 7.2.
- Update $B^+ \rightarrow \bar{D}^{*0}\ell^+\nu_\ell$ control channel checks using the new BDT and corrections (see Section 6.3.4).
- Update double tagged sample (SL FEI + hadronic FEI) using the new BDT and corrections (see Section 6.3.5).

- Update tables in Appendix A.b with the correct FEI reconstruction efficiency.
- Fixed typos / made more clarifications on multiple sections based on additional RC reviewers comments.

Version pre3.0-2

Feb B2GM plenary version.

- Update fit results and new systematics.

Version pre3.0-1

Post WGR, addressing RC reviewers comments.

- Double tagged sample (SL FEI + hadronic FEI) checks for hadronic channels (see Section 6.3.5).
- Added Section 4.3 to see the signal distribution shape on the fit variables.
- Removed 2 least important features from leptonic BDT, updated everything accordingly (see Section 4.1).
- The extra track sample is now used for ROE corrections (see Section 5.2).
- The FEI calibration factors are now applied to only correctly reconstructed B_{tag} (see Section 5.1).
- Update Figure 64 with more points.
- Made NPs distribution Figure 75 readable.
- Added pull plot of all NPs on Asimov datasets in Appendix F.c.
- Fixed typos / made more clarifications on multiple sections based on RC reviewers comments.

Version 2.0

Post FSR, addressing WG reviewers and additional comments.

Major changes since last version:

- Removed Btag decay mode from BDT variables in Section 4.1. Updated everything post-BDT accordingly.
- Corrections are now done after the BDT. The BDT sidebands is now used for ROE and $\tau_{p_{\text{CMS}}}$ (see Sections 5.1 and 5.2). The FEI calibration factors are now used correctly, and continuum is corrected using off-resonance data (see Section 5.1). Updated all related sections and plots.
- Fit is now using `sysvar` for all the correction calculations (see Section 8). Updated all plots and tables with the new fit settings and results.

Minor changes since last version:

- Added additional plots for $\tau^+\tau^-$ background pre-selection cuts in Appendix A.a.
- Increased NP kept in the fit for BF systematics in Section 8.7.1.
- Rare background calculation was updated in Section 8.7.2. Also removed the added rare background from generic MC.
- Updated Figure 2 with the latest $|V_{ub}|$ exclusive value updated.
- Checked: removing ρ channel doesn't increase significance.
- Fixed typos and added more explanations when relevant.

Version 1.4

Note version for the FSR. Continue replying to WG conveners comments.

- Added $B^+ \rightarrow \bar{D}^{*0}\ell^+\nu_\ell$ FF systematic uncertainty in Section 8.8.
- Updated wrongly scaled Table 7 and in Appendix A.b.
- Updated PID systematic uncertainty taking into account the light-release issue in Section 8.5.2.
- Updated fit results with data/MC scaling from blind on-resonance data in Section 6.3.6.

Version 1.3

Replying to WG conveners comments.

- 100 • Clarified Btag BCS selection in Section 3.1.
- 101 • Clarified ROE mask notation in Section 3.2.
- 102 • Updated yrange of plots in Section 6.3.3.
- 103 • Added Figure 65 showing the π ID efficiency correction.
- 104 • Changed uncertainties in Tables 20 and 21.
- 105 • Better explanations of $K_L^0 K_L^0 \ell \nu$ BF in Section 8.7.2.
- 106 • Updated in Tables 22 and 23 the expected yields.
- 107 • Added Figure 72 showing the fit variables distribution for rare $B\bar{B}$ decays.

108 **Version 1.2**

109 Stable version, send to the WG for comments.

- 110 • Removed $D_{1\text{st daughter}} dr$ variable from BDT, updated everything accordingly.
- 111 • Fit:
 - 112 – All NPs now split into norm+shape (normsys+histosys).
 - 113 – Added rare $B\bar{B}$ decays section and added a new template in the fit.
 - 114 – Signal and SCF templates now taken from signal-only sample.
 - 115 – $E_{\text{extra}}^{\text{ROE}}$ corrections applied to signal, $\tau_{p_{\text{CMS}}}$ corrections removed from signal.
 - 116 – Complete list of systematic uncertainties.

117 **Version 1.1**

118 Addressing Ishikawa-san's comments from the last version.

119

120 Major changes since last version:

- 121 • Added $D * \ell \nu$ control sample study.
- 122 • Removed irrelevant `sigProb` cuts, updated the syst corrections accordingly.
- 123 • Added blind on-resonance data checks.

124 • Updated fit checks.

125 Minor changes since last version:

126 • Updated $\mathcal{B}$ and f_B values.

127 • Updated luminosity numbers and introduction section.

128 • Harmonized the notation of $\tau_{p_{\text{CMS}}}$.

129 • BDT variables explained.

130 • Fixed typos, added more text when relevant.

131 **Version 1.0**

132 • Initial version with all sections filled.

133 ANALYSIS SOFTWARE AND DATASETS

134 The ntuples, produced with the basf2 [1] release light-2506-deimos, are kept
135 on KEKCC at the following path:

`/gpfs/group/belle/users/tfilling/b2tv/grid/merged/`

136 The gitlab repository of the analysis is located at the following path:

`https://gitlab.desy.de/tristan.fillinger/b2taunu/`

137 The document server repository of the analysis is located at the following path:

`https://docs.belle2.org/pub.data/analyses/SLME-016/`

138 Code dependencies:

- 139 • `plothist>=1.2.6` [2]
- 140 • `awkward==2.6.6`
- 141 • `b2luigi==1.2.2`
- 142 • `boost-histogram==1.4.0`
- 143 • `cabinetry @ git+https://github.com/MoAly98/cabinetry@5d441aacb35705c`
144 `5028ed198d17813fb160ca252`
- 145 • `matplotlib==3.8.4`
- 146 • `numpy==1.26.0`
- 147 • `optuna==3.6.1`
- 148 • `pandas==2.0.3`
- 149 • `paramiko==3.2.0`
- 150 • `particle==0.25.3`
- 151 • `psutil==5.9.0`
- 152 • `pyarrow==11.0.0`
- 153 • `pyhf[minuit]==0.7.2`

- 154 • PyYAML==6.0
- 155 • scikit-learn==1.5.2
- 156 • scikit-optimize==0.10.2
- 157 • scipy==1.12.0
- 158 • scp==0.14.5
- 159 • seaborn==0.13.2
- 160 • tabulate==0.8.9
- 161 • uncertainties==3.1.6
- 162 • uproot==5.3.10
- 163 • xgboost==2.0.0
- 164 • zfit==0.24.3
- 165 • prek==0.1.6
- 166 • awkward-pandas==2023.8.0
- 167 • jax==0.4.13
- 168 • jaxlib==0.4.13
- 169 • numexpr==2.8.4
- 170 • iminuit==2.24.0
- 171 • shap==0.48.0
- 172 • tensorflow==2.18

173 1. INTRODUCTION

174 In this paper, we present the Branching Fraction (BF) measurement of $B^+ \rightarrow \tau^+ \nu_\tau$
 175 decays, a rare process in the Standard Model (SM) with a theoretical branching
 176 fraction of $\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau) = (0.80 \pm 0.06) \times 10^{-4}$ [3]. This measurement uses a data
 177 sample of 387 million $B\bar{B}$ meson pairs recorded by the Belle II detector at the
 178 SuperKEKB electron-positron collider between 2019 and 2022. It corresponds to a
 179 luminosity of $(365.4 \pm 1.7) \text{ fb}^{-1}$ of on-resonance data, corresponding to a number of
 180 produced $\Upsilon(4S)$ estimated to be $n_{\Upsilon(4S)} = (387 \pm 6) \times 10^6$, and 42 fb^{-1} of off-resonance
 181 data, where the energy is around 60 MeV below the $\Upsilon(4S)$ resonance peak.

182 Two scenarios can arise with the $B^+ \rightarrow \tau^+ \nu_\tau$ decay: a measurement of the SM
 183 with better precision or the discovery of New Physics (NP).

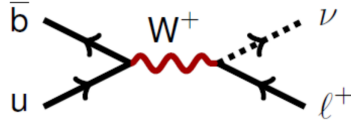


FIG. 1. Feynman diagram of the $B^+ \rightarrow \tau^+ \nu_\tau$ decay.

184 The branching fraction of $B^+ \rightarrow \tau^+ \nu_\tau$ can be expressed as:

$$\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau) = \frac{G_F^2 m_B m_\tau^2}{8\pi} \left(1 - \frac{m_\tau^2}{m_B^2}\right)^2 f_B^2 |V_{ub}|^2 \tau_B, \quad (1)$$

185 with the values of the input parameters used in the calculation of the branching
 186 fraction of $B^+ \rightarrow \tau^+ \nu_\tau$ decays are shown in Table 1.

187 The BF is sensitive to the CKM matrix element $|V_{ub}|$, which is usually measured
 188 with $B \rightarrow \pi l \nu$ decays [6]. The inclusive and exclusive central values of $|V_{ub}|$ have a
 189 slight disagreement. Measuring the BF of $B^+ \rightarrow \tau^+ \nu_\tau$ decays with better precision
 190 can help to clarify this discrepancy and get a better uncertainty on $|V_{ub}|$.

191 In addition, the $B^+ \rightarrow \tau^+ \nu_\tau$ decay is a good probe to search NP effects. The
 192 W boson in Figure 1 can be replaced by a charged Higgs boson in the Two-Higgs-
 193 Doublet Model (2HDM) [7] or various super-symmetric extensions of the SMs. The
 194 NP effects can be parameterized by the ratios [8]:

$$R_{\text{ps}} \propto \frac{\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau)}{\mathcal{B}(B \rightarrow \pi l \nu)} = (0.539 \pm 0.043) \cdot |1 + r_{\text{NP}}|^2, \quad (2)$$

195 and

	Value	Relative Uncertainty
G_F	11.7 TeV^{-2}	$5 \cdot 10^{-7}$
τ_B	1.64 ps	$2 \cdot 10^{-3}$
m_B	5.28 GeV	$1 \cdot 10^{-5}$
m_τ	1.78 GeV	$5 \cdot 10^{-5}$
m_μ	105 MeV	$2 \cdot 10^{-8}$
f_B	190.0 MeV	$1 \cdot 10^{-2}$
$ V_{ub} _{\text{inc}}$	$4.13 \cdot 10^{-3}$	$6 \cdot 10^{-2}$
$ V_{ub} _{\text{exc}}$	$3.70 \cdot 10^{-3}$	$4 \cdot 10^{-2}$
$ V_{ub} _{\text{avg}}$	$3.82 \cdot 10^{-3}$	$5 \cdot 10^{-2}$

TABLE 1. Parameters used in the calculation of the branching fraction of $B^+ \rightarrow \tau^+ \nu_\tau$ decays [4, 5].

$$R_{\text{pl}} = \frac{\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau)}{\mathcal{B}(B^+ \rightarrow \mu^+ \nu_\tau)} = 222 \cdot |1 + r_{\text{NP}}|^2, \quad (3)$$

where r_{NP} is a combination of Wilson coefficients sensitive to the NP contribution. With the full Belle II dataset, we can expect to search for a NP contribution to $B^+ \rightarrow \tau^+ \nu_\tau$ decays with a sensitivity of $r_{\text{NP}} > \mathcal{O}(0.1)$ at 95% confidence level.

Past measurements of the BF of $B^+ \rightarrow \tau^+ \nu_\tau$ decays have been performed by the Belle and BaBar collaborations using two approaches: with a semileptonic tag [9, 10] and with a hadronic tag [11, 12]. The semileptonic tag method has the advantage of a larger branching fraction, but it is more challenging due to the presence of neutrinos in the final state. In 2025, Belle II also measured the BF with a fully reconstructed tag [13]. All the different measurements are shown Figure 2. The current PDG average is compatible within 1σ with the SM prediction [4]. There are currently no 5σ observations of the $B^+ \rightarrow \tau^+ \nu_\tau$ decay, only evidence at the level of $\approx 3\sigma$.

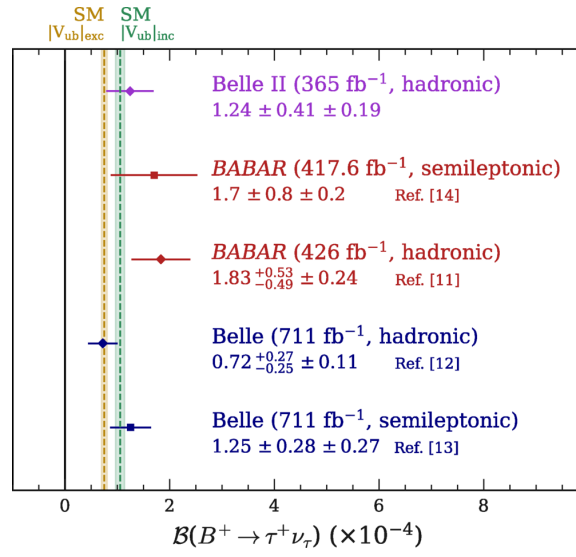


FIG. 2. Summary of the measurements of the branching fraction of $B^+ \rightarrow \tau^+ \nu_\tau$ decays [13].

208 2. DATA SAMPLES

209 This analysis uses:

210 • Monte Carlo (MC) samples:

- 211 – **Signal:** 50M $B^+ \rightarrow \tau^+ \nu_\tau$ signal only: a $\Upsilon(4S)$ decaying to $B^+ \rightarrow$
 212 $\tau^+ \nu_\tau$ and an additional $B^\pm$ that decays generically. This corresponds
 213 to $\approx 400 \text{ ab}^{-1}$ of signal MC (MC15rd).
- 214 – **Generic:** 1444 fb^{-1} $B\bar{B}$ background (MC15rd)
- 215 – **non-BB:** 1444 fb^{-1} continuum ($q\bar{q}$) and $\tau^+ \tau^-$, and 361 fb^{-1} of $\ell\ell XX$
 216 backgrounds (MC15rd)

217 **Control samples:**

- 218 * **Off-resonance:** 170 fb^{-1} off-resonance ($q\bar{q}$ and $\tau^+ \tau^-$) (MC15rd)
- 219 * **Sidebands (SB) Signal:** 50M $B^+ \rightarrow \tau^+ \nu_\tau$ signal only with at least
 220 one track in the rest of event
- 221 * **SB Generic:** 1444 fb^{-1} $B\bar{B}$ background with at least one track in
 222 the rest of event
- 223 * **SB Continuum:** 1444 fb^{-1} continuum ($q\bar{q}$) and $\tau^+ \tau^-$, and 361 fb^{-1}
 224 of $\ell\ell XX$ backgrounds with at least one track in the rest of event
 225 (MC15rd)

226 • Data samples:

- 227 – **On-resonance:** 385 fb^{-1} on-resonance data (LS1 dataset)
- 228 – **Off-resonance:** 42 fb^{-1} off-resonance data (LS1 dataset)

229 **Control samples:**

- 230 * **SB On-resonance:** 385 fb^{-1} on-resonance data with at least one
 231 track in the rest of event (LS1 dataset)
- 232 * **SB Off-resonance:** 42 fb^{-1} off-resonance data with at least one
 233 track in the rest of event (LS1 dataset)
- 234 * **Embedded signal:** ≈ 200 candidates of embedded $B^+ \rightarrow \tau^+ \nu_\tau$
 235 events (see Section 6.1)

236 3. EVENT SELECTION

237 This section describes the reconstruction and the pre-selection of the $B^+ \rightarrow \tau^+ \nu_\tau$
 238 candidates. The reconstruction is done using the Full-Event Interpretation (FEI)
 239 Skim, and the pre-selection is done using a set of loose cuts on the reconstructed
 240 candidates.

241 3.1. Skim

242 This analysis uses the semileptonic (SL) Full-Event Interpretation (FEI) Skim [14].
 243 One of the $B^\pm$ in the decay is reconstructed as one of the following modes shown in
 244 Tables 2 and 3.

Mode ID	Decay mode
0	$B_{\text{SL}}^+ \rightarrow D^0 e^+ \nu_e$
1	$B_{\text{SL}}^+ \rightarrow D^0 \mu^+ \nu_\mu$
2	$B_{\text{SL}}^+ \rightarrow D^{0*} e^+ \nu_e$
3	$B_{\text{SL}}^+ \rightarrow D^{0*} \mu^+ \nu_\mu$
4	$B_{\text{SL}}^+ \rightarrow D^- \pi^+ e^+ \nu_e$
5	$B_{\text{SL}}^+ \rightarrow D^- \pi^+ \mu^+ \nu_\mu$
6	$B_{\text{SL}}^+ \rightarrow D^{*-} \pi^+ e^+ \nu_e$
7	$B_{\text{SL}}^+ \rightarrow D^{*-} \pi^+ \mu^+ \nu_\mu$

TABLE 2. SL FEI Skim modes and their corresponding Mode ID. The $D^{(*)}$ is reconstructed in hadronic modes only.

245 The D is only reconstructed with hadronic modes, so only one neutrino is present
 246 in this $B^\pm$ decay. This correctly SL tagged $B^\pm$ is called B_{tag} . The rest of the event
 247 can be analyzed to reconstruct our signal event $B^+ \rightarrow \tau^+ \nu_\tau$, called B_{sig} . In the FEI
 248 Skim, the events are already filtered imposing very loose selection requirements on
 249 the quality of tracks and clusters and their multiplicities. Additionally, this set of
 250 cut is applied to the B_{tag} :

- 251 • $-4 < \cos \theta_{B, D_l}^* < 3$,
- 252 • $\log_{10}(\text{signal probability}) > -2.4$,
- 253 • $p_l^{\text{CMS}} > 1.0 \text{ GeV}$ (in center-of-mass (CMS) frame).

Mode ID	D^0 decay	D^+ decay	D^{*0} decay	D^{*+} decay
0	$K^-\pi^+$	$K^-\pi^+\pi^+$	$D^0\pi^0$	$D^0\pi^+$
1	$K^-\pi^+\pi^0$	$K^-\pi^+\pi^+\pi^0$	$D^0\gamma$	$D^+\pi^0$
2	$K^-\pi^+\pi^0\pi^0$	$K^-K^+\pi^+$		$D^+\gamma$
3	$K^-\pi^+\pi^+\pi^-$	$K^-K^+\pi^+\pi^0$		
4	$K^-\pi^+\pi^+\pi^-\pi^0$	$\pi^+\pi^0$		
5	$\pi^-\pi^+$	$\pi^+\pi^+\pi^-$		
6	$\pi^-\pi^+\pi^+\pi^-$	$\pi^+\pi^+\pi^-\pi^0$		
7	$\pi^-\pi^+\pi^0$	$K_S^0\pi^+$		
8	$\pi^-\pi^+\pi^0\pi^0$	$K_S^0\pi^+\pi^0$		
9	$K_S^0\pi^0$	$K_S^0\pi^+\pi^+\pi^-$		
10	$K_S^0\pi^+\pi^-$	$K^+K_S^0K_S^0$		
11	$K_S^0\pi^+\pi^-\pi^0$			
12	K^-K^+			
13	$K^-K^+\pi^0$			
14	$K^-K^+K_S^0$			

TABLE 3. Charm meson decay modes and their corresponding Mode ID. The D^0 and D^+ are reconstructed in hadronic modes only, while the D^{*0} and D^{*+} are reconstructed from their charm-meson daughters.

If multiple B_{tag} are found in the event, the one with the highest signal probability is selected as the B_{tag} , and the rest are discarded. No multiple candidate remain after this selection.

In addition, the Rest of Event (ROE) of the B_{tag} is reconstructed using the remaining tracks and clusters that doesn't belong to it. Continuum suppression variables are calculated in the CMS using a standard set of cuts on the tracks and clusters in the ROE recommended by the FEI Task Force (see ref):

- Track cut: $d_r < 0.5$ cm, $|d_z| < 2$ cm, $p_t > 0.2$ GeV/ c and in the CDC acceptance.
- ECL cut: ($E_{\text{cluster}} > 0.08$ GeV in the forward region or $E_{\text{cluster}} > 0.03$ GeV in the barrel region or $E_{\text{cluster}} > 0.06$ in the backward region), `clusterNHits` > 1.5 , `clusterTiming` < 200 ns and in ECL acceptance.

Finally, a cut on $\cos \theta_{\text{thrust}, B_{\text{sig}}, \text{ROE}} < 0.9$ is applied.

3.2. Reconstruction

The B_{sig} is reconstructed with the remaining tracks and clusters in the event. Only the one-prong decays of the τ are reconstructed. This accounts for around 72% of the τ decays (see Table 4).

τ decay	Branching fraction
$\mu\nu\nu$	17%
$e\nu\nu$	18%
$\pi\nu$	11%
$(\rho \rightarrow \pi\pi^0)\nu$	26%
Total BF	72%

TABLE 4. Branching fractions of the τ decays.

The τ decay modes are reconstructed as follows:

- μ channel: the tracks that satisfy the `leptonid_Moriond2023_Official_rel6_v0b` standard muon requirements for `FixedThresh09` selection with global `likelihood` are selected. Then a standard set of cut is applied on the muon candidates: they must be in the CDC acceptance and have $d_r < 0.5$ cm, $|d_z| < 2$ cm, $p_t > 0.1$ GeV/ c and $E < 5.5$ GeV.
- e channel: The tracks that satisfy the `leptonid_Moriond2023_Official_rel6_v0b` standard electron requirements for `FixedThresh09` selection with global BDT are selected. On data, we apply the `tracking_momentumScaleFactor_global` scaling factor to each track candidate. We then apply the Belle-like bremsstrahlung correction to the electrons. Finally, the candidates must pass the same standard set of cut as the muons.
- π channel: the pion candidates must also follow the same set of cuts as for the electrons or muons, and we also require that they have a `pionID_noSVD > 0.6`. If we run on data, we apply the `tracking_momentumScaleFactor_global` scaling factor to each track candidate.
- ρ channel: a ρ candidate is reconstructed by combining a pion candidate with a π^0 candidate. For the pion candidate, the same list created for the pion channel is used. The π^0 candidate is reconstructed using the `pi0:eff40_May2020` requirements. If running on data, the energies of the photons are corrected

291 using the
 292 `PhotonEnergyBiasCorrection_MC15rd_June2023` table. The photons are
 293 combined to form a π^0 candidate if they have $0.120 < M < 0.145 \text{ GeV}/c^2$.
 294 The τ candidate is then formed by combining the pion and the π^0 candidates
 295 if they are around the ρ mass $0.625 < M < 0.925 \text{ GeV}/c^2$.

296 The B_{sig} is reconstructed with a single τ using the four channels described above.
 297 Additionally, a ROE of the B_{sig} is reconstructed using the remaining tracks and
 298 clusters that doesn't belong to it. Continuum suppression variables are calculated in
 299 the CMS using a common set of cuts on the tracks and clusters in the ROE (ref):

- 300 • Track cut: $p_{\text{CMS}} \leq 3.2 \text{ GeV}/c$.
- 301 • ECL cut: $E \geq 0.05 \text{ GeV}/c$ and $E_{\text{CMS}} \leq 3.2 \text{ GeV}/c$.

302 Once the B_{sig} and the B_{tag} are reconstructed, they are combined together to form
 303 a $\Upsilon(4S)$. If multiple candidates are found for the B_{sig} , the following criteria are used
 304 to select the best candidate:

- 305 • μ channel: the candidate with the highest `muonID_noSVD` is selected.
- 306 • e channel: the candidate with the highest `pidChargedBDTScore_e` is selected.
- 307 • π channel: the candidate with the highest `pionID_noSVD` is selected.
- 308 • ρ channel: the candidate which has a mass closest to the nominal ρ mass is
 309 selected

310 In some events, two $\Upsilon(4S)$ candidates remain. For 99.3% of the case, it's between
 311 π and ρ channels. For 0.5% of the case, it's between μ and π channels, and for 0.1%
 312 of the case between e and ρ channel. We select at random one of the two candidates.
 313 This happens in around 5% for the signal events.

314 Only one $\Upsilon(4S)$ candidate remains after this selection. We then reconstruct the
 315 ROE of the $\Upsilon(4S)$ by all the remaining tracks and clusters that doesn't belong to
 316 $B^+ \rightarrow \tau^+ \nu_\tau$ or the B_{tag} . To remove most of the background without losing signal
 317 efficiency, we clean the ROE using the following criteria:

- 318 • No extra tracks coming from the IP:
 - 319 – `nTracks in ROE with ( $d_r < 0.5 \text{ cm}$ ,  $|d_z| < 2 \text{ cm}$ , in CDC acceptance`
 320 `) = 0`.
- 321 • No neutrals in the ROE:

- 322 – Number of K_S^0 in ROE = 0; with standard K_S^0 :merged list.
- 323 – Number of Λ^0 in ROE = 0; with standard Λ^0 :merged list.
- 324 – Number of π^0 in ROE = 0; with pi0:eff20_May2020 list.

325 The first cut removes more than 99% of the expected background if no cut on the
 326 tracks of the ROE is applied. The second set of cuts further reduces the background
 327 by 82% on the remaining candidates after the first cut, while only losing 7% on the
 328 signal efficiency.

329 In addition, we also have a control sample, referred as Extra track SB, where the
 330 no extra track cut is inverted to select events with at least one extra track in the
 331 ROE (nTracks in ROE > 0). No significant signal is expected in this sample, and it
 332 is used to get some of the corrections in Section 5.

333 The event shape variables for the $\Upsilon(4S)$ are calculated in the CMS using the
 334 photons that are in the ECL acceptance, have $E > 0.05$ GeV, `minC2TDist` > 20 cm,
 335 `|cluster timing/error|` < 2.0 and `|cluster timing|` < 200 ns, and the pions that
 336 are in the CDC acceptance, have $d_r < 0.5$ cm, $|d_z| < 2.0$ cm, $p_t > 0.1$ GeV/c and
 337 $E < 5.5$.

338 The energy of the ROE $E_{\text{extra}}^{\text{ROE}}$ (often seen as $E_{\text{extra}}^{\text{ROE}}$ c2_bbS_32 in the plots) used
 339 as one of our final fitting variable is using photons that follows:

- 340 • Energy cut: $E_{\text{cluster}} > 0.100$ GeV in the forward region or $E_{\text{cluster}} > 0.055$ GeV
 341 in the barrel region or $E_{\text{cluster}} > 0.080$ in the backward region
- 342 • Additional cut: `minC2TDist` > 30 cm and `beamBackgroundSuppression` >
 343 0.15.

344 The photons in data are corrected using the
 345 `PhotonEnergyBiasCorrection_MC15rd_June2023` table. The signal is peaking at 0
 346 while the different backgrounds follow a Gaussian around $\approx 2 - 3$ GeV, except for
 347 the $\tau^+\tau^-$ and $\ell\ell XX$ backgrounds which also peak at 0 (see Section 3.3 on how
 348 it has been removed). The values of the additional cut have been optimized, see
 349 Appendix B. In the rest of the note, the signal region (SR) is defined as $E_{\text{extra}}^{\text{ROE}} < 0.2$
 350 GeV, corresponding to the first two bins of the $E_{\text{extra}}^{\text{ROE}}$ distribution.

351 The reconstruction efficiency on the signal is calculated using the $B^+ \rightarrow \tau^+\nu_\tau$
 352 signal only sample of 50M events. We find an efficiency of $0.6000 \pm 0.0011\%$. For
 353 the background, we use the $B\bar{B}$ generic sample of 1444 fb^{-1} .

354 **3.3. Pre-selection**

355 A small set of loose cuts is applied on the B candidates to further reduce back-
 356 ground without losing signal efficiency. Some pre-selection cuts are motivated by the

old $B^+ \rightarrow \tau^+ \nu_\tau$ Belle analysis [9]. The selected variables are (see Figure 3 to see the distributions):

- B_{tag} decays: the decay mode of the B_{tag} (see Table 2). All the $D^{-(*)}\pi^+\ell^+\nu_\ell$ modes are removed as these modes have low statistics and the calibration factors are not reliable (see Figure 3(a)).
- $\tau_{p_{\text{CMS}}}$: the CMS momentum of the τ (*e.g.* of the μ , e , π or $\pi\pi^0$) in the B_{sig} to reduce miss-reconstructed B_{sig} candidates (see Figure 3(b)). A loose cut inspired from the Belle analysis [9] is applied.
- $p_{D,\text{tag}}^*$: the CMS momentum of the D in the B_{tag} to further reduce miss-reconstructed B_{tag} candidates (see Figure 3(c)). A loose cut inspired from the Belle analysis [9] is applied.
- $\cos \theta_{B,D_l}^*$: the cosine of the angle in CMS between the Dl system and the B_{tag} . If only a mass-less particle like a neutrino is missing in the reconstruction, it is expected to have a value between -1 and 1 (see Figure 3(d)). A loose cut inspired from the Belle analysis [9] is applied.
- R_2 based on ROE of $\Upsilon(4S)$: the second Fox-Wolfram moment, using the ROE of the $\Upsilon(4S)$ (event shape), to reduce the background from $\tau^+\tau^-$ and $\ell\ell XX$ events, which is peaking in the signal region (see Figures 3(e) and 4(a)). The value of the cut has been optimized to remove as much $\tau^+\tau^-$ background as possible while keeping a high signal efficiency.
- $B_{\text{tag}} R_2$: the second Fox-Wolfram moment of the B_{tag} , using the ROE of the B_{tag} , (continuum suppression, see here the difference with above), to further reduce the background from $\tau^+\tau^-$ and $\ell\ell XX$ events (see Figures 3(f) and 4(b)). The value of the cut has been optimized to remove as much $\tau^+\tau^-$ background as possible while keeping a high signal efficiency.
- $\cos \theta_{\text{thrust},B_{\text{tag}},z}$: the cosine of the angle between the thrust axis of the B_{tag} and the z axis to remove a peaking structure seen in data and not modeled in the MC (see Figures 3(g) and 5)¹. The value of the cut has been optimized to remove the peaking structure while keeping a high signal efficiency.

¹ In all the data/MC comparison plots, the MC normalization is first scaled using the luminosity ratio with data. Then, to compare the shape of the distributions, the ratio of the number of entries between MC and data is computed, and the MC is rescaled accordingly. If used, the ratio used is displayed in the y-label of the bottom panel.

386 Furthermore, τ channel specific pre-selection cuts are applied to the B candidates.
387 For the hadronic channels, a tighter cut on $p_{B_{\text{sig}}} = \tau_{p_{\text{CMS}}}$ is applied. This cut
388 greatly reduces the background (see Figures 6(a) and 6(b)), which will help the
389 BDT to focus on the remaining background candidates without using momentum
390 information as features (see Section 4).

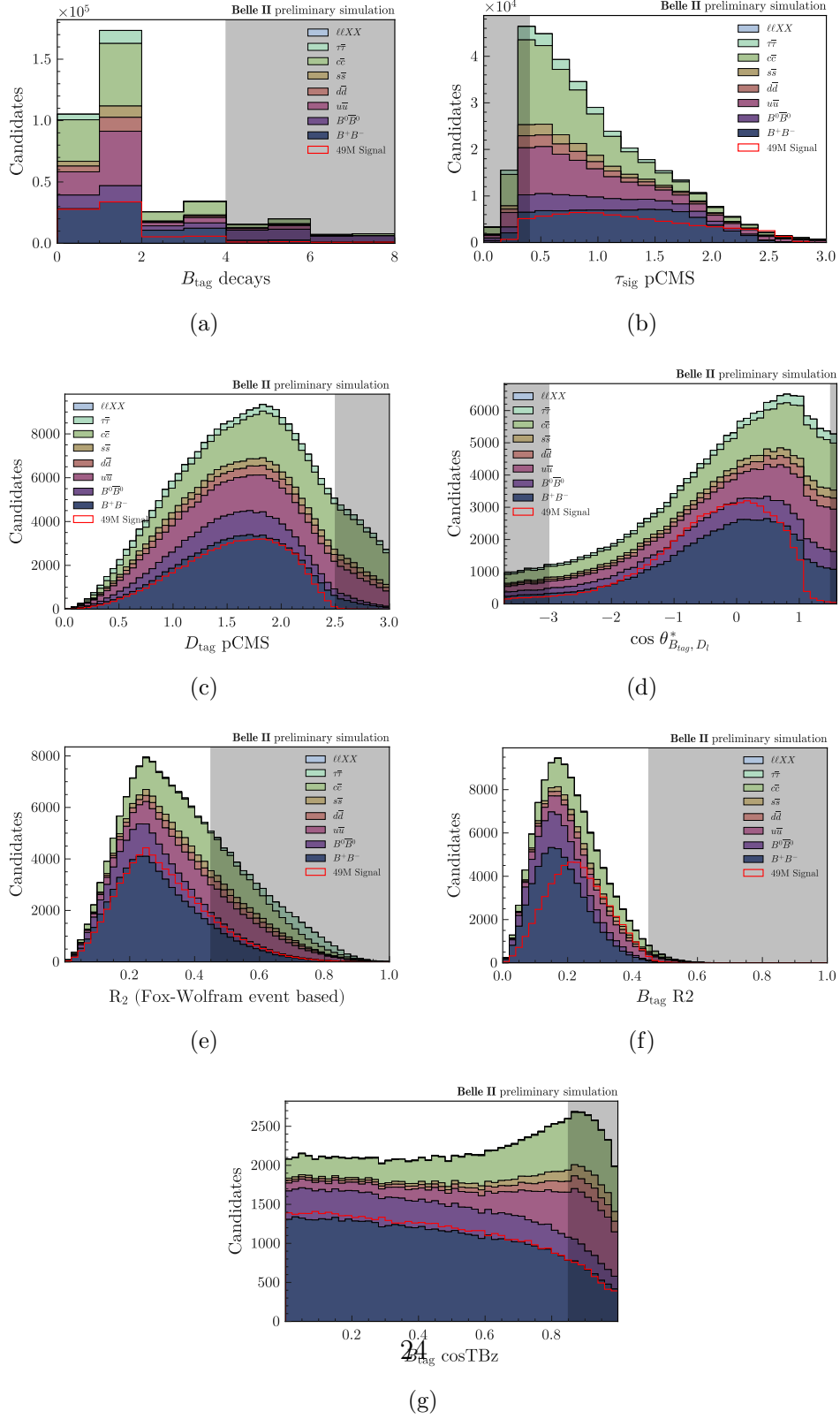


FIG. 3. Distributions of the pre-selection variables before the cut is applied. The gray area represents the discarded region. Each variable is shown after the previous pre-selection cuts have been applied.

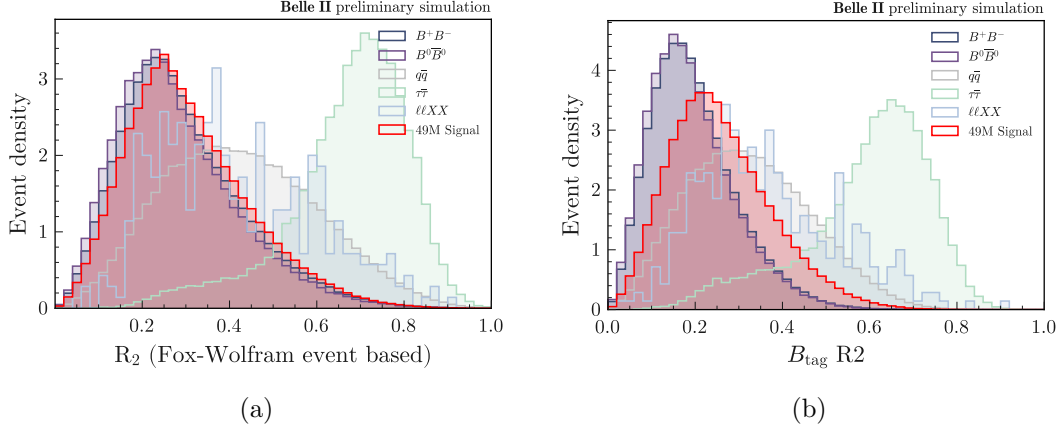


FIG. 4. Showing the $\tau^+\tau^-$ background in the R_2 and $B_{\text{sig}} R_2$ variables. The pre-selection cut on both variables removes the peaking structure.

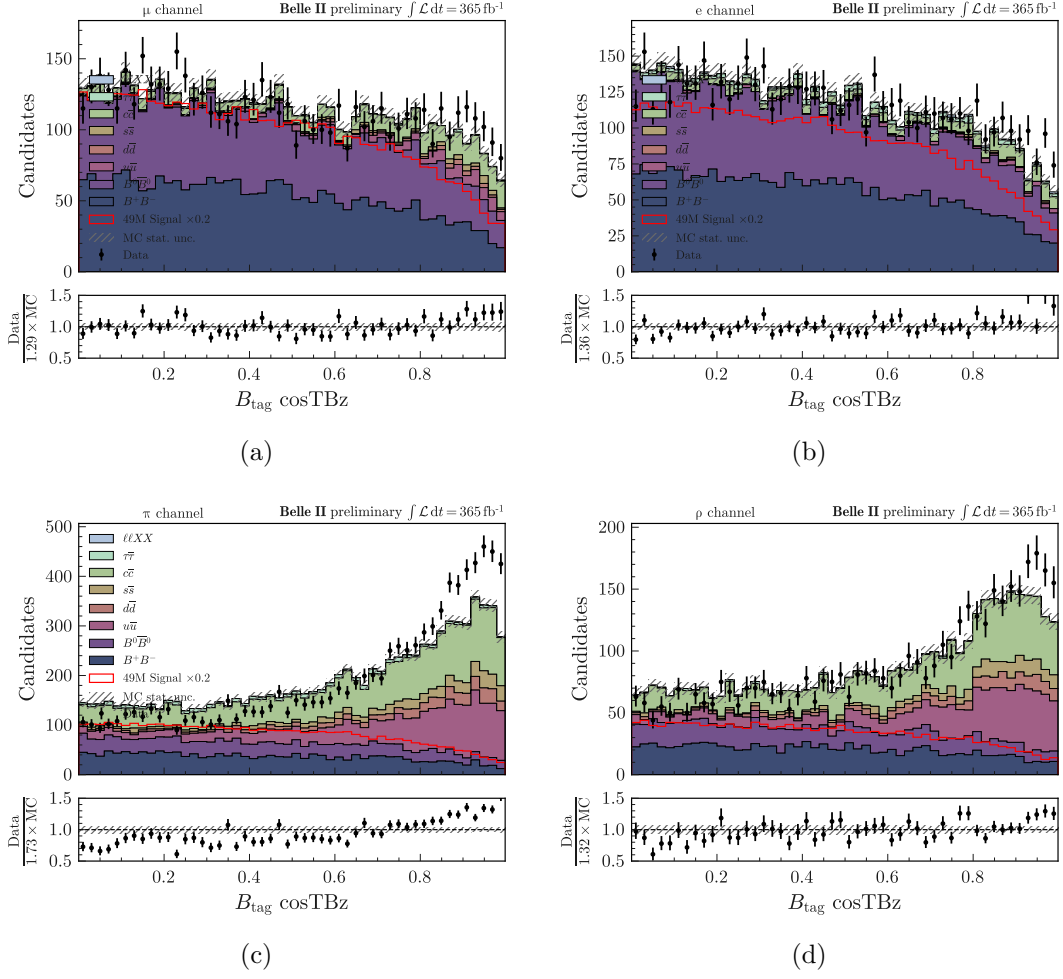


FIG. 5. Distributions of the pre-selection cuts for the $\cos \theta_{\text{thrust}, B_{\text{tag}}, z}$ variable for the μ (a), e (b), π (c) and ρ (d) events in a signal enriched region. We can see a peaking structure in data that is not modeled in the MC. The structure is removed by the pre-selection cut.

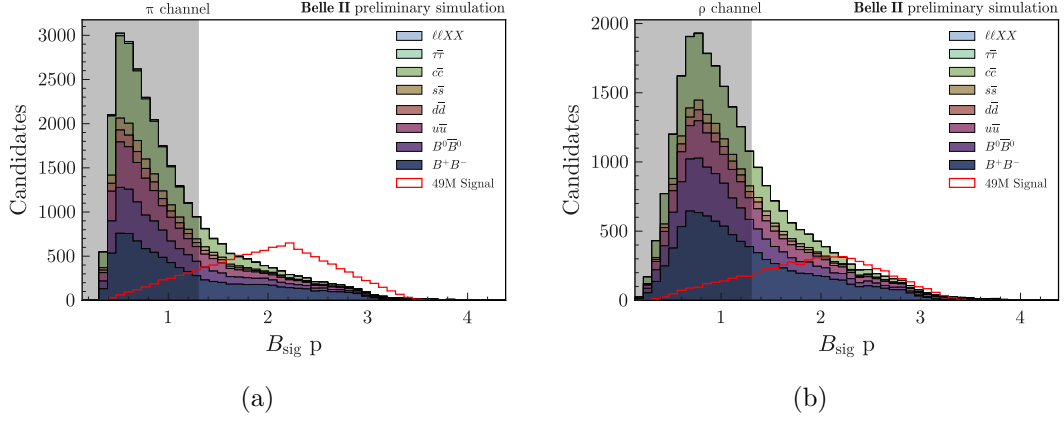
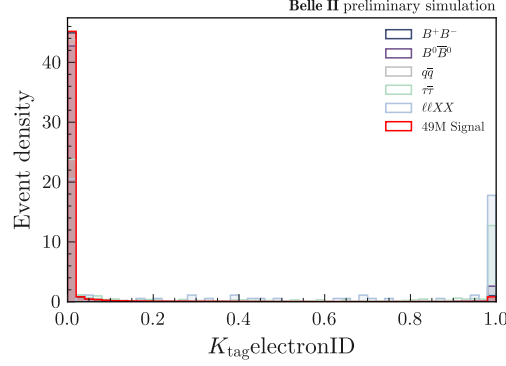


FIG. 6. Distributions of the pre-selection cuts for the $p_{B_{\text{sig}}}$ variable for only the π (a) and ρ (b) events. The Gray area represents the discarded region.

391 For the e channel, a peaking structure in the signal region from $\tau^+\tau^-$ backgrounds
 392 remains after applying all the cuts above. This background is coming from ISR/FSR
 393 photons converting into e^+e^- pairs, which are then wrongly reconstructed as particles
 394 from the B_{tag} or the electron from the B_{sig} . To remove it, we apply a set of cuts
 395 using tag-side final state particle information:

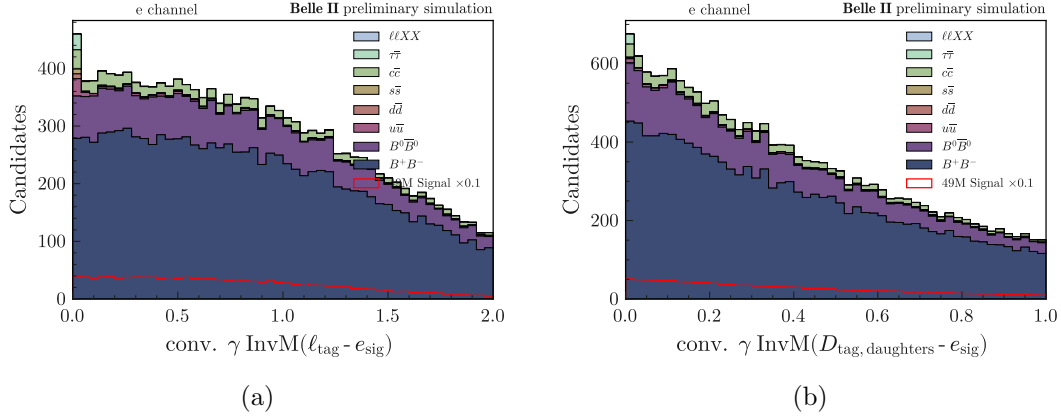
- 396 • K_{tag} electronID: an electron from the ISR/FSR photon conversion can be
 397 wrongly reconstructed as a kaon by the FEI. By cutting on the electronID, we
 398 can remove $49.7 \pm 2.3\%$ of the $\tau^+\tau^-$ background while keeping $96.18 \pm 0.21\%$ of
 399 the signal events (see Figure 7(a)). If multiple kaons are present, we take the
 400 kaon with the highest electronID. If the B_{tag} does not have a kaon, the event
 401 is not impacted by the cut.
- 402 • Converted γ InvM($\ell_{\text{tag}} - e_{\text{sig}}$): the invariant mass of the $\ell_{\text{tag}} - e_{\text{sig}}$ system as
 403 defined in this presentation assuming it's a converted photon. If the system is
 404 from ISR/FSR photon conversion, the invariant mass is expected to be 0. This
 405 cut removes an additional $44.0 \pm 3.2\%$ of the $\tau^+\tau^-$ background while keeping
 406 $98.64 \pm 0.13\%$ of the signal events (see Figure 8(a)). If the charge of the two
 407 particles is not opposite, the event is not impacted by the cut.
- 408 • Converted γ InvM($D_{\text{tag, daughters}} - e_{\text{sig}}$): for each daughter of the D_{tag} , the
 409 invariant mass of the $D_{\text{tag, daughter}} - e_{\text{sig}}$ system, assuming it's a converted photon,
 410 is calculated. Then, the minimum of these invariant masses is taken as the value
 411 for the B candidate. If the system is from ISR/FSR photon conversion, the
 412 invariant mass is expected to be 0. This cut removes an additional $72 \pm 4\%$
 413 of the $\tau^+\tau^-$ background while keeping $96.40 \pm 0.21\%$ of the signal events (see
 414 Figure 8(b)). If the charge of the two particles is not opposite, the event is not
 415 impacted by the cut.

416 These three cuts are applied only on the e channel and remove $92.1 \pm 1.2\%$ of the
 417 $\tau^+\tau^-$ background. With the addition of the pre-selection cuts, $99.20 \pm 0.01\%$ of the
 418 $\tau^+\tau^-$ background is removed.



(a)

FIG. 7. Distributions of the K_{tag} electronID variable for only the e events after applying the $p_{B_{\text{sig}}}$ cut. $\tau^+\tau^-$ and $\ell\ell XX$ events are peaking at 1, while the signal is peaking at 0.



(a)

(b)

FIG. 8. Distributions of the converted γ $\text{InvM}(\ell_{\text{tag}} - e_{\text{sig}})$ (a) and converted γ $\text{InvM}(D_{\text{tag, daughters}} - e_{\text{sig}})$ (b) variables for only the e events after applying the K_{tag} electronID cut. $\tau^+\tau^-$ and $\ell\ell XX$ events are peaking at 0, while the signal is flat.

419 The pre-selection cuts values are shown in Table 5, the signal efficiency and back-
 420 ground retention are shown in Table 6. The cumulative version of this table, with the
 421 number of signal and background candidate, is shown in Table 7. The cumulative
 422 tables for each individual channel are shown in Appendix A.b.

Channel	Cut value
All	B_{tag} decay mode ID < 4
All	$p_{\tau,\text{sig}}^* > 0.4 \text{ GeV}/c$
All	$p_{D_{\text{tag}}}^* < 2.5 \text{ GeV}/c$
All	$-3 < \cos \theta_{B,D_l}^* < 1.5$
All	$R_2(\text{Event based}) < 0.45$
All	$B_{\text{tag}} R_2 < 0.45$
All	$\cos \theta_{\text{thrust},B_{\text{tag}},z} < 0.85$
For π channel: $p_{B_{\text{sig}}} > 1. \text{ GeV}/c$	
For ρ channel: $p_{B_{\text{sig}}} > 1. \text{ GeV}/c$	
For e channel: $K_{\text{tag}} \text{electronID} < 0.2$	
For e channel: $\text{conv. } \gamma \text{ InvM}(\ell_{\text{tag}} - e_{\text{sig}}) > 0.02 \text{ GeV}/c^2$	
For e channel: $\text{conv. } \gamma \text{ InvM}(D_{\text{tag, daughters}} - e_{\text{sig}}) > 0.02 \text{ GeV}/c^2$	

TABLE 5. Pre-selection cuts values.

Cut	Sgn. eff. [%]	Bkg. ret. [%]	FOM
Reconstruction (w/ FEI)	0.6138 ± 0.0011	-	-
B_{tag} decay mode ID	94.61 ± 0.04	87.092 ± 0.027	0.85
$p_{\tau, \text{sig}}^*$	94.65 ± 0.04	85.547 ± 0.030	0.87
$p_{D_{\text{tag}}}^*$	99.870 ± 0.007	81.09 ± 0.04	0.96
$\cos \theta_{B, D_l}^*$	97.237 ± 0.031	69.95 ± 0.05	1.12
R_2 (Event based)	84.02 ± 0.07	68.93 ± 0.06	1.13
$B_{\text{tag}} R_2$	99.358 ± 0.017	98.202 ± 0.020	1.14
$\cos \theta_{\text{thrust}, B_{\text{tag}}, z}$	92.10 ± 0.06	83.22 ± 0.06	1.15
For π channel: $p_{B_{\text{sig}}}$	$*92.31 \pm 0.13$	$*40.72 \pm 0.15$	$*0.69$
For ρ channel: $p_{B_{\text{sig}}}$	$*90.07 \pm 0.19$	$*52.53 \pm 0.17$	$*0.34$
For e channel: K_{tag} electronID	$*96.04 \pm 0.08$	$*95.03 \pm 0.07$	$*0.73$
For e channel: conv. γ InvM($\ell_{\text{tag}} - e_{\text{sig}}$)	$*98.66 \pm 0.05$	$*98.70 \pm 0.04$	$*0.73$
For e channel: conv. γ InvM($D_{\text{tag, daughters}} - e_{\text{sig}}$)	$*96.71 \pm 0.07$	$*96.78 \pm 0.06$	$*0.71$
Total pre-selection	0.3879 ± 0.0009	16.648 ± 0.030	1.30

TABLE 6. Pre-selection cuts efficiency, calculated on the 50M signal only sample, and background retention, obtained on the 1444 fb⁻¹ MC sample. The Figure Of Merit (FOM) is calculated as the ratio of the signal efficiency and the square root of the background retention. The values with a star only apply to the specific τ channel.

Cut	#Sgn.	Sgn. eff. cumul [%]	#Bkg	Bkg. ret. cumul [%]
Reconstruction (w/ FEI)	263	0.6138 ± 0.0011	389385	-
B_{tag} decay mode ID	249	0.581 ± 0.009	339121	87.09 ± 0.05
$p_{\tau,\text{sig}}^*$	235	0.550 ± 0.012	290106	74.50 ± 0.07
$p_{D_{\text{tag}}}^*$	235	0.549 ± 0.012	235248	60.42 ± 0.08
$\cos \theta_{B,D_l}^*$	228	0.534 ± 0.013	164545	42.26 ± 0.08
R_2 (Event based)	192	0.448 ± 0.017	113415	29.13 ± 0.07
$B_{\text{tag}} R_2$	191	0.446 ± 0.017	111377	28.60 ± 0.07
$\cos \theta_{\text{thrust}, B_{\text{tag}}, z}$	176	0.410 ± 0.018	92686	23.80 ± 0.07
π channel: $p_{B_{\text{sig}}}$	173	0.403 ± 0.018	77179	19.82 ± 0.06
ρ channel: $p_{B_{\text{sig}}}$	171	0.399 ± 0.018	66857	17.17 ± 0.06
e channel: K_{tag} electronID	168	0.394 ± 0.018	65762	16.89 ± 0.06
e channel: conv. γ InvM($\ell_{\text{tag}} - e_{\text{sig}}$)	168	0.392 ± 0.018	65490	16.82 ± 0.06
e channel: conv. γ InvM($D_{\text{tag, daughters}} - e_{\text{sig}}$)	166	0.388 ± 0.018	64824	16.65 ± 0.06

TABLE 7. Cumulative signal efficiency and background retention after each pre-selection cut. The number of truth-matched signal candidates and the number of background candidates is obtained on the 1444 fb^{-1} MC sample and has been scaled expected LS1 data luminosity.

4. SELECTION OPTIMIZATION

To optimize the signal-to-background ratio, the following steps are taken:

- Background suppression: Training of a BDT to suppress the background (Section 4.1).
- Signal optimization: Optimize the cut value on the BDT outputs to maximize the signal-to-background ratio (see Section 4.2).

4.1. Background suppression

For the μ , e and hadronic channels of the B_{sig} , a BDT is trained on variables that have the most discriminating power between signal and background. All background process are considered as background, including $B\bar{B}$ (B^+B^- , $B^0\bar{B}^0$), continuum ($c\bar{c}$, $s\bar{s}$, $d\bar{d}$, $u\bar{u}$) and $\tau^+\tau^-$. For the training and testing, we use 66% and 34% of the signal only sample and background from the generic MC sample respectively. The resulting signal efficiency is cross-checked later using the signal contained in the generic MC sample.

The BDT is trained using **XGBoost** package [15] on GPUs, and the hyperparameters are optimized using the **optuna** package [16]. Since the signal sample is higher than the background sample, a scaling factor is applied to the background sample to balance the signal and background samples. The scaling factor is calculated as the ratio of the number of background candidates to the number of signal candidates.

Reducing the false positive rate, not having significant overtraining and not having correlations between the output of the BDT and the two fitting variables are the main goals of the optimization. Each training uses two objectives to optimize the hyperparameters. The first one is defined as:

$$\text{objective}_1 = \text{precision}_{\text{test}} \times \left(\frac{\text{precision}_{\text{test}}}{\text{precision}_{\text{train}}} \right)^2 \times \left(\frac{\text{logloss}_{\text{train}}}{\text{logloss}_{\text{test}}} \right)^2 \quad (4)$$

where $\text{precision}_{\text{test}}$ and $\text{precision}_{\text{train}}$ are the precision of the test and train samples respectively, and $\text{logloss}_{\text{test}}$ and $\text{logloss}_{\text{train}}$ are the logloss of the test and train samples respectively. The precision is defined as:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (5)$$

where TP and FP are the true positive and false positive rates respectively. A cut on the BDT output at 0.5 is applied to calculate the precision. The goal of

451 this objective is to maximize the precision of the test sample while minimizing the
 452 overtraining. The second objective is defined as:

$$\text{objective}_2 = \sqrt{\text{r_excess}_A^2 + \text{r_excess}_B^2} \quad (6)$$

453 where r_excess_A and r_excess_B are defined as:

$$\text{r_excess}_A = \max(0, |r(\text{XGBoost output}, E_{\text{extra}}^{\text{ROE}})| - 0.1) \quad (7)$$

$$\text{r_excess}_B = \max(0, |r(\text{XGBoost output}, \tau_{p_{\text{CMS}}})| - 0.1) \quad (8)$$

$$(9)$$

454 where r is the Pearson correlation coefficient between the output of the BDT
 455 and the two fitting variables, defined in Equation (31). The goal of this objective
 456 is to minimize the correlations between the output of the BDT and the two fitting
 457 variables. The correlation is considered as acceptable if its absolute value is below
 458 0.1. The best set of objectives is the one that maximizes objective_1 while minimizing
 459 objective_2 . The optimal set is taken following the Pareto front. The trial with the
 460 lowest objective_2 is prioritized, then the one with the highest objective_1 is selected.

461 We use 15 and 18 variables for the leptonic and hadronic channels respectively.
 462 Here is a brief description of them:

- 463 • aplanarity: event shape variable describing the flatness of the event, defined as
 464 the $3/2$ of the third sphericity eigenvalue.
- 465 • $\cos\text{TBz}$: cosine of the angle between the thrust axis of the B and the z axis
- 466 • $\cos\text{TBTO}$: cosine of the angle between the thrust axis of the B and the thrust
 467 axis of the rest of the event.
- 468 • deltaE : difference between the energy of the B_{tag} and half of the center-of-mass
 469 energy.
- 470 • τ_{sig} decays: decay mode of the B_{sig} (in this case, π or ρ channel).
- 471 • miss_E : missing energy in the event, calculated as the difference between the
 472 center-of-mass energy and the sum of the energies of all reconstructed particles
 473 in the event.
- 474 • CleoConeCS.i : CLEO Cones [17] calculated using the final state particles as-
 475 sociated to the B_{tag} .

- 476 • KSFVVariables_hso01: modified Fox-Wolfram moment [18, 19].
- 477 • $\cos \theta_{B_{tag}, D_l}^*$: the cosine of the angle in CMS between the Dl system and the
- 478 B_{tag} .
- 479 • thrustBm: magnitude of the B thrust axis.
- 480 • Δz : distance along the z axis between the decay vertices of the B_{tag} and B_{sig} .
- 481 • R_2 : ratio of the second to zeroth Fox-Wolfram moment calculated using all
- 482 particles in the event.
- 483 • harmonicMomentThrust_i: harmonic moments of the i -th order calculated us-
- 484 ing the thrust axis of the event.
- 485 • miss_theta: polar angle of the missing momentum in the event.
- 486 • nROETracksGood: number of remaining tracks in the rest of the event that
- 487 pass the mask $d_r < 2$ cm, $||d_z|| < 4$ cm, θ in CDC acceptance and $n_{\text{CDC hits}} >$
- 488 20.
- 489 • nRemainingTracks: number of all the remaining tracks in the event minus the
- 490 tracks used in the B_{tag} and B_{sig} candidates.
- 491 • thrustAxisCosTheta: cosine of the polar angle of the thrust axis.

492 The list of variables used in the training is shown in Table 8. Table 9 shows the
 493 hyperparameters used in the optimization. The Data/MC agreement of each variable
 494 is shown Appendix C.

Feature	μ	e	hadronic
B_{tag} cosTBz	✓	✓	
B_{sig} cosTBTO	✓	✓	
B_{tag} cosTBTO			✓
B_{tag} deltaE			✓
τ_{sig} decays			✓
miss_E			✓
harmonicMomentThrust3			✓
B_{tag} CleoConeCS_1	✓	✓	✓
B_{tag} CleoConeCS_2	✓	✓	✓
B_{tag} CleoConeCS_3	✓	✓	✓
B_{tag} KSFVVvariables_hso01	✓	✓	✓
$\cos \theta_{B_{tag}, D_l}^*$	✓	✓	✓
B_{tag} thrustBm	✓	✓	✓
Δz	✓	✓	✓
R_2 (Fox-Wolfram event based)	✓	✓	✓
harmonicMomentThrust1	✓	✓	✓
miss_theta	✓	✓	✓
nROETracksGood	✓	✓	✓
nRemainingTracks	✓	✓	✓
thrustAxisCosTheta	✓	✓	✓

TABLE 8. Variables used in the training of the BDT for each BDTs.

Hyperparameter	Type	Range	Sampling
<code>n_estimators</code>	Integer	[500, 2000]	Uniform
<code>max_depth</code>	Integer	[1, 10]	Uniform
<code>min_child_weight</code>	Integer	[1, 10]	Uniform
<code>learning_rate</code>	Float	[0.01, 0.3]	Logarithmic
<code>subsample</code>	Float	[0.1, 1.0]	Step size 0.1
<code>colsample_bytree</code>	Float	[0.1, 1.0]	Step size 0.1
<code>gamma</code>	Float	[0.01, 0.5]	Logarithmic
<code>lambda</code>	Float	[1e-3, 1]	Logarithmic
<code>alpha</code>	Float	[1e-3, 1]	Logarithmic

TABLE 9. Hyperparameters optimized by Optuna for XGBoost.

495 The hyperparameters are optimized over 1000 trials. After the training, good
496 performance with no significant overtraining is observed. The BDT outputs and
497 the comparison between training and testing samples are shown in Figure 9. The
498 variable importance is shown in Figures 10 and 11.

499 As for the performance, the ROC curve is shown in Figure 12 and the logloss is
500 shown in Figure 13.

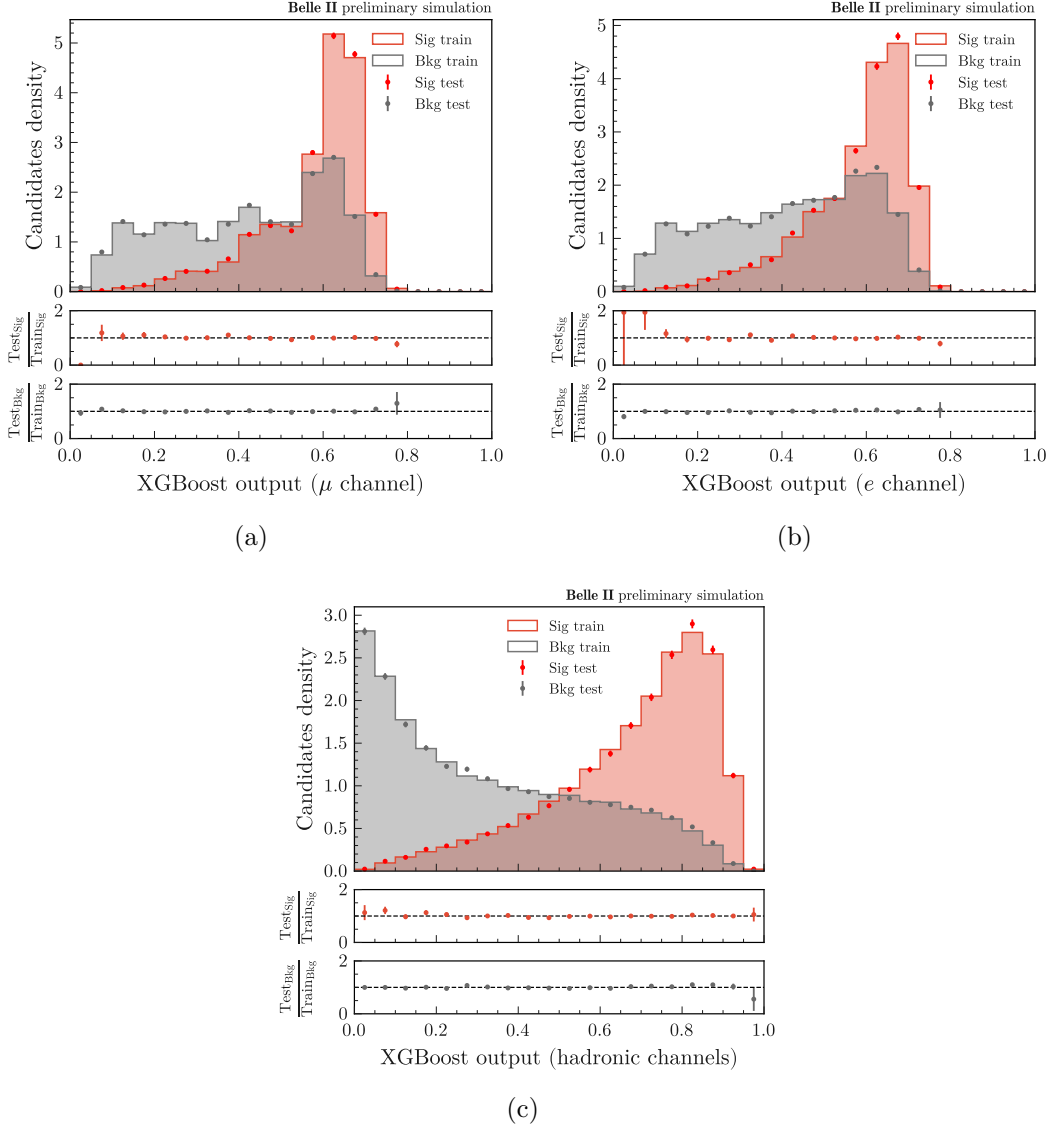
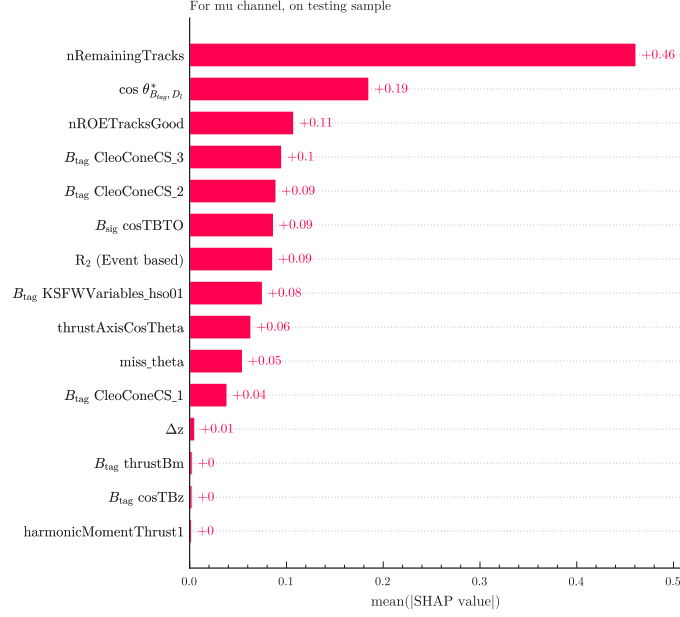
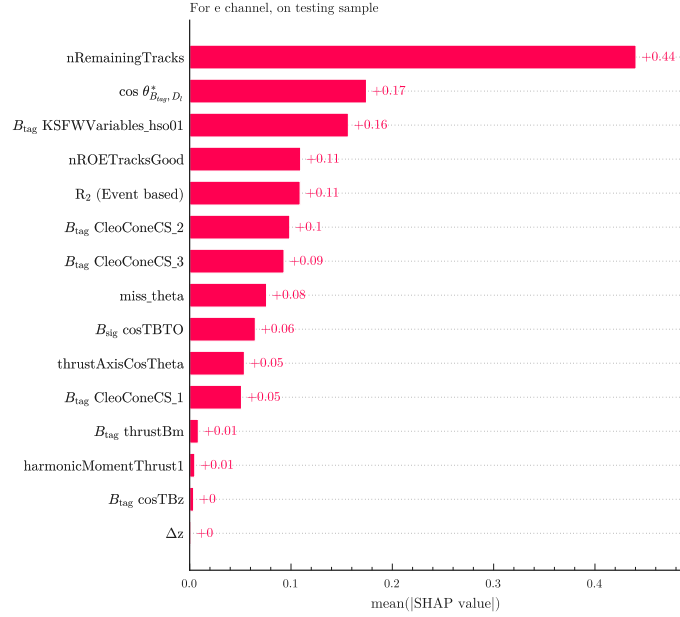


FIG. 9. BDT outputs for each channels. The Signal and Background are the signal only and background from the generic MC sample respectively. The pull distribution between the training and testing samples is shown in the bottom panel. No overtraining is observed.



(a)



(b)

FIG. 10. The shapley value is used to evaluate the importance of each variable (see here). The higher the absolute value of the shapley value, the more important the variable is to separate signal from background.

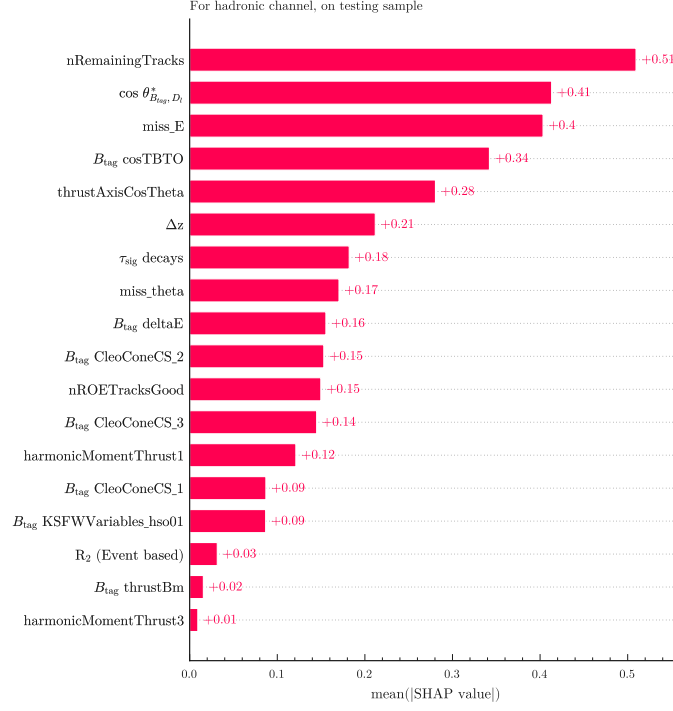
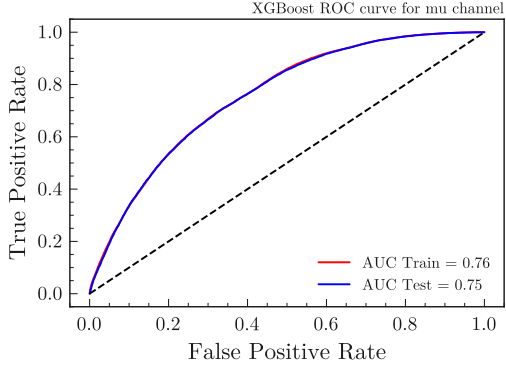
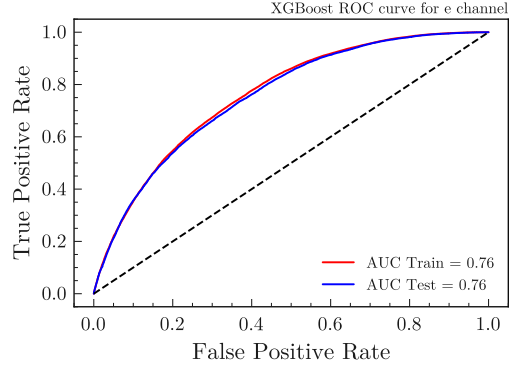


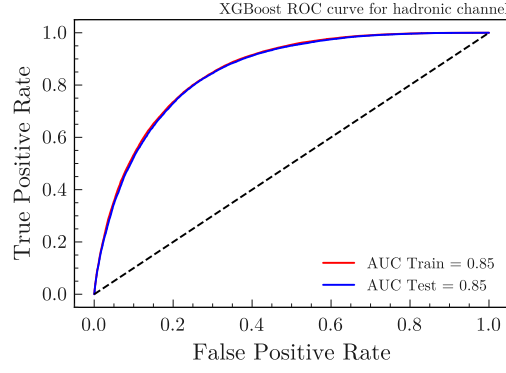
FIG. 11. The shapley value is used to evaluate the importance of each variable (see here). The higher the absolute value of the shapley value, the more important the variable is to separate signal from background.



(a)

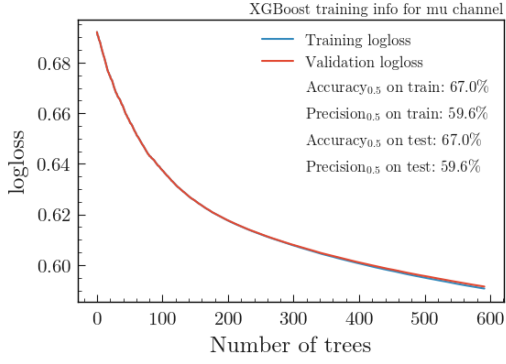


(b)

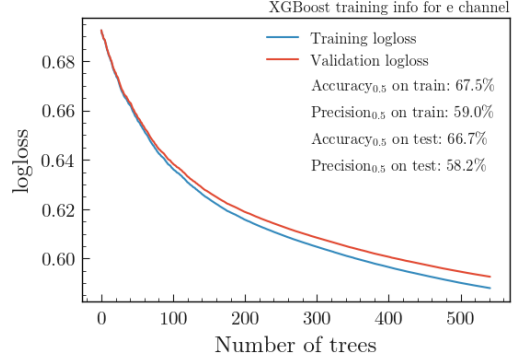


(c)

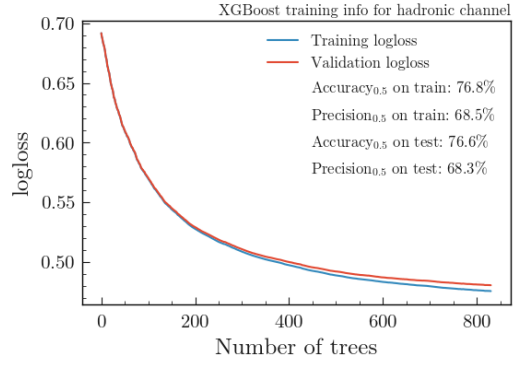
FIG. 12. ROC curve for each channels. No overtraining is observed.



(a)



(b)



(c)

FIG. 13. Logloss for each channels. No overtraining is observed.

501 **4.2. Signal optimization**

502 A cut on the output of each BDT is applied to reduce the remaining background
 503 while keeping a high signal efficiency. Only the candidates in the SR are considered
 504 for the optimization process, to keep the shape information of $E_{\text{extra}}^{\text{ROE}}$. The following
 505 steps are taken to optimize the cut value for each channels:

- 506 1. The signal efficiency as a function of the BDT output is calculated for the
 507 signal only sample. The plots are shown in Figure 14.
- 508 2. The FOM is calculated as a function of the BDT output. The FOM is defined
 509 as:

$$\text{FOM} = \frac{S^*}{\sqrt{S^* + B}} \quad (10)$$

510 where B is the number of background candidates, and S^* corresponds to the
 511 number of signal event from the signal only sample normalized to the signal
 512 expected for the luminosity of the background sample (1444 fb^{-1}). The plots
 513 are shown in Figure 15.

- 514 3. The BDT output is flattened to have a uniform distribution of the signal effi-
 515 ciency using the cumulative distribution function of the signal efficiency shown
 516 in Figure 14. This way, the area around the maximum of the FOM in Figure 15
 517 is enhanced, and a cut on the flattened BDT output will directly correspond
 518 to a cut on the signal efficiency. The plots are shown in Figure 16.
- 519 4. The FOM is calculated as a function of the signal efficiency. The maximum of
 520 the FOM is found, and the corresponding signal efficiency is used as the cut
 521 value. The plots are shown in Figure 17.

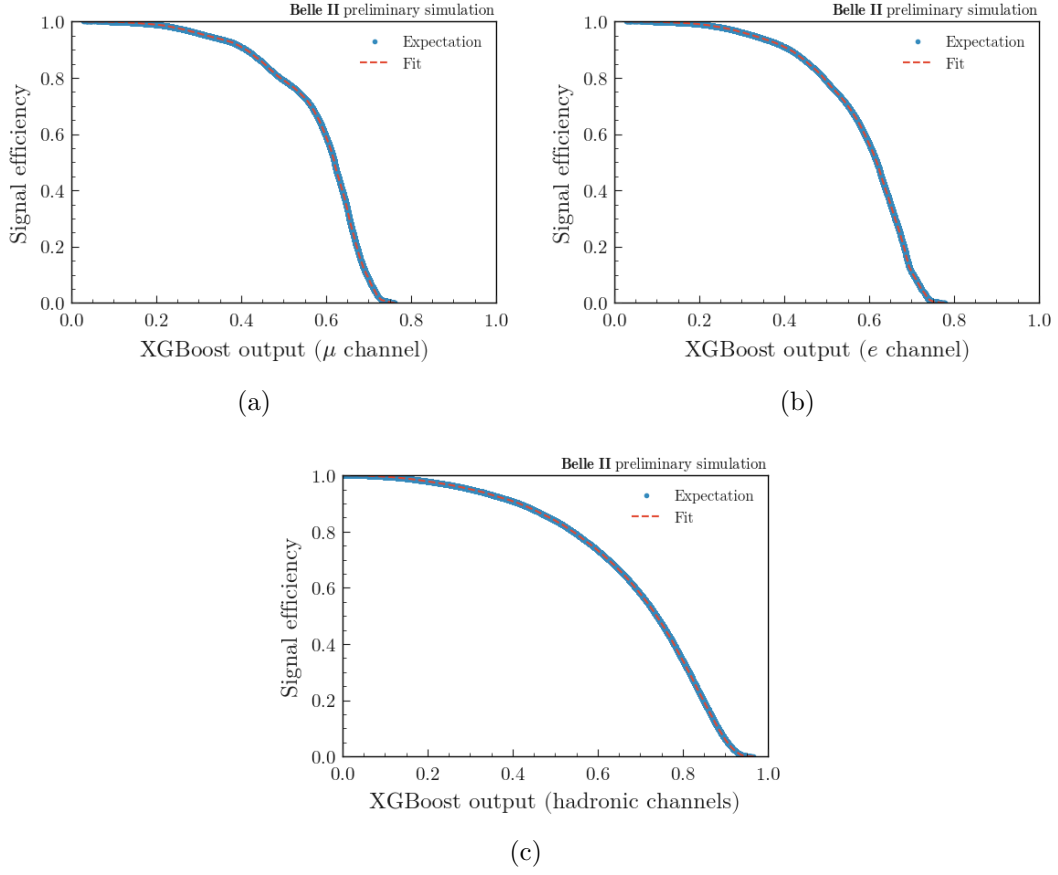


FIG. 14. Signal efficiency as a function of the BDT output for each channels.

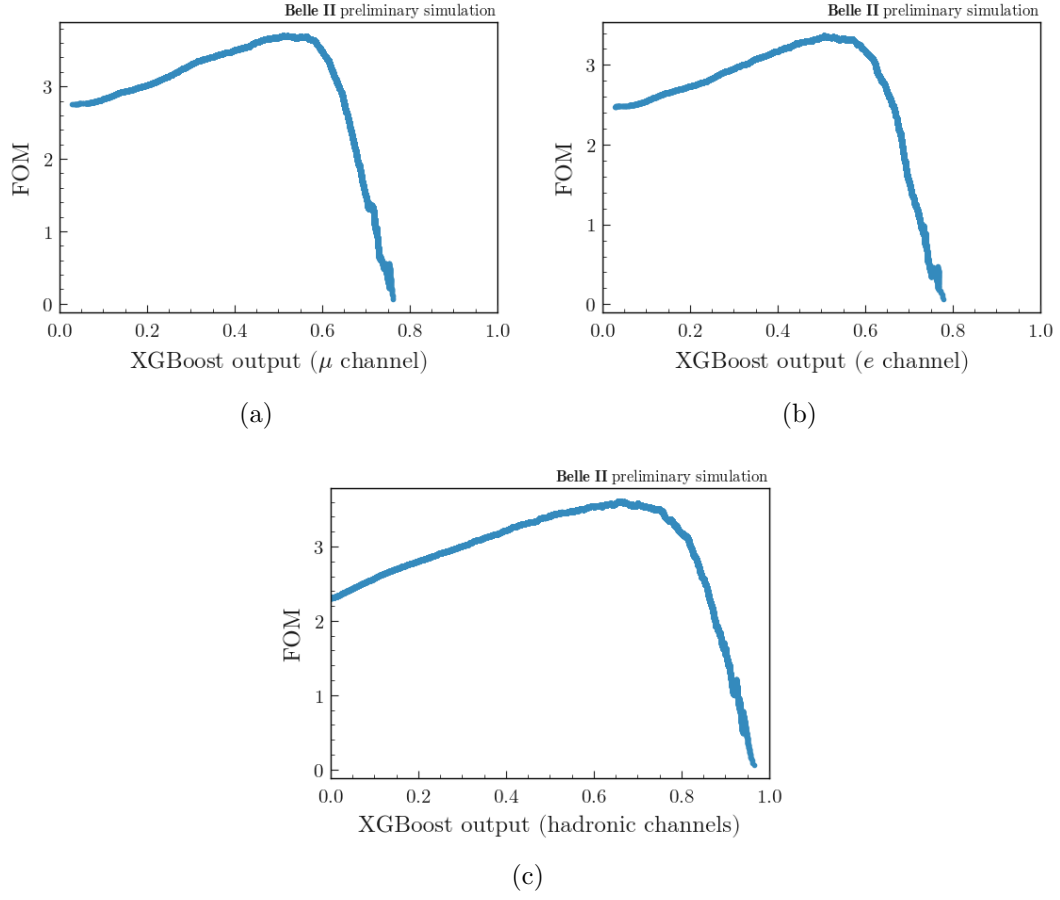


FIG. 15. FOM as a function of the BDT output for each channels. Luminosity is four times the LS1 dataset (1444 fb^{-1}).

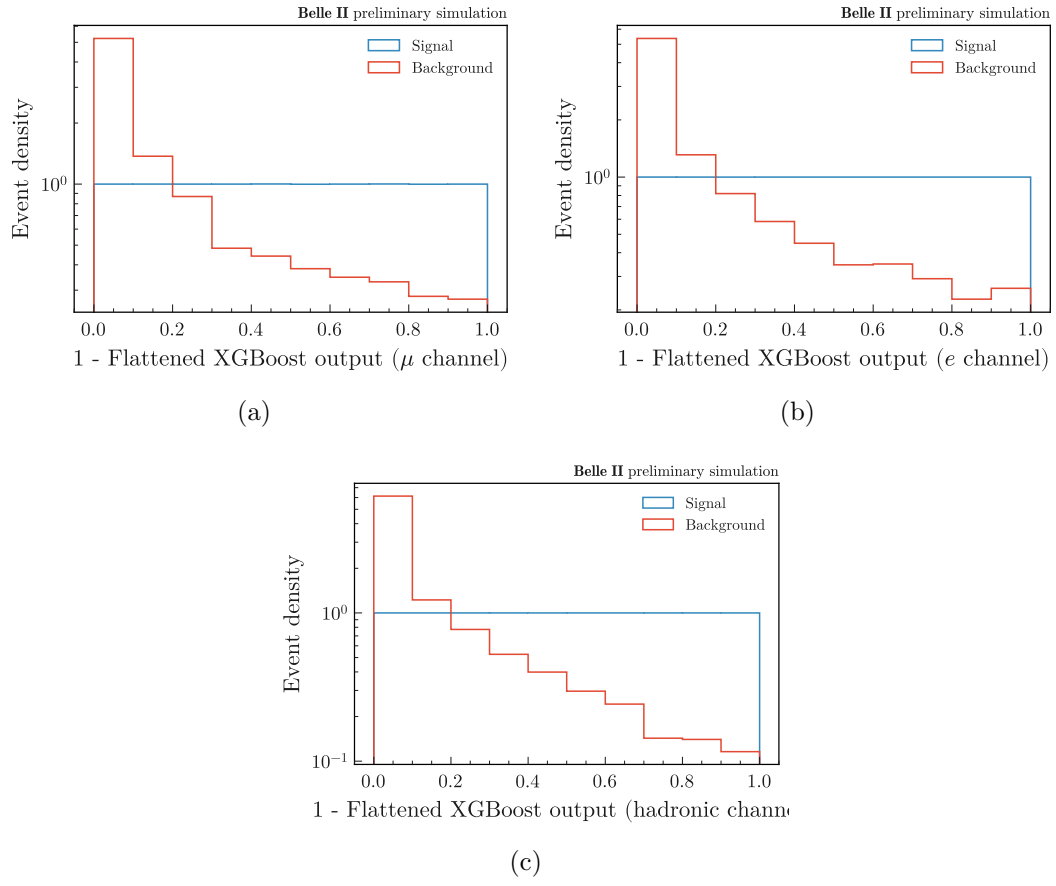


FIG. 16. Flattened BDT output for each channels.

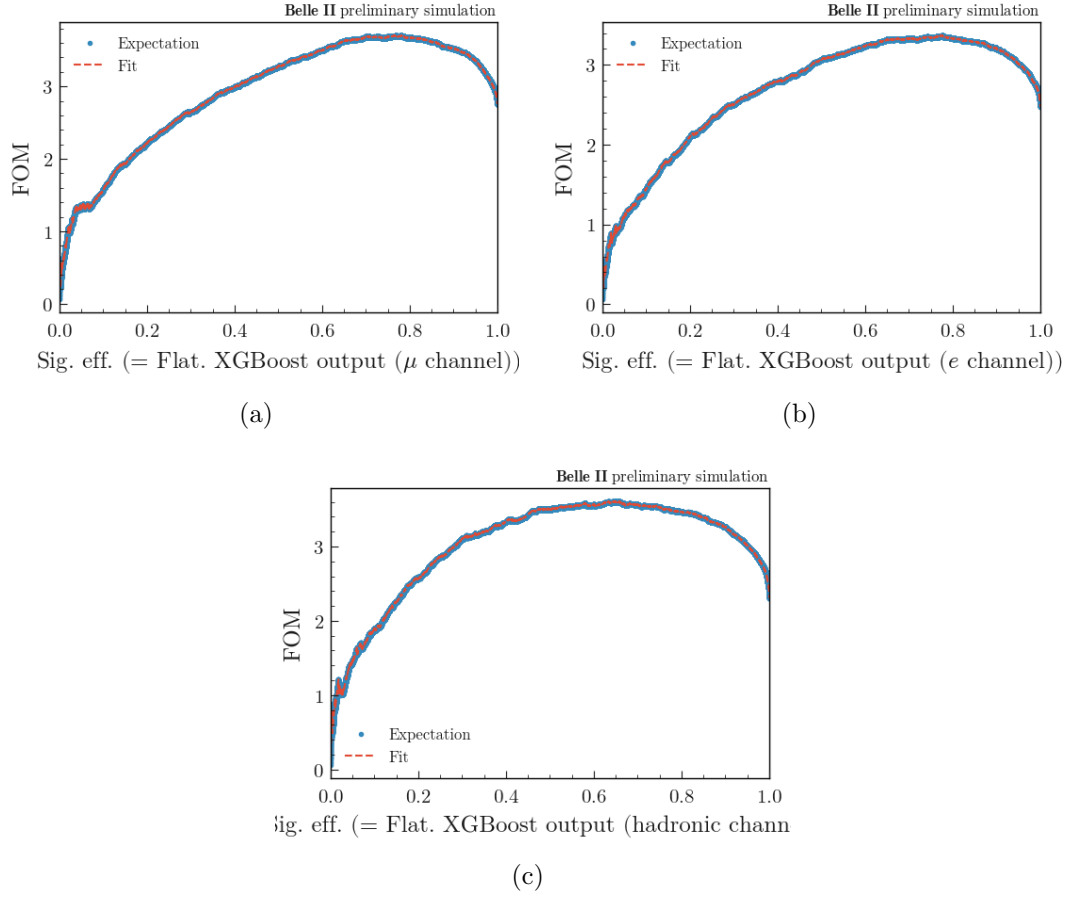


FIG. 17. FOM as a function of the signal efficiency for each channels.

522 The signal efficiency (corresponding to the cut value on the flattened BDT output)
523 and background retention for each channels is shown in Table 10. The background
524 composition for each channels is shown in Table 11, and in the fit region in Table 12.
525 The total background retention of this step is equal to $30.99 \pm 0.09\%$. The total
526 signal efficiency on the signal only sample and the signal found in the generic MC
527 sample is equal to $72.01 \pm 0.10\%$ and $71.0 \pm 1.7\%$ respectively. With the pre-selection,
528 the total background retention, signal efficiency on the signal only sample and the
529 signal found in the generic MC sample is equal to $5.159 \pm 0.018\%$, $0.2793 \pm 0.0008\%$
530 and $0.290 \pm 0.009\%$ respectively.

τ channel	Sgn Eff [%]	Bkg Ret [%]
μ	76.87 ± 0.31	40.43 ± 0.33
e	76.52 ± 0.35	39.33 ± 0.35
hadronic	62.3 ± 0.4	13.70 ± 0.23

TABLE 10. Relative signal efficiency (corresponding to the cut value on the flattened BDT output) and background retention for each channels after applying the BDT cuts.

	μ	e	π	ρ	Total
Signal	43	46	20	5	115
$B^0 \bar{B}^0$	514	490	80	46	1132
$B^+ B^-$	1828	1699	302	141	3973
$u \bar{u}$	48	17	41	12	120
$d \bar{d}$	13	3	11	4	31
$s \bar{s}$	16	2	10	2	32
$c \bar{c}$	203	128	92	25	451
$\tau \bar{\tau}$	13	4	11	1	29
$\ell \ell X X$	12	12	2	0	27
Total	2690	2401	569	236	5910
FOM	0.83	0.94	0.84	0.33	1.50

TABLE 11. Expected number of candidates with LS1 dataset for each channels in the fit region after the optimization.

	μ	e	π	ρ	Total
Signal	30	35	11	4	82
$B^0\bar{B}^0$	69	67	9	3	149
B^+B^-	135	134	40	12	322
$u\bar{u}$	11	7	11	4	35
$d\bar{d}$	5	2	1	0	9
$s\bar{s}$	3	0	2	0	7
$c\bar{c}$	37	29	20	5	93
$\tau\bar{\tau}$	9	2	6	1	20
$\ell\ell XX$	12	6	2	0	20
Total	311	282	102	29	737
FOM	1.70	2.08	1.09	0.74	3.02

TABLE 12. Expected number of candidates with LS1 dataset for each channels in the signal region after the optimization.

531 4.3. Fitting variables before/after optimization

532 The distributions of the fitting variables ($E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$) before and after the
533 optimization for each channels are shown in Figures 18 to 20.

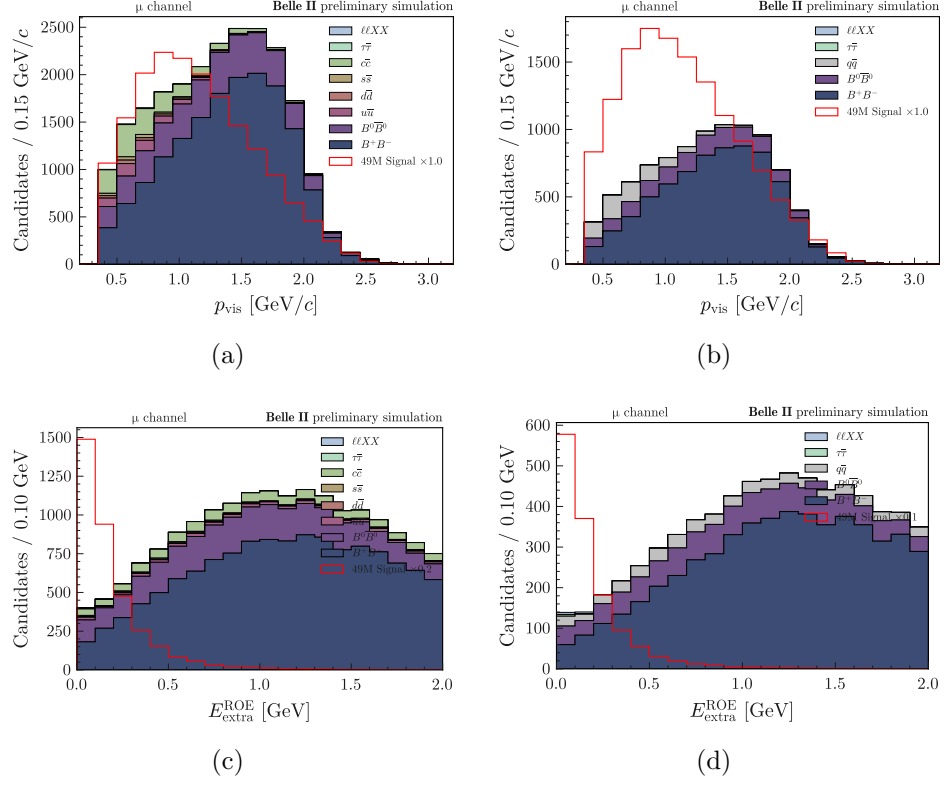


FIG. 18. Distributions of the fitting variables for the μ channel before and after the optimization.

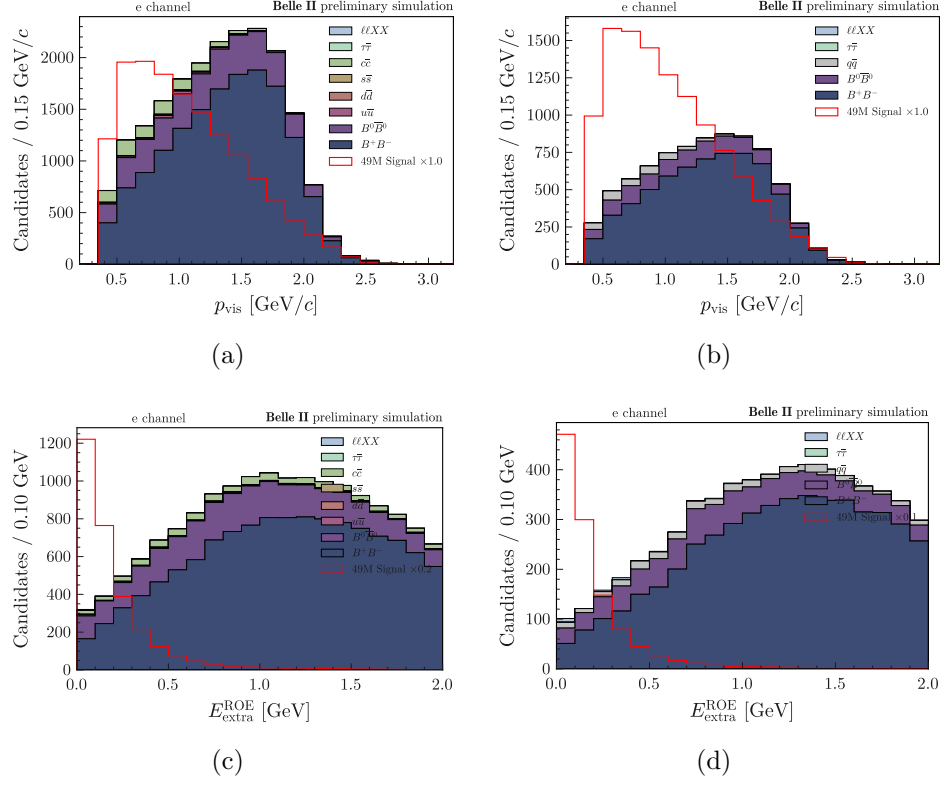


FIG. 19. Distributions of the fitting variables for the e channel before and after the optimization.

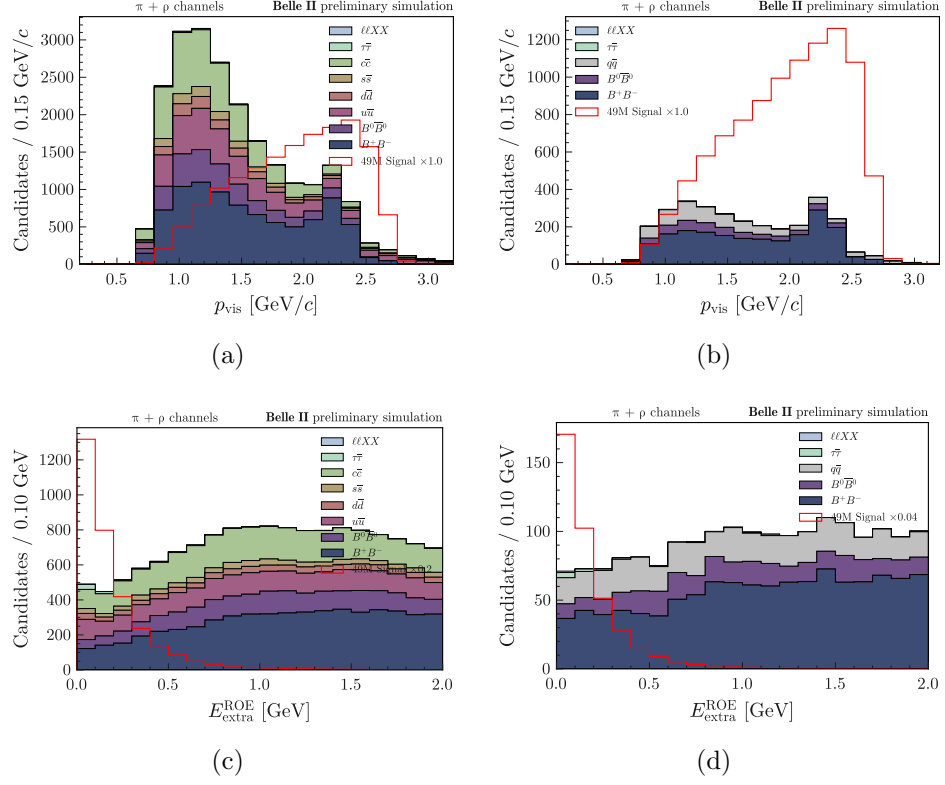


FIG. 20. Distributions of the fitting variables for the hadronic channel before and after the optimization.

534 5. CORRECTIONS

535 We describe in this section a set of corrections applied to important variables used
536 in the analysis to improve the data/MC agreement. Some corrections (like the PIDs,
537 BF, form factors...) are described in details in Section 8.

538 5.1. FEI corrections

539 A discrepancy between the efficiency of the different FEI decay modes is seen
540 between Data and MC after the pre-selection, as seen in Figure 21.

541 Calibrations factors corresponding to the FEI signal probability cut used in the re-
542 construction are available for the B^+B^- and $B^0\bar{B}^0$ samples (see SLFEI_rd_cal_factors_Bp_004).
543 After applying the calibration factors corresponding to the FEI signal probability cut
544 used in the reconstruction to the correctly reconstructed B_{tag} , the data/MC agree-
545 ment is improved as shown in Figure 22, but some discrepancy remains, specially for
546 the hadronic τ decays which have a larger fraction of $q\bar{q}$ background.

547 To correct the continuum sample, we therefore derive additional correction factors
548 based on the FEI decay mode distributions using off-resonance data and MC. The
549 small energy shift between off-resonance data and MC is taken into account. The
550 correction factors are derived as the ratio of the data and MC distributions of the
551 FEI decay modes. The off-resonance data distribution before correction can be seen
552 in Figure 23, and the final data/MC agreement after applying all corrections is shown
553 in Figure 24.

554 In addition, an efficiency discrepancy of the D_{tag}^0 channels is seen after the selection
555 on the $E_{\text{extra}}^{\text{ROE}} > 0.3$ GeV sideband region for the μ channel (see Figure 25). We decided
556 to correct the first four D_{tag}^0 channels efficiencies. An additional systematic has been
557 included in the fit to account for this effect.

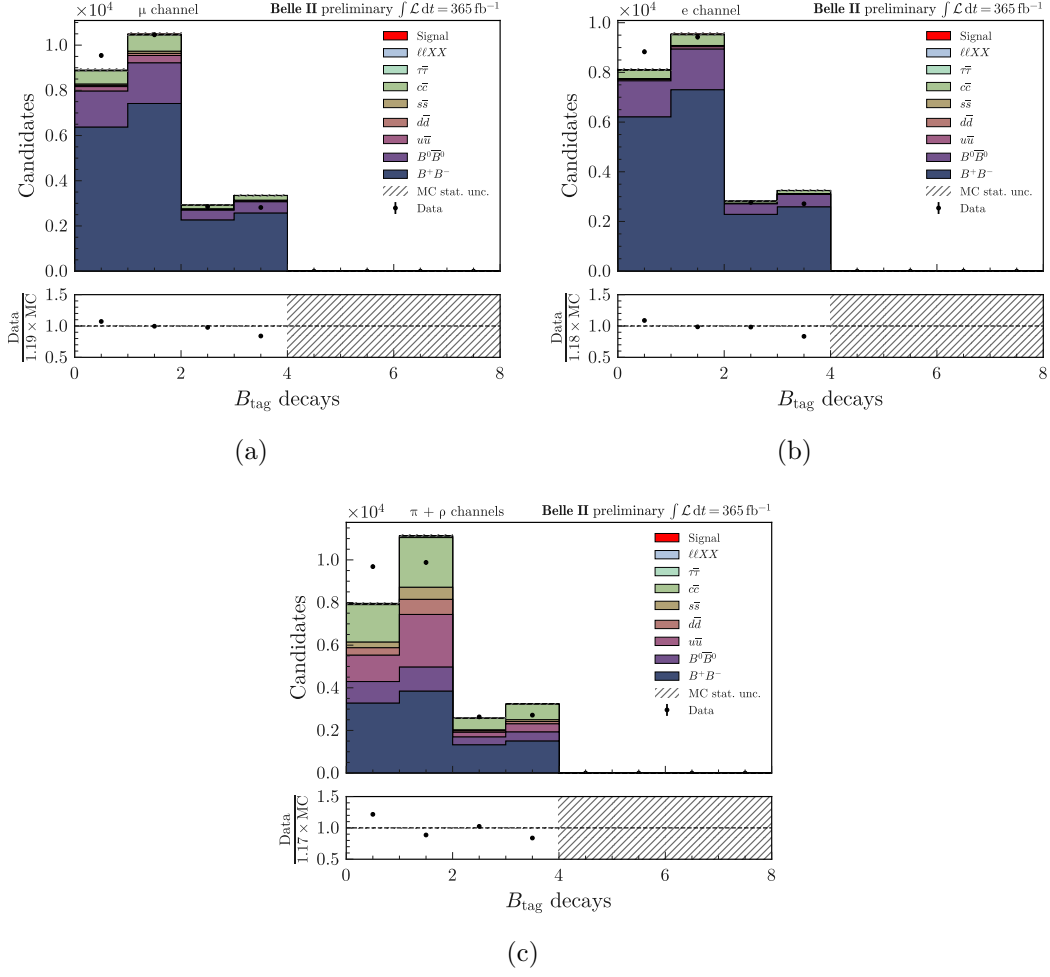


FIG. 21. Comparison between data and MC of the FEI decay mode for the different τ decays before any FEI corrections.

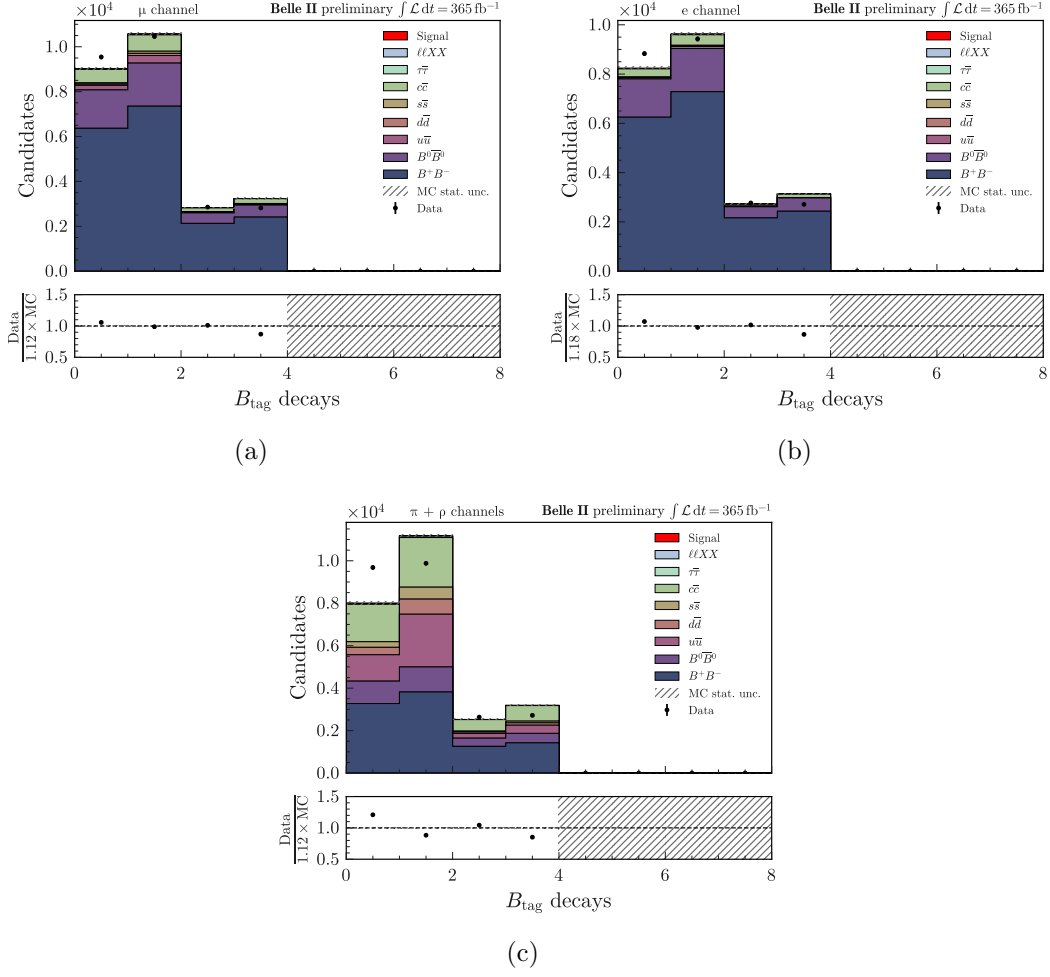


FIG. 22. Comparison between data and MC of the FEI decay mode for the different τ decays after applying the FEI calibration correction.

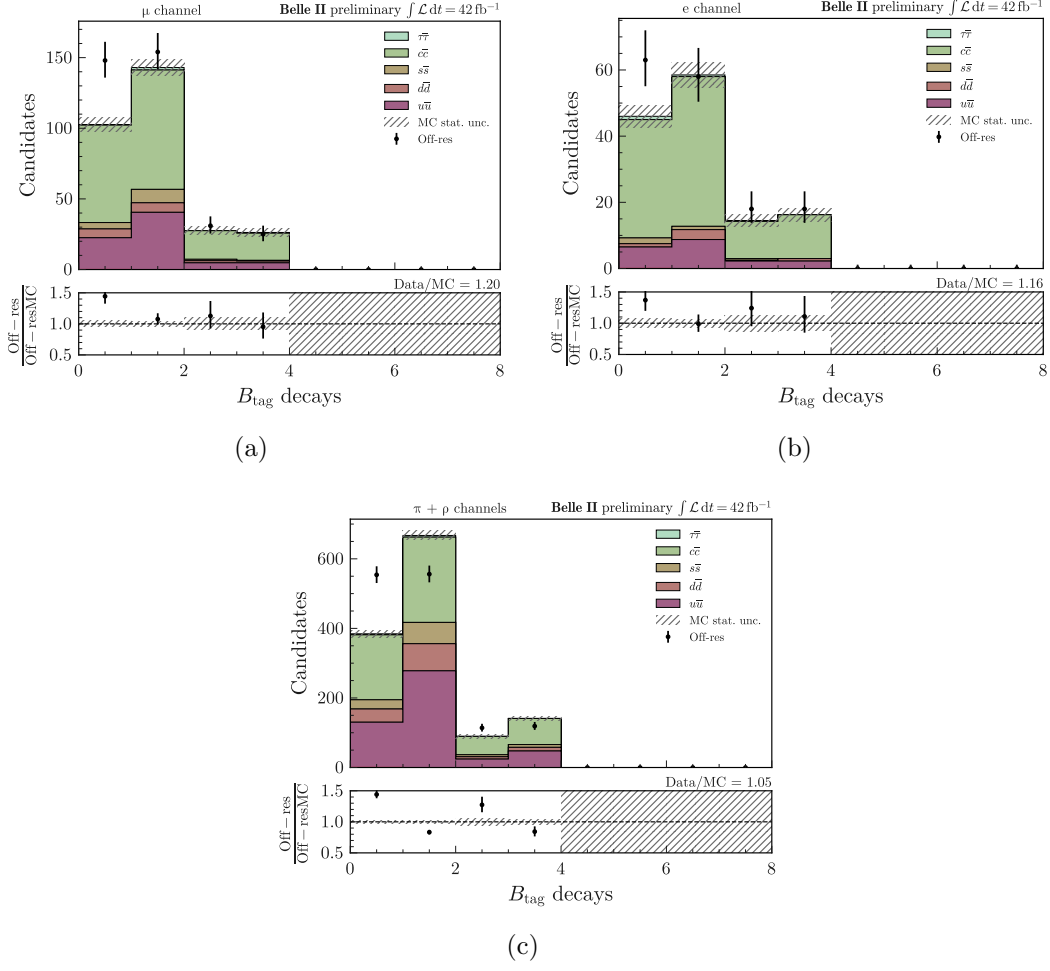


FIG. 23. Comparison between off-resonance data and MC of the FEI decay mode for the different τ decays before any FEI corrections.

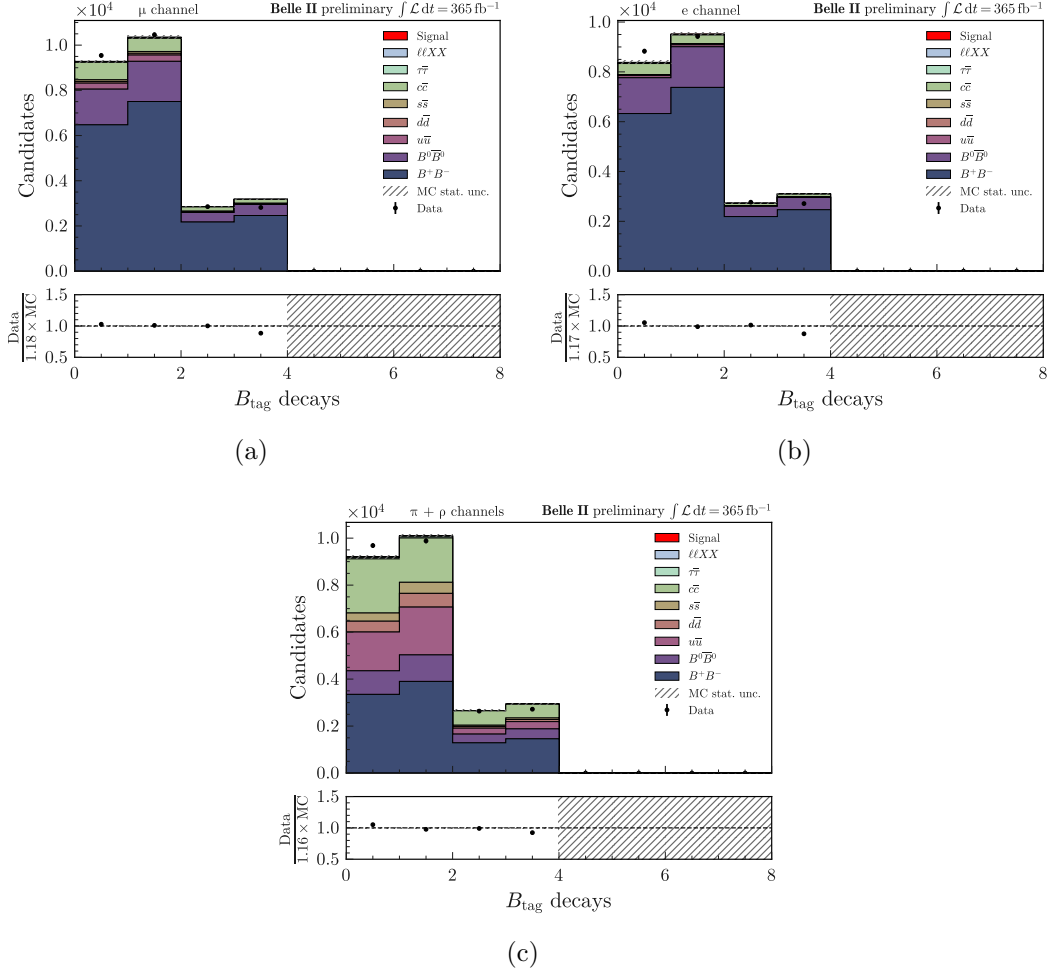


FIG. 24. Comparison between data and MC of the FEI decay mode for the different τ decays after applying all FEI corrections.

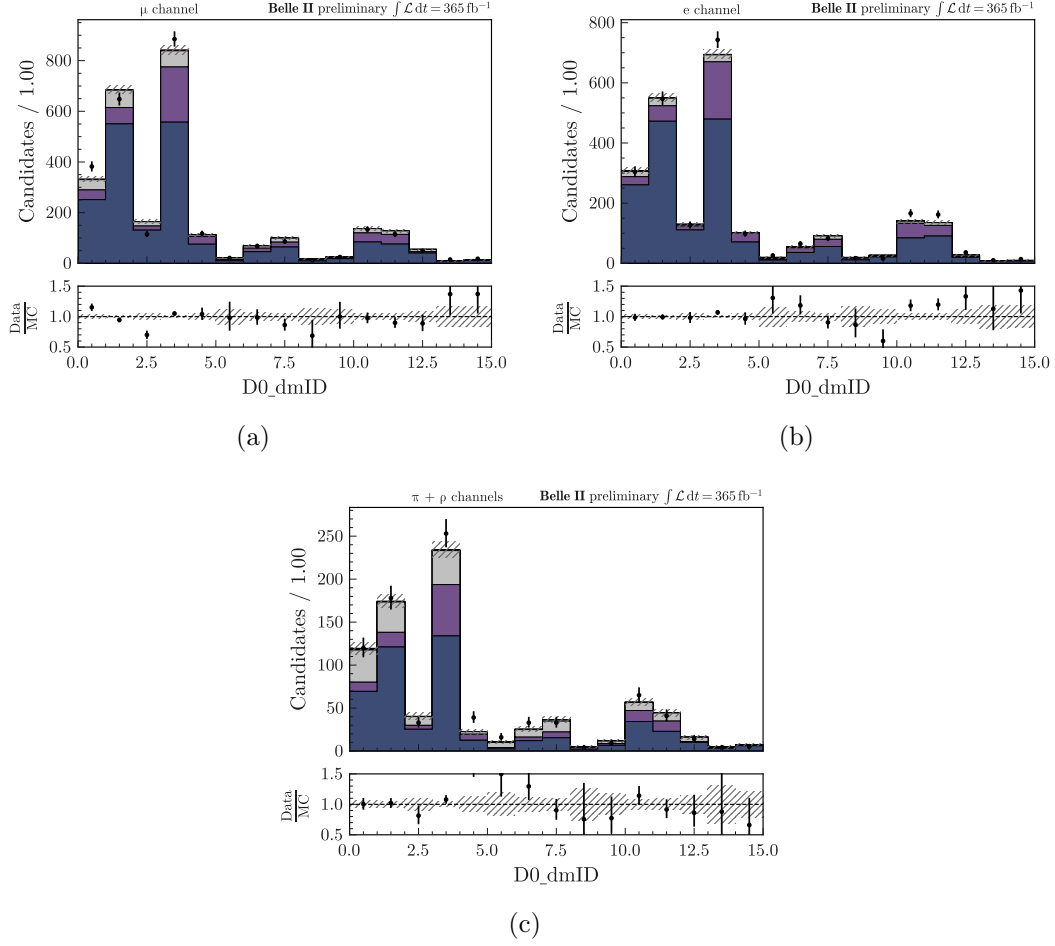


FIG. 25. Comparison between data and MC of the D_{tag}^0 decay mode for the different τ decays after the selection and in the $E_{\text{extra}}^{\text{ROE}} > 0.3 \text{ GeV}$ sideband selection. Each bin corresponds to a mode ID in Table 3.

5.2. Extra energy corrections

559 A shift can be seen between data and MC on the energy of the rest of event for
 560 each τ channel after the pre-selection (see Figure 26).

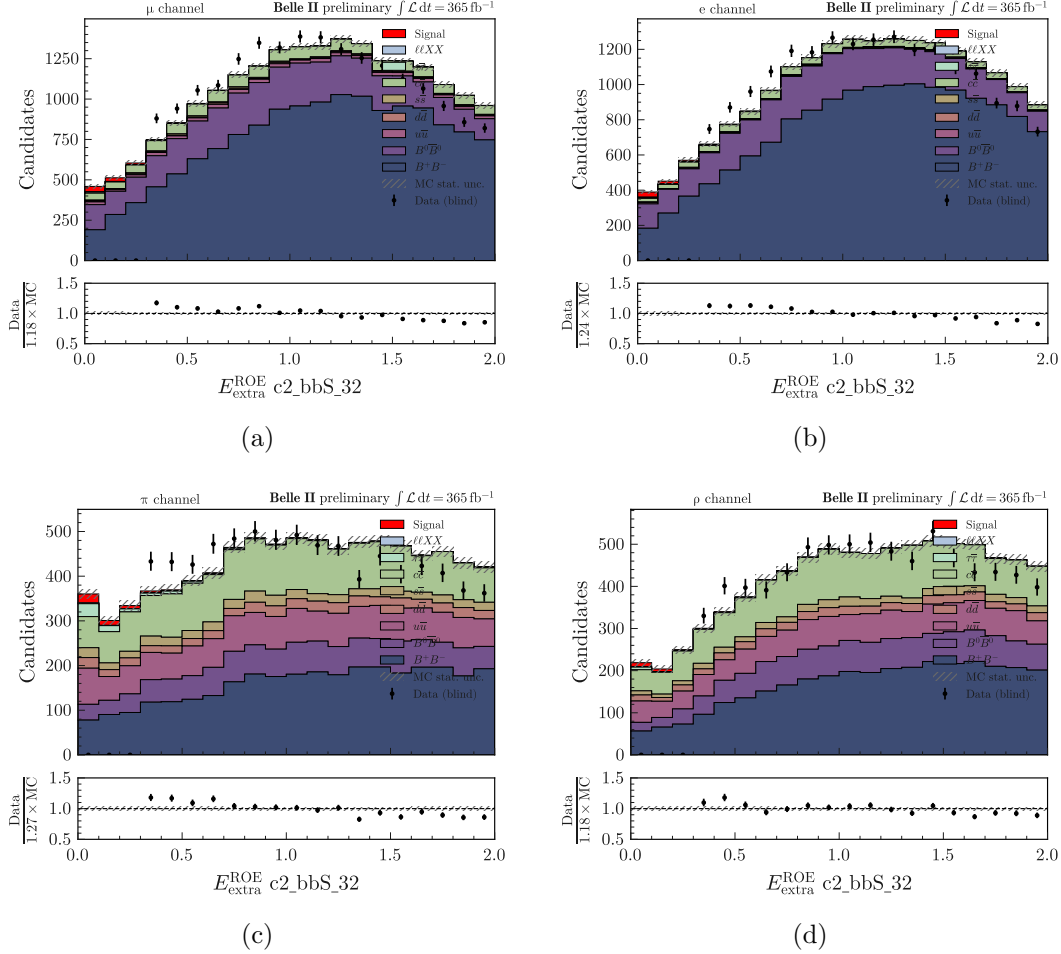


FIG. 26. Comparison between data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay.

In order to correct this discrepancy, three steps are performed:

1. Apply the photon efficiency data/MC ratio correction from neutral group study [ref], which gives a weight for each ROE photons given its energy, ϕ and θ . Then, for each photon, a random number in $[0, 1]$ is uniformly generated and compared to the weight. If the random number is smaller than the weight, the

photon is kept, otherwise it is removed from the ROE. The distribution of the number of photon in the ROE N_{γ}^{ROE} and the $E_{\text{extra}}^{\text{ROE}} = \sum E_{\gamma}$ are updated. The weight table used is PhotonEfficiencyDataMCRatio_Run1MC15rd_April2024_rev_1.

2. The shape further corrected using off and on-resonance data sidebands:

- (a) Continuum correction: using off-resonance MC and data, each bin of N_{γ}^{ROE} is corrected by the ratio data/MC ration. The shape of the $E_{\text{extra}}^{\text{ROE}}$ distribution is directly impacted by this correction.
- (b) Signal correction: using the data/MC ratio of the N_{γ}^{ROE} distributions for the control channels ($B^+ \rightarrow \bar{D}^{*0} \ell^+ \nu_{\ell}$ for leptonic channels and double tagged for hadronic channels), we derive correction factors for each bin of N_{γ}^{ROE} . The shape of the $E_{\text{extra}}^{\text{ROE}}$ distribution is directly impacted by this correction. More information about the control channels can be found in Sections 6.3.4 and 6.3.5.
- (c) $B\bar{B}$ background correction: using the BDT sidebands, removing candidates above 0.51 and 0.62 BDT score for leptonic and hadronic channel respectively. We subtract from data the continuum corrected MC. The resulting distribution corresponds to the $B\bar{B}$ contribution of data. By comparing $B\bar{B}$ distribution of MC and data, each bin of N_{γ}^{ROE} is then corrected by the data/MC ratio. Since the hadronic channels have low statistics and a single control channel, they are also merged together to have better statistics for the correction. More information on the composition of the BDT sidebands can be found in Appendix D.c.

The distributions before and after step 1 are shown in Figure 27. The photon multiplicity N_{γ}^{ROE} and the $E_{\text{extra}}^{\text{ROE}}$ before and after step 2 (a) are shown in Figures 28 and 29. The final distributions with all corrections applied on the BDT sideband and on the full region (blind) are shown in Figures 32 and 33. No data/MC discrepancy is observed in the $E_{\text{extra}}^{\text{ROE}}$ distribution after all the corrections.

The corrections is also checked using the extra track SB control sample, where we don't expect signal. Thus, the signal region is not blinded for this control sample. The distributions can be seen in Appendix D.

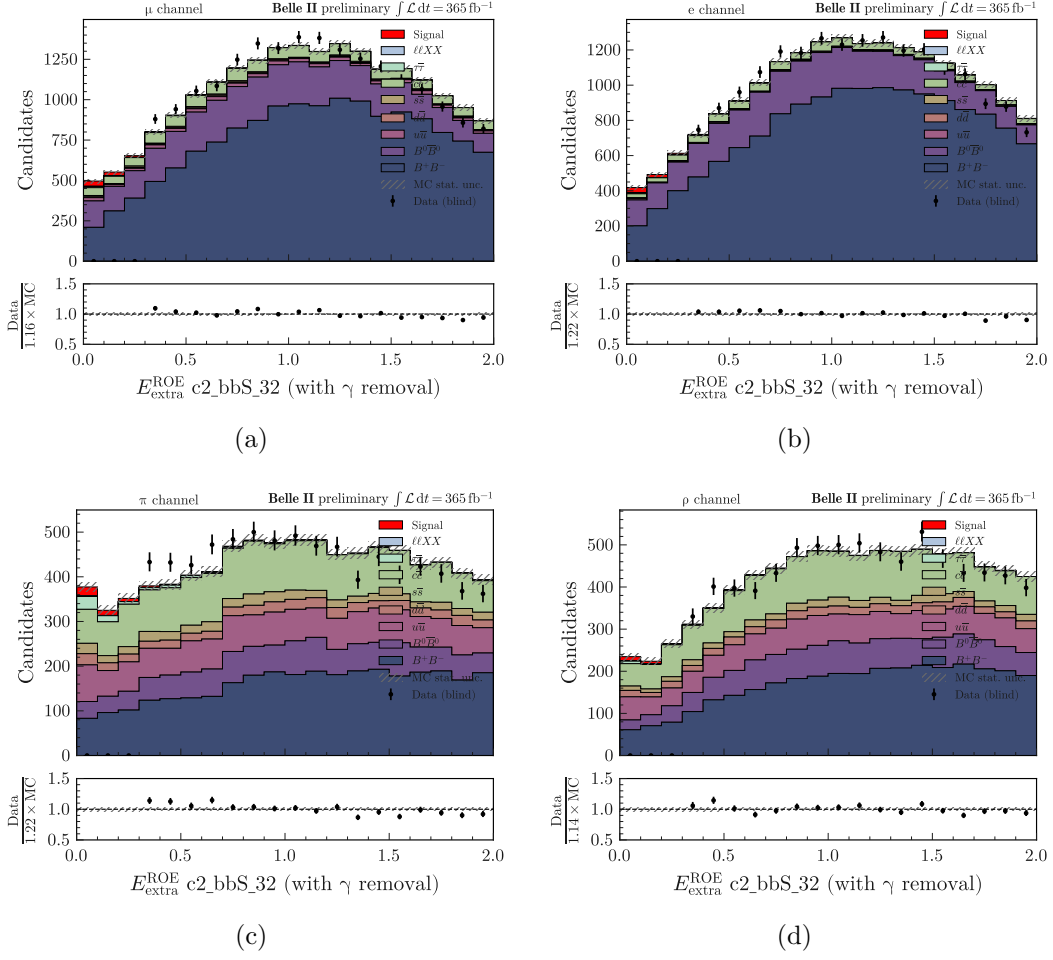


FIG. 27. Comparison between data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay after step 1 of the correction.

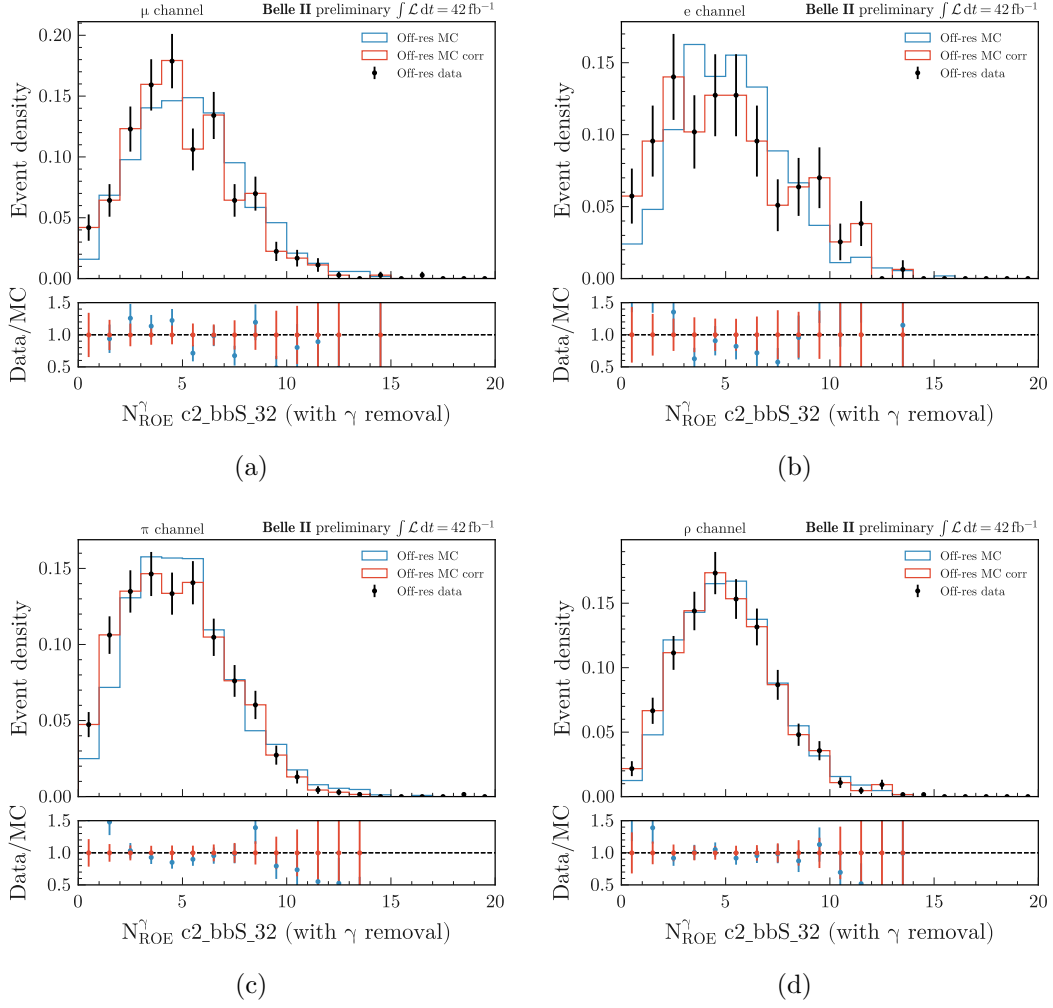


FIG. 28. Comparison between off-resonance data and MC of N_{γ}^{ROE} for each τ decay.

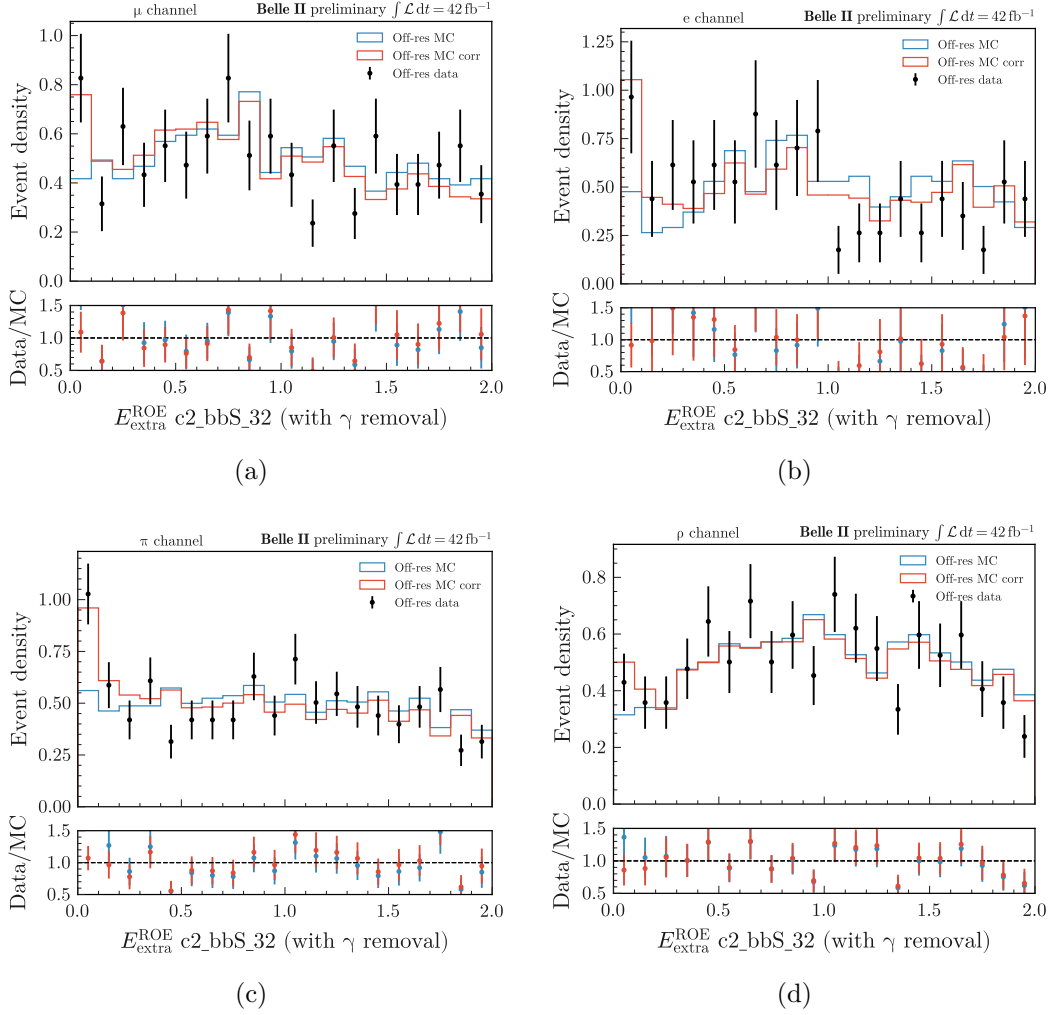


FIG. 29. Comparison between off-resonance data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay after reweighting each bin of the N_{γ}^{ROE} distribution to its data/MC ratio.

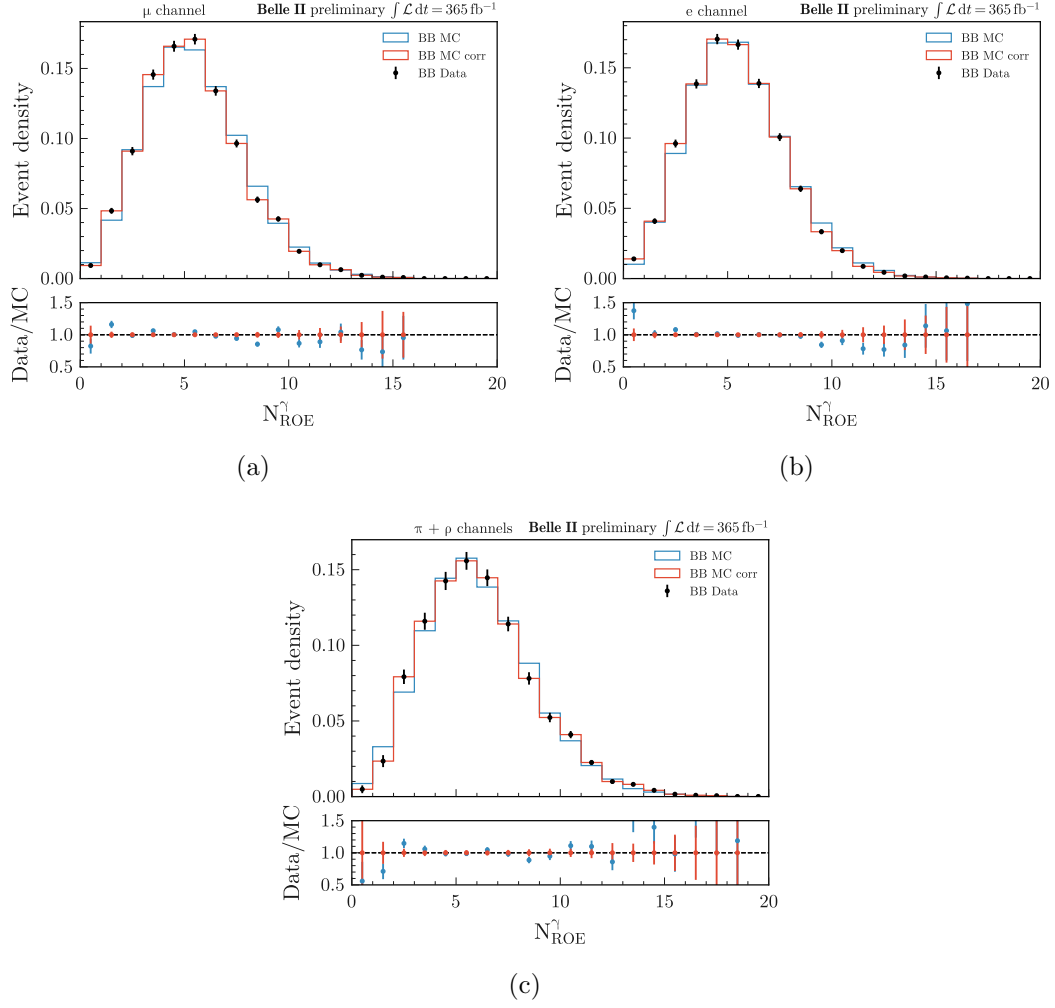


FIG. 30. Comparison between on-resonance $B\bar{B}$ data and MC of N_{γ}^{ROE} for each τ decay for the BDT sideband.

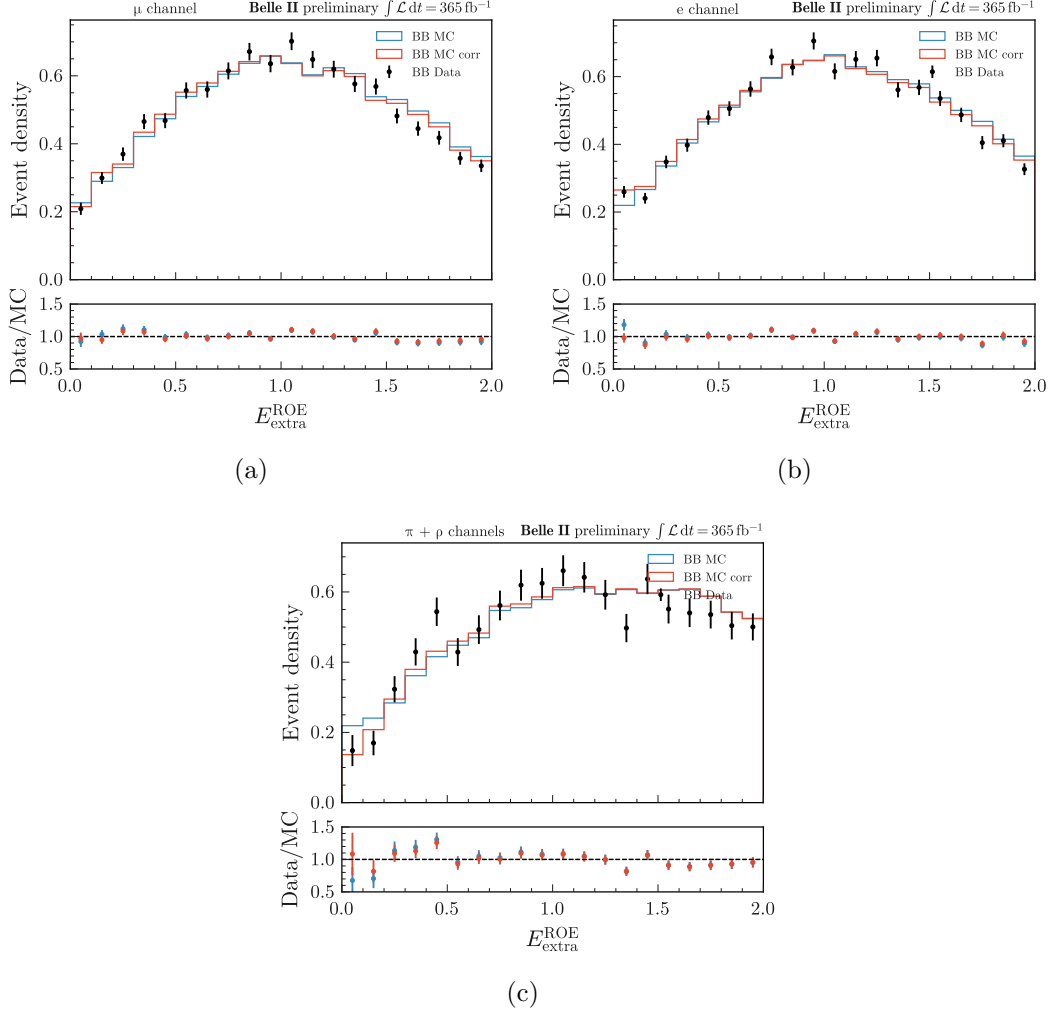


FIG. 31. Comparison between on-resonance $B\bar{B}$ data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay of the BDT sideband after reweighting each bin of the N_{γ}^{ROE} distribution to its data/MC ratio.

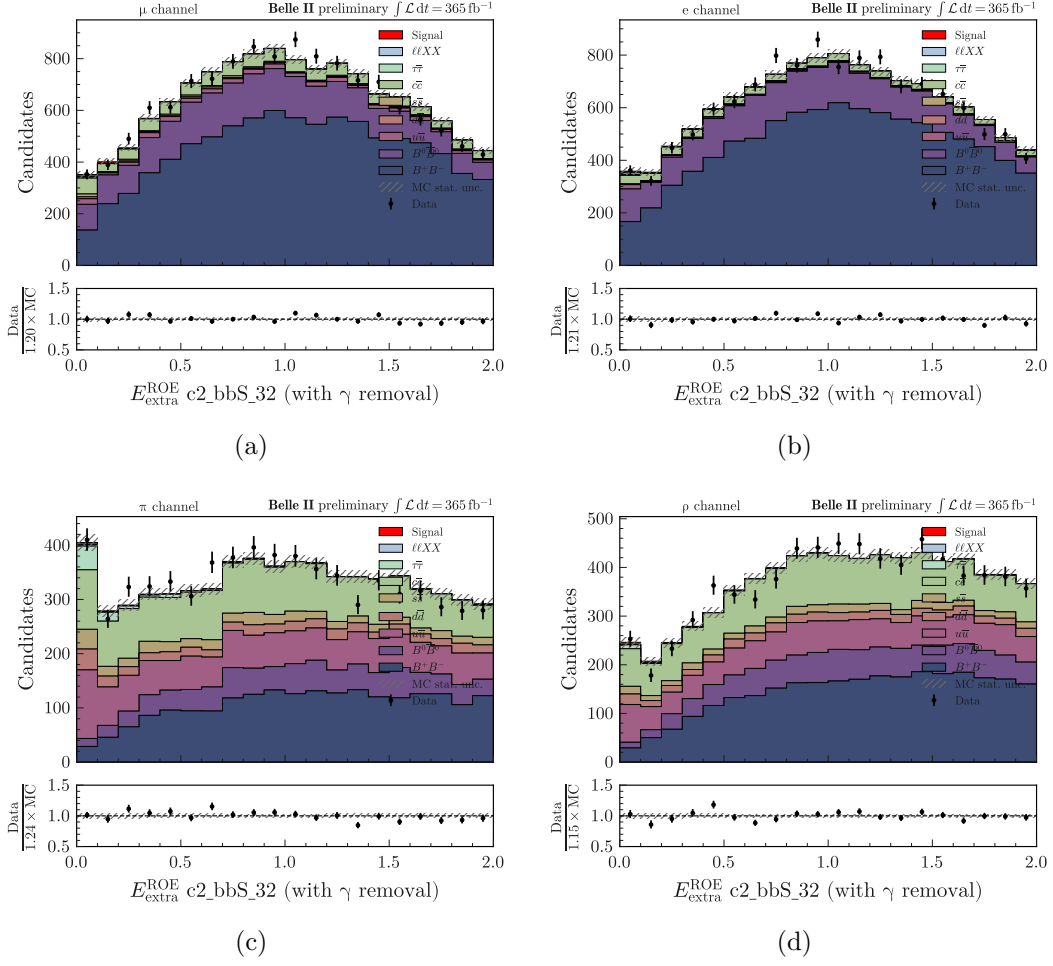


FIG. 32. Comparison between data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay on the sideband of the BDT output after step 2 of the correction.

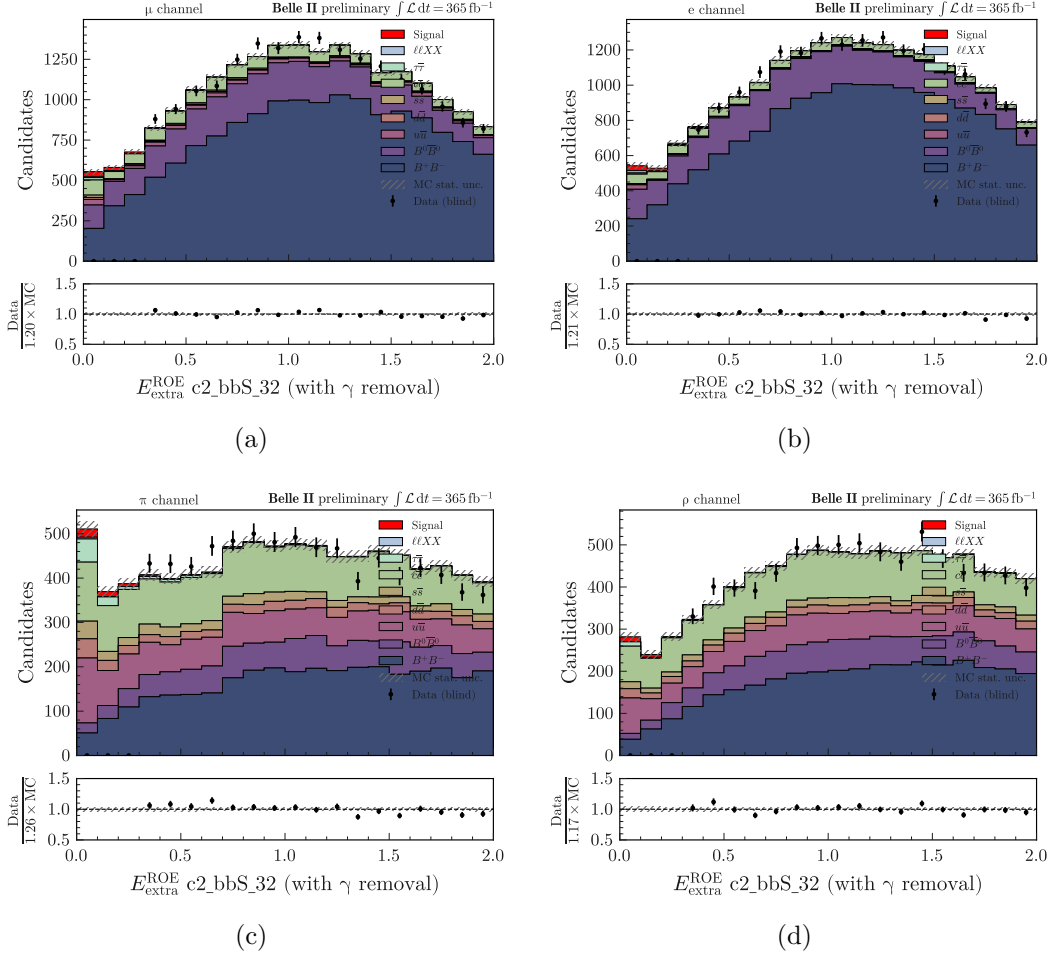


FIG. 33. Comparison between data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay after step 2 of the correction.

5.3. p_{CMS} corrections

The momentum of the τ distribution does not agree between data and MC after the pre-selection, as shown in Figure 34. Since this is one of the fit variable, the distribution has to be corrected. We observe a good agreement between off-resonance data and MC after the pre-selection (see Figure 35), thus the p_{CMS} correction is only calculated using the BB background. We use the BDT sideband to correct the MC. We remove the non- BB samples from MC, and the same amount is removed from the data distribution. Then, for each τ channel, each bin of p_{CMS} is corrected to match the data distribution. Figure 36 shows the data/MC agreement before and after applying the corrections on the BB MC. Figure 37 shows the final data/MC agreement after all corrections are applied. No significant discrepancy remains after the correction.

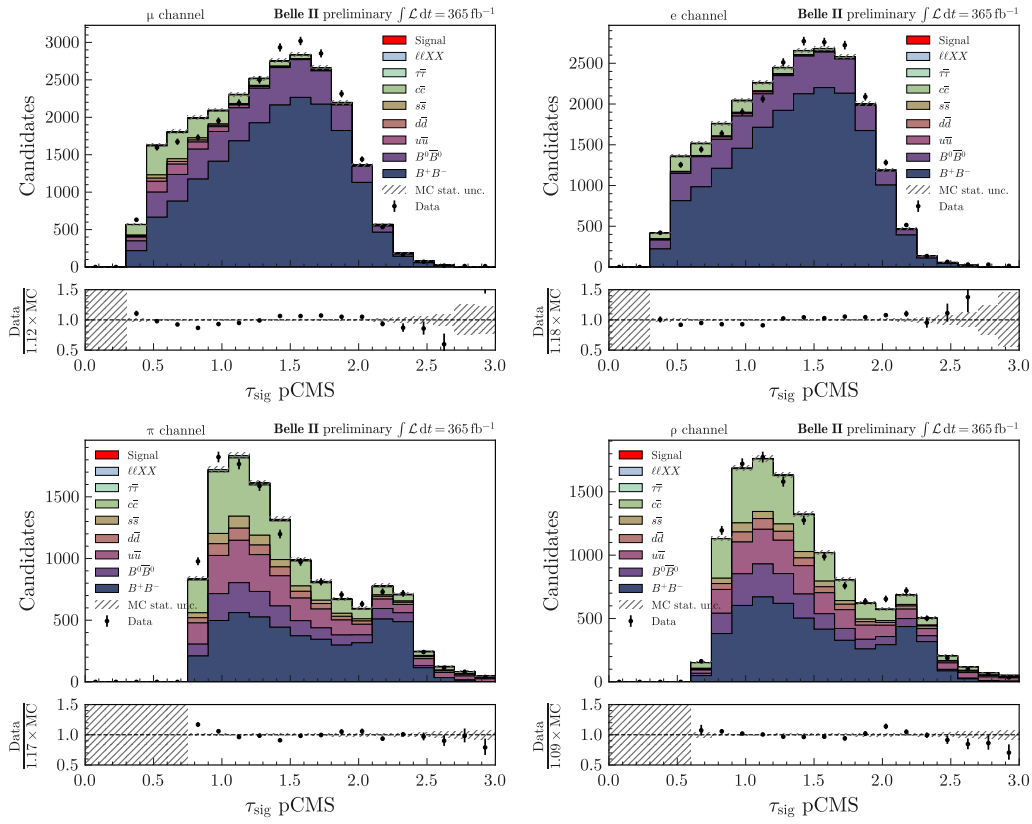


FIG. 34. Comparison between data and MC of the momentum of the τ in the CMS frame for each τ decay after the pre-selection without correction applied.

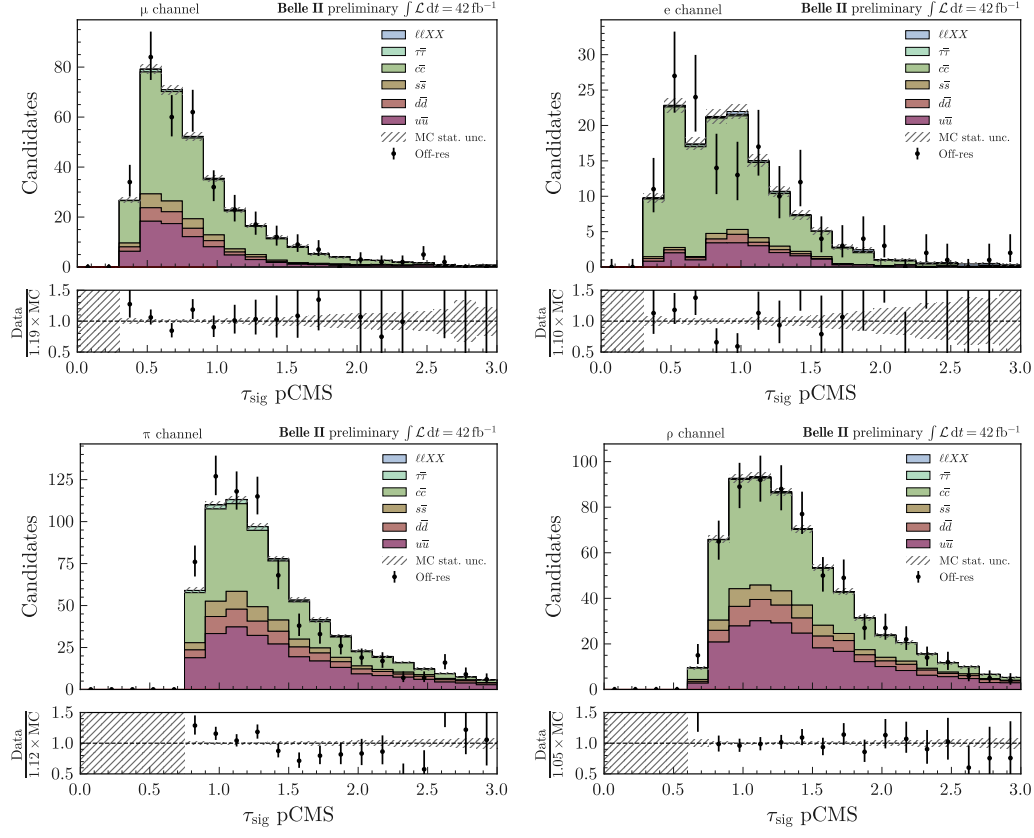


FIG. 35. Comparison between off-resonance data and MC of the momentum of the τ in the CMS frame for each τ decay on the sideband of the BDT output.

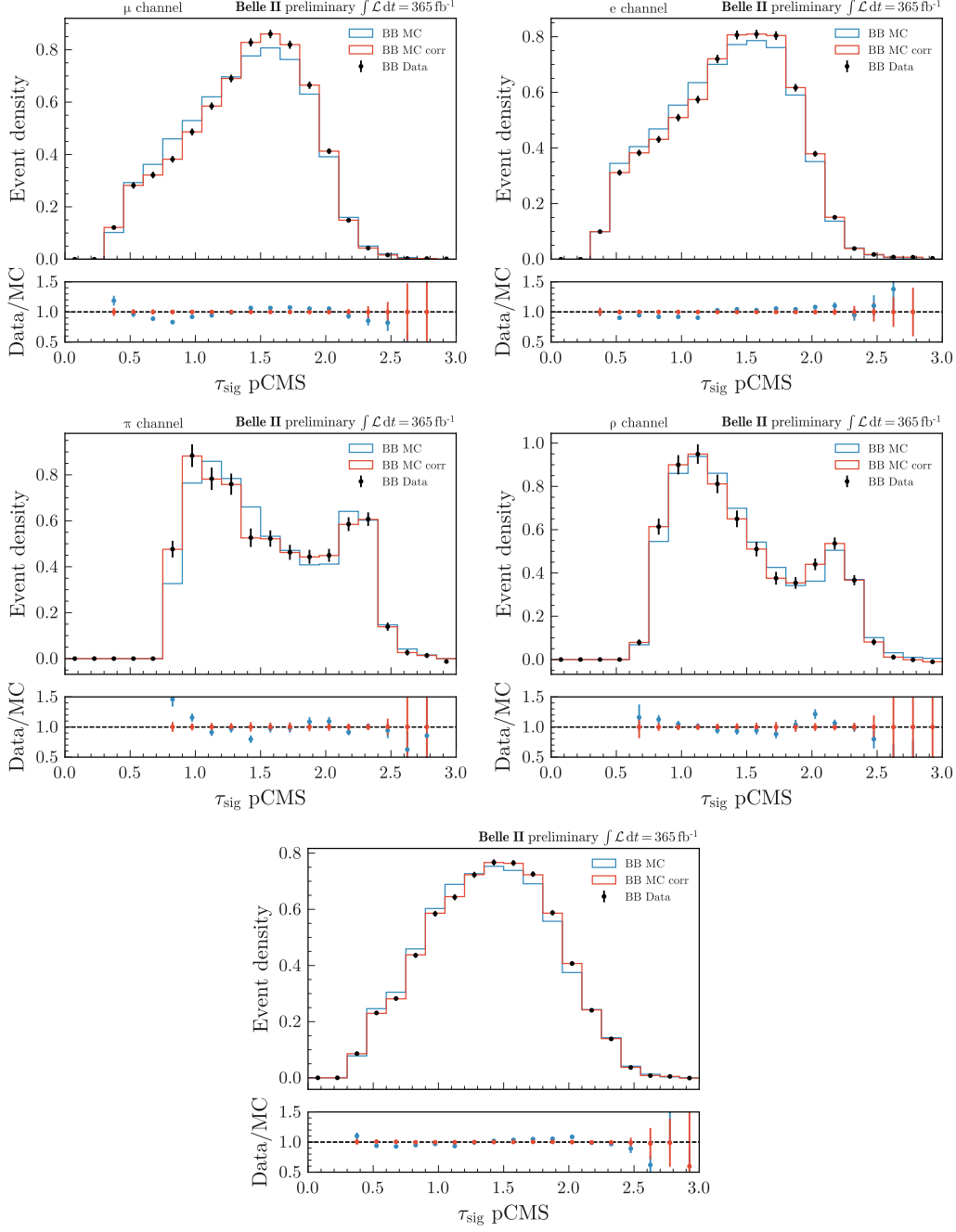


FIG. 36. Comparison between data and MC of the momentum of the τ in the CMS frame for each τ decay on the sideband of the BDT output, before (blue) and after (red) applying the p_{CMS} correction.

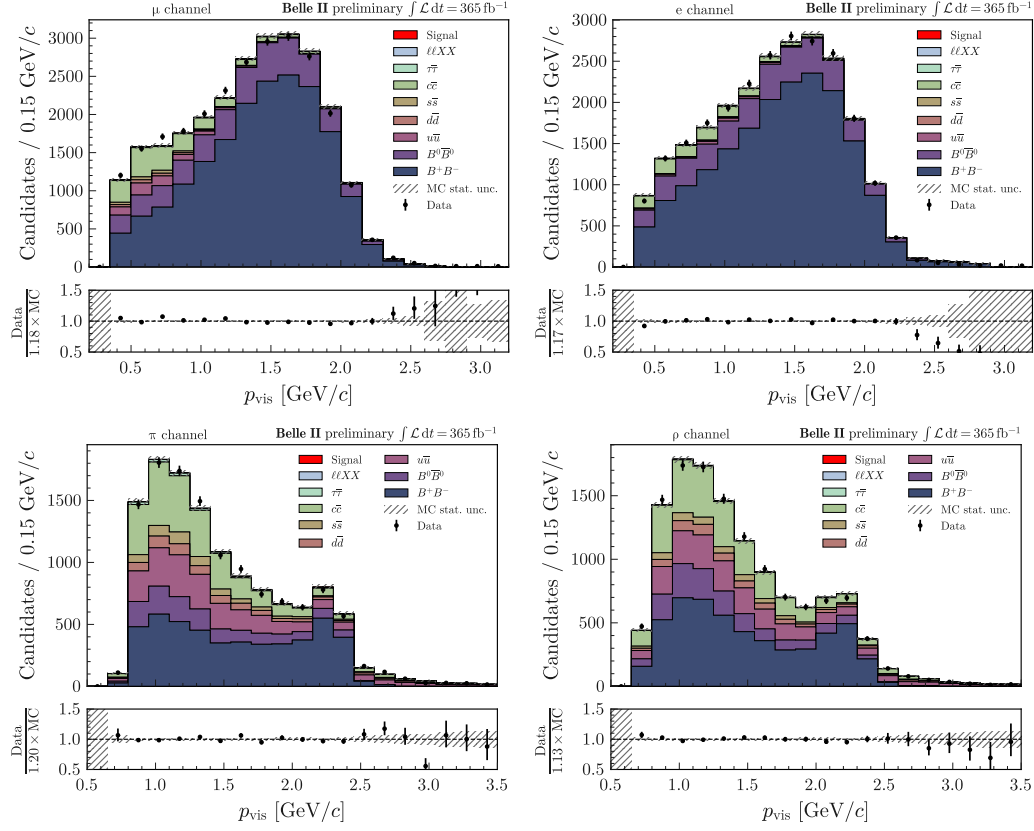


FIG. 37. Comparison between data and MC of the momentum of the τ in the CMS frame for each τ decay with the τ_{PCMS} correction applied.

608 6. DATA/MC COMPARISON AND VALIDATION

609 This section shows some data/MC comparison plots and validation checks.

610 6.1. Embedded sample efficiencies

611 The embedded sample is used to validate the analysis procedure and check the
612 different efficiencies and data/MC agreements. Each event of the embedded sample
613 is created by taking an on-resonance data event where the tag-side B meson is fully
614 reconstructed and replacing the signal-side B meson by a simulated $B^+ \rightarrow \tau^+ \nu_\tau$
615 decay. The ROE is kept from data. This way, the embedded sample has the same
616 background conditions as data, but with a known signal decay. The same method
617 used in the $B \rightarrow K \nu \nu$ ITA analysis is used. Here are the different steps to achieve
618 this:

- 619 • Create $B^+ \rightarrow \tau^+ \nu_\tau$ signal MC **udst** skim using very loose cuts.
- 620 • Create fully hadronic **udst** skims of the on-resonance data using $B^+ \rightarrow (K^* \rightarrow$
621 $K\pi)(J/\psi \rightarrow \mu\mu)$ and $B^+ \rightarrow K^+(J/\psi \rightarrow \mu\mu \text{ or } ee)$ decays. All the final states
622 particles of the B^+ are required to be reconstructed and identified.
- 623 • Use the embedding tool to remove the $KJ\psi$ system from the **udst** data skims.
- 624 • Use the embedding tool to remove the tag-side B meson and the ROE from
625 the **udst** signal MC skims.
- 626 • Merge the two **udst** skims together to create the embedded **udst** skims. The
627 sample is now composed of a tag-side B meson and ROE from data, and a
628 simulated signal-side $B^+ \rightarrow \tau^+ \nu_\tau$ decay.
- 629 • Run the reconstruction script on the embedded **udst** skims to create the final
630 embedded **mdst** sample.

631 The same procedure is run on MC samples to create an embedded MC sample,
632 which is used to check the data/MC agreement of the embedding procedure.

633 After the reconstruction, 257 signal candidates remain in the embedded sample.
634 The same pre-selection is applied. The resulting data/MC efficiency ratios for the
635 different pre-selection cuts are shown in Table 13. All efficiencies are compatible
636 between data and MC within the statistical uncertainty.

637 The same BDT cuts as on for the main analysis are applied on the embedded
638 sample. The resulting data/MC efficiency ratios for the different BDT selection cuts

Selection step	Data/MC ratio
Reconstruction	0.96 ± 0.06
$p_{l,\text{tag}}^* > 0.3 \text{ GeV}/c$	1.000 ± 0.004
$p_{\tau,\text{sig}}^* > 0.4 \text{ GeV}/c$	1.015 ± 0.017
$p_{D\text{tag}}^* < 2.5 \text{ GeV}/c$	1.004 ± 0.004
$-3 < \cos \theta_{B,D_l}^* < 1.5$	1.007 ± 0.017
$R_2(\text{Event based}) < 0.45$	1.026 ± 0.028
$B_{\text{tag}} R_2 < 0.45$	0.994 ± 0.007
$\cos \theta_{\text{thrust}, B_{\text{tag}}, z} < 0.85$	0.967 ± 0.026
B_{tag} decay mode ID < 4	1.006 ± 0.016
π : $p_{B_{\text{sig}}} > 1. \text{ GeV}/c$	1.04 ± 0.08
ρ : $p_{B_{\text{sig}}} > 1. \text{ GeV}/c$	1.10 ± 0.08
e : K_{tag} electronID < 0.2	0.98 ± 0.04
e : conv. γ $\text{InvM}(\ell_{\text{tag}} - e_{\text{sig}}) > 0.02 \text{ GeV}/c^2$	0.990 ± 0.023
e : conv. γ $\text{InvM}(D_{\text{tag, daughters}} - e_{\text{sig}}) > 0.02 \text{ GeV}/c^2$	0.981 ± 0.033
Total	
Pre-selection	1.03 ± 0.06
Reco + pre-selection	0.99 ± 0.08

TABLE 13. Data/MC efficiency ratios for the embedded sample during reconstruction and pre-selection.

for each τ decays are shown in Table 14. The data/MC efficiency ratio in bins of $\tau_{p_{\text{CMS}}}$ are also shown Figure 38. The data/MC efficiency ratios are compatible with 1 within the statistical uncertainty.

We decide to correct the measured BF value by 10% and assign a 10% systematic uncertainty to the signal efficiency to cover the observed data/MC discrepancy in the embedded sample.

Channel	Data/MC ratio
μ	1.02 ± 0.08
e	1.19 ± 0.09
hadronic	1.12 ± 0.13
Total	
Selection	1.11 ± 0.06
Reco + pre-sel + selection	1.10 ± 0.11

TABLE 14. Data/MC BDT efficiency ratios for the embedded samples.

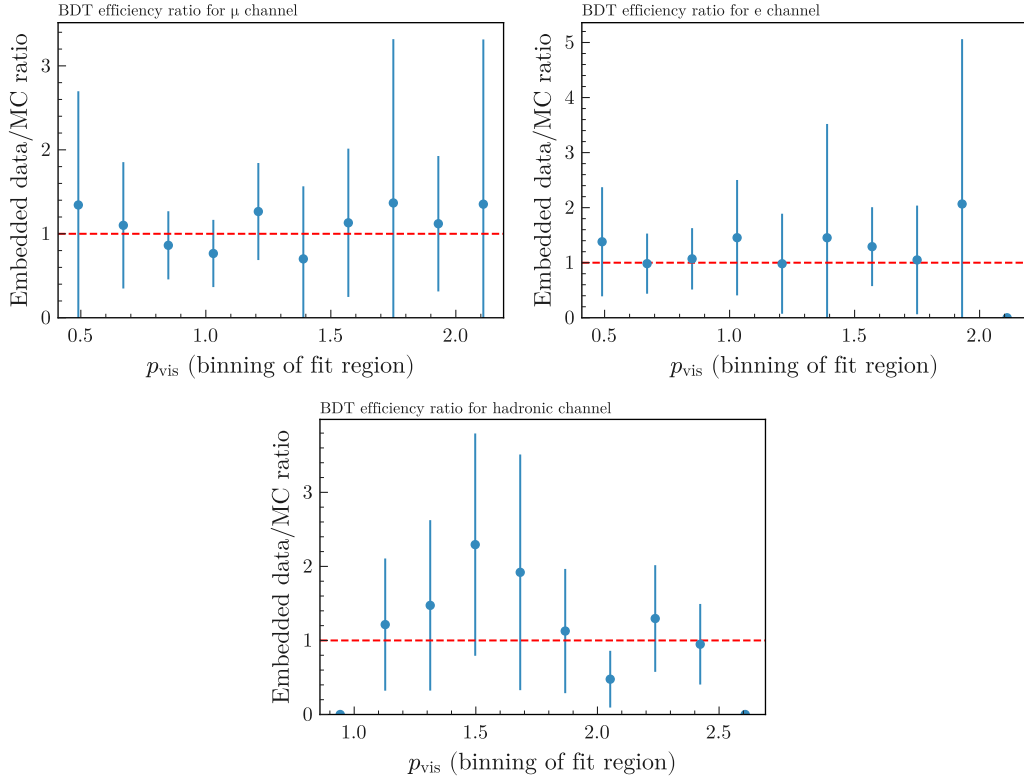


FIG. 38. Data/MC BDT efficiency ratio in bins of $\tau_{p_{\text{CMS}}}$ for the embedded samples.

645

6.2. BDT output checks

646 To check if the output of the BDT is agreeing between data and MC, we compare
 647 the BDT output distributions after the pre-selection for each τ channels on the off-
 648 resonance, on the on-resonance data by cutting away the BDT signal region (BDT
 649 sideband), and on the embedded data sample. The results are shown in Figures 39
 650 to 41. A good agreement between data and MC is observed.

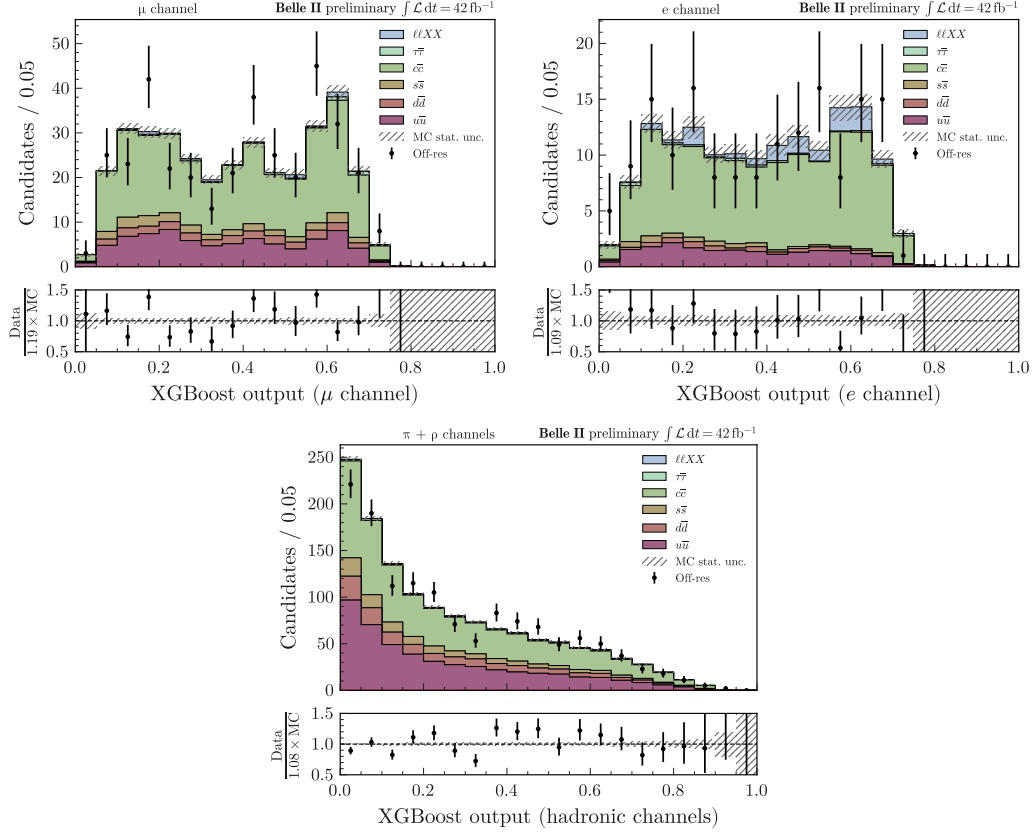


FIG. 39. Distribution of the BDT output after the pre-selection for the μ (left), e (right) and hadronic (bottom) channels on the off-resonance data.

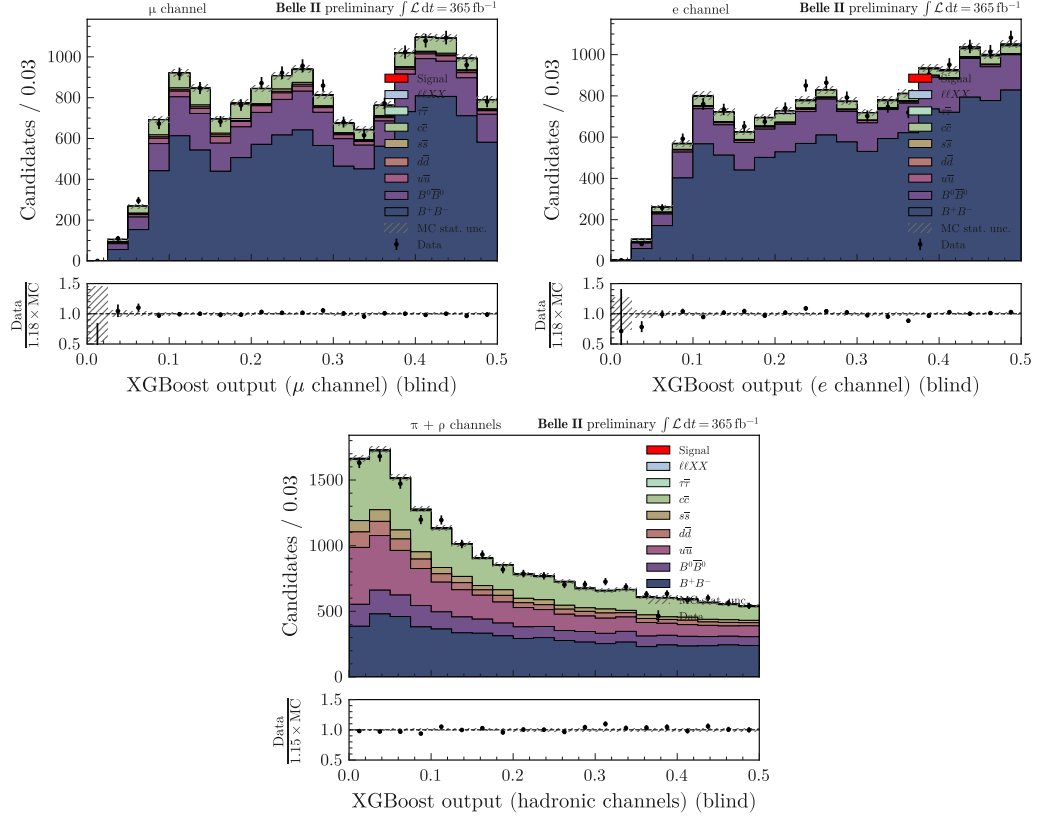


FIG. 40. Distribution of the BDT output after the pre-selection for the μ (left), e (right) and hadronic (bottom) channels on the on-resonance data by cutting away the BDT signal region.

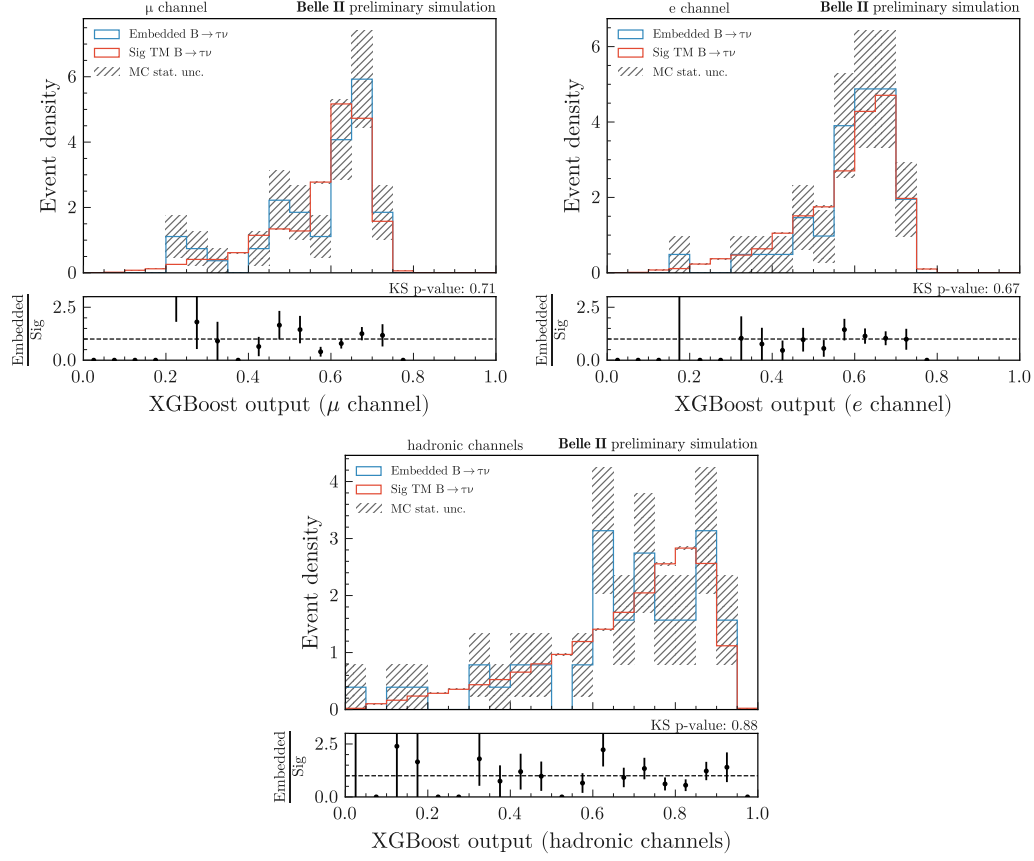


FIG. 41. Distributions of the fitting variables in the embedded sample after the full selection for the μ channel. The Kolmogorov-Smirnov (KS) test p-value is used to quantify the agreement between data and MC.

6.3. Fitting variables after full selection

The distributions of $E_{\text{extra}}^{\text{ROE}}$ and τ_{pCMS} after the full selection for each τ channels are checked using the different data sample available.

6.3.1. SB control sample

To check if the selection cuts are well modeled by MC, we use the SB control sample defined in Section 2 to compare the fitting variables. No signal is expected in the signal region. The $E_{\text{extra}}^{\text{ROE}}$ and τ_{pCMS} distributions after the full selection for each τ channels are shown in Figures 42 to 44. No significant data/MC discrepancies are observed.

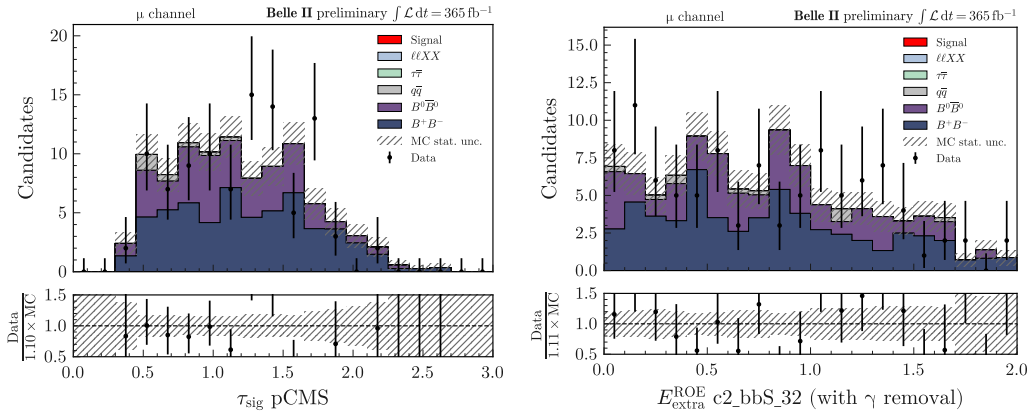


FIG. 42. Distributions of the fitting variables in the SB control sample after the full selection for the μ channel

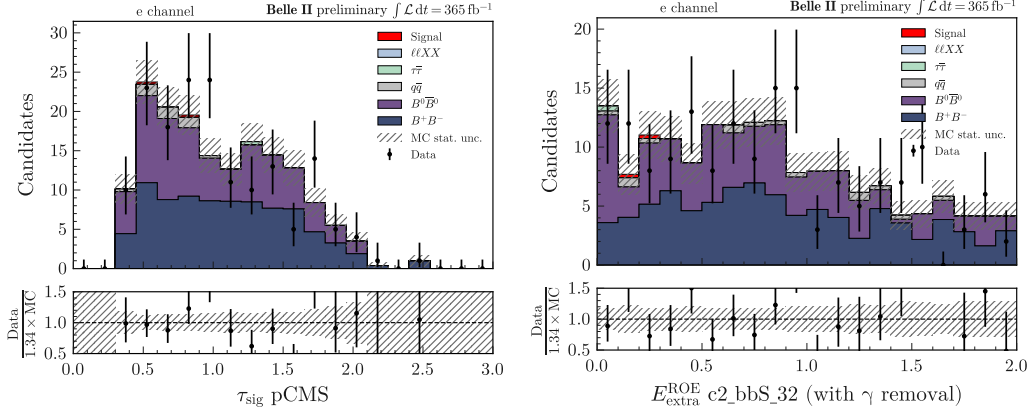


FIG. 43. Distributions of the fitting variables in the SB control sample after the full selection for the e channel

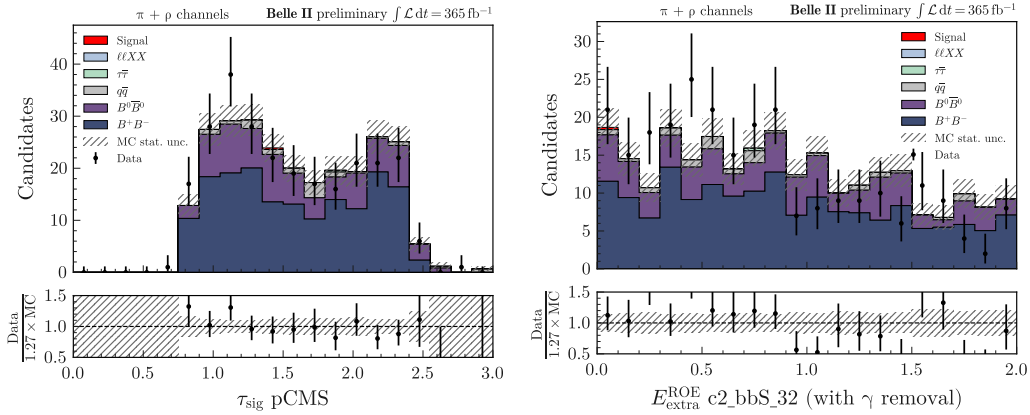


FIG. 44. Distributions of the fitting variables in the SB control sample after the full selection for the hadronic channels

6.3.2. Off-resonance data

The off-resonance data is used to validate the non- $B\bar{B}$ background modeling. The $E_{\text{extra}}^{\text{ROE}}$ and τ_{pCMS} distributions after the full selection for each τ channels are shown in Figures 45 to 47. No significant data/MC discrepancies are observed.

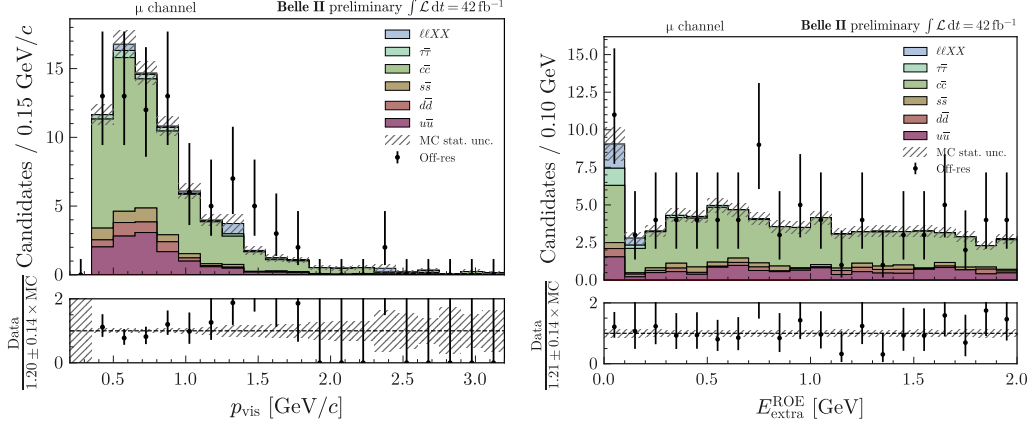


FIG. 45. Distributions of the fitting variables in the off-resonance data after the full selection for the μ channel

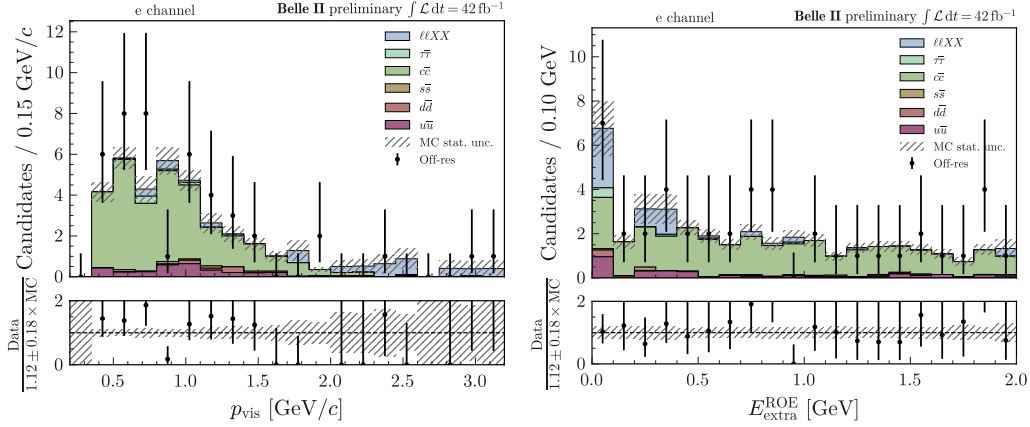


FIG. 46. Distributions of the fitting variables in the off-resonance data after the full selection for the e channel

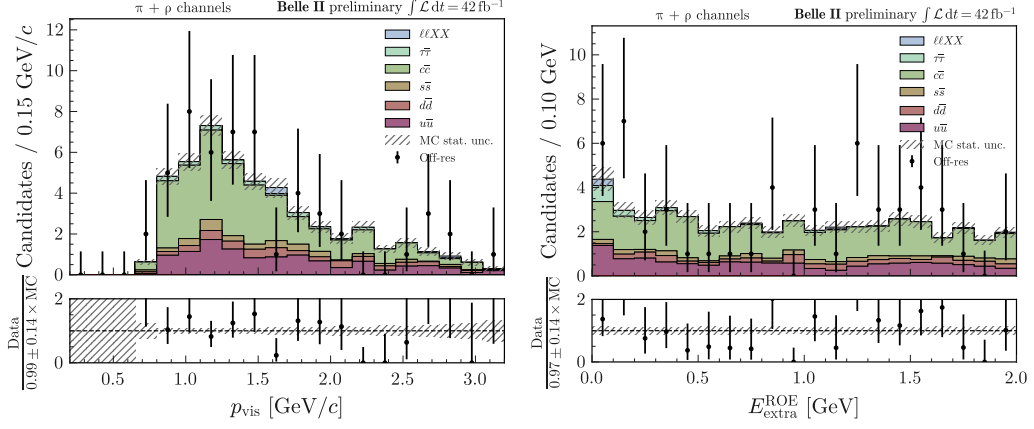


FIG. 47. Distributions of the fitting variables in the off-resonance data after the full selection for the hadronic channels

664 *6.3.3. Embedded sample*

665 The embedded sample defined in Section 6.1 is used to check the modeling of the
 666 fitting variables for signal candidates. The $E_{\text{extra}}^{\text{ROE}}$ and τ_{pCMS} distributions after the
 667 full selection for each τ channels are shown in Figures 48 to 50. Despite the low
 668 statistics, no significant data/MC discrepancies are observed.

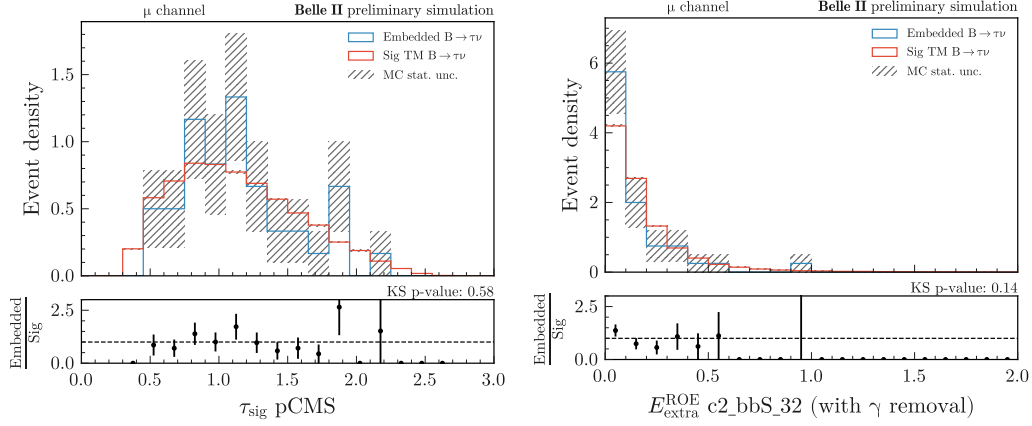


FIG. 48. Distributions of the fitting variables in the embedded sample after the full selection for the μ channel. The Kolmogorov-Smirnov (KS) test p-value is used to quantify the agreement between data and MC.

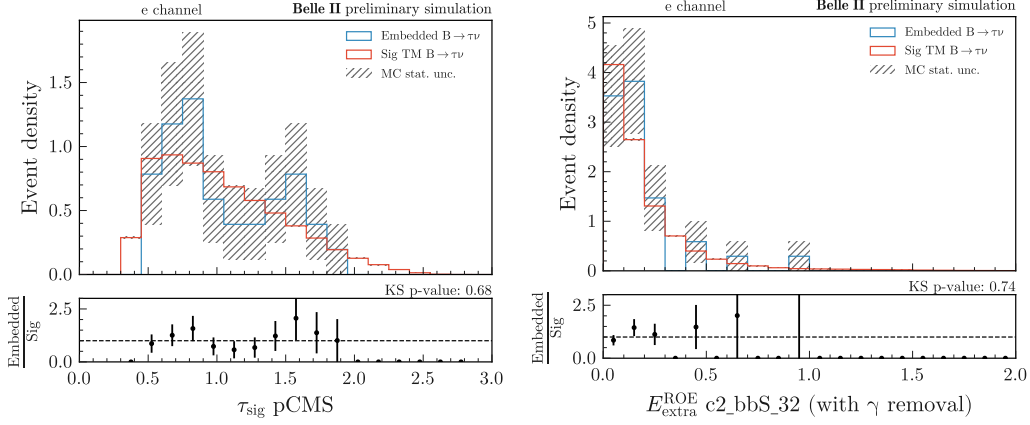


FIG. 49. Distributions of the fitting variables in the embedded sample after the full selection for the e channel. The Kolmogorov-Smirnov (KS) test p-value is used to quantify the agreement between data and MC.

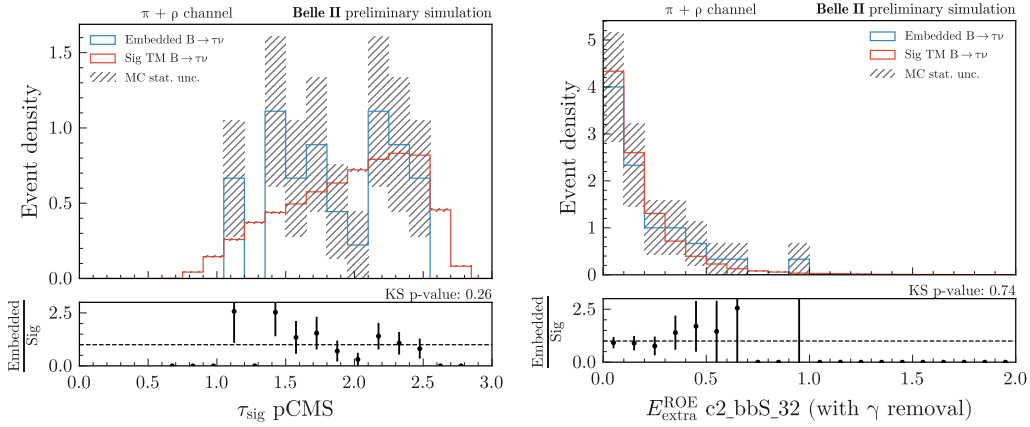


FIG. 50. Distributions of the fitting variables in the embedded sample after the full selection for the hadronic channels. The Kolmogorov-Smirnov (KS) test p-value is used to quantify the agreement between data and MC.

669 6.3.4. $B^+ \rightarrow \bar{D}^{*0} \ell^+ \nu_\ell$ control sample

670 We select a control sample of $B^+ \rightarrow \bar{D}^{*0} \ell^+ \nu_\ell$ ($\ell = e, \mu$) decay to check data/MC
671 consistency and validate the analysis procedure. The control sample is similar with
672 the signal $B^+ \rightarrow \tau^+ \nu_\tau$, with a charged lepton and neutrino in the final state and the

extra energy is also expected peaking at zero. To select the control sample, we re-construct a semi-leptonic B_{tag} candidate per event with semi-leptonic FEI algorithm and a signal B_{sig} decaying to $\bar{D}^{*0}\ell^+\nu$. The D^{*0} is reconstructed from $D^{*0} \rightarrow D^0\gamma$ and $D^{*0} \rightarrow D^0\pi^0$. The D^0 is reconstructed from $D^0 \rightarrow K^-\pi^+$, $D^0 \rightarrow K^-\pi^+\pi^+\pi^-$ and $D^0 \rightarrow K_S^0\pi^+\pi^-$. The selection criterion are same with $B^+ \rightarrow \tau^+\nu_\tau$ when applicable. Other selection criterion are summarized in Table 15.

Variable	requirement
$K^\pm$	$d_r < 0.5 \text{ cm}$ and $ d_z < 2 \text{ cm}$ and kaonID_noSVD > 0.6
K_S^0	K_S^0 standard list from ksSelector
$M(D^0)$	(1.855, 1.875) GeV/ c^2
$M(D^{*0}) - M(D^0)$	(0.120, 0.155) GeV/ c^2
γ in $D^{*0} \rightarrow D^0\gamma$	$E_\gamma > 0.1 \text{ GeV}$, fakePhotonSuppression > 0.15
$\cos\theta_{B, D\ell}^*$	(-1, 1)
MissingMass square	(-1, 1)
Background Suppression	Apply the same BDT of $B^+ \rightarrow \tau^+\nu_\tau$ and same cut
BCS	Best D^0 mass; best D^{*0} mass; then the highest PID of the lepton
ROE	No extra tracks coming from the IP. No extra K_S^0 , Λ^0 , π^0

TABLE 15. Selection criterion for control sample $B^+ \rightarrow \bar{D}^{*0}\ell^+\nu_\ell$, ($\ell = e, \mu$).

The extra energy distribution of the control sample is expected to be same with signal mode $B^+ \rightarrow \tau^+\nu_\tau$. We compare the extra energy distributions of correctly reconstructed $B^+ \rightarrow \bar{D}^{*0}\ell^+\nu_\ell$ and $B^+ \rightarrow \tau^+\nu_\tau$, shown in Figure 51. The distributions are consistent.

After final event selection, the continuum backgrounds in $B^+ \rightarrow \bar{D}^{*0}\ell^+\nu_\ell$ are negligible (expect 0 events in $B^+ \rightarrow D^{*0}e^+\nu$ and 1.1 events in $B^+ \rightarrow D^{*0}\mu^+\nu$). We applied the same FEI correction and pCMS correction factors for $B^+ \rightarrow \bar{D}^{*0}\ell^+\nu_\ell$ as that of $B^+ \rightarrow \tau^+\nu_\tau$. We also applied the extra energy correction following the same procedure of $B^+ \rightarrow \tau^+\nu_\tau$. The extra energy correction factors are obtained by comparing the distributions of number of photons in the ROE in data and MC. After all corrections, the fit variables (i.e. momentum of the lepton in CMS frame and extra energy) of the control sample are shown in Figure 52. The data/MC ratio for each bin of N_γ^{ROE} distribution is shown in Table 16. These weights are applied to the signal MC sample of the leptonic decays to correct its N_γ^{ROE} distribution. The background breakdown is shown and also overlaid with data points. The shapes between data and MC are consistent. The ratio of data and MC events is 0.912 ± 0.046 for electron mode and 0.954 ± 0.044 for muon mode, respectively.

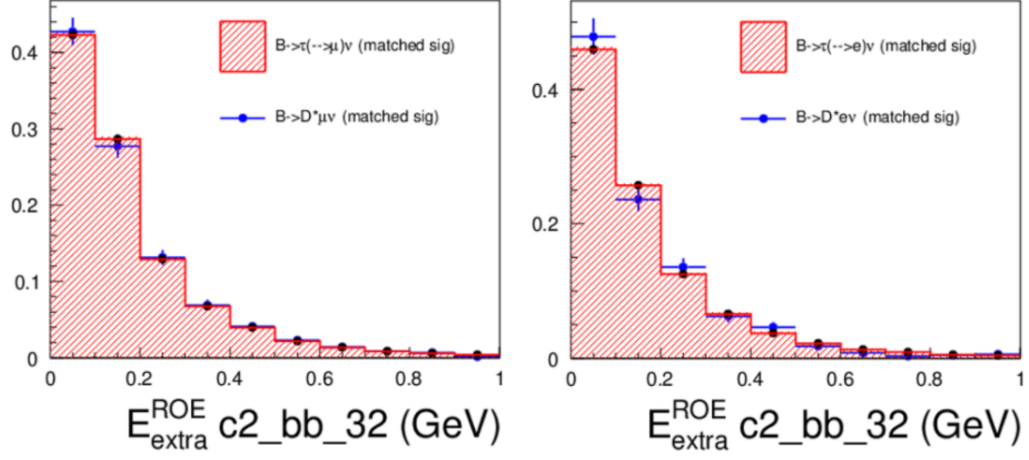


FIG. 51. Extra energy distribution of $B^+ \rightarrow \bar{D}^{*0} \ell^+ \nu_\ell$ compared to that of $B^+ \rightarrow \tau^+ \nu_\tau$, only MC truth-matched signal events are shown.

bin	μ channel	e channel
0	1.07 ± 0.10	0.99 ± 0.10
1	1.11 ± 0.09	1.05 ± 0.09
2	0.97 ± 0.10	0.95 ± 0.10
3	0.71 ± 0.12	1.09 ± 0.16
4	1.09 ± 0.27	0.89 ± 0.22
5	0.46 ± 0.24	0.65 ± 0.35
6	0.29 ± 0.30	1.27 ± 1.02
7	0 ± 0	0 ± 0

TABLE 16. Data/MC ratio in each bin of N_γ^{ROE} distribution of the control sample $B^+ \rightarrow \bar{D}^{*0} \ell^+ \nu_\ell$.

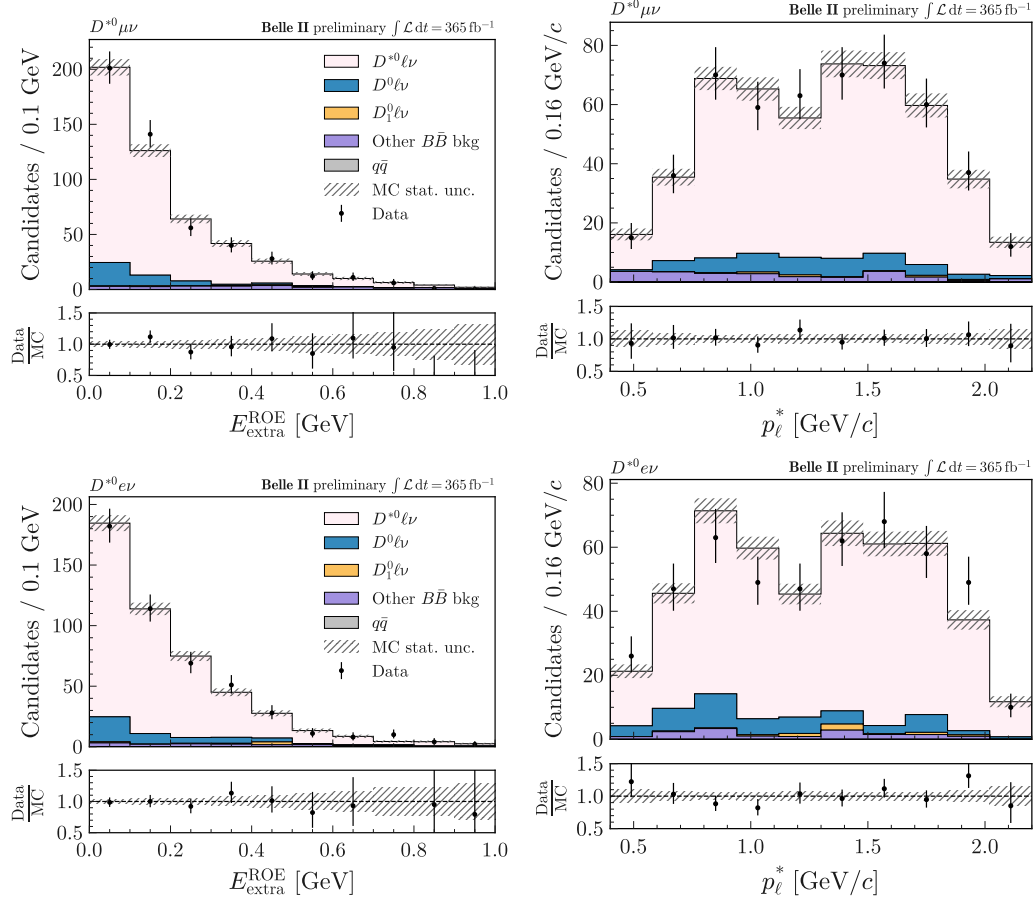


FIG. 52. Lepton pCMS and extra energy distributions of the control sample $B^+ \rightarrow \bar{D}^{*0} \ell^+ \nu_\ell$, the MC histograms are scaled to number of events in data.

696 6.3.5. Double tagged control sample

697 To check the hadronic channels, we use a double tagged control sample where the
698 signal side is reconstructed using hadronic FEI. The tag side is still reconstructed
699 using the SL FEI. The selection criterion are same with $B^+ \rightarrow \tau^+ \nu_\tau$ when applicable.
700 Other selection criterion are summarized in Table 17.

Variable	requirement
sigProb (or extraInfo(dmID))	> 0.001 (or $= 25$)
M_{bc}	$> 5.27 \text{ GeV}/c^2$
ΔE	$(-0.15, 0.1) \text{ GeV}$
R2	< 0.6
Keep the best Btag candidate highest sigProb	
Background Suppression	Apply the same BDT of $B^+ \rightarrow \tau^+ \nu_\tau$ and same cut
ROE	No extra tracks coming from the IP
ROE	No extra K_S^0, Λ^0, π^0

TABLE 17. Selection criterion for double tagged control sample.

701 After the final event selection, the continuum backgrounds in double tagged con-
702 trol sample are negligible. The same method as for the $B^+ \rightarrow \bar{D}^{*0} \ell^+ \nu_\ell$ control sample
703 is used to correct the FEI efficiency and extra energy. The $E_{\text{extra}}^{\text{ROE}}$ distribution before
704 and after the extra energy correction is shown in Figure 53. The data/MC ratio for
705 each bin of N_γ^{ROE} distribution is shown in Table 18. These weights are applied to
706 the signal MC sample of the hadronic decays to correct its N_γ^{ROE} distribution.

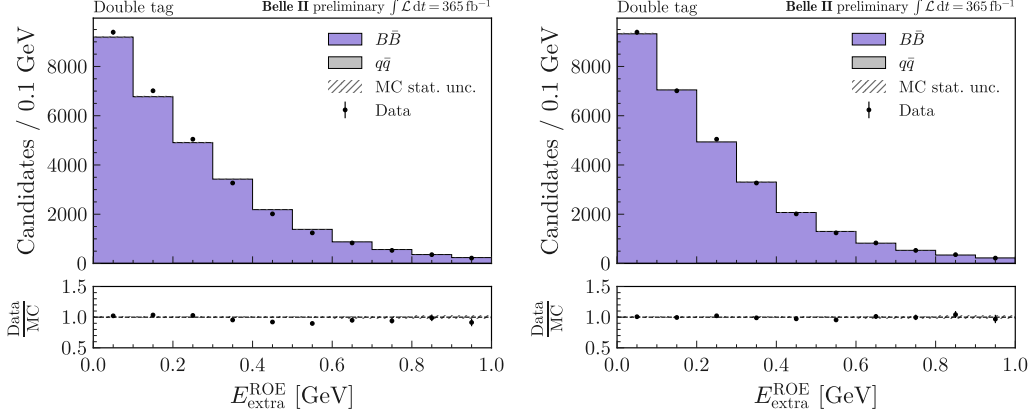


FIG. 53. Extra energy distribution of double tagged control sample before (left) and after (right) the ROE correction.

bin	Data/MC ratio
0	1.10 ± 0.02
1	1.06 ± 0.02
2	0.99 ± 0.02
3	0.89 ± 0.02
4	0.86 ± 0.03
5	0.84 ± 0.04
6	0.80 ± 0.06
7	0.96 ± 0.12
8	0.55 ± 0.16
9	0 ± 0

TABLE 18. Data/MC ratio in each bin of $E_{\text{extra}}^{\text{ROE}}$ distribution of the double tagged control sample.

6.3.6. Blind on-resonance data

We also check the fitting variables on the blind on-resonance data. An enlarged SR cut is applied to fully remove the signal events. The $E_{\text{extra}}^{\text{ROE}}$ and τ_{PCMS} distributions after the full selection for each τ channels in the fit region are shown in Figures 54 to 56. No significant data/MC discrepancies are observed. We observe a slight

712 data/MC scaling difference. The μ , e and hadronic channels will be scaled by a
 713 factor 1.19, 1.21 and 1.38 respectively in the final fit.

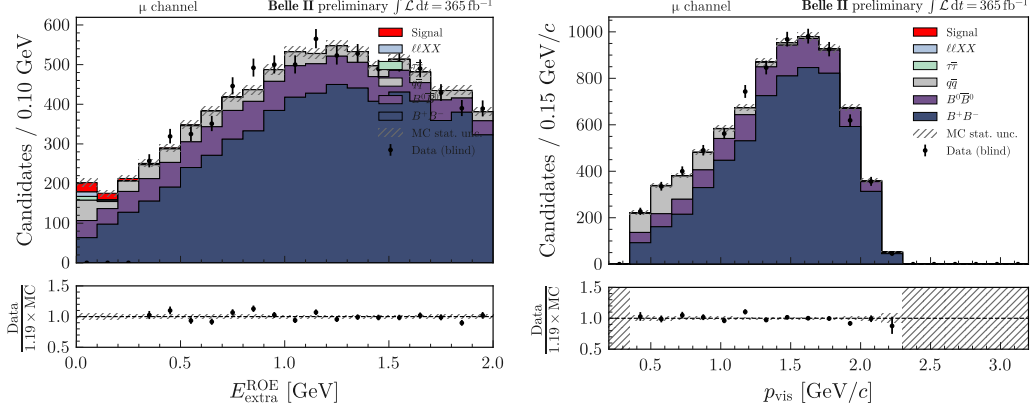


FIG. 54. Distributions of the fitting variables in the blind on-resonance data after the full selection for the μ channel

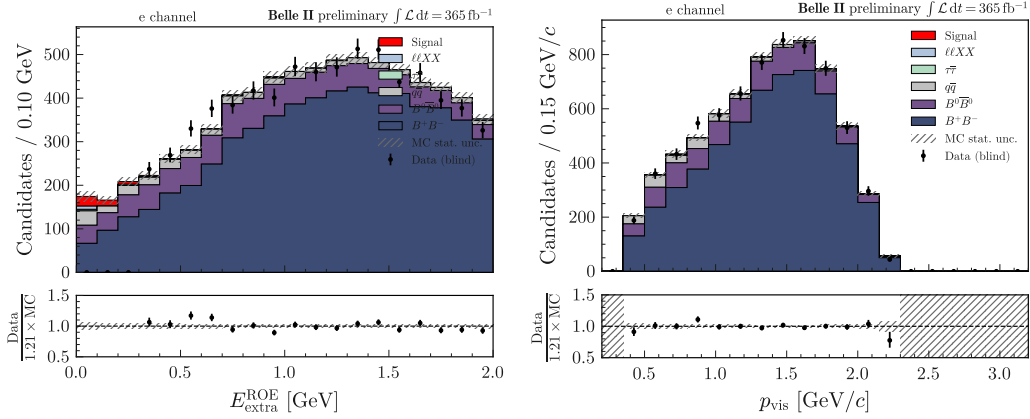


FIG. 55. Distributions of the fitting variables in the blind on-resonance data after the full selection for the e channel

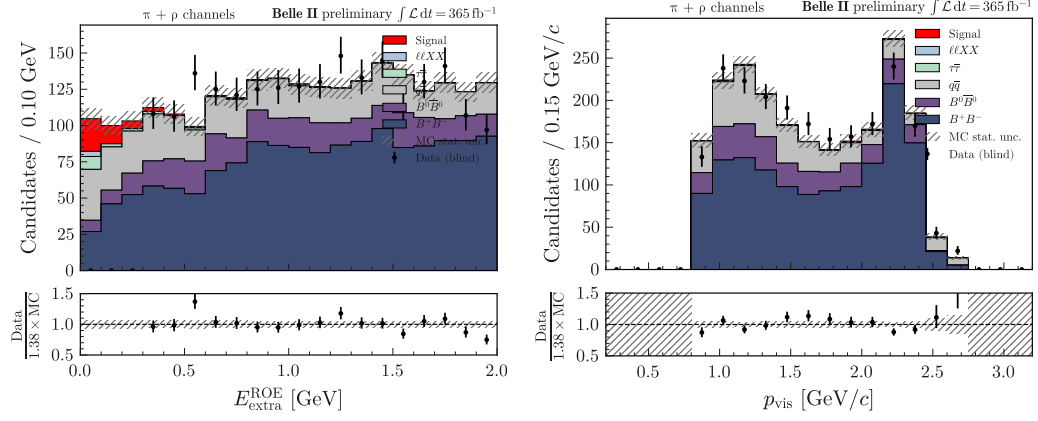


FIG. 56. Distributions of the fitting variables in the blind on-resonance data after the full selection for the hadronic channels

7. BR EXTRACTION

This section describes the fitting procedure and the validation of the fit results. The fitting procedure is based on the `pyhf` package, which is a pure Python implementation of the HistFactory model [20], with the help of the `cabinetry` package [21].

7.1. Fitting procedure

The fit is performed using the `pyhf` package, which allows for the construction of a statistical model based on the HistFactory framework. The fit is performed using the `cabinetry` package, which provides a convenient interface for constructing and fitting models. The model used in the fit is a binned likelihood function that includes the signal and background contributions, along with their associated statistical and systematic uncertainties. Then, `cabinetry` can directly compute the significance of the signal. To learn more about the `pyhf` package, see here.

To fit the signal, $E_{\text{extra}}^{\text{ROE}}$ and p_{CMS} ($= \tau_{p_{\text{CMS}}} = p_{\text{vis}}$) are used. The fit is performed simultaneously in 3 orthogonal regions: one is for the μ channel, one is for the e channel, and one is for the two hadronic channel combined due to the low statistics. For the μ , e and hadronic channels, we create a 2D histogram of the $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ variables. By default, `pyhf` doesn't support 2D histograms, so a flattening of the 2D histogram is performed. The $E_{\text{extra}}^{\text{ROE}}$ is binned into 10 bins in the range $[0, 1.0]$ GeV, while $\tau_{p_{\text{CMS}}}$ has a custom range for the leptonic and hadronic modes in order to avoid empty bins in the edges. For μ and e , $\tau_{p_{\text{CMS}}}$ has 10 bins in the range $[0.4, 2.2]$ GeV/ c , for the hadronic modes, they have 10 bins in the range $[0.85, 2.7]$ GeV/ c . $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ variables are then flattened into a 1D histogram with a total of 100 bins, which is used as the input to the fit. The new variable is called `global_index`.

The fit is composed of 6 components:

- Signal: the signal template is taken from truth-matched signal events from the signal-only sample.
- Self-crossfeed (SCF): the SCF corresponds to miss-reconstructed signal events. The template is taken from non truth-matched signal events from the signal-only sample. The SCF is also present and removed from the charged background using the information from `GenMCTagTool` (see here). More study on the SCF component is described in Appendix E.b.
- non- $B\bar{B}$ background: the non- $B\bar{B}$ background is composed of $q\bar{q}$, $\tau^+\tau^-$ and $\ell\ell XX$ background. The template is taken from the MC15rd.

- $B\bar{B}$ background: the $B\bar{B}$ background is composed of $B^0\bar{B}^0$ and B^+B^- background, where the SCF has been removed. The template is taken from the MC15rd.
- Rare $B\bar{B}$ background: the rare $B\bar{B}$ background described in Section 8.7.2. The template is taken from privately generated samples.
- Data: all the tests and fits are performed on Asimov dataset, which is taken from the MC samples. The final fit on data will be performed after unblinding.

The signal and the SCF have the same `normfactor` parameter, which is used to scale without constraints the contribution in the fit. The signal normalization factor is called μ , and represents the signal fraction in the fit. The expected value of μ is 1, and we let it float in the range $[-0.5, 7.0]$. For $B\bar{B}$, non- $B\bar{B}$ and rare $B\bar{B}$ backgrounds, a `normsys` parameter is used to scale the contribution, which varying the template by a factor $[0.5, 2]$, $[0.83, 1.21]$ (according to off-resonance/MC agreement) and $[0.80, 1.25]$ (according to the uncertainty on the rare $B\bar{B}$ branching fractions) respectively.

The plots of $E_{\text{extra}}^{\text{ROE}}$, $\tau_{p_{\text{CMS}}}$ and the flattened 2D histogram of the 3 regions before the fit can be seen in Figures 57 to 62.

Each contribution template plots are shown in Appendix F.

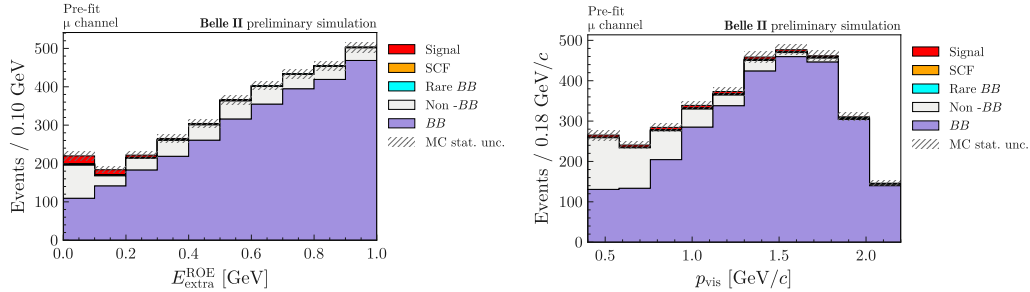


FIG. 57. The $E_{\text{extra}}^{\text{ROE}}$ and p_{CMS} distribution for the μ channel before the fit.

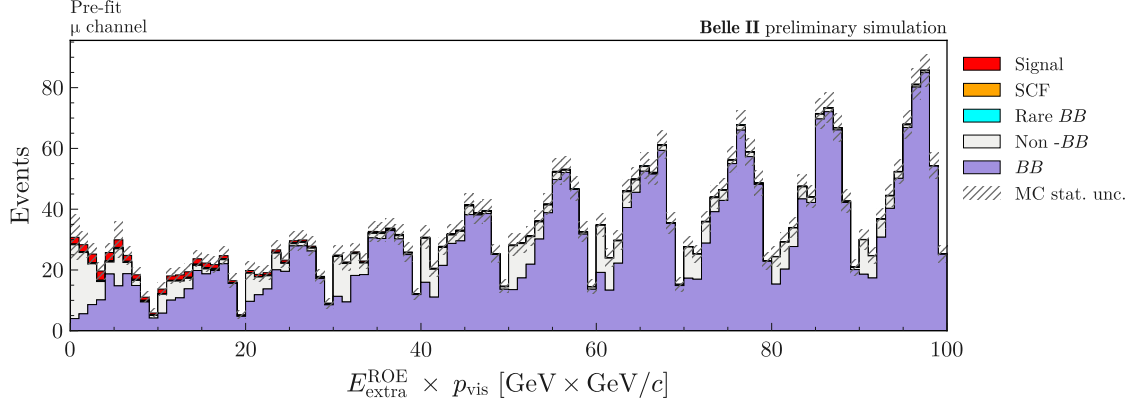


FIG. 58. The $E_{\text{extra}}^{\text{ROE}}$ and p_{CMS} flattened distribution for the μ channel before the fit.

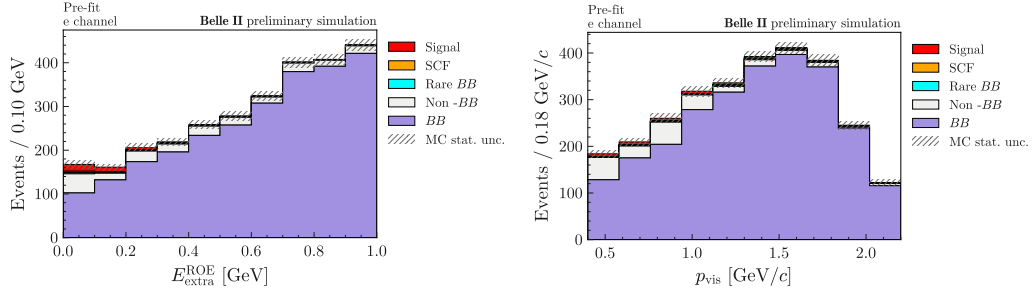


FIG. 59. The $E_{\text{extra}}^{\text{ROE}}$ and p_{CMS} distribution for the e channel before the fit.

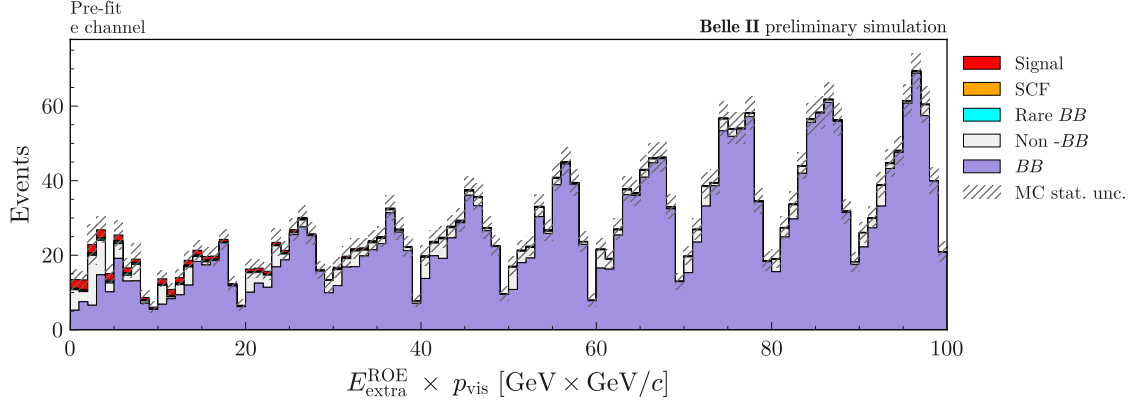


FIG. 60. The $E_{\text{extra}}^{\text{ROE}}$ and p_{CMS} flattened distribution for the e channel before the fit.

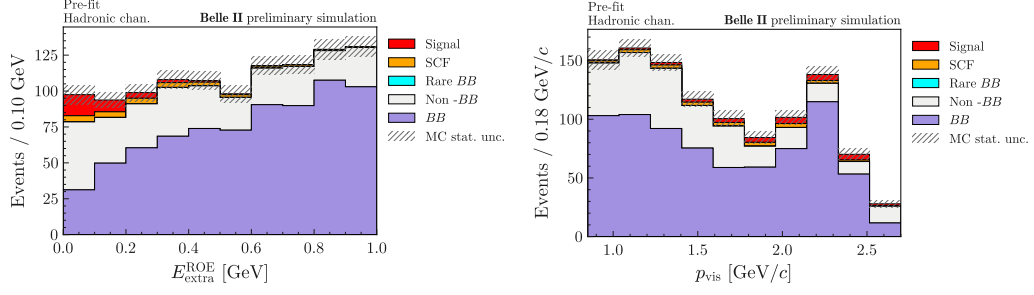


FIG. 61. The $E_{\text{extra}}^{\text{ROE}}$ and p_{CMS} distribution for the hadronic channel before the fit.

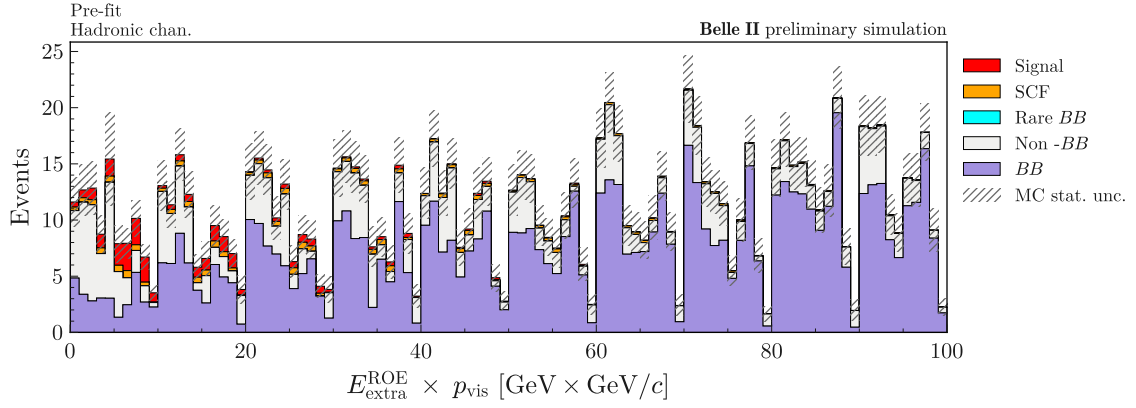


FIG. 62. The $E_{\text{extra}}^{\text{ROE}}$ and p_{CMS} flattened distribution for the hadronic channel before the fit.

765 The result of the fit, μ , is then directly used to compute the branching fraction
 766 of $B^+ \rightarrow \tau^+ \nu_\tau$ decay using:

$$\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau) = \mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau)_{\text{PDG}} \times \mu \quad (11)$$

767 with $\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau)_{\text{PDG}} = (1.09 \pm 0.24) \times 10^{-4}$ [4].

768 We can also directly get the $|V_{ub}|$, using the values from Table 19 and the fitted
 769 value of μ :

$$|V_{ub}| = \sqrt{\frac{8\pi\mu\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau)_{\text{PDG}}}{G_F^2 m_{B^+} m_\tau^2 \left(1 - \frac{m_\tau^2}{m_{B^+}^2}\right)^2 f_B^2 \tau_{B^+}}} \quad (12)$$

Parameter	Value	Source
G_F	$(1.1663785 \pm 0.0000006) \times 10^{-5} \text{ GeV}^{-2}$	[4]
m_{B^+}	$5.27941 \pm 0.00007 \text{ GeV}$	[4]
m_τ	$1.77693 \pm 0.00009 \text{ GeV}$	[4]
f_B	$0.190 \pm 0.0013 \text{ GeV}$	[5]
τ_{B^+}	$(2.489 \pm 0.006) \times 10^{12} \text{ GeV}^{-1}$	[4]

TABLE 19. Inputs used to compute $|V_{ub}|$ from the fitted signal strength μ .

770 The fit is returning an up and down uncertainty on μ from the `minos` method and a
 771 symmetric uncertainty from `minuit`. The uncertainty is then split into statistical and
 772 systematic uncertainties. The statistical uncertainty is obtained in Section 7.2.1 using
 773 a toy study, while the systematic uncertainty is obtained in Section 8 by quadratically
 774 adding all the systematic uncertainties. The total uncertainty of the fit can be
 775 retrieved by quadratically adding the statistical and systematic uncertainties.

776 The expected significance of the signal is also computed directly by `cabinetry`
 777 by fitting the Asimov dataset with the background-only hypothesis ($\mu = 0$) and
 778 comparing it to the nominal fit.

779 With all the systematics included by fitting on the Asimov dataset, the expected
 780 uncertainty on μ is found to be:

$$\mu = 1.000_{-0.404}^{+0.424} \Rightarrow \text{Significance} = 2.6 \sigma \quad (13)$$

781 7.2. Fit validation

782 Multiple checks are performed to validate the fitting procedure, which are de-
 783 scribed in the next sections. For each checks, all the systematics described in Sec-

784 tion 8 are included in the fit as nuisance parameters.

785 7.2.1. Toy study

786 A toy study is performed to check the validity of the fit under Poisson fluctuations.
787 This allows us to get the statistical uncertainty on μ from the fit. The procedure is
788 as follows:

- 789 • Get the Asimov dataset from MC.
- 790 • Fit the Asimov dataset to extract the best fit parameters.
- 791 • Generate 1500 toy datasets from the Asimov dataset by fluctuating each bin of
792 each region according to a Poisson distribution. For the pulls studies, the aux-
793 iliary measurements for the nuisance parameters are also fluctuated according
794 to a Gaussian distribution.
- 795 • Fit each toy dataset with the expected MC, all the parameters (normalization
796 factors and nuisance parameters) are free in the fit.
- 797 • Extract the values of μ .

798 We compute the pull for μ defined as:

$$\text{pull} = \frac{\mu_{\text{fit}} - \mu_{\text{true}}}{\sigma_{\mu_{\text{fit}}}} \quad (14)$$

799 where μ_{fit} is the fitted value of μ , μ_{true} is the true value of μ used to generate the
800 fake data sample, and $\sigma_{\mu_{\text{fit}}}$ is the total uncertainty on the fitted value of μ .

801 Results of the toy study can be seen in Figure 63. The statistical uncertainty on
802 μ is obtained from the width of the distribution of the fitted values of μ , which is
803 expected to be:

$$\sigma_{\text{stat}} = 0.32 \pm 0.01 \quad (15)$$

804 The pull distribution of μ is found to be centered around 0 with a width of
805 compatible with 1 within 2σ .

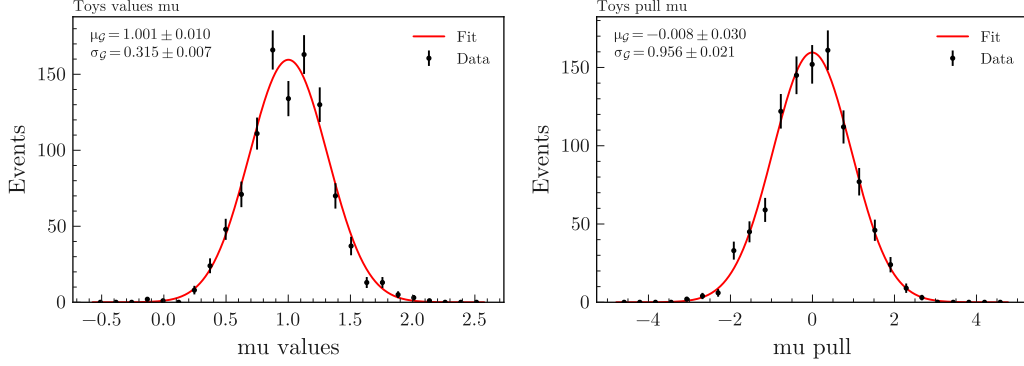


FIG. 63. Distribution of the fitted values of μ from the toy study with a Gaussian fit (left) and distribution of the pulls of μ with a Gaussian fit (right).

7.2.2. Bootstrap study and linearity check

To further check the validity of the fit, we perform a study using the bootstrap method. The procedure is as follows:

- For each sample used in the fit, generate a fake data sample by sampling with replacement from the original sample. The number of sampled candidates in the fake data sample is a Poisson fluctuation of the sum of the weights of the original sample multiplied by the Data/MC scale factor. The sampling uses the event weights as probabilities.
- Fit the fake data sample with the expected MC, with all the parameters (normalization factors and nuisance parameters) free in the fit.
- Extract the values of μ .

This procedure is repeated 250 times for different signal scaling factors, varying from 0.0 to 3.5, corresponding to a significance of the signal varying from 0 to around 5σ .

The mean of the values of μ as a function of the signal factor can be seen in Figure 64. A linear fit is performed to check the linearity of the fit. A good agreement is observed.

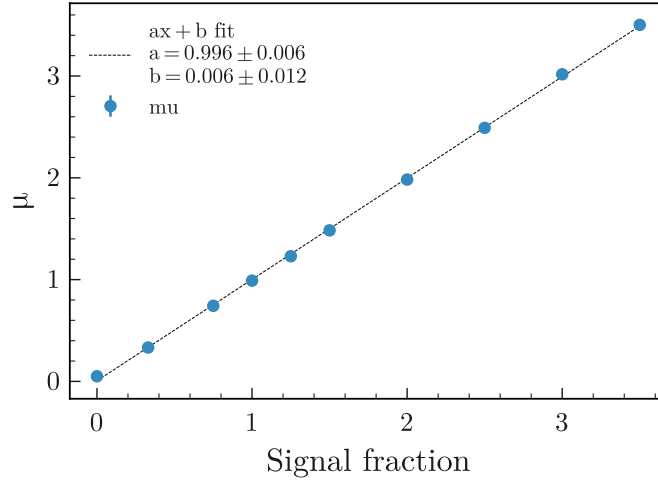


FIG. 64. Mean of the fitted values of μ as a function of the signal factor, with a linear fit.

823 8. SYSTEMATICS

824 An additional set of corrections is applied to the MC samples, all leading to
825 systematic uncertainties in the final measurement, added as nuisance parameters
826 (NPs) in the fit. These corrections are described in the following sections.

827 8.1. ROE corrections

828 To implement the systematic uncertainties in the fit, we use the eigenvector
829 method developed in the `sysvar` package. The covariance matrix is built by draw-
830 ing 500 random variations of the correction weights according to their uncertainties.
831 The covariance matrix is then diagonalized to obtain the eigenvectors and eigen-
832 values. Each eigenvector represents a direction in the parameter space along which
833 the weights can be varied to account for the systematic uncertainties. After some
834 tests to see the impact of the number of eigenvectors, we decided to keep either as
835 many eigendirections as fully correlated NPs as needed to reconstruct the original
836 covariance matrix with differences smaller than 10^{-4} , or a maximum between 1 and 9
837 eigendirections per correction table according to their impact on the POI, whichever
838 is smaller. In addition, all the remaining eigendirections are quadratically merged
839 into an additional NP. This results in a total of 26 NPs for the ROE corrections
840 systematics. These variations are then implemented in the fit as `normsys+histosys`
841 NPs, where the normalization variation accounts for the overall change in yield due
842 to the PID correction variation, and the histogram variation accounts for the shape
843 change in the templates. The two components are kept fully correlated.

844 We can provide detailed information for each individual systematic uncertainty
845 using `sysvar` visualization API per request during the review. `sysvar` visualization
846 API automatically plots the correction weight variations, the covariance matrix of
847 the corrections, either explicitly defined or build from the different set of uncertain-
848 ties, the varied templates across all reconstruction channels and the full correlation
849 matrix across all bins, templates, reconstruction channels. We avoid providing this
850 information for all systematics by default in this document since this would result
851 to an extremely large number of plots. The setup used to generate the eigenvectors
852 is available on gitlab.

853 8.2. FEI corrections

854 Using the same method as described in Section 8.1, the FEI corrections described
855 in Section 5.1 are implemented in the fit as `normsys+histosys` NPs in the model.

Each channel has 4 NPs associated with the continuum corrections, the $B\bar{B}$ calibration corrections and the D_{tag}^0 corrections for the μ channel. This results in a total of 40 NPs for the FEI corrections systematics.

8.3. p_{CMS} corrections

Using the same method as described in Section 8.1, the p_{CMS} corrections described in Section 5.3 are implemented in the fit as **normsys+histosys** NPs in the model. Each channel can have up to 30 NPs associated with the p_{CMS} correction, affecting only the $B\bar{B}$ candidates. This results in a total of 51 NPs for the p_{CMS} corrections systematics.

8.4. Tracking efficiency

The tracking efficiency uncertainty is applied as a percentage per track. The current correction is 0.27%/track. Since we only have one charged track from the signal τ decay, the total uncertainty is 0.27% for all τ channels. This uncertainty is applied as a single **histosys** NP affecting all templates in all channels, varying the event weights up and down by 0.27%.

8.5. Charged particle identification

8.5.1. Corrections and implementation in the fit

We correct for charged-particle identification efficiencies using the **syscorr** framework and apply the weights using the **sysvar** package. The PID correction tables used are:

- μ efficiency correction for the **FixedThresh09** PID selection for the muon channel.
- e efficiency correction for the **FixedThresh09** PID selection for the electron channel.
- π efficiency correction for the **pionID_noSVD** > 0.6 selection for the hadronic channels.
- $K \rightarrow \mu$ fake rate correction for the **FixedThresh09** PID selection for the muon channel.

- $K \rightarrow e$ fake rate correction for the `FixedThresh09` PID selection for the electron channel.
- $K \rightarrow \pi$ fake rate correction for the `pionID_noSVD > 0.6` selection for the hadronic channels.
- $\pi \rightarrow \mu$ fake rate correction for the `FixedThresh09` PID selection for the muon channel.
- $\pi \rightarrow e$ fake rate correction for the `FixedThresh09` PID selection for the electron channel.

The correction tables provide weights binned in track charge, momentum and polar angle. In each bin the weight is defined as the ratio of the efficiency (or fake rate) in data to the efficiency (or fake rate) in MC, measured on dedicated calibration samples. For each bin, the statistical and systematic uncertainties are provided.

All the corrections are already provided in `sysvar` from the performance group, except the pion ID corrections, which are manually computed using `syscorr`. Figure 65 shows the pion ID efficiency correction table used for the hadronic channels.

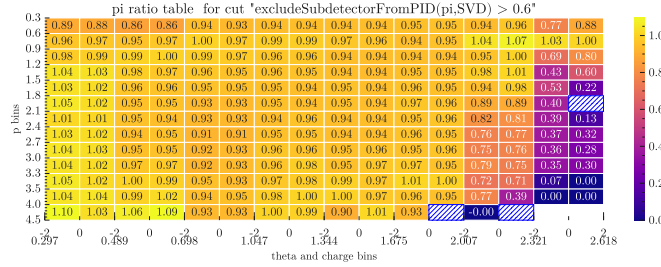


FIG. 65. Pion ID efficiency correction table used for the hadronic channels, computed using `syscorr`. The weights are binned in pion momentum, θ and charge.

We use `sysvar` to implement this systematic. We decided to keep either as many eigendirections as fully correlated NPs as needed to reconstruct the original covariance matrix with differences smaller than 10^{-4} , or a maximum of 2 to 3 eigendirections per correction table according to the impact on the POI, whichever is smaller. This results in a total of 27 NPs for the charged PID corrections systematics. These variations are then implemented in the fit as `normsys+histosys` NPs, where the normalization variation accounts for the overall change in yield due to the PID correction variation, and the histogram variation accounts for the shape change in the templates.

908 8.5.2. *light-release issue correction*

909 It was reported during the <https://indico.belle2.org/event/16060/timetable/> that
910 some charged PID corrections provided with old light-release versions were incor-
911 rectly computed with new light-release versions due to changes in how the input
912 parameters definitions. This issue affects the muon and electron PID corrections
913 used in this analysis. To correct for this issue, the performance group recommended
914 to compare the old and new light-release efficiencies and, if not too large, derive
915 a systematic uncertainty from the difference. For the background candidates, we
916 found that the difference in efficiency is less than 1% across the different samples.
917 For the signal sample, we found a difference of 1.6%. To be conservative, we sum
918 this difference linearly with the systematic uncertainty we get from the eigenvector
919 method described previously.

920 8.6. Neutral particle identification

921 For the ρ channel, we apply a correction for the π^0 identification efficiency for the
922 `eff40` selection using `syscorr` and `sysvar`. The correction table provides weights
923 binned in π^0 momentum and $\cos\theta$. In each bin the weight is defined as the ratio
924 of the efficiency in data to the efficiency in MC, measured on dedicated calibration
925 samples. For each bin, the statistical and systematic uncertainties are provided.

926 Using the same eigenvector method as described in Section 8.5, we implement the
927 systematic uncertainties in the fit. This results in a total of 3 NPs for the neutral
928 PID corrections systematics.

929 8.7. Branching fractions of background decays

930 8.7.1. *Main background composition*

931 The $B\bar{B}$ background template is our main background contribution in the fit. The
932 composition of this background in the signal region can be seen in Figures 66 and 67.
933 The dominant contributions come from semileptonic B decays, mainly $D^{(*)}\ell\nu$, $D\ell\nu$,
934 $\pi^0\ell\nu$. We can also find some rare background modes generated by `PYTHIA`, which is
935 mostly $X_u\ell\nu$ decaying to charged or neutral pions. The composition of these rare
936 background modes from `PYTHIA` are not well modeled in the generic MC generation,
937 thus they are studied in detail in Section 8.7.2.

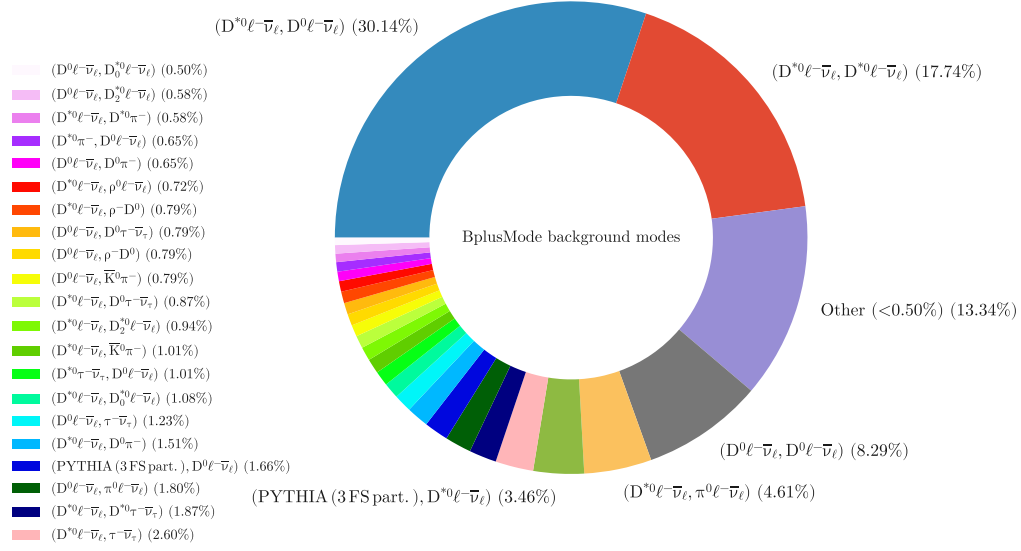


FIG. 66. Composition of the B^+B^- background in the signal region after the full selection.

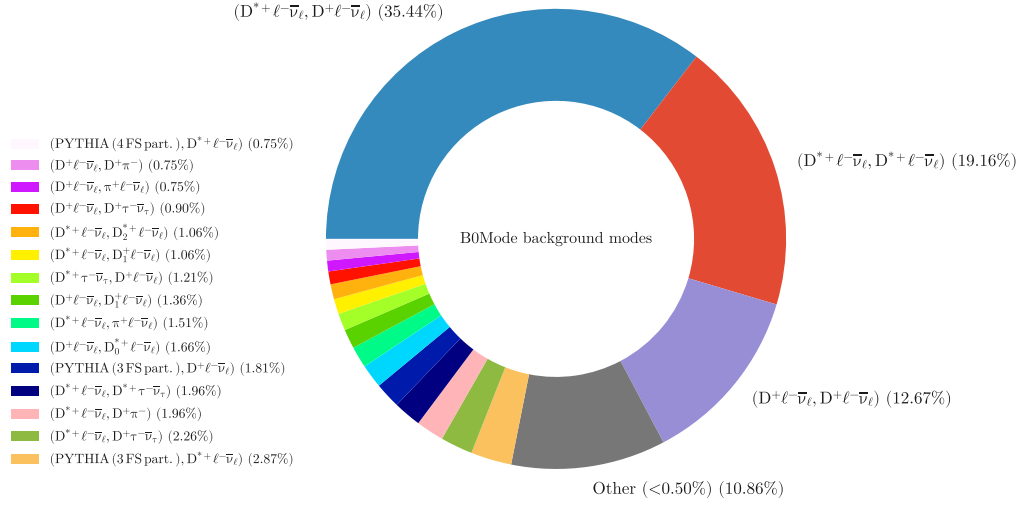


FIG. 67. Composition of the $B^0\bar{B}^0$ background in the signal region after the full selection.

938 The uncertainties on the branching fractions of the various background modes
939 can impact the shape and normalization of the $B\bar{B}$ template. To account for these
940 uncertainties, each event has a new weight corresponding to the ratio of the branching
941 fraction in the PDG [4] to the branching fraction in the MC generation. For the
942 shape variations, **sysvar** is used to compute the upper and lower variations of the
943 $B\bar{B}$ template by varying the branching fractions up and down by their uncertainties
944 in the PDG. These variations are then implemented in the fit as **normsys+histosys**
945 NPs. The branching fractions that are corrected and their uncertainties are listed in
946 Tables 20 and 21. For all the decay modes, the HFLAV averages are taken if available.
947 For some decay modes, the uncertainties are quite large due to the imprecise values
948 in generic MC generation. In these cases, we use an arbitrary 100% uncertainty,
949 indicated by a †. This uncertainty does not impact significantly the final result due
950 to the small contribution of these modes in the signal region.

951 After some tests to see the impact of the number of eigenvectors, we decided to
952 keep either as many eigendirections as fully correlated NPs as needed to reconstruct
953 the original covariance matrix with differences smaller than 10^{-4} , or a maximum of
954 6 eigendirections, whichever is smaller. We correct both B_{sig} and B_{tag} branching
955 fractions where applicable, on the $B\bar{B}$, SCF and signal templates. This results in a
956 total of 28 NPs for the branching fraction systematics.

Decay mode PDG value $\pm$ unc.		Dec File value
$D^{*0}\ell\nu$	0.0560 ± 0.0010	0.0549
$D^0\ell\nu$	0.0226 ± 0.0007	0.0231
$\pi^0\ell\nu$	0.000078 ± 0.0000027	0.000078
$X_u\ell\nu$	$0.00150 \pm 0.00024^{\text{a}}$	0.001724
$D^0\pi^+$	0.00461 ± 0.00010	0.00467517
$D^{*0}\tau\nu$	0.0188 ± 0.0020	0.0141642
$D_0^{*0}\ell\nu$	0.0013 ± 0.0019	0.00389
$D_2^{*0}\ell\nu$	0.0032 ± 0.0003	0.00373
$D^0\tau\nu$	0.0077 ± 0.0025	0.0069069
$K^0\pi^+$	0.0000239 ± 0.0000006	0.00002368
$D^0\rho^+$	0.0097 ± 0.0011	0.00939

^a Partial to full BR + removed resonant contributions

TABLE 20. List of B^+B^- branching fractions of background modes corrected in the fit and their uncertainties taken from the PDG [4] and $D^*\ell\nu$ analysis [22].

Decay mode	PDG value $\pm$ unc.	Dec File value
$D^{*+}\ell\nu$	0.0487 ± 0.0009	0.0511
$D^+\ell\nu$	0.0210 ± 0.0007	0.0214
$D^+\tau\nu$	0.0098 ± 0.0021	0.0063986
$X_u\ell\nu$	0.00132 ± 0.00022 ^a	0.001701
$D^+\pi^-$	0.00251 ± 0.00008	0.00252113
$D_0^{*+}\ell\nu$	0.0012 ± 0.0020	0.00362
$D^{*+}\tau\nu$	0.0148 ± 0.0009	0.0131838
$\pi^+\ell\nu$	0.000150 ± 0.000005	0.00015
$D_1^+\ell\nu$	0.00637 ± 0.00067 ^b	0.00704
$D_2^{*+}\ell\nu$	0.0032 ± 0.0003	0.00347

^a Partial to full BR + removed resonant contributions

^b Combination of multiple BF from the PDG: $\Gamma_{16} + \Gamma_{17} + \Gamma_{18}$

TABLE 21. List of $B^0\bar{B}^0$ branching fractions of background modes corrected in the fit and their uncertainties taken from the PDG [4] and $D^*\ell\nu$ analysis [22].

957 8.7.2. Rare $B\bar{B}$ background

958 The composition of the rare background modes generated by PYTHIA in the signal
959 region is shown in Figures 68 to 71.

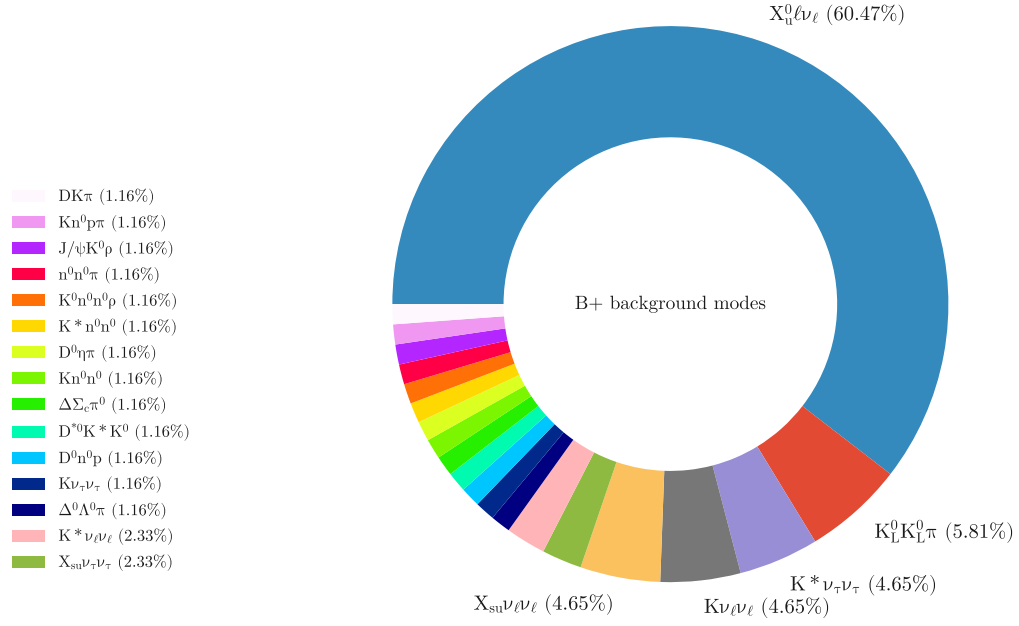


FIG. 68. Composition of the $B^+ B^-$ background from PYTHIA in the signal region after the full selection.

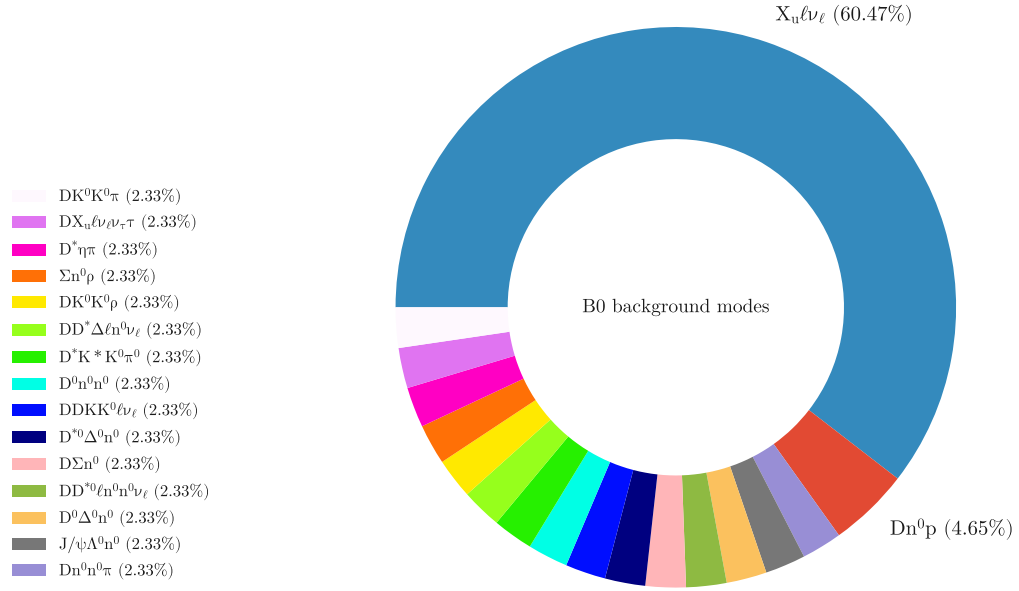


FIG. 69. Composition of the $B^0\bar{B}^0$ background from PYTHIA in the signal region after the full selection.

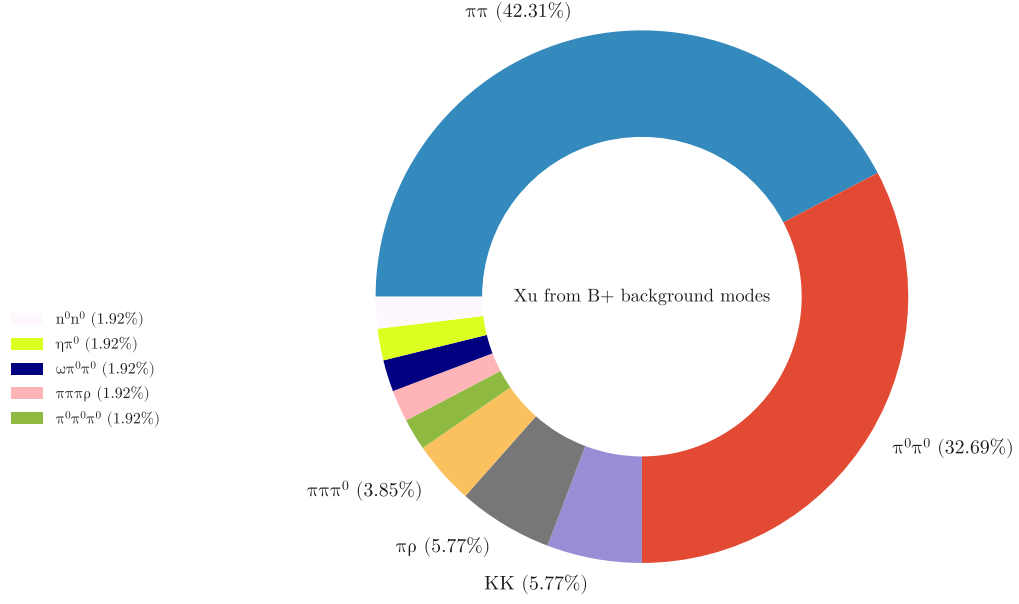


FIG. 70. Composition of the $X_u\ell\nu$ background from PYTHIA in the signal region after the full selection for B^+B^- .

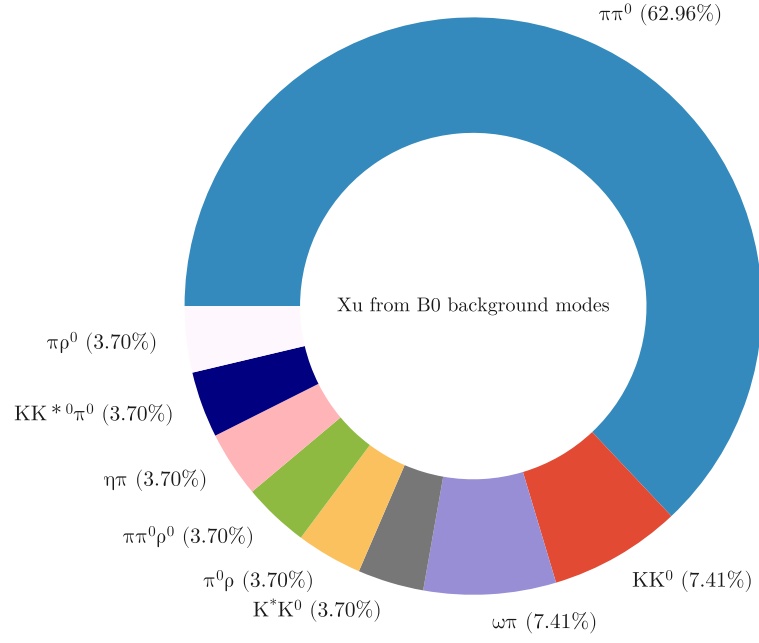


FIG. 71. Composition of the $X_u\ell\nu$ background from PYTHIA in the signal region after the full selection for $B^0\bar{B}^0$.

960 In generic MC generation, the X_u resonances part are simulated with EvtGen,
 961 while the non-resonant part and the contribution of broad higher resonances are
 962 covered by Pythia. To have a better understanding and simulation of the non-
 963 resonant part, we generated additional large MC samples of these modes to study
 964 the efficiency of the selection, and see if some remains in the signal region after the
 965 full selection. The modes with $p\bar{p}$, $n\bar{n}$, K^+K^- and $K_L^0K_L^0$ in the final state, the
 966 form factors used in PYTHIA show non-uniform distribution [23, 24]. We generate
 967 10M events for each of these modes to have a good estimate of their contribution
 968 in the signal region. We also save the invariant mass of the hadronic system to
 969 do the reweighting due to the threshold enhancement. For the other modes, we
 970 generate 2M events each. Table 22 summarizes the different modes that could be a
 971 peaking background in the signal region, along with the number of events generated,
 972 reconstructed, and remaining after the full selection. The total efficiency of each
 973 mode is also provided.

974 Table 23 shows the expected number of events in the signal region for each mode
 975 for LS1 dataset, taking into account the branching fractions from different sources
 976 and the non-uniform distribution of the hadronic system invariant mass where ap-
 977 plicable. For $n\bar{n}\ell\nu$, we use the same BF as for $p\bar{p}\ell\nu$ from the PDG [4]. No mea-
 978 surement is available for the $K_L^0K_L^0\ell\nu$, so we estimate it from the measured BF of
 979 $B^+ \rightarrow K^+K^-\ell\nu$ [24]. For the estimation of the branching fraction $\mathcal{B}(K_LK_L\ell\nu)$
 980 relative to the measured $\mathcal{B}(K^+K^-\ell\nu)$, we consider the following isospin relations:

- 981 • For a pure 0^{++} resonance, the branching fractions satisfy

$$\mathcal{B}(K^+K^-) : \mathcal{B}(K_SK_S) : \mathcal{B}(K_LK_L) = 2 : 1 : 1,$$

982 leading to $\mathcal{B}(K_LK_L\ell\nu) = \frac{1}{2} \mathcal{B}(K^+K^-\ell\nu)$.

- 983 • For a pure 1^{--} resonance,

$$\mathcal{B}(K^+K^-) : \mathcal{B}(K_SK_L) = 1 : 1,$$

984 implying no K_LK_L contribution.

- 985 • In the absence of a resonance, a non-resonant $K\bar{K}$ system yields approximately
 986 $\frac{1}{4} \mathcal{B}(K^+K^-\ell\nu)$.

987 A conservative assumption of $(1/4 \pm 1/4) \mathcal{B}(K^+K^-\ell\nu)$ is therefore adopted for
 988 $\mathcal{B}(K_LK_L\ell\nu)$.

989 For most of the modes, the efficiency after the full selection is quite small, of the
 990 order of 10^{-3} or less. The modes with the highest efficiency are the ones with $n\bar{n}$ and
 991 $K_L^0K_L^0$ in the final state, reaching up to 1%. These two rare background modes are

Decay mode	# gen.	# after reco.	# after full sel. (# in SR)	Tot. eff. (in SR) (%)
$B^+ \rightarrow (a, b, f, h)^a e\nu_e$	1992205	4247	709.0 (38.4)	0.0356 (0.0019)
$B^+ \rightarrow (a, b, f, h)^a \mu\nu_\mu$	1888220	4217	644.4 (17.0)	0.0341 (0.0009)
$B^0 \rightarrow (a, b)^b e\nu_e$	1879042	2860	181.4 (7.1)	0.0097 (0.0004)
$B^0 \rightarrow (a, b)^b \mu\nu_\mu$	1992213	2954	151.5 (11.2)	0.0076 (0.0006)
$B^+ \rightarrow p\bar{p}e\nu_e$	9960825	3239	313.4 (51.9)	0.0031 (0.0005)
$B^+ \rightarrow p\bar{p}\mu\nu_\mu$	9684430	3192	333.9 (63.5)	0.0034 (0.0007)
$B^+ \rightarrow n\bar{n}e\nu_e$	9960867	154031	61406 (11229)	0.6164 (0.1127)
$B^+ \rightarrow n\bar{n}\mu\nu_\mu$	9960876	149943	59442 (9955)	0.5968 (0.0999)
$B^+ \rightarrow K^+ K^- e\nu_e$	9175586	9061	2192.0 (181.9)	0.0239 (0.0020)
$B^+ \rightarrow K^+ K^- \mu\nu_\mu$	9960818	9430	1994.8 (184.5)	0.0200 (0.0019)
$B^+ \rightarrow K_L^0 K_L^0 e\nu_e$	9960858	163261	83884 (14113)	0.8421 (0.1417)
$B^+ \rightarrow K_L^0 K_L^0 \mu\nu_\mu$	9960857	159840	83830 (12850)	0.8416 (0.1290)

^a With $a = a_{00}, a_{10}, a_{20}$; $b = b_{10}$; $f = f_0, f_1, f_2$; $h = h_1$.

^b With $a = a_{00}, a_{10}, a_{20}$; $b = b_{10}$.

TABLE 22. List of potentially peaking background modes from $X_u \ell \nu$ decays and the number of events generated in dedicated signal only samples to study their contribution in the signal region.

also expected to peak in the signal region. We decided to include these two modes in the fit as separate **rare** templates, each with a single **normsys** NP to account for the uncertainty on their branching fractions. The $E_{\text{extra}}^{\text{ROE}}$ and p_{CMS} distributions of these two modes after the full selection can be seen in Figure 72. In addition, the shape of the two modes can vary independently thanks to two additional **histosys** NPs in the fit, one for each mode. The uncertainty on the shape is set to 10%, estimated from the $B \rightarrow K \nu \nu$ analysis (see Appendix V and R). Finally, if these modes are found in generic MC samples, we remove them to avoid double counting.

For the other modes, their expected contribution in the signal region is negligible (less than 1 event each), so we do not include them in the fit.

Decay mode	BF	Expected # in SR
$B^+ \rightarrow (a, b, f, h)e\nu_e$	0.0001 ^a	0.7
$B^+ \rightarrow (a, b, f, h)\mu\nu_\mu$	0.0001 ^a	0.4
$B^0 \rightarrow (a, b)e\nu_e$	0.0002 ^a	0.3
$B^0 \rightarrow (a, b)\mu\nu_\mu$	0.0002 ^a	0.4
$B^+ \rightarrow p\bar{p}e\nu_e$	0.0000058 [4]	0.01
$B^+ \rightarrow p\bar{p}\mu\nu_\mu$	0.00000532 [4]	0.01
$B^+ \rightarrow n\bar{n}e\nu_e$	0.0000058 ^b	2.6
$B^+ \rightarrow n\bar{n}\mu\nu_\mu$	0.00000532 ^b	2.1
$B^+ \rightarrow K^+K^-e\nu_e$	0.0000305 [24]	0.24
$B^+ \rightarrow K^+K^-\mu\nu_\mu$	0.0000305 [24]	0.22
$B^+ \rightarrow K_L^0K_L^0e\nu_e$	0.000007625 ^c	4.3
$B^+ \rightarrow K_L^0K_L^0\mu\nu_\mu$	0.000007625 ^c	3.9

^a Belle decay.dec (inflated)

^b Same as $p\bar{p}\ell\nu$

^c $\mathcal{B}_{B^+ \rightarrow K^+K^-\ell\nu} \times (1/4 \pm 1/4)$

TABLE 23. Expected number of events in the signal region for each potentially peaking background mode from $X_u\ell\nu$ decays for LS1 dataset, taking into account the branching fractions and the non-uniform distribution of the hadronic system invariant mass where applicable.

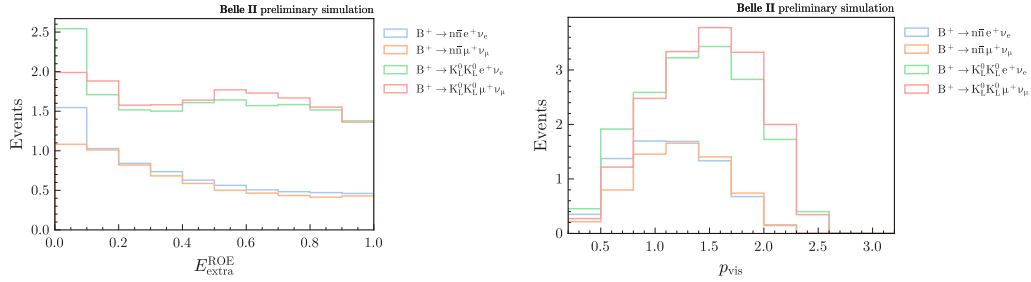


FIG. 72. $E_{\text{extra}}^{\text{ROE}}$ (left) and p_{CMS} (right) distributions of the two rare background modes kept in the fit. The number of events is normalized to LS1 dataset.

1002 8.8. $D^{(*)}\ell\nu_\ell$ form factors

1003 Since $D^{(*)}\ell\nu_\ell$ decays are the dominant background contribution, we include in
 1004 the fit a systematic linked to the form factors (FF) used to generate these decays.
 1005 FF model and its parametrization can alter the kinematic distributions of the decay
 1006 products, which can impact the shape of the fit templates. In Belle II, $D^{(*)}\ell\nu_\ell$
 1007 decays assumes the BGL FF parametrization [25, 26]. We use the Helicity Amplitude
 1008 Module for Matrix Elements Reweighting (HAMMER) [27] software to reweight the
 1009 the dataset from an assumed FF parametrization to a target one. To find more
 1010 detailed information on HAMMER, please refer to [28, 29]. We chose the BLPRXP
 1011 FF parametrization [30] as target model to reweight both $D^*\ell\nu_\ell$ and $D\ell\nu_\ell$ decays.
 1012 Figures 73 and 74 show the set of parameters that have been used for the production
 1013 of the MC samples in Belle II and the target model respectively.

1014 The output of HAMMER can directly be used to create the up and down vari-
 1015 ations of the $D^{(*)}\ell\nu_\ell$ templates in the fit using `sysvar`. BLPRXP is composed of
 1016 9 parameters, resulting in 9 NPs for each B meson decay, giving 18 NPs in total.
 1017 The variations are then implemented in the fit as `normsys+histosys` NPs on $B\bar{B}$
 1018 samples, where the normalization variation accounts for the overall change in yield
 1019 due to the FF variation, and the histogram variation accounts for the shape change
 1020 in the templates.

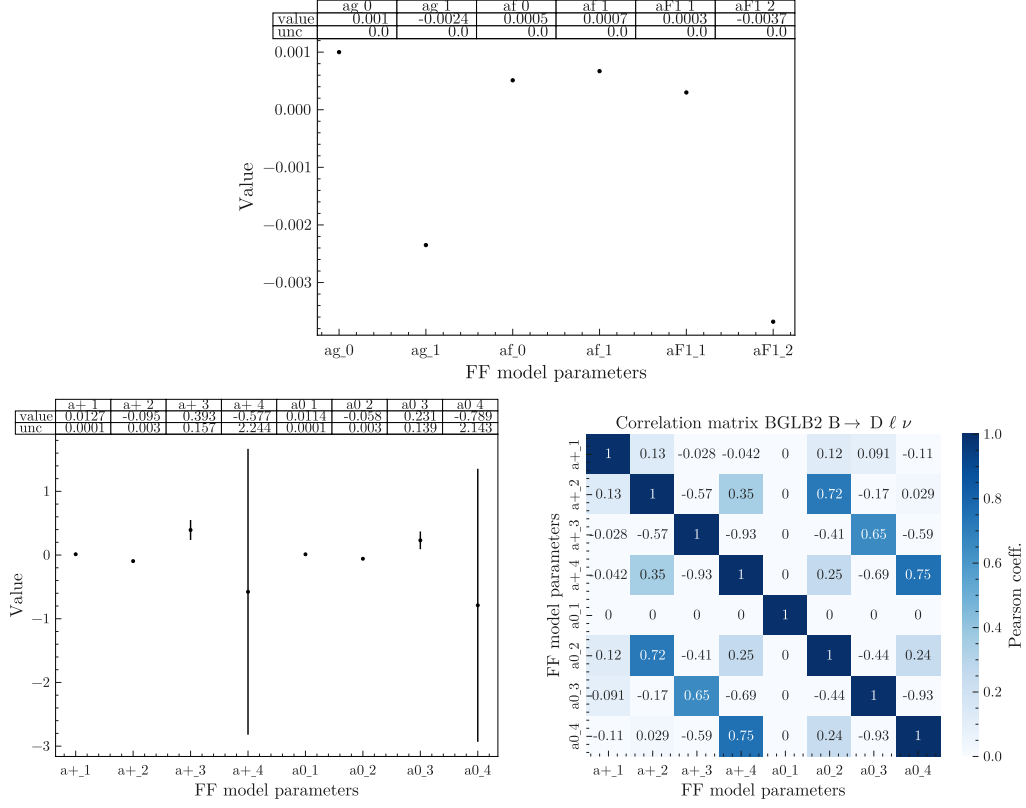


FIG. 73. Parameters used in the BGL parametrization for $D^{(*)} \ell \nu_\ell$ (top) and $D \ell \nu_\ell$ (bottom) decays in the Belle II MC.

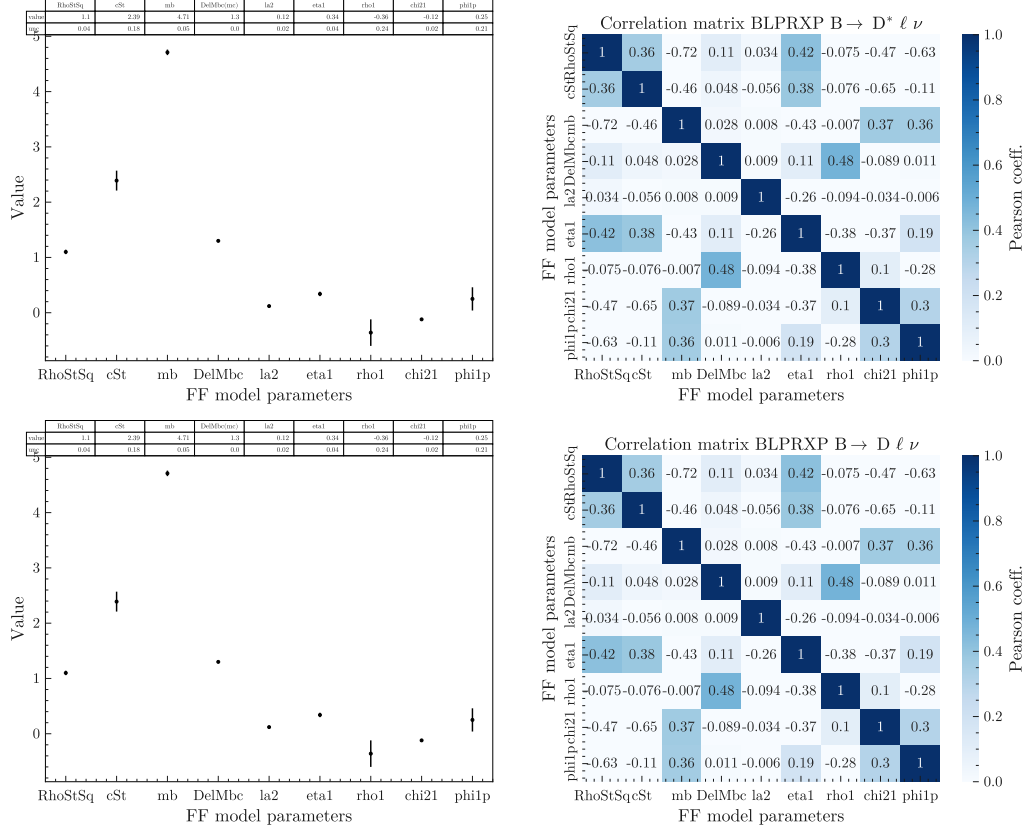


FIG. 74. Target parameters used in the BLPRXP parametrization for $D^{(*)} \ell \nu_\ell$ (top) and $D \ell \nu_\ell$ (bottom) decays for the reweighting with HAMMER.

1021 **8.9. MC statistics**

1022 The number of events in MC samples being finite, the associated statistical needs
 1023 to be taken into account. To do so, we use the `statererror` modifier in `pyhf`, which
 1024 is calculated as:

$$\sigma_{\text{bin}} = \sqrt{\sum_i w_i^2} \quad (16)$$

1025 where w_i is the weight of each event in the bin. This creates one NP per bin of
 1026 each template to account for the statistical fluctuations. This results in 300 NPs for
 1027 the MC statistics systematics.

1028 **8.10. f_{+-} and f_{00}**

1029 We vary the fraction of $\Upsilon(4S) \rightarrow B^+B^-/\Upsilon(4S) \rightarrow B^0\bar{B}^0$ according to the value
 1030 obtained by HFLAV [31]: $f_{+-/00} = 51.1 \pm 1.1\%$.

1031 **8.11. Number of produced $\Upsilon(4S)$**

1032 The total number of produced $\Upsilon(4S)$ between 2019 and 2021 is estimated to be
 1033 $n_{\Upsilon(4S)} = (387.1 \pm 5.6) \times 10^6$.

1034 **8.12. Summary of and impact of the systematics in the fit**

1035 A summary of the systematic uncertainties considered in this analysis is given in
 1036 Table 24 along with the number of associated NPs for each source. The summary
 1037 of the NPs distribution per sample and channel can be seen in Figure 75, and the
 1038 correlation matrix of all the NPs is provided in Appendix F.b.

Source	Number of NPs
MC statistics	300
ROE corrections	26
p_{CMS} corrections	51
Branching fractions of background decays	28
FEI corrections	40
Tracking efficiency	1
Neutral particle identification	3
Charged particle identification	27
$D^{(*)}\ell\nu_\ell$ form factors	18
$B\bar{B}$ normalization	1
non- $B\bar{B}$ normalization	1
Rare $B\bar{B}$ normalization	1
Rare background shape	2
SCF shape	1
Signal efficiency	-
Fit bias	-
Fraction of B^+B^- pairs	-
Number of produced $\Upsilon(4S)$	-

TABLE 24. Summary of systematic uncertainties considered in the analysis and number of associated nuisance parameters (NPs).

8.12.1. Method to evaluate the impact

To estimate the impact of each source of systematic uncertainty on the signal strength μ , we use the same method as described in the $B^0 \rightarrow K^{*0} \tau^+ \tau^-$ analysis [32], which follows the methodology described in Ref. [33]:

- Every NP's auxiliary measurement is shifted up and down by one standard deviation, not fixed, and allowed to vary within the same constraint.
- A fit (with all NPs free) is performed for each variation, and the resulting signal strength μ is saved.
- The difference between the nominal signal strength and the one obtained for each variation is taken as the impact of that NP on μ .
- For each source of systematic uncertainty, the individual impacts of all associated NPs are summed in quadrature to obtain the total impact of that source on μ .

This method has been validated (see here) and its `cabinetry` implementation is used.

8.12.2. Impact of the systematics on the signal strength

The pull and the impact of each systematic can be found in Appendices F.c and F.d. The impact of the different sources of systematic uncertainties on the signal strength μ is summarized in Table 25. As expected, the dominant sources of systematic uncertainties are the MC statistics, followed by the ROE corrections. These two systematics are limited by statistics and are reducible in the future. The total systematic uncertainty is obtained by summing in quadrature the individual contributions. The final result for the signal strength including statistical and systematic uncertainties is:

$$\mu = 1.0 \pm 0.32 \pm 0.29 \quad (17)$$

which result in a expected branching fraction of:

$$\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau) = (1.09 \pm 0.35(\text{stat}) \pm 0.32(\text{syst})) \times 10^{-4} \quad (18)$$

and an expected value of $|V_{ub}|$ of:

$$|V_{ub}| = (4.14 \pm 0.66(\text{stat}) \pm 0.60(\text{syst})) \times 10^{-3} \quad (19)$$

Source	Systematics (= impact on μ) (%)
MC statistics	16.9
ROE corrections	17.6
p_{CMS} corrections	5.8
Branching fractions of background decays	4.1
FEI corrections	6.3
Tracking efficiency	0.3
Neutral particle identification	0.5
Charged particle identification	3.8
$D^{(*)}\ell\nu_\ell$ form factors	< 0.1
$B\bar{B}$ normalization	0.2
non- $B\bar{B}$ normalization	5.8
Rare $B\bar{B}$ normalization	0.8
Rare $B\bar{B}$ shape	0.1
SCF shape	1.4
Signal efficiency	10.0
Fit bias	1.6
Number of produced $\Upsilon(4S)$	1.5
Fraction of $B\bar{B}$ pairs	2.1
Total	29.1 ± 1.0

TABLE 25. Impact of the different sources of systematic uncertainties on the signal strength μ , with the total systematic uncertainty obtained by summing in quadrature the individual contributions. The uncertainty on the total is obtained by summing the positive and negative impacts separately.

9. UNBLINDING PROCEDURE

This section presents the unblinding procedure and the results for each step.

9.1. Unblinding steps

The unblinding procedure consists of the following steps (inspired by the $B^+ \rightarrow \tau^+ \nu_\tau$ with hadronic tagging analysis [13]):

1. We check the μ channel, in which we expect the highest uncertainty. The fitted POI is kept blinded (e.g. by adding a hidden offset). We first check the pulls and pre-fit distributions. If everything looks good, we check the post-fit distributions and compare the uncertainty to the expected one with Asimov dataset.
2. We check the e channel. The fitted POI is kept blinded. The same procedure as for the μ channel is applied.
3. We check the hadronic channels. The fitted POI is kept blinded. The same procedure as for the μ channel is applied. In addition, we compare the consistency between the 3 blinded POIs.
4. We split the data into 3 independent sub-samples and perform the simultaneous fit on the 3 channels on each sample. We check the consistency of the POI in the different sub-samples. The fitted POIs are kept blinded.
5. We perform the final fit, unblind the central value of the fit and report the final result.

9.2. Results of the unblinding steps

9.2.1. Step 1: μ channel

The μ channel is the one with the highest uncertainty. The pre-fit distributions are shown in Figure 76. Good agreement is observed, but a slight increase in the second bin of $E_{\text{extra}}^{\text{ROE}}$ is observed.

The pulls of each NPs are shown in Figure 77. The x-axis shows the pulls of the NPs, where $\hat{\theta}$ is the post-fit value of the NPs, θ_0 is the pre-fit central value, and $\Delta\theta$ is the pre-fit uncertainty. The pulls of the NPs are all within 1σ . The non- $B\bar{B}$ background normalization is pulled down by a factor 0.83, which is still compatible

1094 with the expected distributions. Additionally, the trend is seen on the other channel.
 1095 The small uncertainty on the $B\bar{B}$ background normalization is due to the fact that
 1096 the `normsys` modifier is Gaussian constraining the normalization, and we have a
 1097 large range for this parameter (see Section 7.1).

1098 The post-fit distributions are shown in Figure 78. The p-values of the fit is
 1099 $p = 0.318$, which is compatible with the expected distribution. Good agreement is
 1100 observed. The fitted POI is kept blind, and we find an uncertainty of:

$$\sigma_{\mu} = 1.01^{+0.07}_{-0.03} \quad (20)$$

$$\sigma_{\mu_{\text{Asimov}}} = 0.88^{+0.04}_{-0.01} \quad (21)$$

1101 where the central value comes from MINUIT, while the upper and lower uncer-
 1102 tainties come from the MINOS algorithm. The uncertainty is slightly higher than the
 1103 expected one, but still compatible within 2σ .

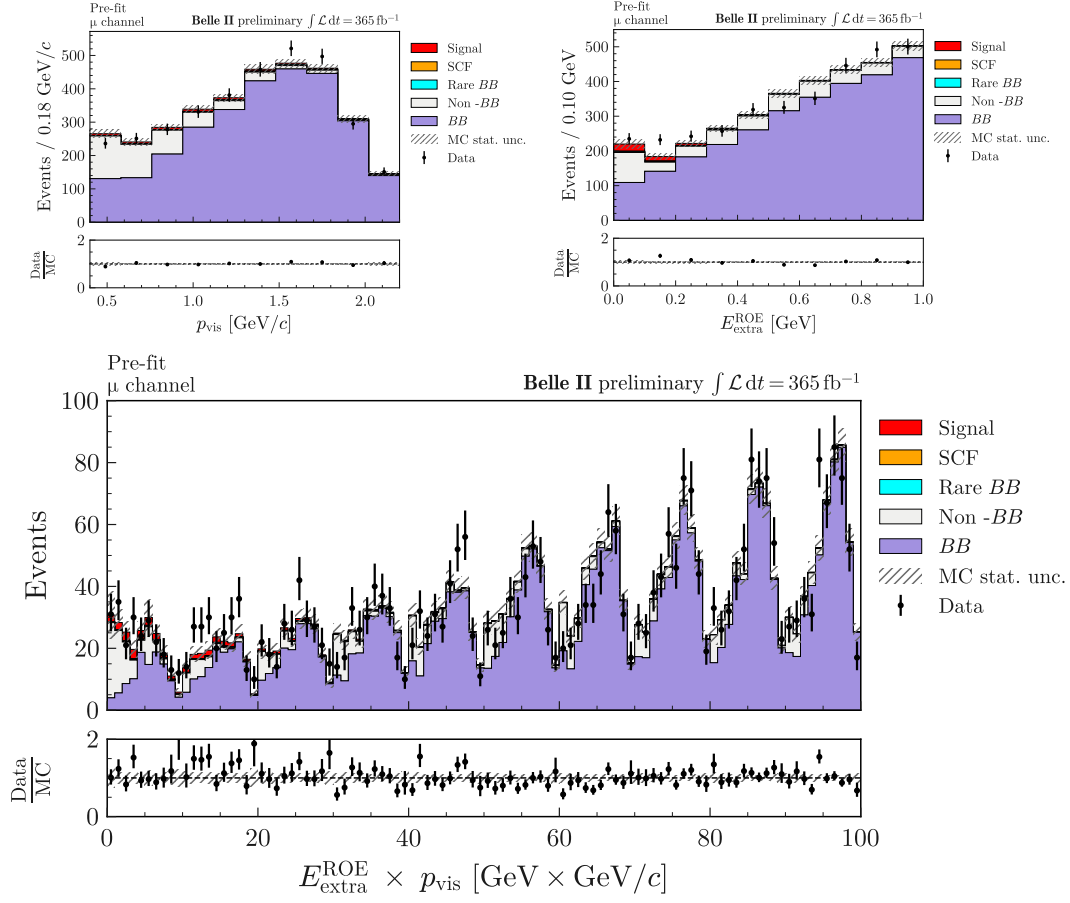


FIG. 76. Pre-fit distributions for the μ channel. The top left plot shows the p_{CMS} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

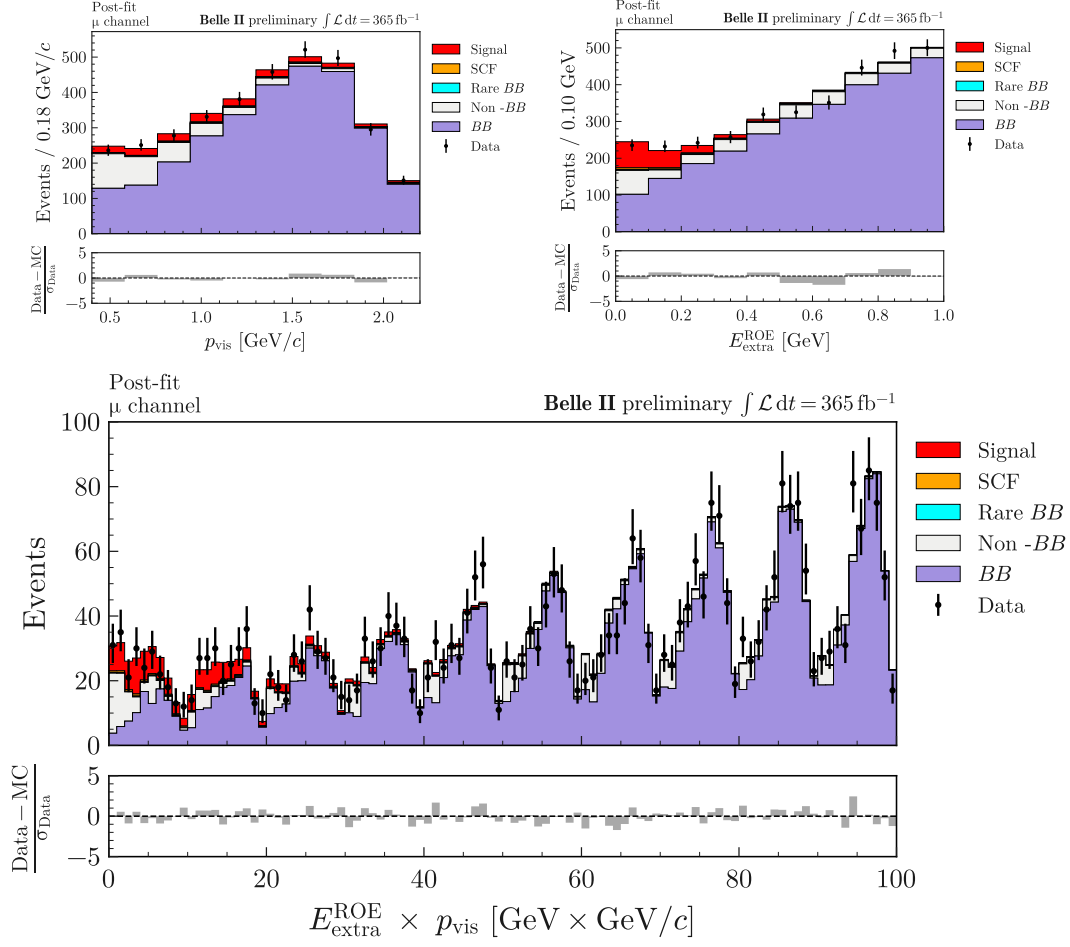


FIG. 78. Post-fit distributions for the μ channel. The top left plot shows the p_{CMS} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

1104 *9.2.2. Step 2: e channel*

1105 The pre-fit distributions for the e channel are shown in Figure 79. Good agreement
 1106 is observed. The pulls of each NPs are shown in Figure 80. The pulls of the NPs
 1107 are all within 1σ . The post-fit distributions are shown in Figure 81. The p-values
 1108 of the fit is $p = 0.460$, which is compatible with the expected distribution. Good
 1109 agreement is observed. The fitted POI is kept blind, and we find an uncertainty of:

$$\sigma_{\mu} = 0.85^{+0.05}_{-0.02} \quad (22)$$

$$\sigma_{\mu_{\text{Asimov}}} = 0.84^{+0.05}_{-0.01} \quad (23)$$

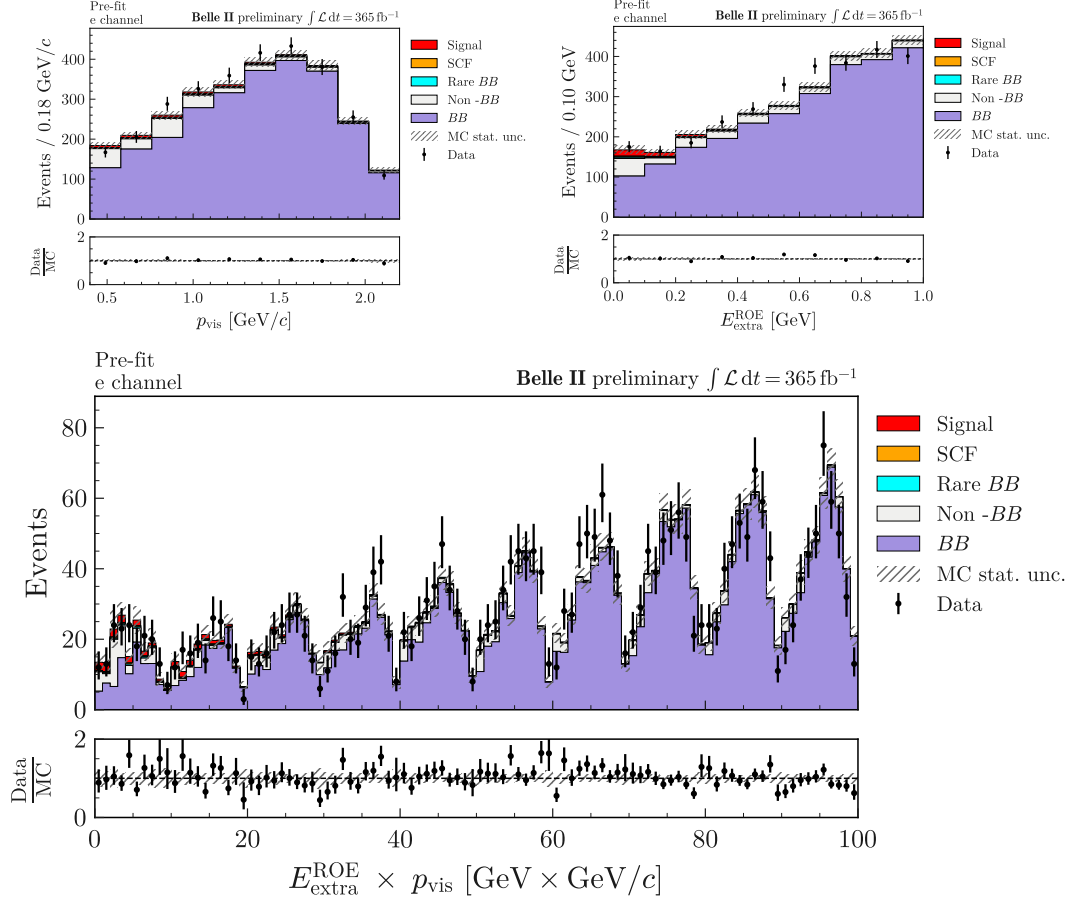


FIG. 79. Pre-fit distributions for the e channel. The top left plot shows the p_{CMS} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

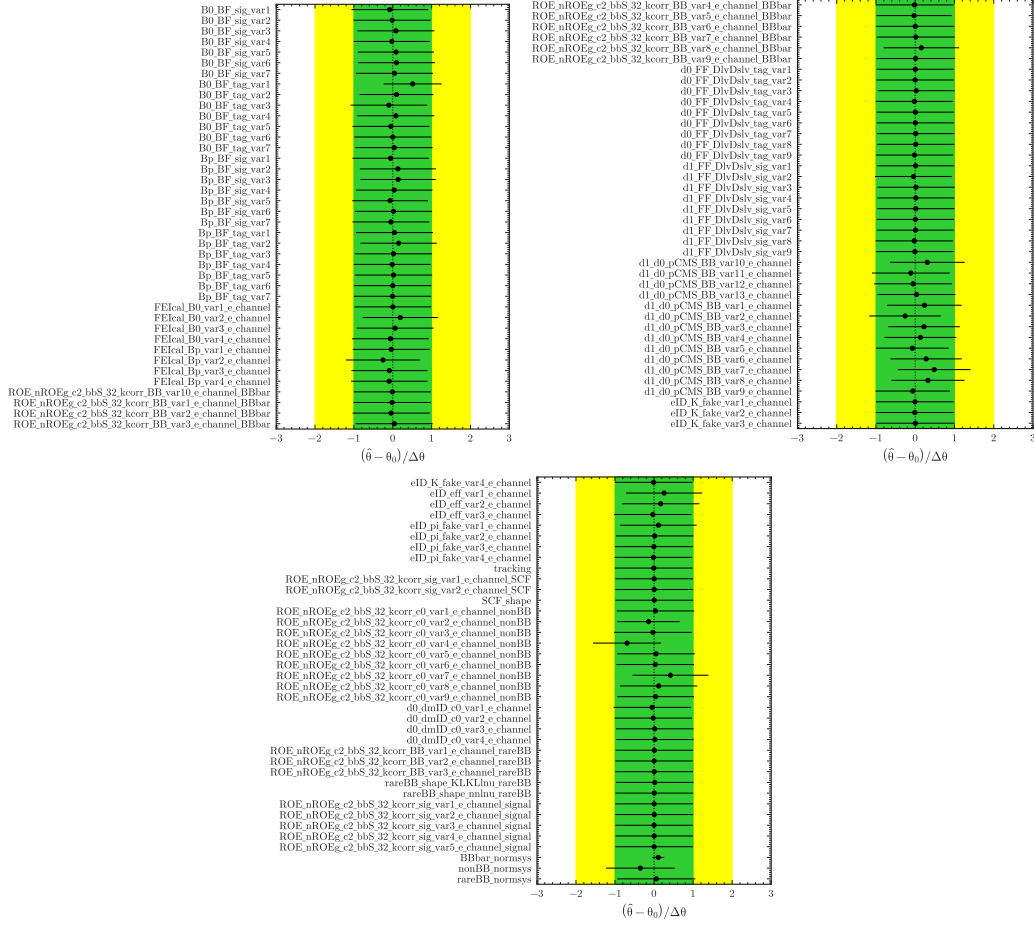


FIG. 80. Pulls of all the NPs included in the fit for the e channel.

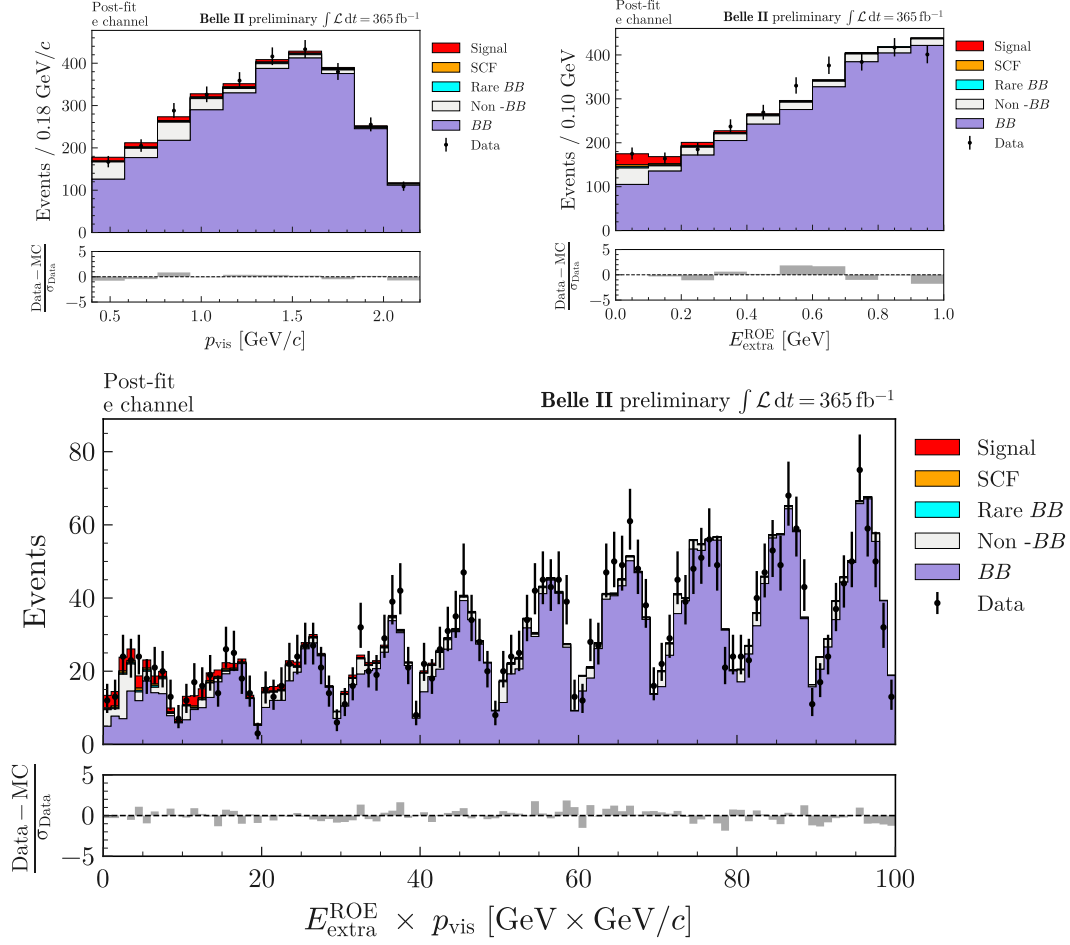


FIG. 81. Post-fit distributions for the e channel. The top left plot shows the p_{CM} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

1110 *9.2.3. Step 3: hadronic channels and blinded POI comparison*

1111 The pre-fit distributions for the hadronic channels are shown in Figure 82. Good
1112 agreement is observed.

1113 The pulls of each NPs are shown in Figure 83. The pulls of the NPs are all
1114 within 1σ . The post-fit distributions are shown in Figure 84. The first pCMS NP's
1115 uncertainty is small

1116 The p-values of the fit is $p = 0.473$, which is compatible with the expected distri-
1117 bution. Good agreement is observed. The fitted POI is kept blind, and we find an
1118 uncertainty of:

$$\sigma_{\mu} = 0.61^{+0.04}_{-0.02} \quad (24)$$

$$\sigma_{\mu_{\text{Asimov}}} = 0.55^{+0.03}_{-0.02} \quad (25)$$

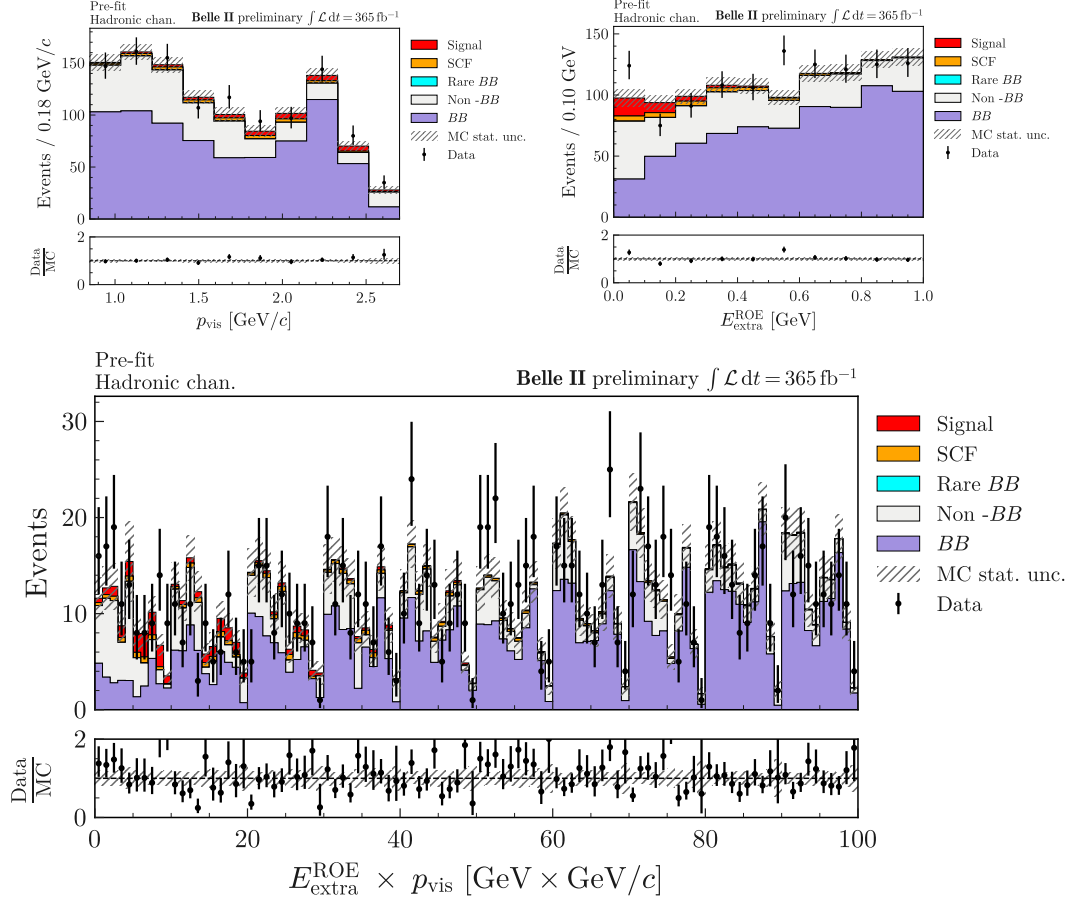


FIG. 82. Pre-fit distributions for the hadronic channels. The top left plot shows the p_{CM} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

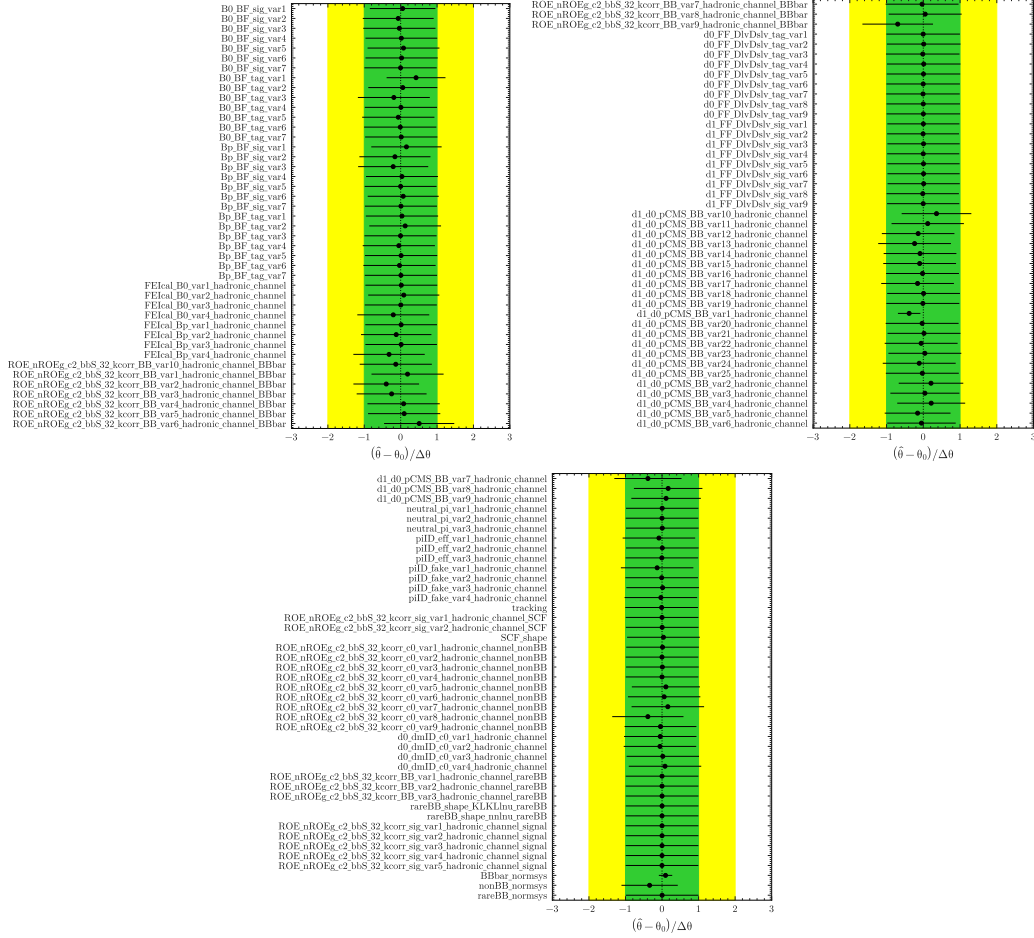


FIG. 83. Pulls of all the NPs included in the fit for the hadronic channels.

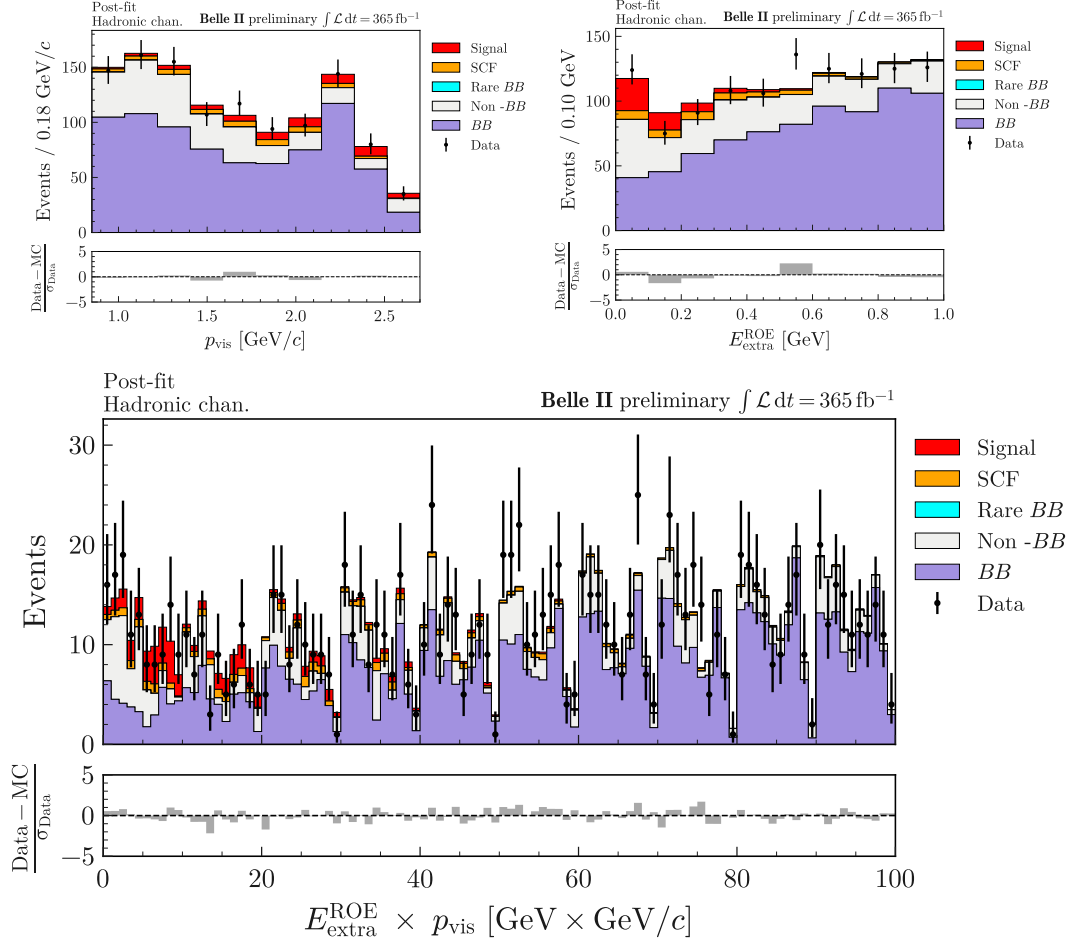


FIG. 84. Post-fit distributions for the hadronic channels. The top left plot shows the p_{CMS} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

1119 The compatibility between the 3 blinded POIs is shown in Figure 85 and Figure 86.
 1120 No discrepancy is observed between the 3 channels.

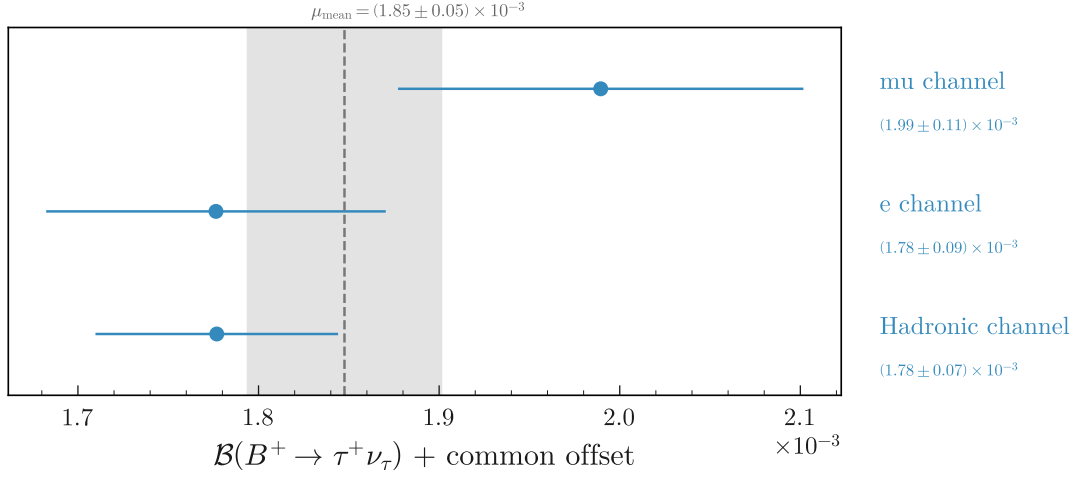


FIG. 85. Comparison of the 3 blinded POIs for the 3 channels.

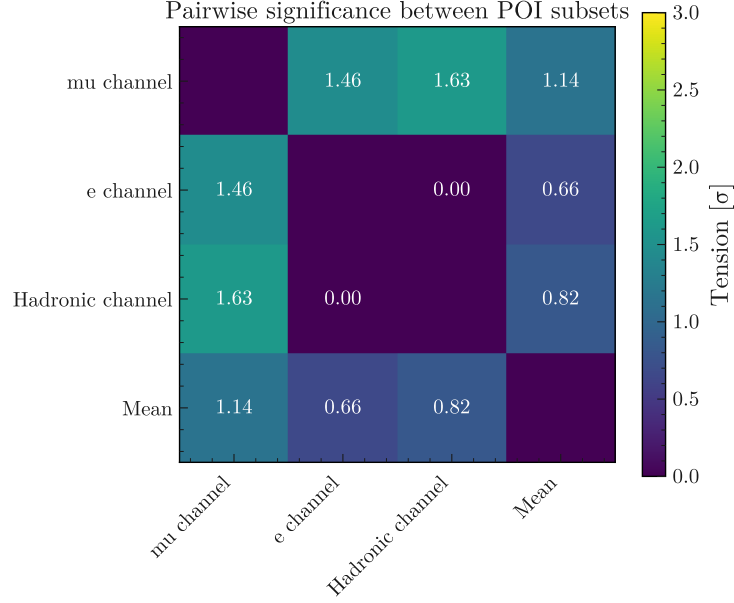


FIG. 86. Significance between the 3 blinded POIs for the 3 channels, assuming independent measurements.

1121 9.2.4. Step 4: consistency checks on sub-samples

1122 We split the data into 3 independent sub-samples and perform the simultaneous fit
 1123 on the 3 channels on each sample. The results are shown in Figure 87 and Figure 88.
 1124 No discrepancy is observed between the 3 sub-samples.

1125 We also compare the 3 sub-samples with the 3 channels fits, and no discrepancy
 1126 is observed. The results are shown in Figure 89 and Figure 90.

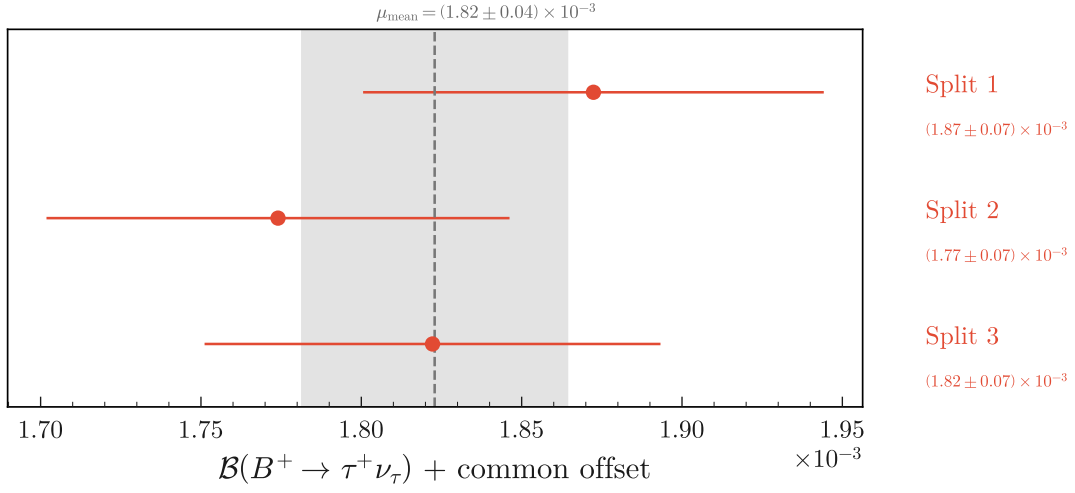


FIG. 87. Comparison of the 3 blinded POIs for the 3 sub-samples.

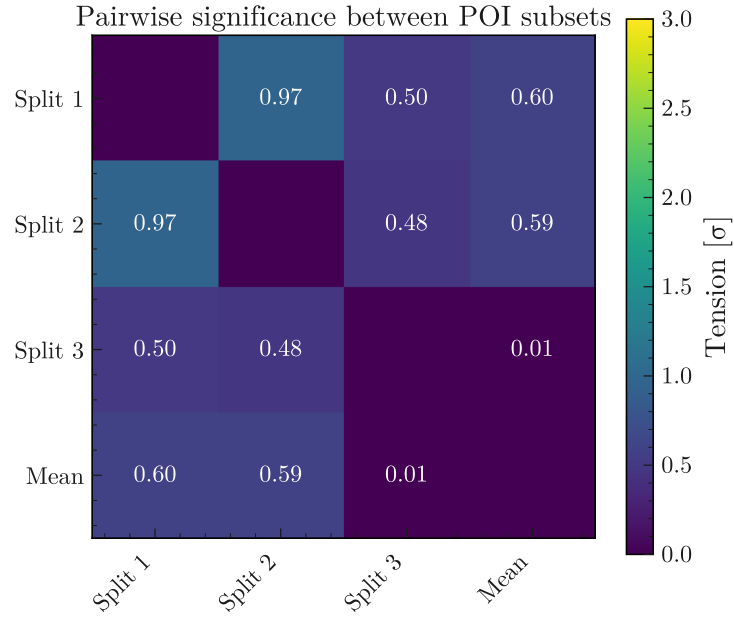


FIG. 88. Significance between the 3 blinded POIs for the 3 sub-samples, assuming independent measurements.

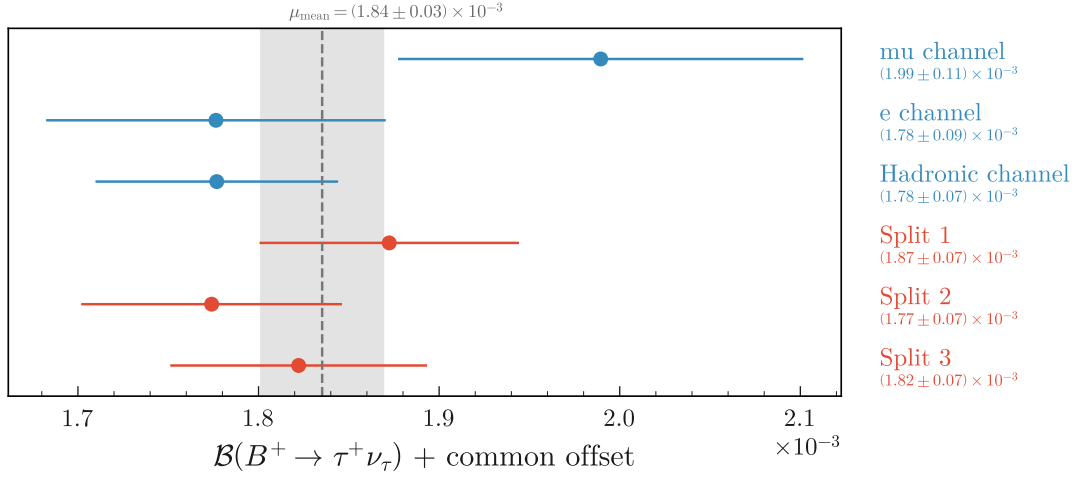


FIG. 89. Comparison of the 6 blinded POIs for the 3 sub-samples and 3 channels.

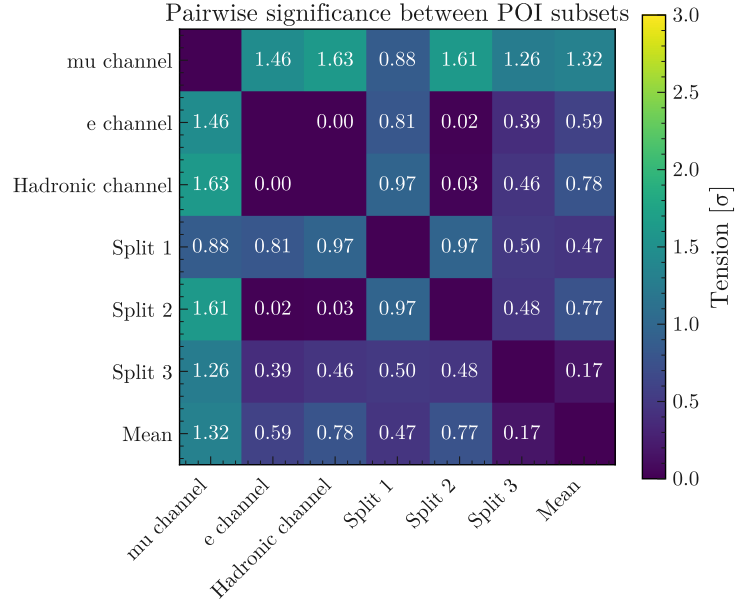


FIG. 90. Significance between the 6 blinded POIs for the 3 sub-samples and 3 channels, assuming independent measurements.

1127 **10. RESULTS**

1128 Once each unblinding steps have been successful, the simultaneous fit of the 3
1129 channels on the full LS1 dataset is performed. All the additional checks and studies
1130 concerning the full unblinding are shown in Appendix G.

1131 **10.1. Fit distributions**

1132 The fit distributions for the μ , e and hadronic channels are shown in Figures 91
1133 to 93. Good agreement between the data and the fit model is observed in all channels.

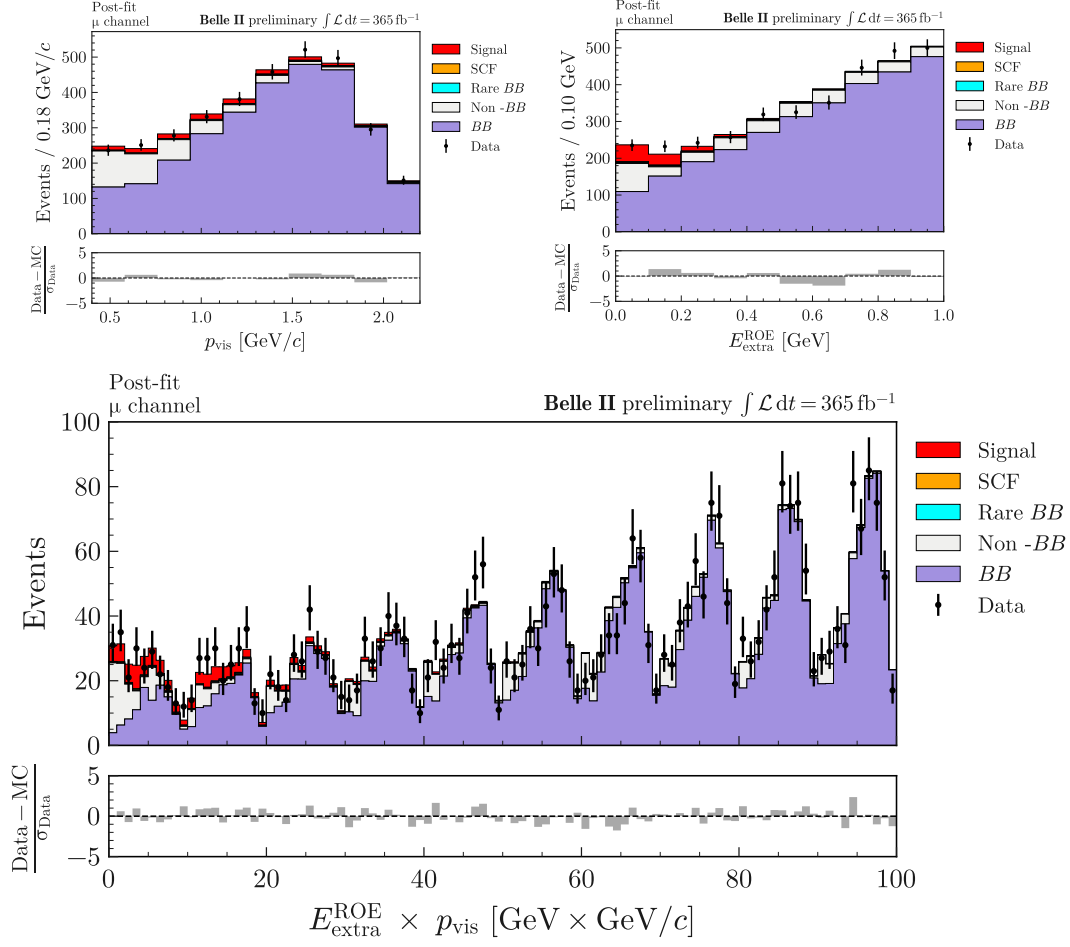


FIG. 91. Post-fit distributions for the μ channel. The top left plot shows the p_{CMS} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

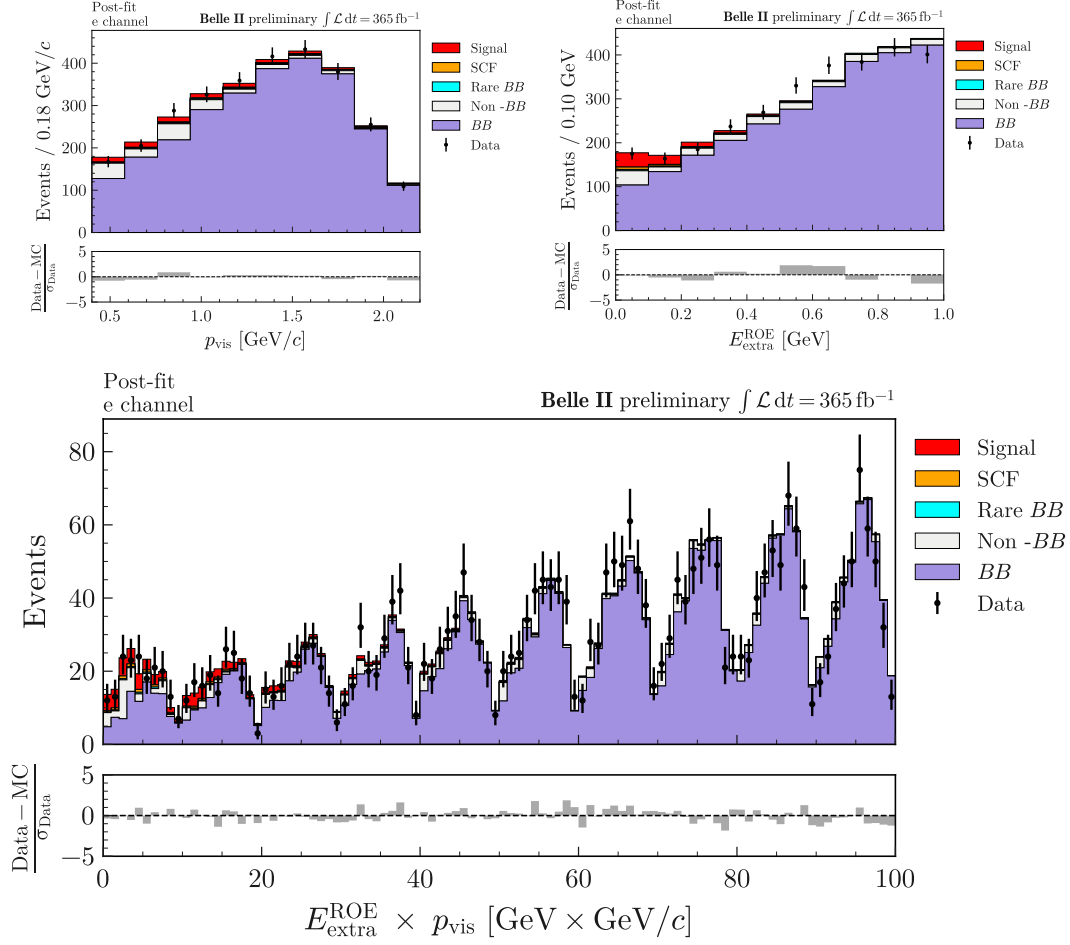


FIG. 92. Post-fit distributions for the e channel. The top left plot shows the p_{CMS} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

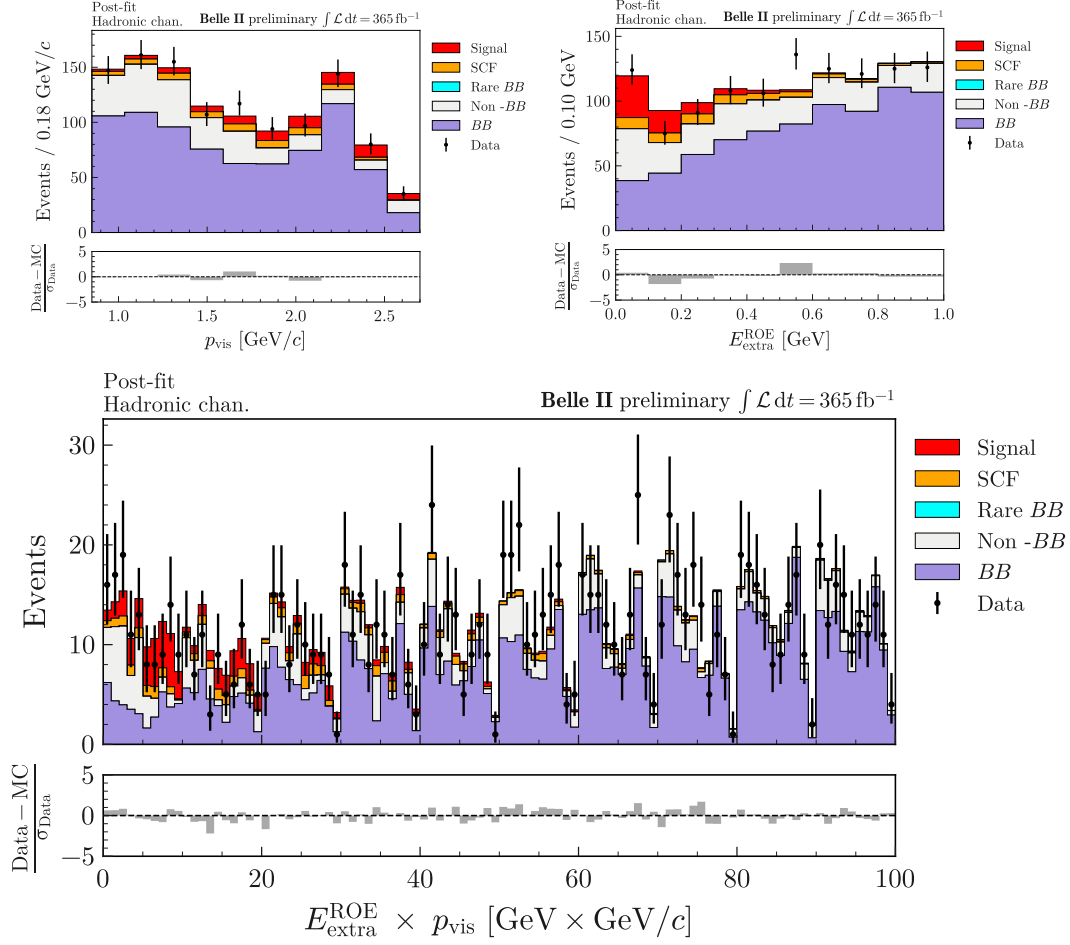


FIG. 93. Post-fit distributions for the hadronic channels. The top left plot shows the p_{CM} distribution, the top right plot shows the $E_{\text{extra}}^{\text{ROE}}$ distribution, and the bottom plot shows the flattened 2D distribution of both variables.

1134 **10.2. Fit results**

1135 We obtain the following signal strength:

$$\mu = 2.19^{+0.490}_{-0.462} \quad (26)$$

1136 Where the uncertainty is statistical + systematics. The p-value of the fit is $p =$
1137 0.388.

1138 The systematic table have been reproduced using the same method described in
1139 Section 8.12.1. The table is shown Table 26. We then get the statistical uncertainty
1140 from toy study using the same method described in Section 7.2.1. The result is:

$$\mu = 2.19 \pm 0.35(\text{stat}) \pm 0.35(\text{syst}) \quad (27)$$

Source	Systematics (= impact on μ) (%)
MC statistics	21.6
ROE corrections	18.6
p_{CMS} corrections	7.9
Branching fractions of background decays	4.4
FEI corrections	12.6
Tracking efficiency	0.7
Neutral particle identification	0.8
Charged particle identification	4.8
$D^{(*)}\ell\nu_\ell$ form factors	0.2
$B\bar{B}$ normalization	0.6
non- $B\bar{B}$ normalization	7.3
Rare $B\bar{B}$ normalization	1.5
Rare $B\bar{B}$ shape	0.5
SCF shape	3.1
Signal efficiency	10.0
Fit bias	1.6
Number of produced $\Upsilon(4S)$	1.5
Fraction of $B\bar{B}$ pairs	2.1
Total	35.4 ± 1.1

TABLE 26. Impact of the different sources of systematic uncertainties on the signal strength μ , with the total systematic uncertainty obtained by summing in quadrature the individual contributions. The uncertainty on the total is obtained by summing the positive and negative impacts separately.

10.3. Branching fraction

Taking into account the corrections from Section 6.1, the resulting branching fraction is equal to:

$$\mathcal{B}(B^+ \rightarrow \tau^+ \nu_\tau) = (2.17 \pm 0.35(\text{stat}) \pm 0.35(\text{syst})) \times 10^{-4} \quad (28)$$

corresponding to a significance of 4.38σ .

We can also get the CKM matrix element $|V_{ub}|$, according to Equation (12):

$$|V_{ub}| = (5.83 \pm 0.47(\text{stat}) \pm 0.47(\text{syst})) \times 10^{-3} \quad (29)$$

1146 The compatibility of this result with the previous measurements is shown in Fig-
1147 ure 94.

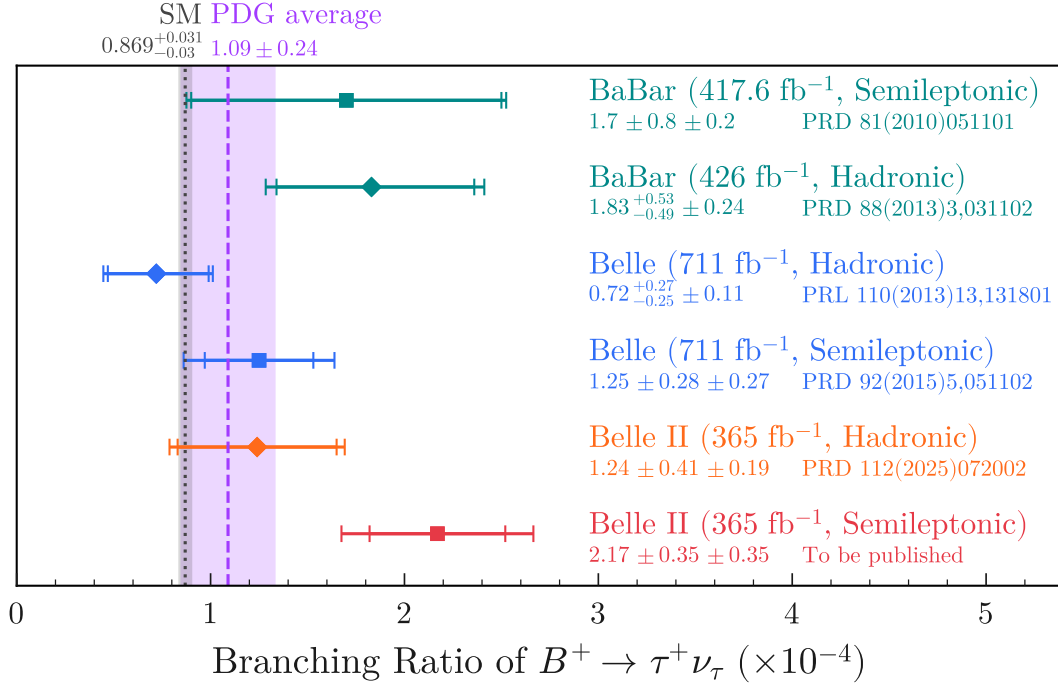


FIG. 94. Comparison of the measured branching fraction with previous measurements.

APPENDIX

A. Event selection, additional material

This section shows additional material related to the event selection.

A.a. Additional pre-selection plots

Here are shown the data/MC agreement of the three variables used in the pre-selection on the electron channel to remove the $\tau^+\tau^-$ background described in Section 3.3. Good agreement is observed between data and MC.

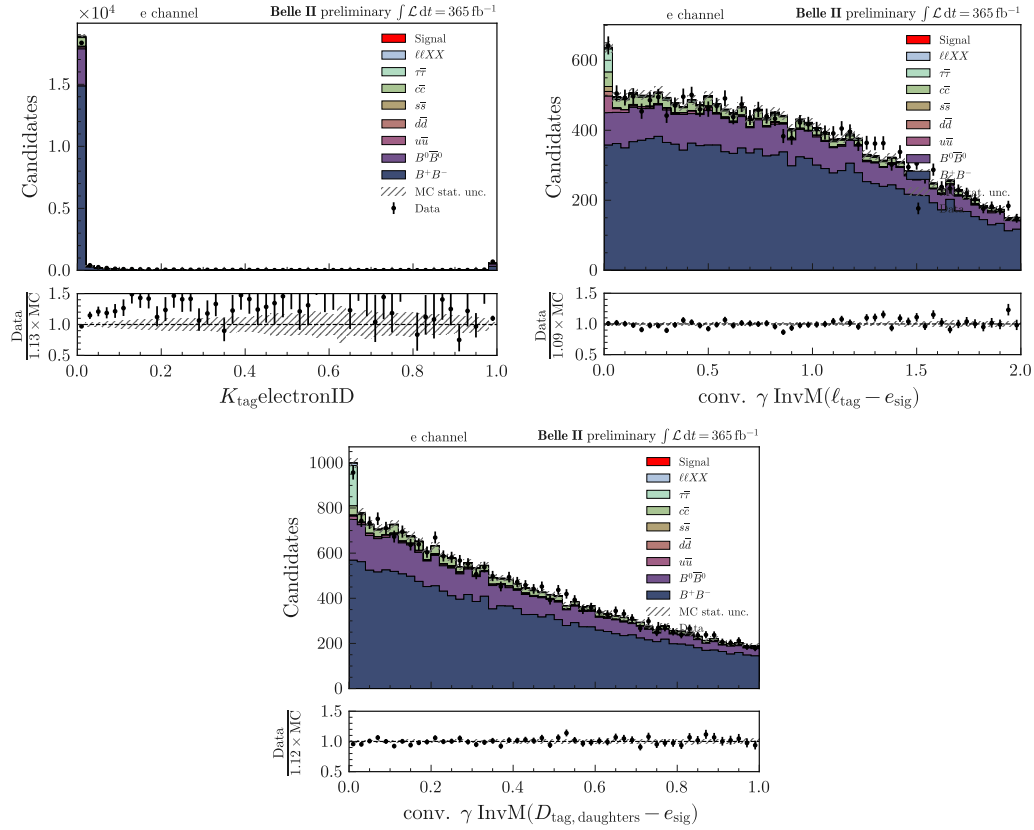


FIG. 95. Comparison between data and MC of the pre-selection variables used to remove the $\tau^+\tau^-$ background on the electron channel, described in Section 3.3.

1155 *A.b. Cumulative pre-selection tables for each channel*

1156 Here are shown the cumulative pre-selection tables for each channel.

1157

Cut (μ channel)	#Sgn.	Sgn. eff. cumul [%]	#Bkg	Bkg. ret. cumul [%]
Reconstruction (w/ FEI)	91	1.227 ± 0.004	63912	-
B_{tag} decay mode ID	86	1.161 ± 0.029	52117	81.54 ± 0.15
$p_{\tau, \text{sig}}^*$	81	1.10 ± 0.04	47670	74.59 ± 0.17
$p_{D_{\text{tag}}}^*$	81	1.10 ± 0.04	42419	66.37 ± 0.19
$\cos \theta_{B, D_l}^*$	79	1.07 ± 0.04	32775	51.28 ± 0.20
R_2 (Event based)	67	0.90 ± 0.06	25893	40.51 ± 0.19
$B_{\text{tag}} R_2$	66	0.89 ± 0.06	25676	40.17 ± 0.19
$\cos \theta_{\text{thrust}, B_{\text{tag}}, z}$	61	0.82 ± 0.06	22759	35.61 ± 0.19
π channel: $p_{B_{\text{sig}}}$	61	0.82 ± 0.06	22759	35.61 ± 0.19
ρ channel: $p_{B_{\text{sig}}}$	61	0.82 ± 0.06	22759	35.61 ± 0.19
e channel: K_{tag} electronID	61	0.82 ± 0.06	22759	35.61 ± 0.19
e channel: conv. γ InvM($\ell_{\text{tag}} - e_{\text{sig}}$)	61	0.82 ± 0.06	22759	35.61 ± 0.19
e channel: conv. γ InvM($D_{\text{tag, daughters}} - e_{\text{sig}}$)	61	0.82 ± 0.06	22759	35.61 ± 0.19

TABLE 27. Cumulative signal efficiency and background retention after each pre-selection cut for the μ channel. The number of truth-matched signal candidates and the number of background candidates is obtained on the 1444 fb^{-1} MC sample and has been scaled expected LS1 data luminosity.

Cut (e channel)	#Sgn.	Sgn. eff. cumul [%]	#Bkg	Bkg. ret. cumul [%]
Reconstruction (w/ FEI)	87	1.139 ± 0.004	55218	-
B_{tag} decay mode ID	82	1.077 ± 0.028	44114	79.89 ± 0.17
$p_{\tau,\text{sig}}^*$	74	0.97 ± 0.04	40941	74.14 ± 0.19
$p_{D\text{tag}}^*$	74	0.97 ± 0.04	37938	68.71 ± 0.20
$\cos \theta_{B,D_l}^*$	72	0.94 ± 0.05	30026	54.38 ± 0.21
R_2 (Event based)	60	0.79 ± 0.06	24529	44.42 ± 0.21
$B_{\text{tag}} R_2$	60	0.78 ± 0.06	24386	44.16 ± 0.21
$\cos \theta_{\text{thrust}, B_{\text{tag}}, z}$	55	0.72 ± 0.06	22025	39.89 ± 0.21
π channel: $p_{B_{\text{sig}}}$	55	0.72 ± 0.06	22025	39.89 ± 0.21
ρ channel: $p_{B_{\text{sig}}}$	55	0.72 ± 0.06	22025	39.89 ± 0.21
e channel: K_{tag} electronID	53	0.69 ± 0.06	20930	37.90 ± 0.21
e channel: conv. γ InvM($\ell_{\text{tag}} - e_{\text{sig}}$)	52	0.69 ± 0.06	20659	37.41 ± 0.21
e channel: conv. γ InvM($D_{\text{tag, daughters}} - e_{\text{sig}}$)	50	0.66 ± 0.06	19993	36.21 ± 0.20

TABLE 28. Cumulative signal efficiency and background retention after each pre-selection cut for the e channel. The number of truth-matched signal candidates and the number of background candidates is obtained on the 1444 fb^{-1} MC sample and has been scaled expected LS1 data luminosity.

Cut (π channel)	#Sgn.	Sgn. eff. cumul [%]	#Bkg	Bkg. ret. cumul [%]
Reconstruction (w/ FEI)	53	1.158 ± 0.005	166520	-
B_{tag} decay mode ID	50	1.10 ± 0.04	150759	90.53 ± 0.07
$p_{\tau,\text{sig}}^*$	50	1.09 ± 0.04	120771	72.53 ± 0.11
$p_{D_{\text{tag}}}^*$	50	1.09 ± 0.04	91858	55.16 ± 0.12
$\cos \theta_{B,D_l}^*$	49	1.06 ± 0.04	59624	35.81 ± 0.12
R_2 (Event based)	42	0.91 ± 0.06	35043	21.04 ± 0.10
$B_{\text{tag}} R_2$	42	0.91 ± 0.07	33918	20.37 ± 0.10
$\cos \theta_{\text{thrust}, B_{\text{tag}}, z}$	38	0.83 ± 0.07	26159	15.71 ± 0.09
π channel: $p_{B_{\text{sig}}}$	35	0.77 ± 0.07	10652	6.40 ± 0.06
ρ channel: $p_{B_{\text{sig}}}$	35	0.77 ± 0.07	10652	6.40 ± 0.06
e channel: K_{tag} electronID	35	0.77 ± 0.07	10652	6.40 ± 0.06
e channel: conv. γ InvM($\ell_{\text{tag}} - e_{\text{sig}}$)	35	0.77 ± 0.07	10652	6.40 ± 0.06
e channel: conv. γ InvM($D_{\text{tag, daughters}} - e_{\text{sig}}$)	35	0.77 ± 0.07	10652	6.40 ± 0.06

TABLE 29. Cumulative signal efficiency and background retention after each pre-selection cut for the π channel. The number of truth-matched signal candidates and the number of background candidates is obtained on the 1444 fb^{-1} MC sample and has been scaled expected LS1 data luminosity.

Cut (ρ channel)	#Sgn.	Sgn. eff. cumul [%]	#Bkg	Bkg. ret. cumul [%]
Reconstruction (w/ FEI)	30	0.2825 ± 0.0015	103732	-
B_{tag} decay mode ID	29	0.268 ± 0.011	92130	88.82 ± 0.10
$p_{\tau,\text{sig}}^*$	28	0.264 ± 0.013	80723	77.82 ± 0.13
$p_{D\text{tag}}^*$	28	0.264 ± 0.013	63032	60.76 ± 0.15
$\cos \theta_{B,D_l}^*$	28	0.257 ± 0.015	42118	40.60 ± 0.15
R_2 (Event based)	22	0.206 ± 0.023	27949	26.94 ± 0.14
$B_{\text{tag}} R_2$	22	0.204 ± 0.023	27395	26.41 ± 0.14
$\cos \theta_{\text{thrust}, B_{\text{tag}}, z}$	20	0.187 ± 0.024	21741	20.96 ± 0.13
π channel: $p_{B_{\text{sig}}}$	20	0.187 ± 0.024	21741	20.96 ± 0.13
ρ channel: $p_{B_{\text{sig}}}$	18	0.168 ± 0.025	11419	11.01 ± 0.10
e channel: K_{tag} electronID	18	0.168 ± 0.025	11419	11.01 ± 0.10
e channel: conv. γ InvM($\ell_{\text{tag}} - e_{\text{sig}}$)	18	0.168 ± 0.025	11419	11.01 ± 0.10
e channel: conv. γ InvM($D_{\text{tag, daughters}} - e_{\text{sig}}$)	18	0.168 ± 0.025	11419	11.01 ± 0.10

TABLE 30. Cumulative signal efficiency and background retention after each pre-selection cut for the ρ channel. The number of truth-matched signal candidates and the number of background candidates is obtained on the 1444 fb^{-1} MC sample and has been scaled expected LS1 data luminosity.

1158 **B. ROE mask optimization**

1159 *B.a. FOM optimization*

1160 To choose which ROE masks to use for the analysis, an optimization study has
 1161 been performed. All the possible combinations of the cuts present in Table 31 have
 1162 been tested on 1 ab^{-1} of generic B^+B^- MC, where our signal is present. For each
 1163 combination, the full selection is applied and the expected signal significance is com-
 1164 puted using the formula:

$$\text{Significance} = \frac{S}{\sqrt{S+B}} \quad (30)$$

1165 where S is the number of signal events and B the number of B^+B^- background
 1166 events, both in the $E_{\text{extra}}^{\text{ROE}} < 1 \text{ GeV}$ region.

1167 In the end, the top 6 combinations with the highest expected significance have
 1168 been selected. The difference in significance between these top 6 combinations is
 1169 very small:

- 1170 1. Energy cut and `minC2TDist` > 30 cm and `beamBackgroundSuppression` >
 1171 0.15
- 1172 2. Energy cut and `minC2TDist` > 30 cm and `beamBackgroundSuppression` >
 1173 0.15 and `|clusterTiming|` < 200 ns
- 1174 3. Energy cut and `minC2TDist` > 30 cm and `beamBackgroundSuppression` > 0.3
- 1175 4. Energy cut and `minC2TDist` > 30 cm and `beamBackgroundSuppression` > 0.1
- 1176 5. Energy cut and `minC2TDist` > 30 cm and `beamBackgroundSuppression` >
 1177 0.15 and `fakePhotonSuppression` > 0.02
- 1178 6. Energy cut and `minC2TDist` > 30 cm and `|clusterTiming|` < 200 ns

1179 with energy cut defined as: $E_{\text{cluster}} > 0.100 \text{ GeV}$ in the forward region or $E_{\text{cluster}} >$
 1180 0.055 GeV in the barrel region or $E_{\text{cluster}} > 0.080$ in the backward region.

1181 After checking the distributions of the different ROE variables for these top 6 com-
 1182 binations, the first one has been chosen as the nominal ROE mask for the analysis,
 1183 as it shown the best data/MC agreement.

TABLE 31. ROE mask cuts values in the optimization study. All possible combinations have been tested.

Variable	operator	Cut values
clusterE9E21	>	0.8, 0.9, 0.96
minC2TDist [cm]	>	20, 30, 40, 50, 60, 70, 80, 90, 100
beamBackgroundSuppression	>	0.02, 0.05, 0.1, 0.15, 0.3, 0.5
fakePhotonSuppression	>	0.02, 0.05, 0.1, 0.15, 0.3, 0.5
clusterTiming [ns]	<	200
clusterTimingClusterErrorTiming	<	2

1184 *B.b. ROE correction uncertainty*

1185 The ROE correction systematic is proportional to the uncertainty of the first
1186 bins of the photon multiplicity corrections (see Section 5.2). The mask optimization
1187 study has been re-run taking into account the FOM from the last section, along with
1188 the uncertainty of the 2 first bins of N_{γ}^{ROE} on the off-resonance data and on the
1189 BDT sidebands for $B\bar{B}$ background. No improvement was found compared to the
1190 nominal ROE mask, which was found to be part of the top combinations in this new
1191 optimization study as well.

1192 C. Background suppression, additional plots

1193 C.a. Correlations

1194 This section shows the correlations between the variables used for $B^+ \rightarrow \tau^+ \nu_\tau$
 1195 background suppression. The method used to compute the correlations is the Pearson
 1196 correlation coefficient. The correlation coefficient r is defined as:

$$r(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (31)$$

1197 where x_i and y_i are the values of the two variables, $\bar{x}$ and $\bar{y}$ are the means of the
 1198 two variables, and n is the number of data points. The correlation coefficient can
 1199 take values between -1 and 1, where -1 indicates a perfect negative correlation, 0
 1200 indicates no correlation, and 1 indicates a perfect positive correlation.

1201 For each channel, the correlation matrix is shown for signal and background. In
 1202 addition, the two variables used in the fit (but not in the training) and the XGBoost
 1203 output are shown below and after the black line.

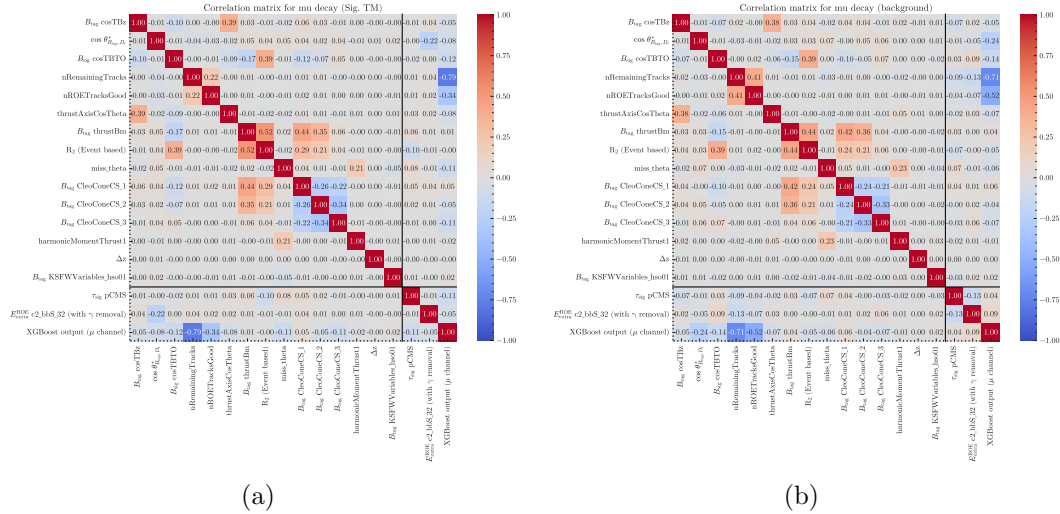


FIG. 96. Correlation matrix for signal (a) and background (b) for the μ channel. The variables used in the fit and the XGBoost output are shown below and after the black line.

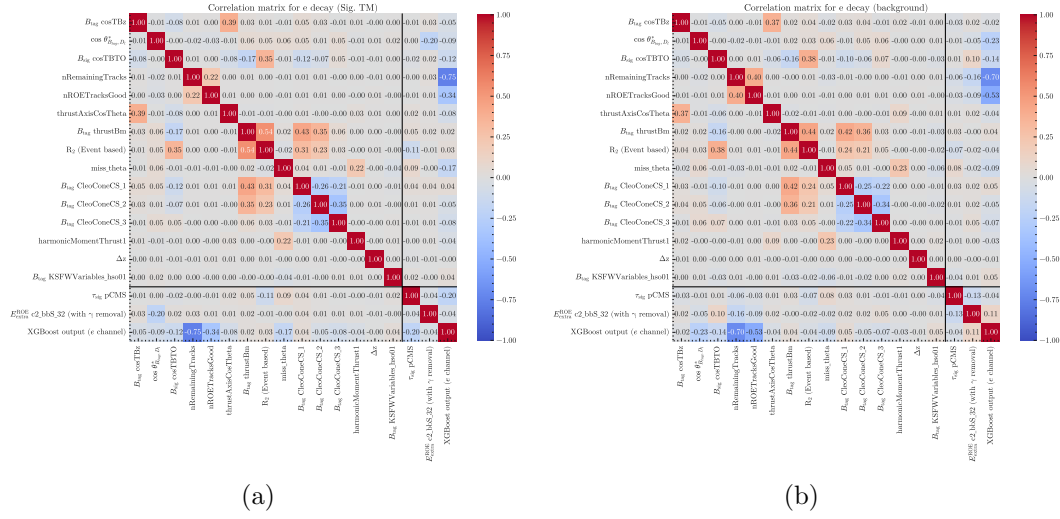


FIG. 97. Correlation matrix for signal (a) and background (b) for the e channel. The variables used in the fit and the XGBoost are shown below and after the black line.

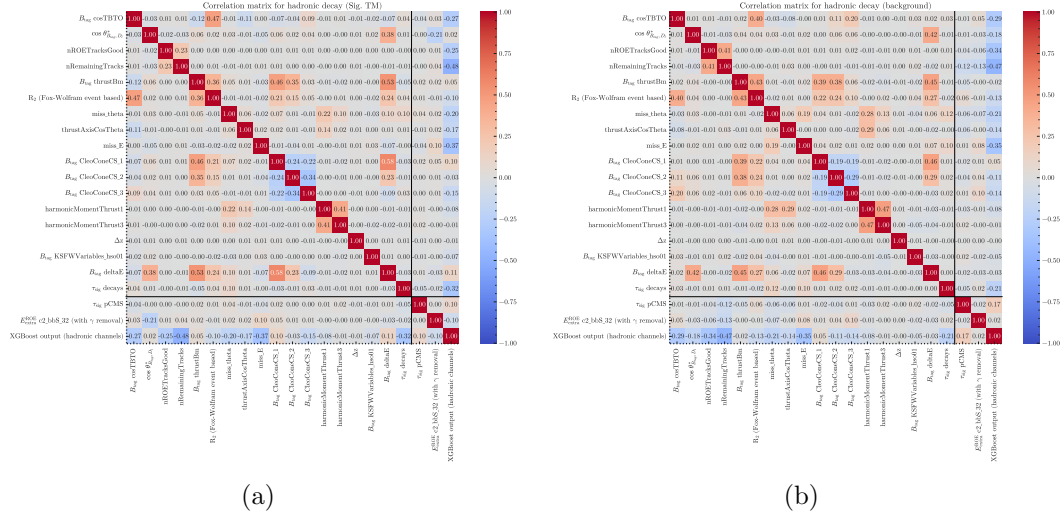


FIG. 98. Correlation matrix for signal (a) and background (b) for the hadronic channels. The variables used in the fit and the XGBoost are shown below and after the black line.

1204

C.b. Variable importance

1205

1206

Below is a more detailed plot of the variable importance for each channel using shapley beeswarm representation.

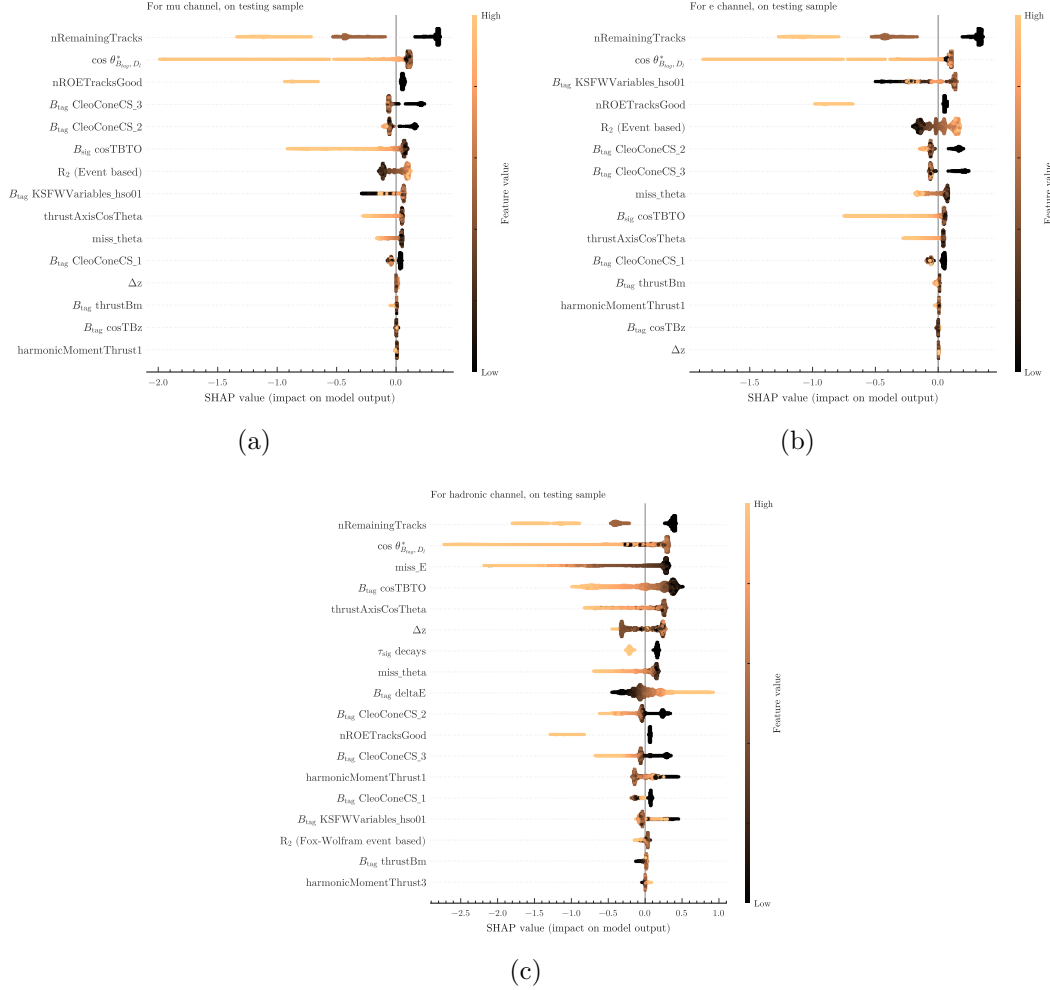


FIG. 99. The shapley value is used to evaluate the importance of each variable (see here). The higher the absolute value of the shapley value, the more important the variable is to separate signal from background. The color represents the value of the variable (orange for high value, black for low value).

1207 *C.c. Data/MC agreement*

1208 This section shows the Data/MC agreement after pre-selection for each variables
 1209 used for $B^+ \rightarrow \tau^+ \nu_\tau$ background suppression.

1210

1211 μ and e channels:

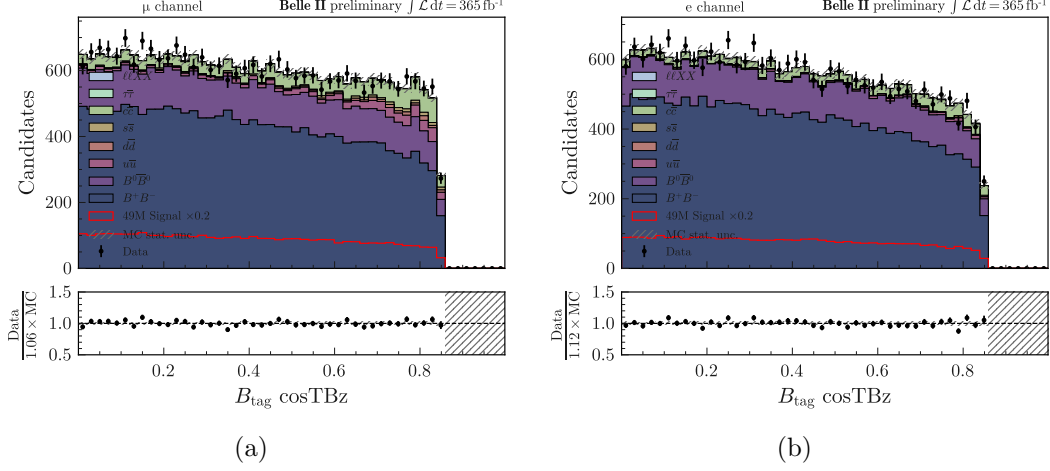


FIG. 100. Data/MC agreement of $B_{tag} \cos TBz$ for mu and e channels.

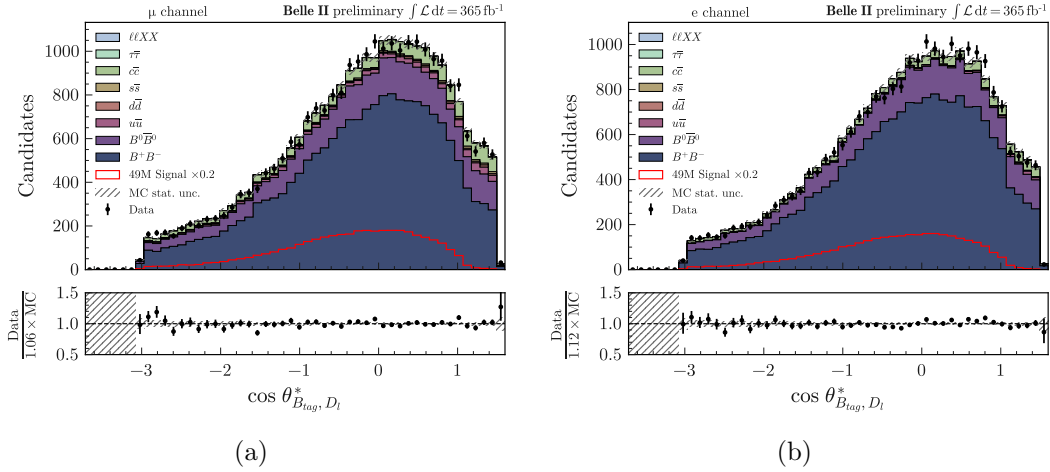


FIG. 101. Data/MC agreement of $\cos \theta_{B_{tag}, D_l}^*$ for mu and e channels.

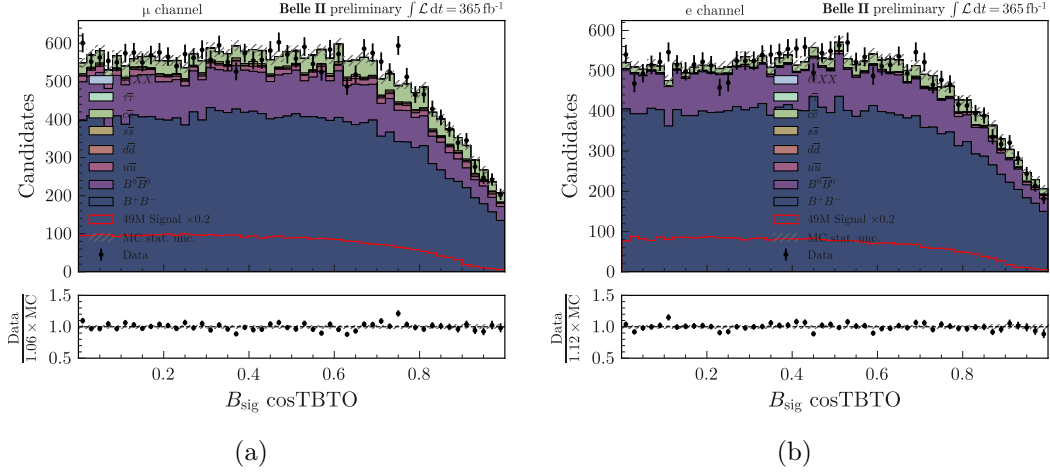


FIG. 102. Data/MC agreement of $B_{sig} \cos TBTO$ for mu and e channels.

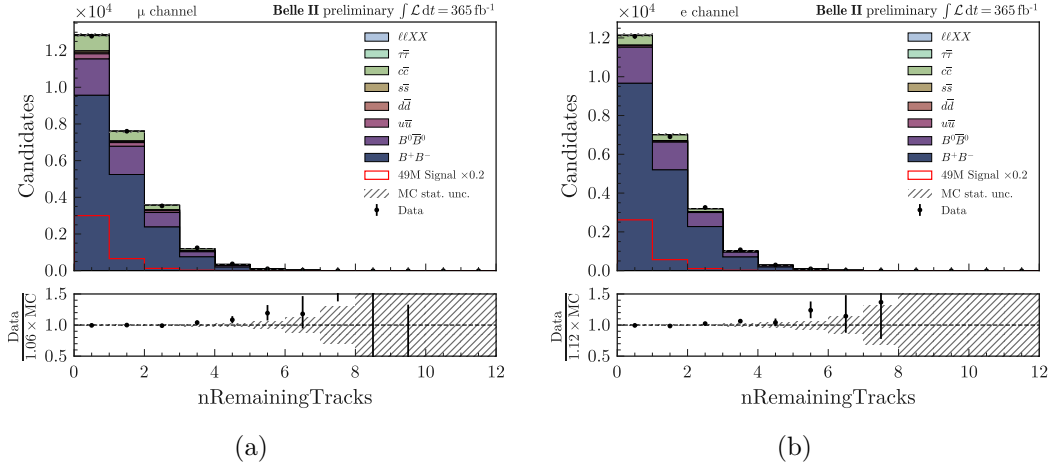


FIG. 103. Data/MC agreement of $nRemainingTracks$ for mu and e channels.

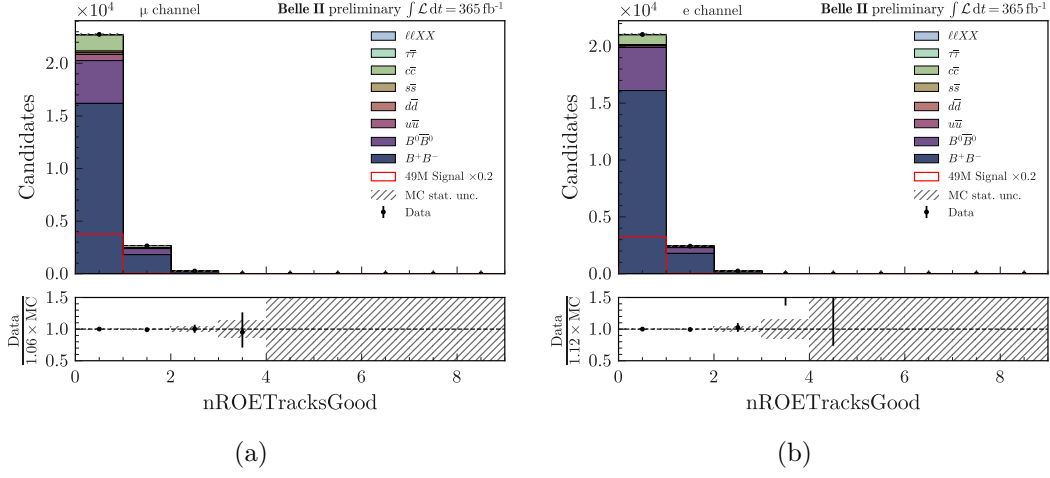


FIG. 104. Data/MC agreement of nROETracksGood for mu and e channels.

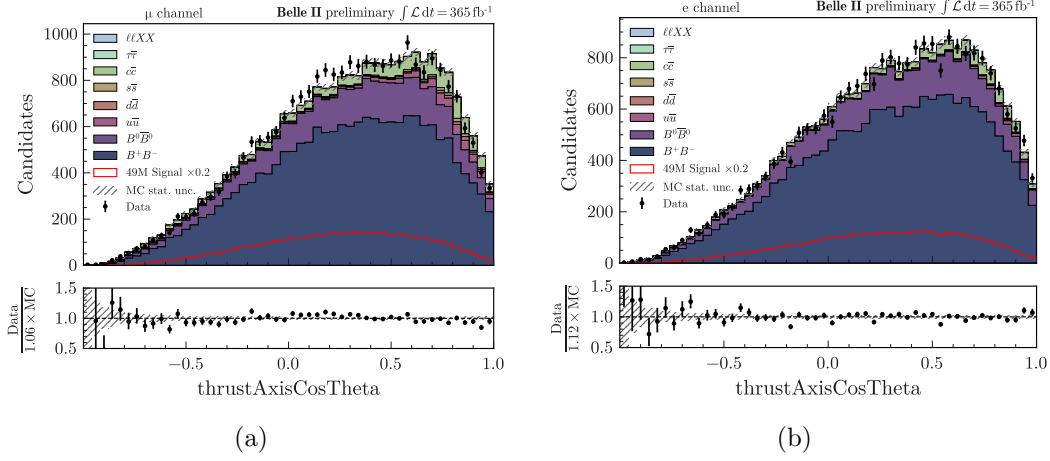


FIG. 105. Data/MC agreement of thrustAxisCosTheta for mu and e channels.

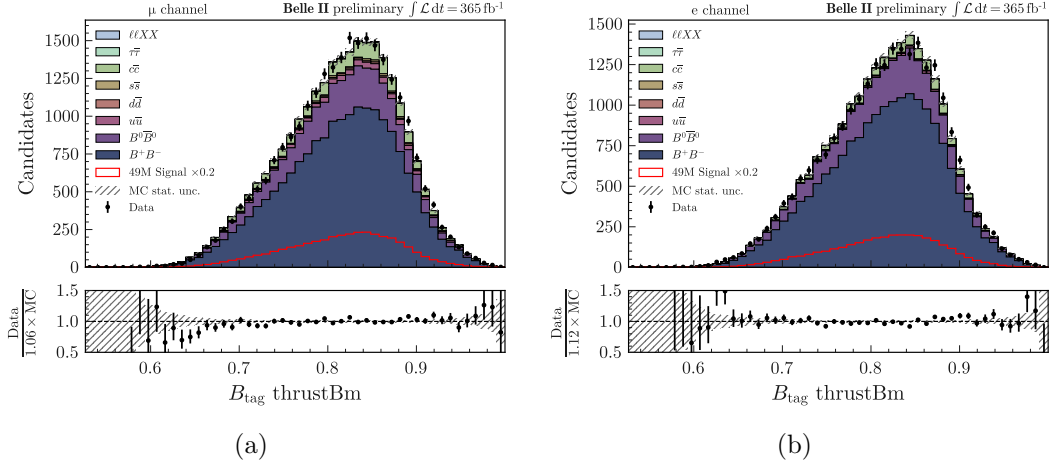


FIG. 106. Data/MC agreement of B_{tag} thrustBm for mu and e channels.

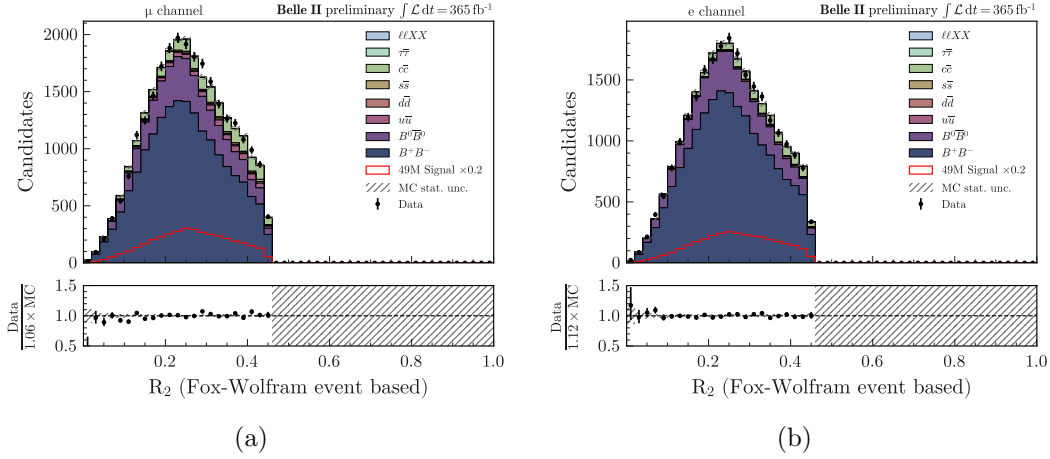


FIG. 107. Data/MC agreement of R_2 (Fox-Wolfram event based) for mu and e channels.

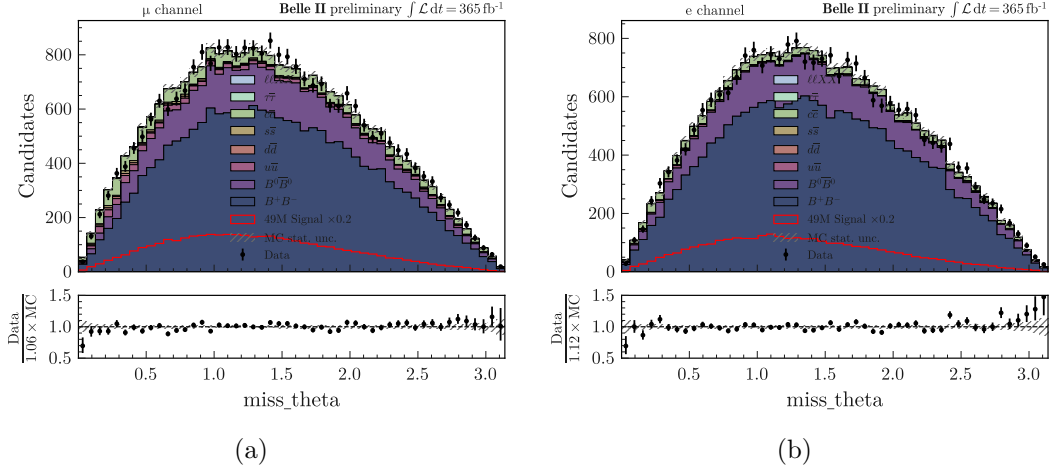


FIG. 108. Data/MC agreement of miss.theta for mu and e channels.

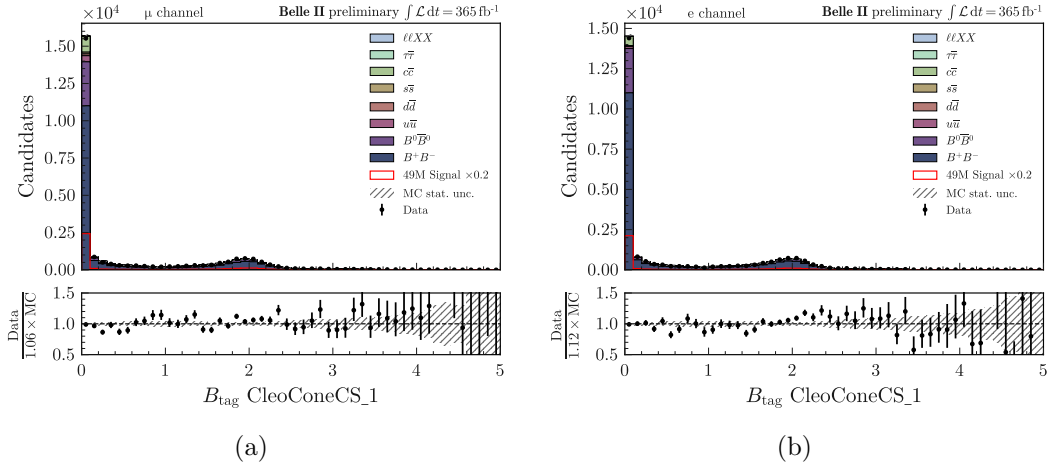


FIG. 109. Data/MC agreement of B_{tag} CleoConeCS_1 for mu and e channels.

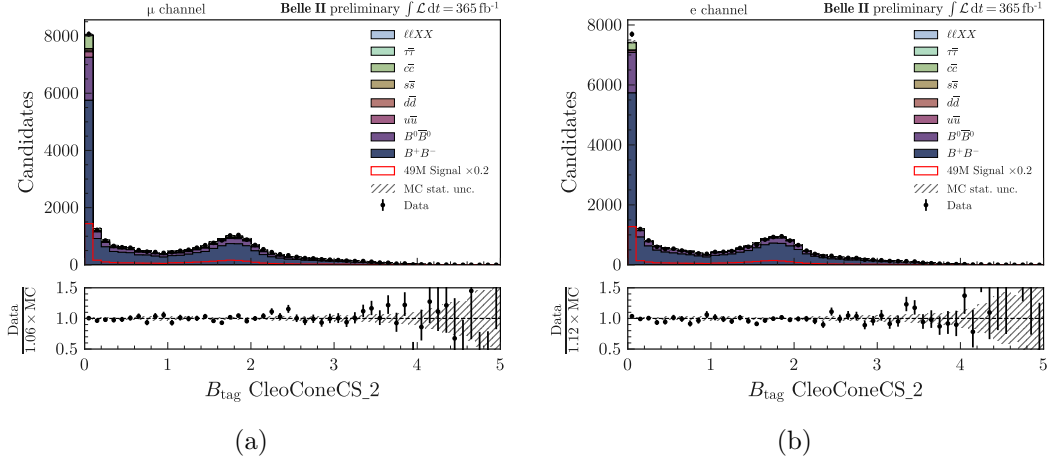


FIG. 110. Data/MC agreement of B_{tag} CleoConeCS_2 for mu and e channels.

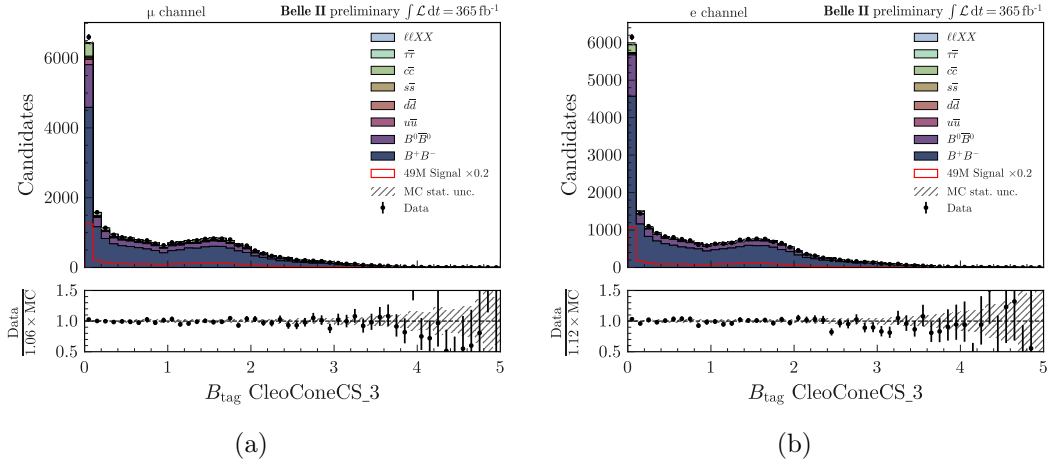


FIG. 111. Data/MC agreement of B_{tag} CleoConeCS_3 for mu and e channels.

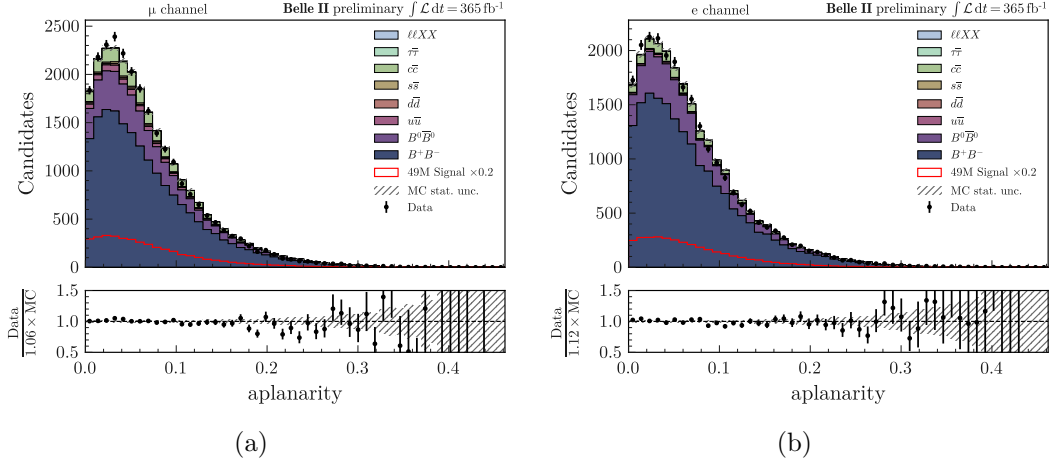


FIG. 112. Data/MC agreement of aplanarity for mu and e channels.

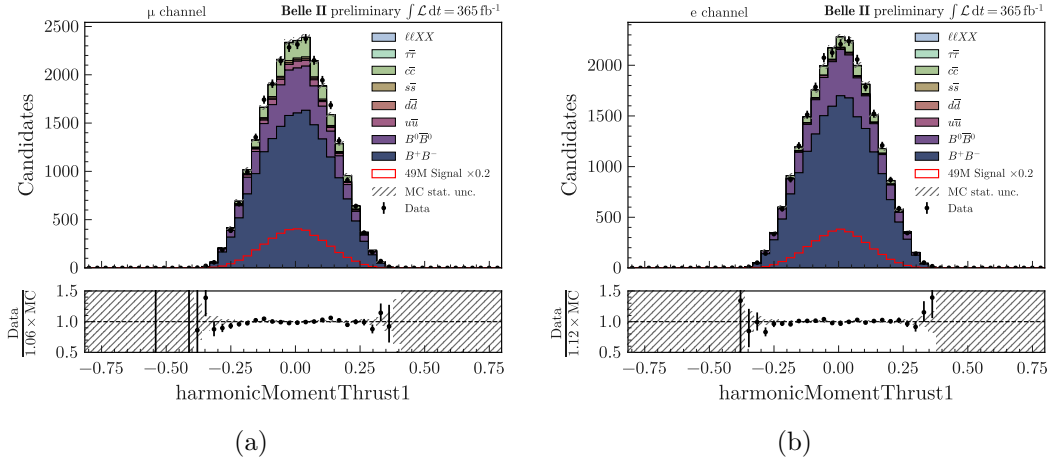


FIG. 113. Data/MC agreement of harmonicMomentThrust1 for mu and e channels.

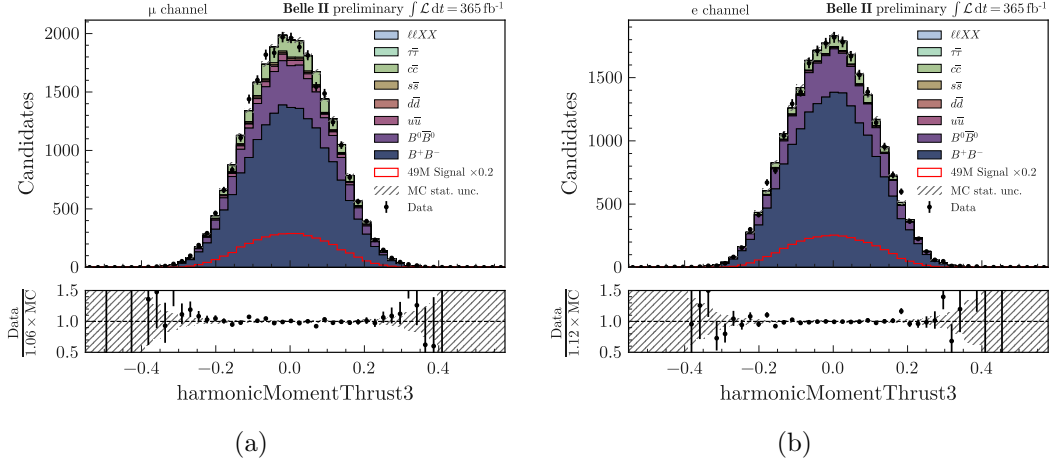


FIG. 114. Data/MC agreement of harmonicMomentThrust3 for mu and e channels.

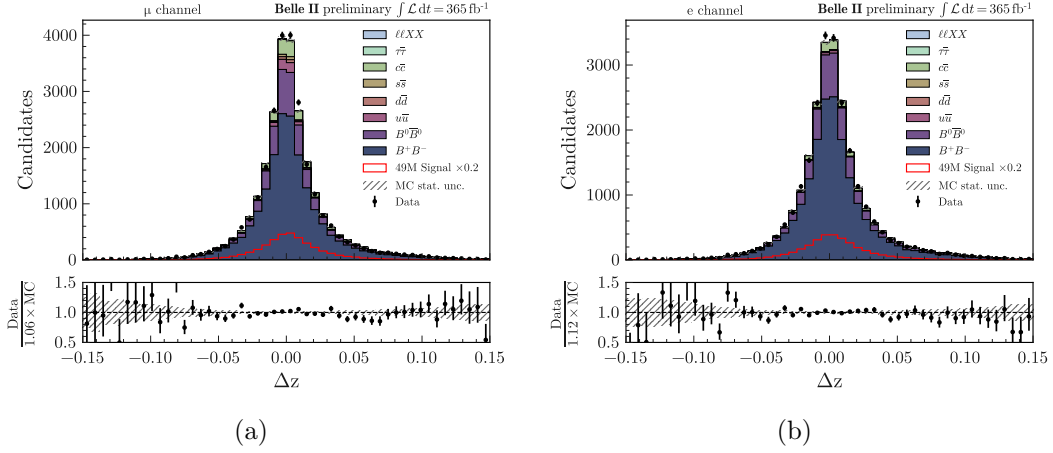


FIG. 115. Data/MC agreement of Δz for mu and e channels.

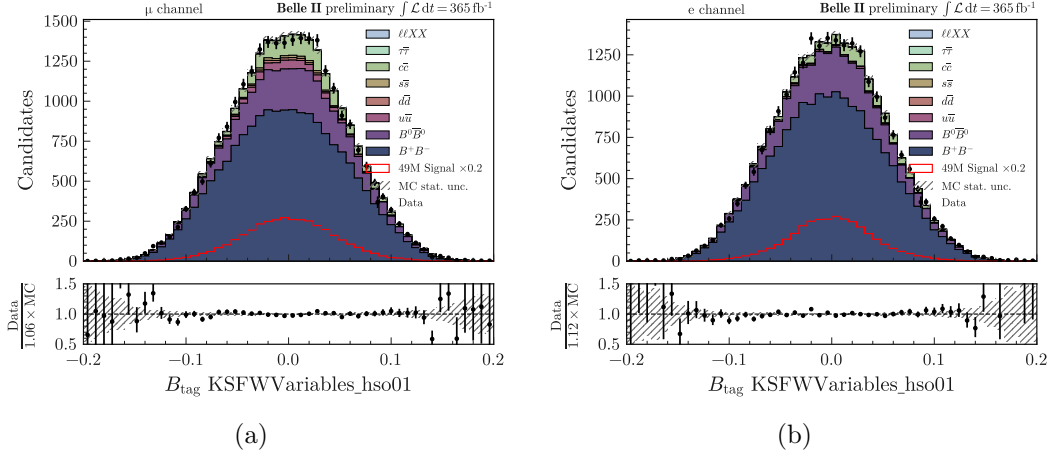
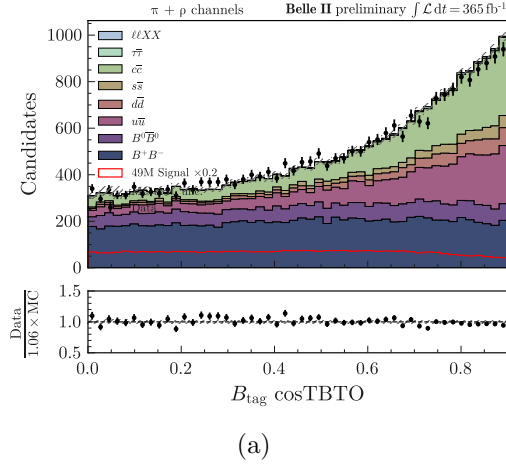
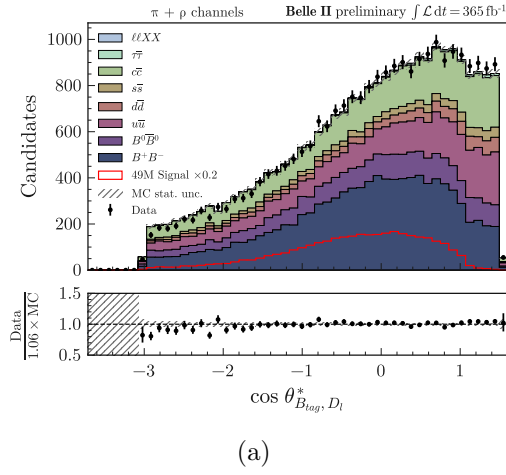
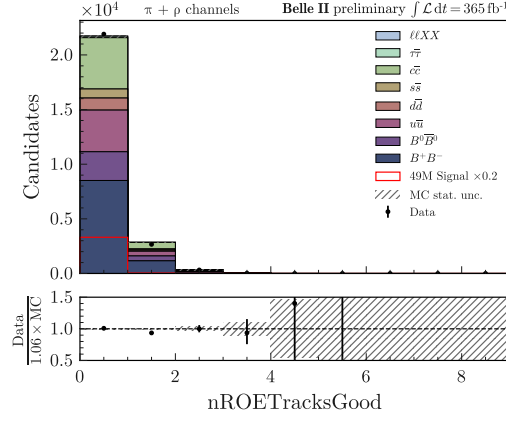


FIG. 116. Data/MC agreement of B_{tag} KSFVVariables_hso01 for mu and e channels.

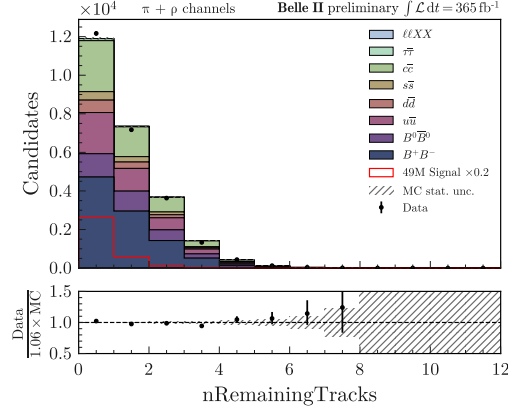
hadronic channels:

FIG. 117. Data/MC agreement of $B_{\text{tag}} \cos \text{TBTO}$ for hadronic channels.FIG. 118. Data/MC agreement of $\cos \theta_{B_{\text{tag}}, D_l}^*$ for hadronic channels.



(a)

FIG. 119. Data/MC agreement of `nROETracksGood` for hadronic channels.



(a)

FIG. 120. Data/MC agreement of `nRemainingTracks` for hadronic channels.

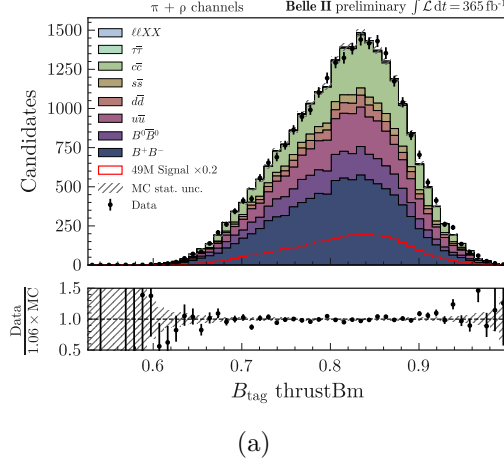


FIG. 121. Data/MC agreement of B_{tag} thrustBm for hadronic channels.

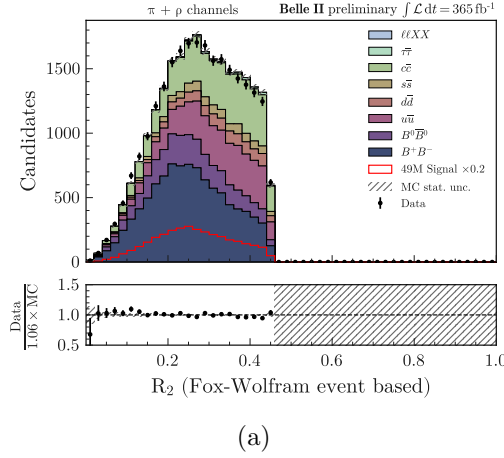
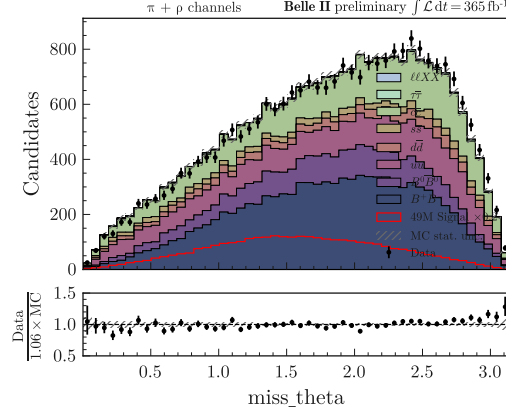
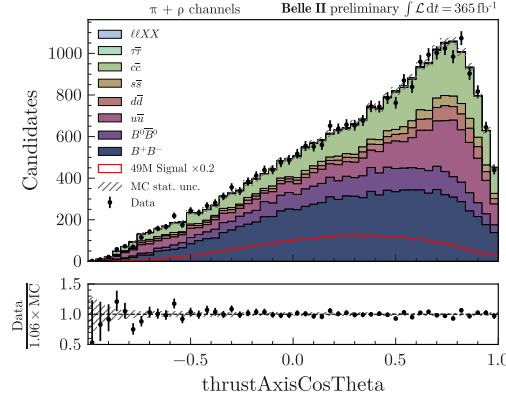


FIG. 122. Data/MC agreement of R_2 (Fox-Wolfram event based) for hadronic channels.



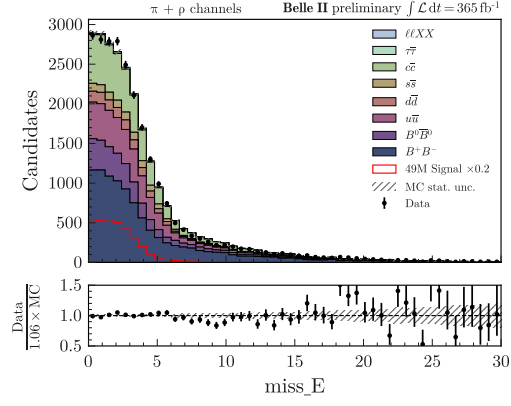
(a)

FIG. 123. Data/MC agreement of miss_theta for hadronic channels.



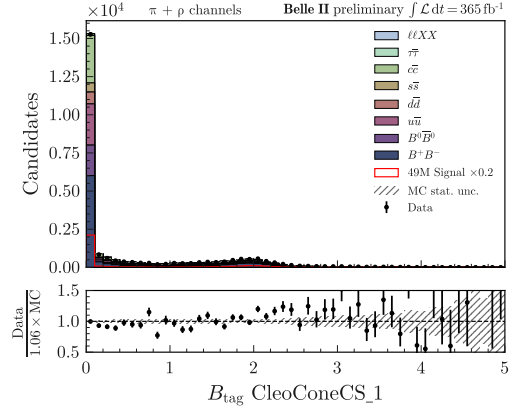
(a)

FIG. 124. Data/MC agreement of thrustAxisCosTheta for hadronic channels.



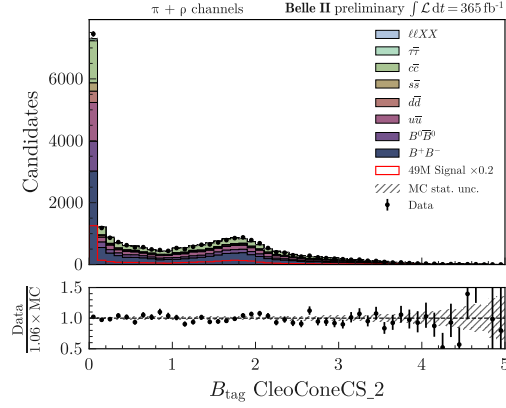
(a)

FIG. 125. Data/MC agreement of miss_E for hadronic channels.



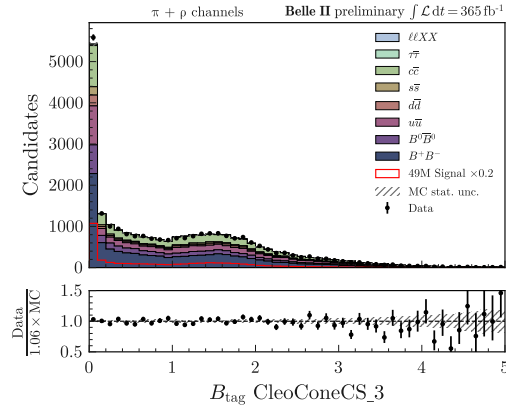
(a)

FIG. 126. Data/MC agreement of B_{tag} CleoConeCS_1 for hadronic channels.



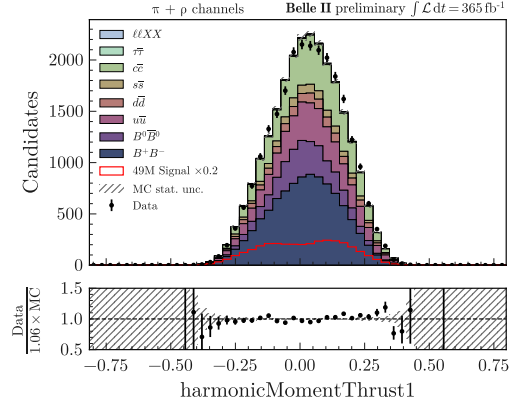
(a)

FIG. 127. Data/MC agreement of B_{tag} CleoConeCS_2 for hadronic channels.



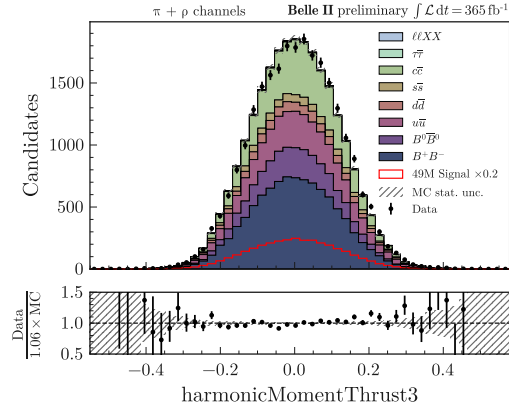
(a)

FIG. 128. Data/MC agreement of B_{tag} CleoConeCS_3 for hadronic channels.



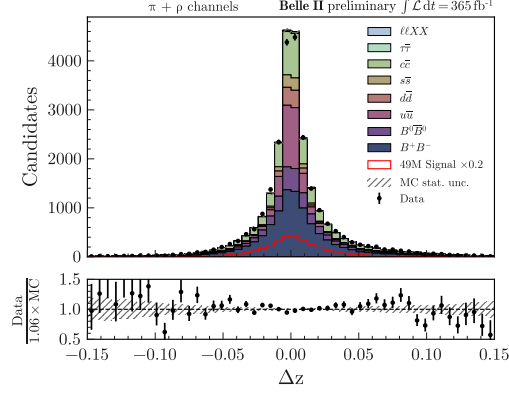
(a)

FIG. 129. Data/MC agreement of harmonicMomentThrust1 for hadronic channels.



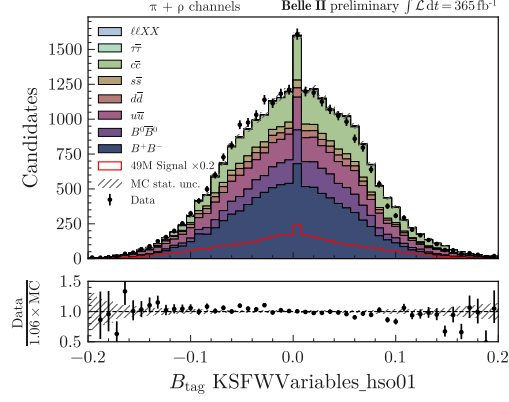
(a)

FIG. 130. Data/MC agreement of harmonicMomentThrust3 for hadronic channels.



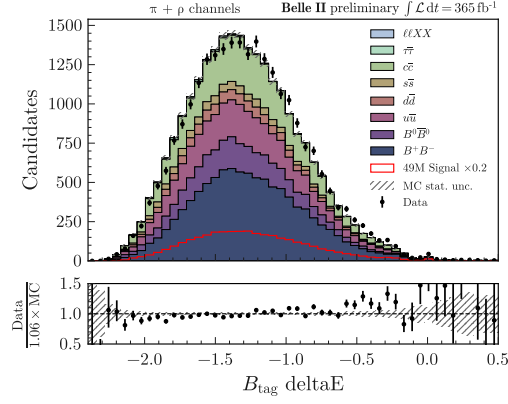
(a)

FIG. 131. Data/MC agreement of Δz for hadronic channels.



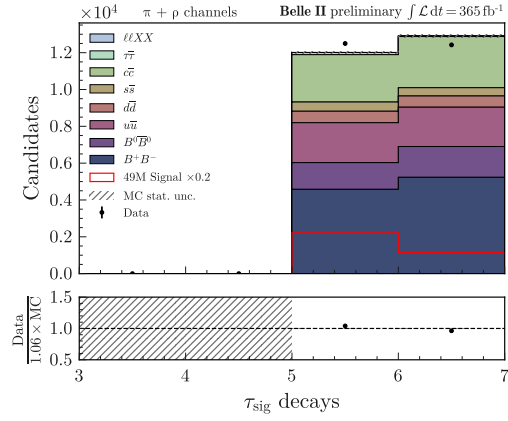
(a)

FIG. 132. Data/MC agreement of B_{tag} KSFVVariables_hso01 for hadronic channels.



(a)

FIG. 133. Data/MC agreement of $B_{tag} \text{ deltaE}$ for hadronic channels.



(a)

FIG. 134. Data/MC agreement of $\tau_{sig} \text{ decays}$ for hadronic channels.

1213 **D. Corrections, additional plots**

1214 *D.a. ROE corrections validation with the extra track SB control sample*

1215 The ROE corrections method is also validated using the extra track SB control
1216 sample (see Section 2) where we don't expect signal. The distributions before and
1217 after step 1 are shown in Figures 135 and 136. The photon multiplicity N_{γ}^{ROE} and
1218 the $E_{\text{extra}}^{\text{ROE}}$ before and after step 2 (a) are shown in Figures 137 and 138, and before
1219 and after the step 2 (b) are shown in Figures 139 and 140. The final distributions
1220 with all corrections applied are shown in Figure 141 for after the pre-selection. No
1221 data/MC discrepancy is observed in the $E_{\text{extra}}^{\text{ROE}}$ distribution after all corrections.

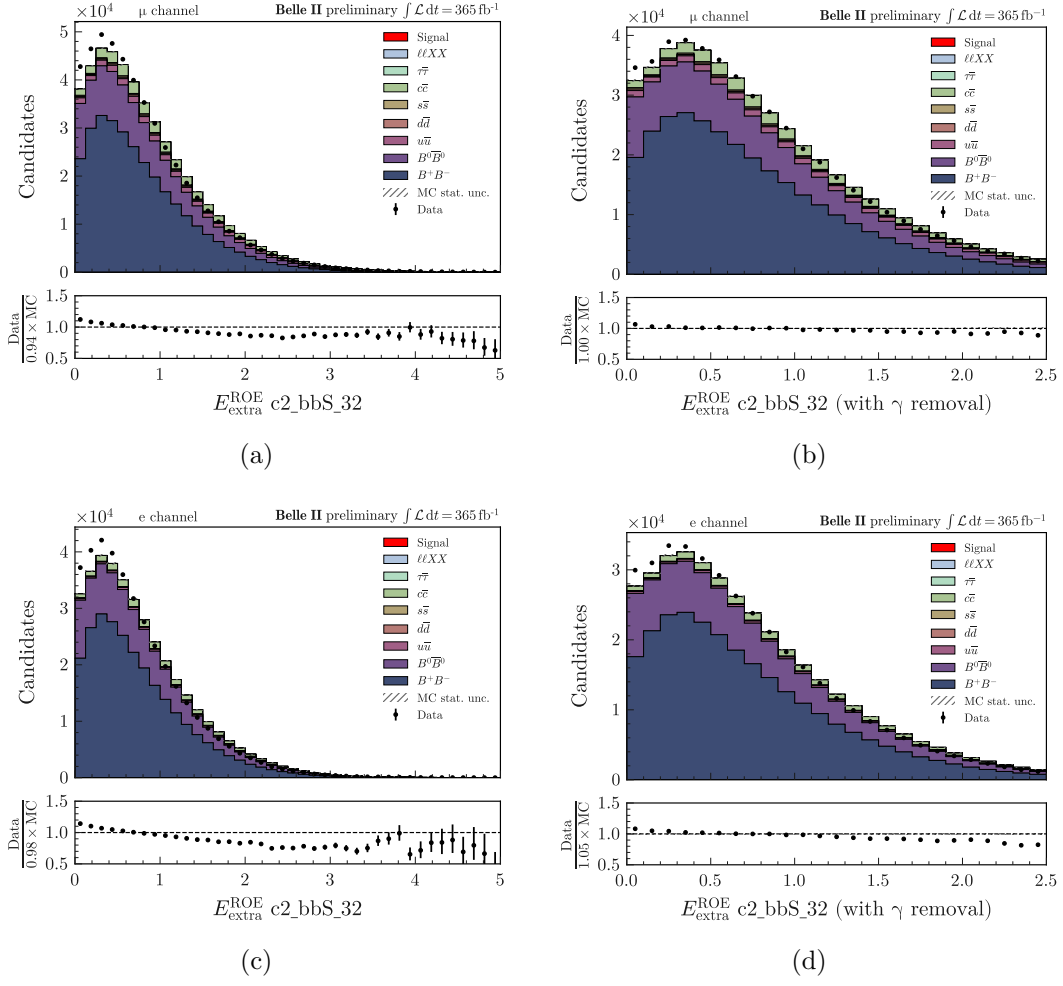


FIG. 135. Comparison between data and MC of $E_{\text{extra}}^{\text{ROE}}$ for μ and e decays before (left) and after (right) first step of the correction.

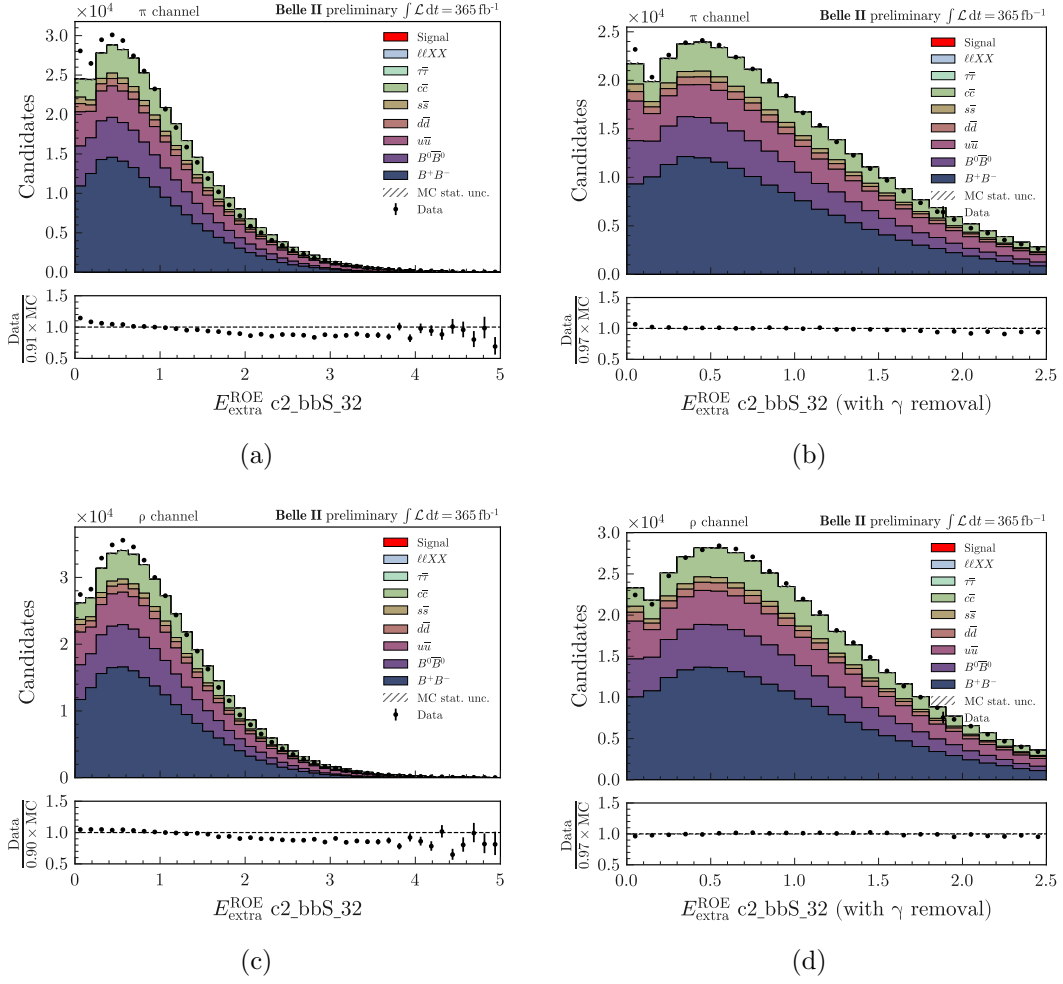


FIG. 136. Comparison between data and MC of $E_{\text{extra}}^{\text{ROE}}$ for π and ρ decays before (left) and after (right) first step of the correction.

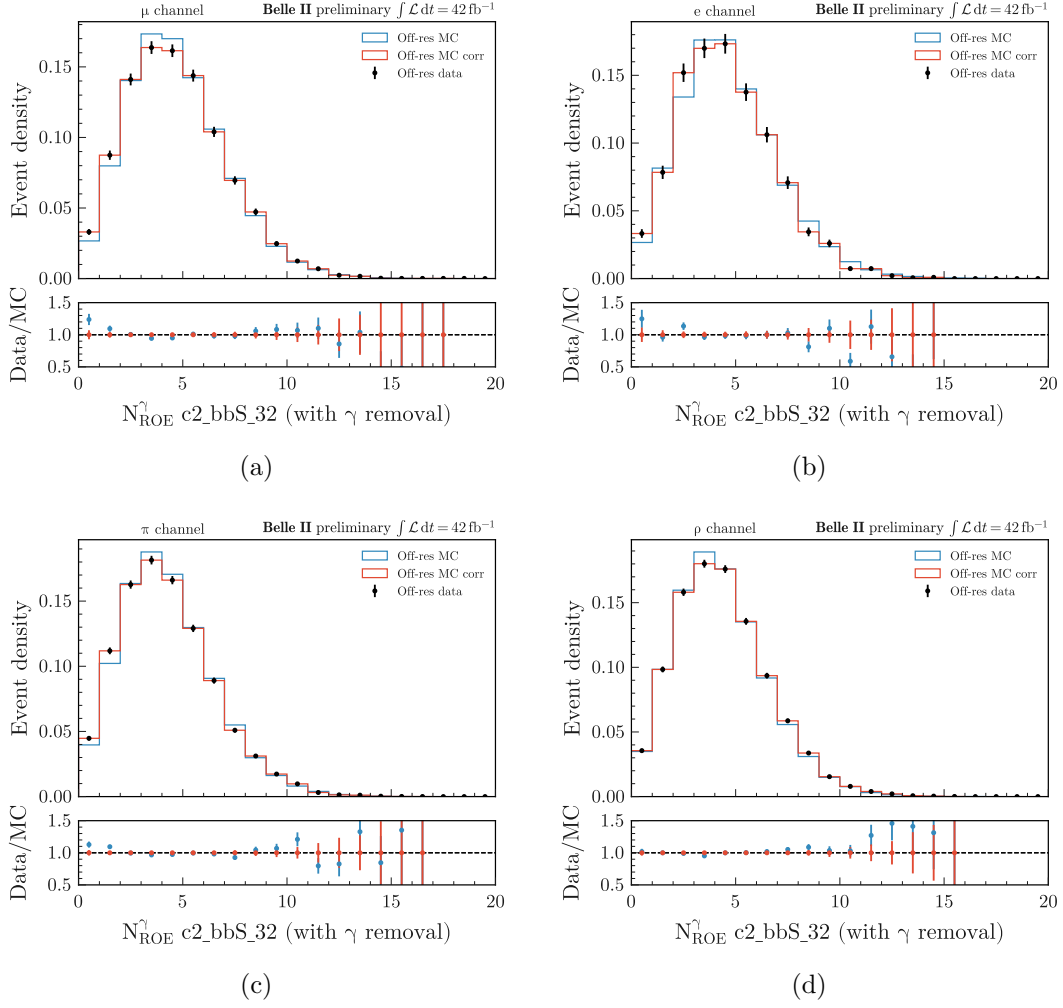
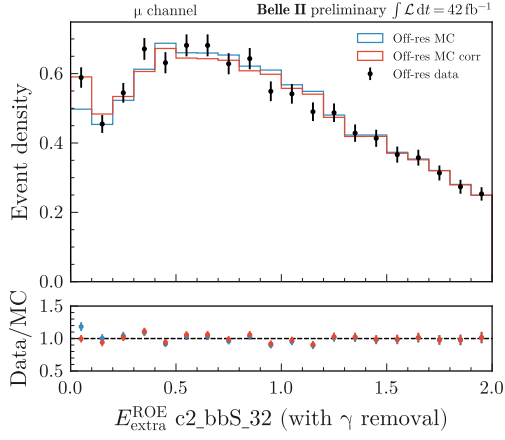
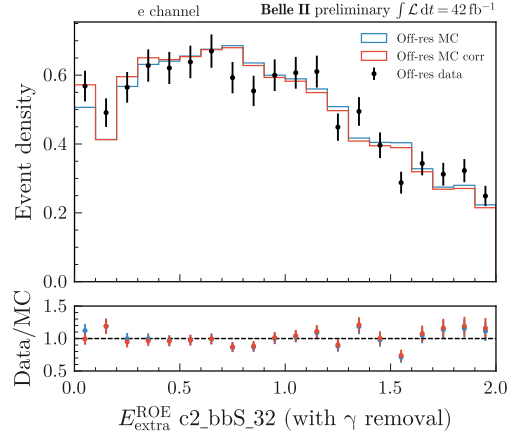


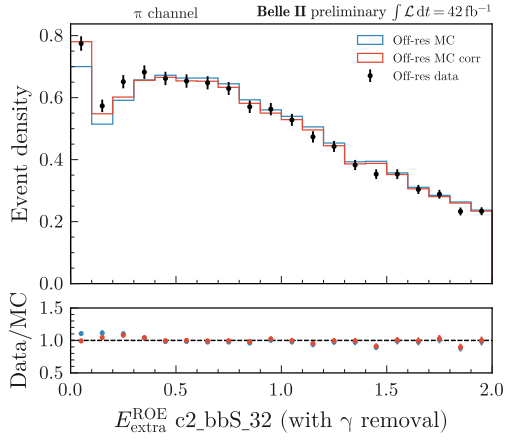
FIG. 137. Comparison between off-resonance data and MC of N_{γ}^{ROE} for each τ decay.



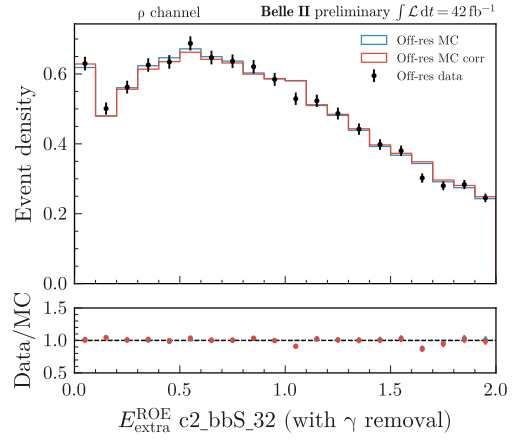
(a)



(b)



(c)



(d)

FIG. 138. Comparison between off-resonance data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay after reweighting each bin of the N_{γ}^{ROE} distribution to its data/MC ratio.

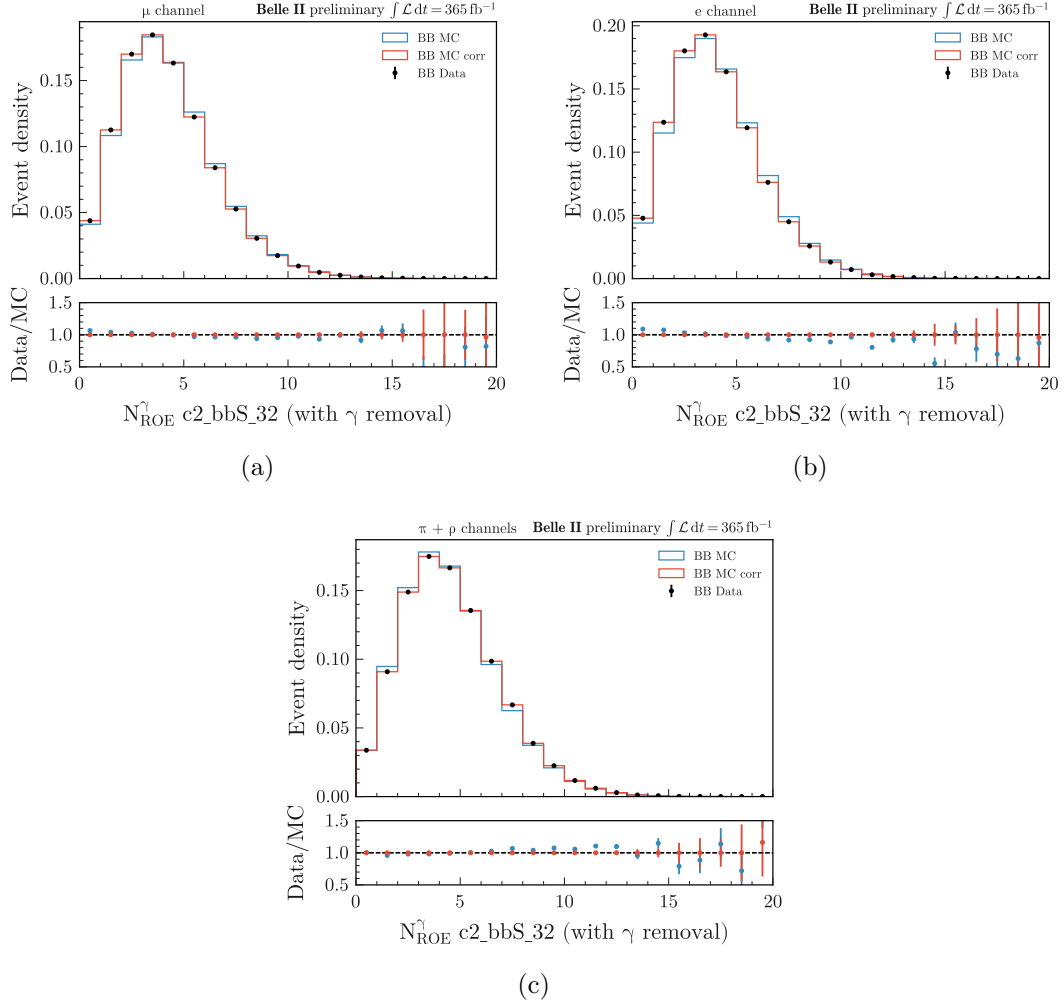
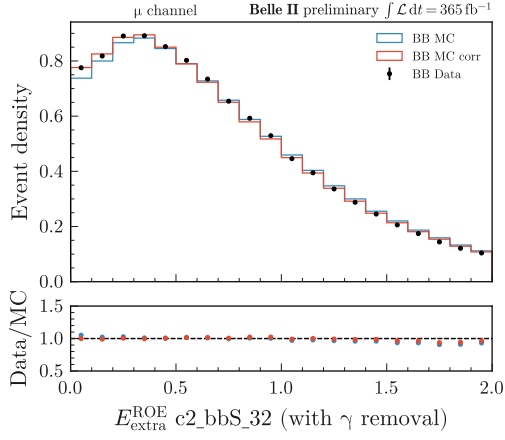
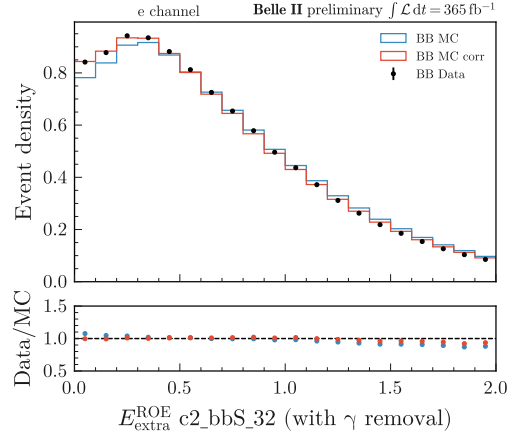


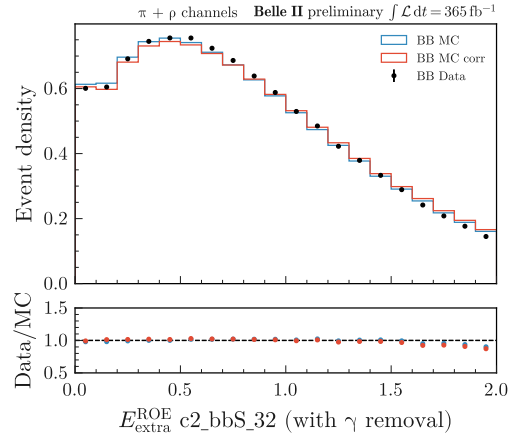
FIG. 139. Comparison between on-resonance $B\bar{B}$ data and MC of N_{γ}^{ROE} for each τ decay.



(a)



(b)



(c)

FIG. 140. Comparison between on-resonance $B\bar{B}$ data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay after reweighting each bin of the N_{γ}^{ROE} distribution to its data/MC ratio.

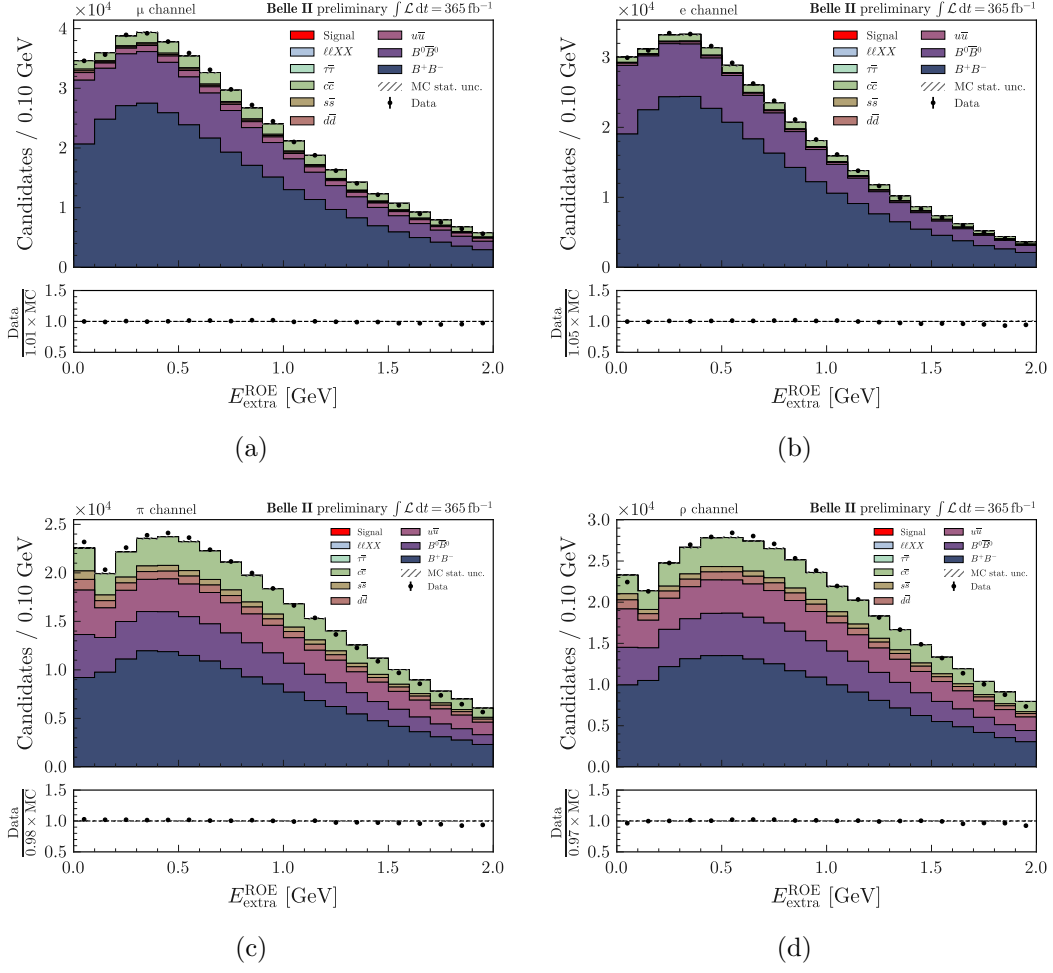


FIG. 141. Comparison between data and MC of $E_{\text{extra}}^{\text{ROE}}$ for each τ decay with all corrections after pre-selection.

1223 To further check the ROE corrections, we split the BDT sidebands into 2 sub-
 1224 samples, one more background like (BDT score < BDT cut /2) and one more signal
 1225 like (BDT score between BDT cut /2 and BDT cut). The $E_{\text{extra}}^{\text{ROE}}$ distributions for
 1226 each τ decay in the two sub-samples are shown in Figures 142 and 143. No data/MC
 1227 discrepancy is observed in the $E_{\text{extra}}^{\text{ROE}}$ distribution in both sub-samples after applying
 1228 all corrections.

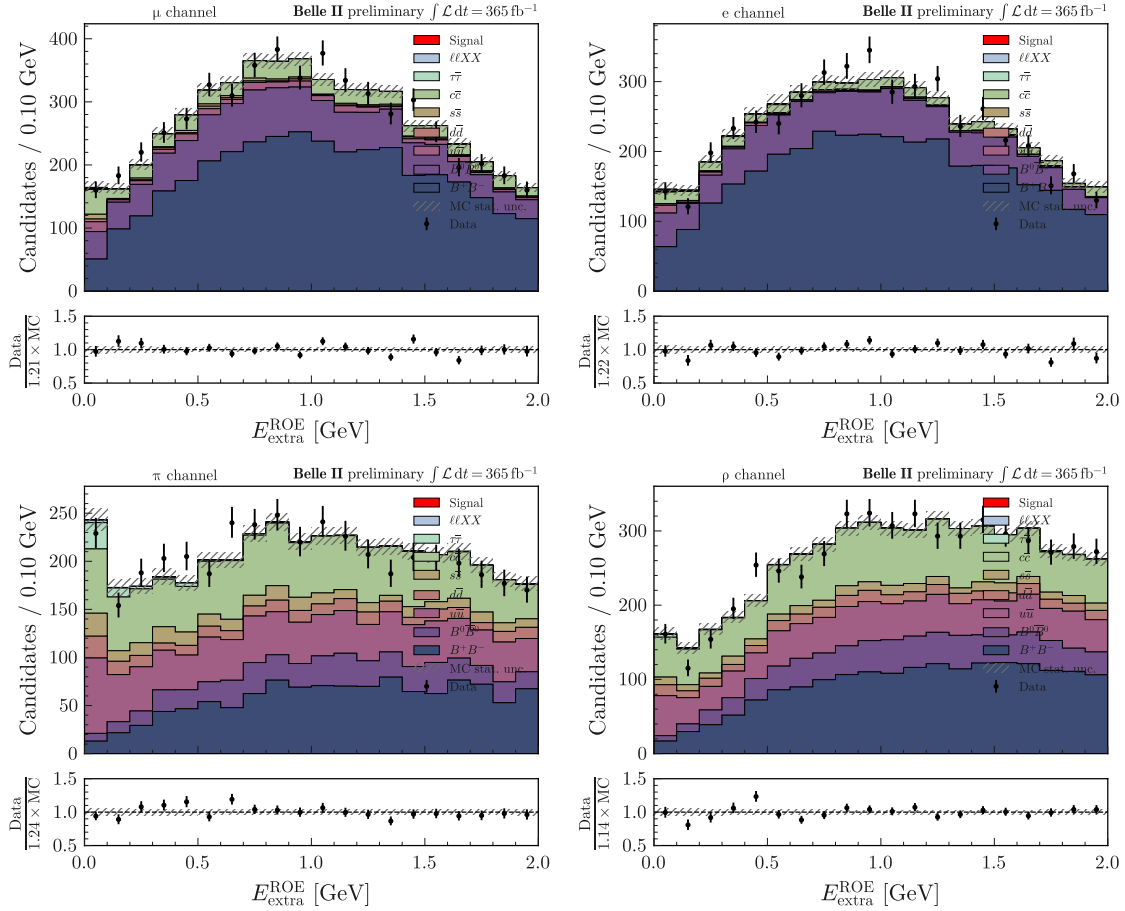


FIG. 142. $E_{\text{extra}}^{\text{ROE}}$ distribution for each τ decay in the low BDT sideband.

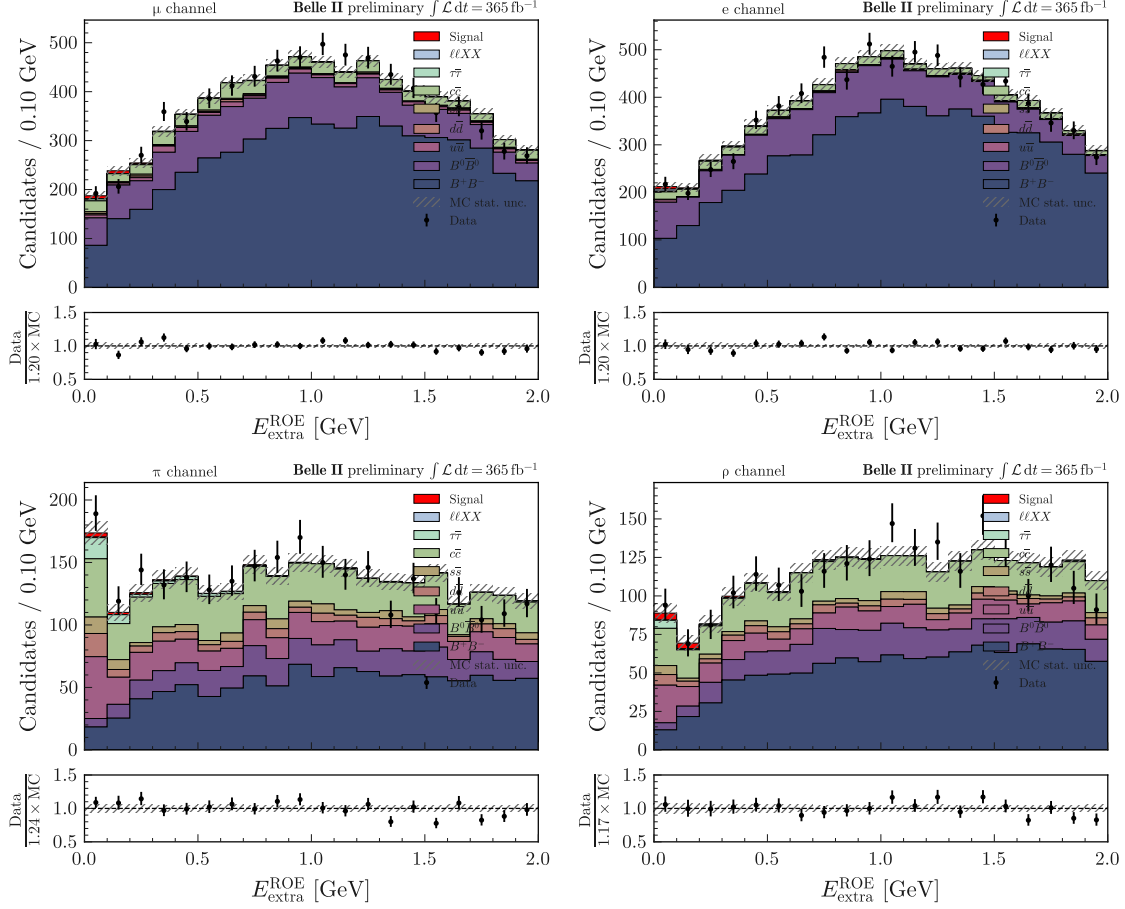


FIG. 143. $E_{\text{extra}}^{\text{ROE}}$ distribution for each τ decay in the high BDT sideband.

D.c. $B\bar{B}$ ROE correction transferability checks

The ROE corrections for the $B\bar{B}$ background can be derived from the BDT sideband or the extra track SB control sample. Fitting toys made of MC templates using the method to get the $B\bar{B}$ ROE corrections using templates from the second method shows a non-negligible bias on the POI. To decide which region is more suitable for deriving the $B\bar{B}$ ROE corrections, we compare the $B\bar{B}$ composition as well as the $E_{\text{extra}}^{\text{ROE}}$ and N_{γ}^{ROE} distributions. The $B\bar{B}$ composition results can be found in Tables 32 and 33 for all channels and in Tables 34 to 39 for each τ decay separately. The $E_{\text{extra}}^{\text{ROE}}$ and N_{γ}^{ROE} distributions are shown in Figure 144. The BDT sideband is found to be more similar to the signal region than the extra track SB control sample,

1239 which have less modes containing a π^0 while having more modes containing D^{**} .
1240 Therefore, the BDT sideband is used for deriving the $B\bar{B}$ ROE corrections in the
1241 nominal fit.

Mode combination	After Sel. (%)	BDT Sideband (%)	Extra Track SB CC (%)
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\ell^-\bar{\nu}_\ell)$	35.51	30.49	17.11
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\ell^-\bar{\nu}_\ell)$	19.01	21.54	17.81
$(D^+\ell^-\bar{\nu}_\ell, D^+\ell^-\bar{\nu}_\ell)$	11.74	6.70	1.83
(Pythia (3 fs parts.), $D^{*+}\ell^-\bar{\nu}_\ell$)	2.61	2.55	2.80
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\tau^-\bar{\nu}_\tau)$	2.24	2.18	1.32
(Pythia (3 fs parts.), $D^+\ell^-\bar{\nu}_\ell$)	2.05	2.07	1.12
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\tau^-\bar{\nu}_\tau)$	2.05	1.78	0.91
$(D^+\ell^-\bar{\nu}_\ell, D_0^{*+}\ell^-\bar{\nu}_\ell)$	1.77	0.76	1.00
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\pi^-)$	1.40	0.80	0.75
(Pythia (4 fs parts.), $D^{*+}\ell^-\bar{\nu}_\ell$)	1.30	0.98	2.29
$(D^+\ell^-\bar{\nu}_\ell, D_1^{*+}\ell^-\bar{\nu}_\ell)$	1.30	1.64	1.45
$(D^{*+}\ell^-\bar{\nu}_\ell, D_1^{*+}\ell^-\bar{\nu}_\ell)$	1.12	1.27	2.66
$(D^{*+}\tau^-\bar{\nu}_\tau, D^+\ell^-\bar{\nu}_\ell)$	1.12	1.27	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, \pi^+\ell^-\bar{\nu}_\ell)$	0.93	1.16	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, D_2^{*+}\ell^-\bar{\nu}_\ell)$	0.93	0.55	1.86
$(D^+\ell^-\bar{\nu}_\ell, D^+\tau^-\bar{\nu}_\tau)$	0.84	1.13	< 0.01
$(D^+\ell^-\bar{\nu}_\ell, \rho^+\ell^-\bar{\nu}_\ell)$	0.75	< 0.01	< 0.01
$(D^+\ell^-\bar{\nu}_\ell, \pi^+\ell^-\bar{\nu}_\ell)$	0.75	< 0.01	< 0.01
$(D^+\ell^-\bar{\nu}_\ell, D^+\pi^-)$	0.56	< 0.01	< 0.01
Remaining (below 0.5%)	11.18	10.44	15.89

TABLE 32. The $B^0\bar{B}^0$ composition in the signal region after full selection for all modes, in the BDT sideband and in the extra track SB control sample. The mode combinations are ordered by their percentage in the signal region.

Mode combination	After Sel. (%)	BDT Sideband (%)	Extra Track SB CC (%)
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\ell^-\bar{\nu}_\ell)$	29.99	29.71	22.02
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\ell^-\bar{\nu}_\ell)$	21.43	25.34	17.17
$(D^0\ell^-\bar{\nu}_\ell, D^0\ell^-\bar{\nu}_\ell)$	8.18	8.00	6.09
$(D^{*0}\ell^-\bar{\nu}_\ell, \pi^0\ell^-\bar{\nu}_\ell)$	3.69	0.74	< 0.01
$(\text{Pythia (3 fs parts.)}, D^{*0}\ell^-\bar{\nu}_\ell)$	2.85	2.52	2.51
$(D^{*0}\ell^-\bar{\nu}_\ell, \tau^-\bar{\nu}_\tau)$	2.27	0.66	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\tau^-\bar{\nu}_\tau)$	2.22	1.89	1.22
$(\text{Pythia (3 fs parts.)}, D^0\ell^-\bar{\nu}_\ell)$	1.38	1.59	1.46
$(D^0\ell^-\bar{\nu}_\ell, \pi^0\ell^-\bar{\nu}_\ell)$	1.34	< 0.01	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\pi^-)$	1.26	1.04	1.22
$(D^{*0}\ell^-\bar{\nu}_\ell, D_0^{*0}\ell^-\bar{\nu}_\ell)$	1.17	1.42	1.46
$(D^0\ell^-\bar{\nu}_\ell, \tau^-\bar{\nu}_\tau)$	1.09	< 0.01	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\tau^-\bar{\nu}_\tau)$	0.96	1.27	0.92
$(D^{*0}\ell^-\bar{\nu}_\ell, \rho^- D^0)$	0.92	1.15	1.34
$(D^{*0}\tau^-\bar{\nu}_\tau, D^0\ell^-\bar{\nu}_\ell)$	0.84	1.36	0.76
$(D^{*0}\ell^-\bar{\nu}_\ell, \bar{K}^0\pi^-)$	0.84	< 0.01	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D_2^{*0}\ell^-\bar{\nu}_\ell)$	0.80	0.74	1.18
$(\text{Pythia (4 fs parts.)}, D^{*0}\ell^-\bar{\nu}_\ell)$	0.80	0.83	1.64
$(D^0\ell^-\bar{\nu}_\ell, D^0\tau^-\bar{\nu}_\tau)$	0.71	0.81	0.51
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\pi^-)$	0.63	0.64	0.77
$(D^0\ell^-\bar{\nu}_\ell, \rho^- D^0)$	0.63	0.59	0.79
$(D^0\ell^-\bar{\nu}_\ell, D_0^{*0}\ell^-\bar{\nu}_\ell)$	0.59	0.61	0.81
$(D^0\ell^-\bar{\nu}_\ell, D^0\pi^-)$	0.55	< 0.01	0.64
$(D^{*0}\ell^-\bar{\nu}_\ell, \rho^0\ell^-\bar{\nu}_\ell)$	0.55	0.59	< 0.01
$(D^0\ell^-\bar{\nu}_\ell, \bar{K}^0\pi^-)$	0.55	< 0.01	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D_{10}\ell^-\bar{\nu}_\ell)$	0.50	1.00	1.67
$(D^{*0}\pi^-, D^0\ell^-\bar{\nu}_\ell)$	0.50	< 0.01	< 0.01
$(\text{Pythia (4 fs parts.)}, D^0\ell^-\bar{\nu}_\ell)$	0.50	< 0.01	0.99
Remaining (below 0.5%)	10.78	7.93	13.26

TABLE 33. The B^+B^- composition in the signal region after full selection for all modes, in the BDT sideband and in the extra track SB control sample. The mode combinations are ordered by their percentage in the signal region.

Mode combination	After Sel. (%)	BDT Sideband (%)	Extra Track SB CC (%)
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\ell^-\bar{\nu}_\ell)$	34.88	34.79	21.70
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\ell^-\bar{\nu}_\ell)$	20.74	24.48	22.97
$(D^+\ell^-\bar{\nu}_\ell, D^+\ell^-\bar{\nu}_\ell)$	14.73	7.17	2.23
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\tau^-\bar{\nu}_\tau)$	2.71	1.84	1.43
(Pythia (3 fs parts.), $D^{*+}\ell^-\bar{\nu}_\ell$)	2.71	2.01	2.31
(Pythia (3 fs parts.), $D^+\ell^-\bar{\nu}_\ell$)	2.13	2.19	0.90
$(D^{*+}\tau^-\bar{\nu}_\tau, D^+\ell^-\bar{\nu}_\ell)$	1.55	1.66	0.51
$(D^+\ell^-\bar{\nu}_\ell, D_0^{*+}\ell^-\bar{\nu}_\ell)$	1.55	0.52	1.20
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\tau^-\bar{\nu}_\tau)$	1.36	1.75	0.98
$(D^+\ell^-\bar{\nu}_\ell, \rho^+\ell^-\bar{\nu}_\ell)$	1.36	< 0.01	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, D_1^+\ell^-\bar{\nu}_\ell)$	1.36	0.70	3.38
$(D^{*+}\ell^-\bar{\nu}_\ell, D_2^{*+}\ell^-\bar{\nu}_\ell)$	1.16	0.70	2.25
$(D^+\ell^-\bar{\nu}_\ell, D^+\tau^-\bar{\nu}_\tau)$	1.16	1.31	< 0.01
$(D^+\ell^-\bar{\nu}_\ell, D_1^+\ell^-\bar{\nu}_\ell)$	0.97	1.84	1.77
(Pythia (4 fs parts.), $D^{*+}\ell^-\bar{\nu}_\ell$)	0.97	0.70	1.42
$(D^+\ell^-\bar{\nu}_\ell, \pi^+\ell^-\bar{\nu}_\ell)$	0.97	< 0.01	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, D_0^{*+}\ell^-\bar{\nu}_\ell)$	0.78	1.31	2.62
(Pythia (4 fs parts.), $D^+\ell^-\bar{\nu}_\ell$)	0.58	0.70	0.78
Remaining (below 0.5%)	7.56	4.28	6.89

TABLE 34. The $B^0\bar{B}^0$ composition in the signal region after full selection for the μ mode, in the BDT sideband and in the extra track SB control sample. The mode combinations are ordered by their percentage in the signal region.

Mode combination	After Sel. (%)	BDT Sideband (%)	Extra Track SB CC (%)
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\ell^-\bar{\nu}_\ell)$	33.90	33.70	28.90
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\ell^-\bar{\nu}_\ell)$	23.36	29.94	22.64
$(D^0\ell^-\bar{\nu}_\ell, D^0\ell^-\bar{\nu}_\ell)$	9.44	9.23	7.91
$(D^{*0}\ell^-\bar{\nu}_\ell, \pi^0\ell^-\bar{\nu}_\ell)$	4.37	1.05	< 0.01
(Pythia (3 fs parts.), $D^{*0}\ell^-\bar{\nu}_\ell$)	2.49	2.14	2.15
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\tau^-\bar{\nu}_\tau)$	2.39	1.67	1.31
$(D^0\ell^-\bar{\nu}_\ell, \pi^0\ell^-\bar{\nu}_\ell)$	1.69	0.57	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D_0^0\ell^-\bar{\nu}_\ell)$	1.29	1.48	1.75
(Pythia (3 fs parts.), $D^0\ell^-\bar{\nu}_\ell$)	1.19	1.38	1.24
$(D^{*0}\ell^-\bar{\nu}_\ell, D_2^{*0}\ell^-\bar{\nu}_\ell)$	1.19	0.67	1.42
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\tau^-\bar{\nu}_\tau)$	1.09	0.76	0.97
$(D^{*0}\tau^-\bar{\nu}_\tau, D^0\ell^-\bar{\nu}_\ell)$	1.09	1.19	0.83
$(D^{*0}\ell^-\bar{\nu}_\ell, \tau^-\bar{\nu}_\tau)$	0.89	< 0.01	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D_{10}\ell^-\bar{\nu}_\ell)$	0.80	0.90	2.16
$(D^{*0}\ell^-\bar{\nu}_\ell, \rho^0\ell^-\bar{\nu}_\ell)$	0.70	0.62	< 0.01
$(D^0\ell^-\bar{\nu}_\ell, D^0\tau^-\bar{\nu}_\tau)$	0.70	0.62	0.54
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\pi^-)$	0.70	< 0.01	< 0.01
$(D^0\ell^-\bar{\nu}_\ell, D_0^{*0}\ell^-\bar{\nu}_\ell)$	0.70	< 0.01	0.96
$(D^0\ell^-\bar{\nu}_\ell, \tau^-\bar{\nu}_\tau)$	0.60	-	< 0.01
Remaining (below 0.5%)	9.54	6.66	7.45

TABLE 35. The B^+B^- composition in the signal region after full selection for the μ mode, in the BDT sideband and in the extra track SB control sample. The mode combinations are ordered by their percentage in the signal region.

Mode combination	After Sel. (%)	BDT Sideband (%)	Extra Track SB CC (%)
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\ell^-\bar{\nu}_\ell)$	42.40	35.73	23.71
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\ell^-\bar{\nu}_\ell)$	20.28	25.57	24.87
$(D^+\ell^-\bar{\nu}_\ell, D^+\ell^-\bar{\nu}_\ell)$	10.60	8.52	2.27
(Pythia (3 fs parts.), $D^+\ell^-\bar{\nu}_\ell$)	2.30	1.34	0.85
$(D^+\ell^-\bar{\nu}_\ell, D_0^{*+}\ell^-\bar{\nu}_\ell)$	2.30	1.15	1.20
(Pythia (3 fs parts.), $D^{*+}\ell^-\bar{\nu}_\ell$)	1.84	1.82	2.11
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\tau^-\bar{\nu}_\tau)$	1.61	1.44	0.94
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\tau^-\bar{\nu}_\tau)$	1.61	1.63	1.42
$(D^+\ell^-\bar{\nu}_\ell, D_1^+\ell^-\bar{\nu}_\ell)$	1.61	1.53	1.90
$(D^{*+}\ell^-\bar{\nu}_\ell, D_1^+\ell^-\bar{\nu}_\ell)$	1.15	2.01	3.56
(Pythia (4 fs parts.), $D^{*+}\ell^-\bar{\nu}_\ell$)	0.92	1.05	1.30
$(D^{*+}\ell^-\bar{\nu}_\ell, D_2^{*+}\ell^-\bar{\nu}_\ell)$	0.92	0.67	2.46
$(D^{*+}\tau^-\bar{\nu}_\tau, D^+\ell^-\bar{\nu}_\ell)$	0.69	< 0.01	0.53
$(D^+\ell^-\bar{\nu}_\ell, D^+\tau^-\bar{\nu}_\tau)$	0.69	< 0.01	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, J/\psi K_L^0)$	0.69	< 0.01	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, \pi^+\ell^-\bar{\nu}_\ell)$	0.69	1.25	0.63
Remaining (below 0.5%)	8.76	4.98	6.63

TABLE 36. The $B^0\bar{B}^0$ composition in the signal region after full selection for the e mode, in the BDT sideband and in the extra track SB control sample. The mode combinations are ordered by their percentage in the signal region.

Mode combination	After Sel. (%)	BDT Sideband (%)	Extra Track SB CC (%)
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\ell^-\bar{\nu}_\ell)$	35.31	35.41	31.24
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\ell^-\bar{\nu}_\ell)$	27.06	29.69	24.85
$(D^0\ell^-\bar{\nu}_\ell, D^0\ell^-\bar{\nu}_\ell)$	8.25	8.70	8.56
$(D^{*0}\ell^-\bar{\nu}_\ell, \pi^0\ell^-\bar{\nu}_\ell)$	4.73	0.70	< 0.01
$(\text{Pythia (3 fs parts.)}, D^{*0}\ell^-\bar{\nu}_\ell)$	2.31	1.89	1.88
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\tau^-\bar{\nu}_\tau)$	1.65	1.09	1.31
$(D^0\ell^-\bar{\nu}_\ell, \pi^0\ell^-\bar{\nu}_\ell)$	1.54	< 0.01	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D_0^{*0}\ell^-\bar{\nu}_\ell)$	1.32	1.59	1.83
$(D^{*0}\ell^-\bar{\nu}_\ell, \tau^-\bar{\nu}_\tau)$	1.32	< 0.01	< 0.01
$(\text{Pythia (3 fs parts.)}, D^0\ell^-\bar{\nu}_\ell)$	1.21	1.19	1.11
$(D^{*0}\ell^-\bar{\nu}_\ell, \eta\ell^-\bar{\nu}_\ell)$	0.88	< 0.01	< 0.01
$(\text{Pythia (4 fs parts.)}, D^{*0}\ell^-\bar{\nu}_\ell)$	0.77	0.80	0.88
$(D^{*0}\tau^-\bar{\nu}_\tau, D^0\ell^-\bar{\nu}_\ell)$	0.77	1.19	0.81
$(D^0\ell^-\bar{\nu}_\ell, D_0^{*0}\ell^-\bar{\nu}_\ell)$	0.66	0.80	1.00
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\tau^-\bar{\nu}_\tau)$	0.66	1.14	0.97
$(D^0\ell^-\bar{\nu}_\ell, \tau^-\bar{\nu}_\tau)$	0.55	< 0.01	< 0.01
Remaining (below 0.5%)	10.12	8.95	12.68

TABLE 37. The B^+B^- composition in the signal region after full selection for the e mode, in the BDT sideband and in the extra track SB control sample. The mode combinations are ordered by their percentage in the signal region.

Mode combination	After Sel. (%)	BDT Sideband (%)	Extra Track SB CC (%)
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\ell^-\bar{\nu}_\ell)$	13.82	11.96	6.74
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\pi^-)$	12.20	3.57	1.79
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\ell^-\bar{\nu}_\ell)$	7.32	8.04	6.47
$(D^{*+}\ell^-\bar{\nu}_\ell, D^+\tau^-\bar{\nu}_\tau)$	6.50	2.50	< 0.01
(Pythia (3 fs parts.), $D^{*+}\ell^-\bar{\nu}_\ell$)	4.88	5.00	3.91
(Pythia (4 fs parts.), $D^{*+}\ell^-\bar{\nu}_\ell$)	4.07	1.43	4.04
$(D^{*+}\ell^-\bar{\nu}_\ell, \pi^+\ell^-\bar{\nu}_\ell)$	4.07	< 0.01	< 0.01
$(D^+\ell^-\bar{\nu}_\ell, D^+\pi^-)$	4.07	< 0.01	< 0.01
$(D^+\ell^-\bar{\nu}_\ell, D^+\ell^-\bar{\nu}_\ell)$	3.25	2.32	1.05
$(D^+\ell^-\bar{\nu}_\ell, \rho^-D^+)$	3.25	1.96	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\tau^-\bar{\nu}_\tau)$	2.44	3.93	1.13
$(D^{*+}\ell^-\bar{\nu}_\ell, D^{*+}\pi^-)$	2.44	1.07	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, \rho^-D^+)$	2.44	4.46	2.41
$(D^{*+}D_s^-, D^{*+}\ell^-\bar{\nu}_\ell)$	1.63	-	< 0.01
$(D^{*+}\ell^-\bar{\nu}_\ell, \rho^+\ell^-\bar{\nu}_\ell)$	1.63	< 0.01	< 0.01
(Pythia (4 fs parts.), $D^+\ell^-\bar{\nu}_\ell$)	1.63	1.43	1.89
$(D^+\ell^-\bar{\nu}_\ell, D_1^+\ell^-\bar{\nu}_\ell)$	1.63	1.43	< 0.01
(Pythia (3 fs parts.), $\pi^+\ell^-\bar{\nu}_\ell$)	1.63	-	< 0.01
Remaining (below 1.0%)	20.33	11.61	6.59

TABLE 38. The $B^0\bar{B}^0$ composition in the signal region after full selection for the hadronic modes, in the BDT sideband and in the extra track SB control sample. The mode combinations are ordered by their percentage in the signal region.

Mode combination	After Sel. (%)	BDT Sideband (%)	Extra Track SB CC (%)
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\ell^-\bar{\nu}_\ell)$	11.30	12.79	6.42
$(D^{*0}\ell^-\bar{\nu}_\ell, \tau^-\bar{\nu}_\tau)$	7.04	2.47	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\ell^-\bar{\nu}_\ell)$	6.40	9.63	4.46
$(D^0\ell^-\bar{\nu}_\ell, D^0\ell^-\bar{\nu}_\ell)$	5.33	4.60	1.93
(Pythia (3 fs parts.), $D^{*0}\ell^-\bar{\nu}_\ell$)	4.69	4.26	3.45
$(D^{*0}\ell^-\bar{\nu}_\ell, \bar{K}^0\pi^-)$	4.26	< 0.01	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\pi^-)$	4.26	3.67	2.92
$(D^0\ell^-\bar{\nu}_\ell, \tau^-\bar{\nu}_\tau)$	3.20	1.36	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, \rho^-D^0)$	2.99	3.75	3.45
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\tau^-\bar{\nu}_\tau)$	2.99	3.67	1.03
$(D^0\ell^-\bar{\nu}_\ell, D^0\pi^-)$	2.35	< 0.01	1.47
$(D^0\ell^-\bar{\nu}_\ell, \bar{K}^0\pi^-)$	2.35	< 0.01	< 0.01
$(D^0\ell^-\bar{\nu}_\ell, \rho^-D^0)$	2.35	1.62	1.95
$(D^{*0}\pi^-, D^0\ell^-\bar{\nu}_\ell)$	2.13	1.28	1.02
(Pythia (3 fs parts.), $D^0\ell^-\bar{\nu}_\ell$)	2.13	2.64	2.00
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\pi^-)$	1.92	1.71	1.80
(Pythia (4 fs parts.), $D^{*0}\ell^-\bar{\nu}_\ell$)	1.71	1.36	3.10
$(D^{*0}\ell^-\bar{\nu}_\ell, D^0\tau^-\bar{\nu}_\tau)$	1.28	2.39	< 0.01
$(D^0\ell^-\bar{\nu}_\ell, D^0\tau^-\bar{\nu}_\tau)$	1.28	1.02	< 0.01
$(D^{*0}\ell^-\bar{\nu}_\ell, \rho^0\ell^-\bar{\nu}_\ell)$	1.07	< 0.01	< 0.01
(Pythia (4 fs parts.), $D^0\ell^-\bar{\nu}_\ell$)	1.07	< 0.01	1.87
$(D^{*0}\ell^-\bar{\nu}_\ell, D^{*0}\rho^-)$	1.07	1.96	1.30
(Pythia (5 fs parts.), $D^{*0}\ell^-\bar{\nu}_\ell$)	1.07	< 0.01	1.32
Remaining (below 1.0%)	24.09	15.17	16.75

TABLE 39. The B^+B^- composition in the signal region after full selection for the hadronic modes, in the BDT sideband and in the extra track SB control sample. The mode combinations are ordered by their percentage in the signal region.

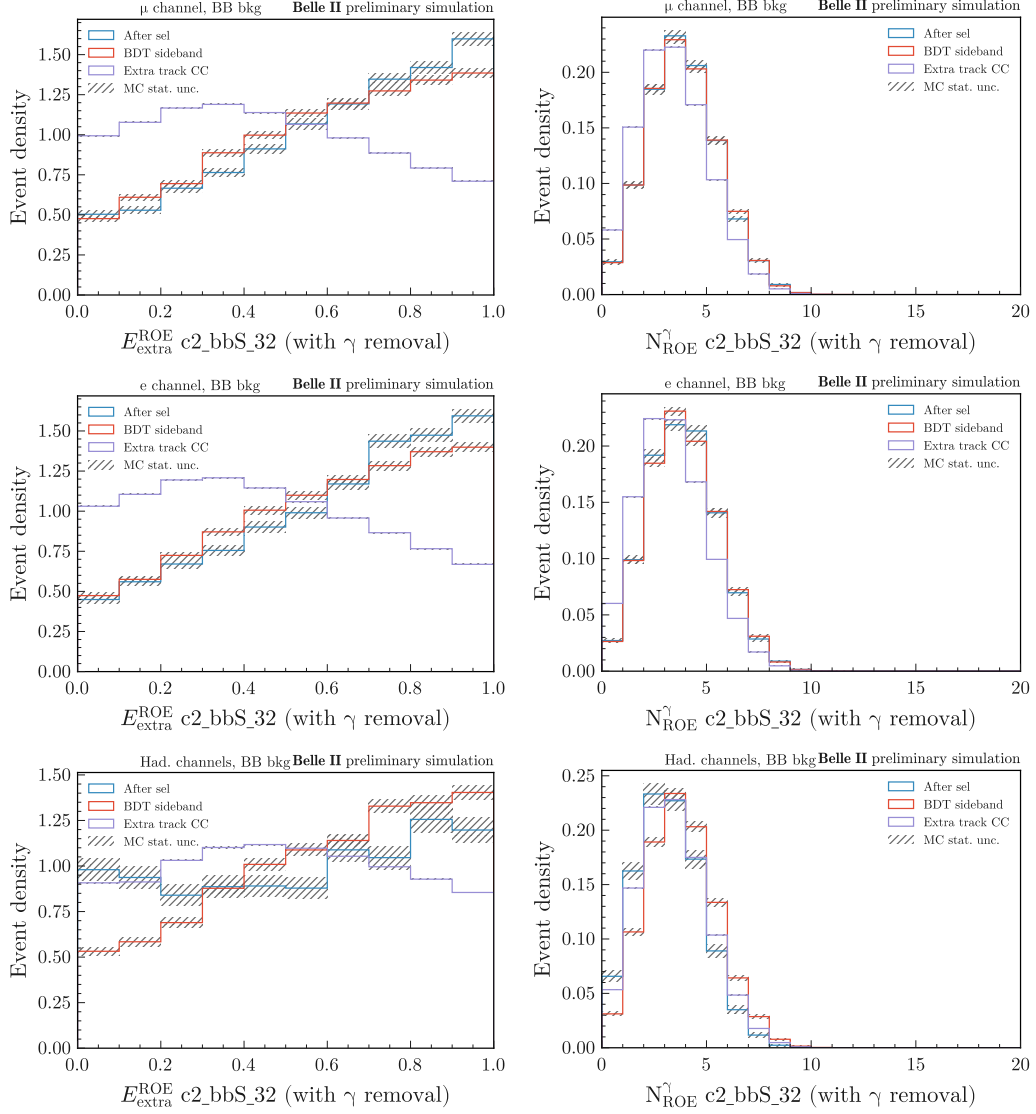


FIG. 144. The $E_{\text{extra}}^{\text{ROE}}$ and N_{ROE}^{γ} distributions of the $B\bar{B}$ background in the fitting region for the nominal sample after the selection, the BDT sideband and the extra track SB control sample.

1242 *D.d. $\tau_{p_{CMS}}$ checks*

1243 Figure 145 show the $\tau_{p_{CMS}}$ distribution in the fourth to ninth bins = 0.1 GeV of
1244 E_{extra}^{ROE} for the hadronic channels, where we don't expect signal. A good agreement is
1245 observed between data/MC.

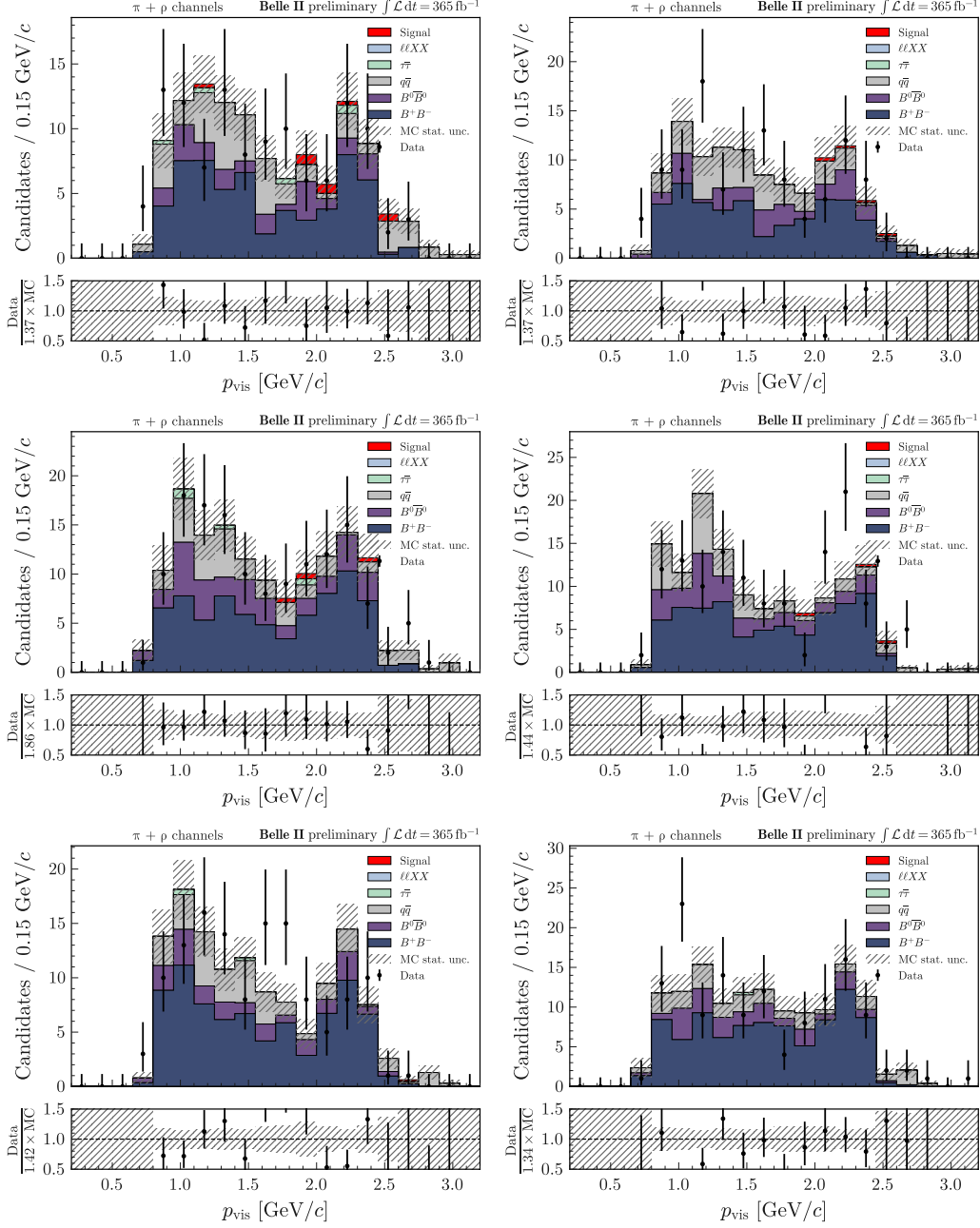


FIG. 145. The $\tau_{p_{\text{CMS}}}$ distribution in the fourth (top left), fifth (top right), sixth (middle left), seventh (middle right), eighth (bottom left) and ninth (bottom right) $E_{\text{extra}}^{\text{ROE}}$ bins for the hadronic channels.

1246 **E. Post-selection, additional material**

1247 This section shows additional material related to the events after the full selection.

1248 *E.a. Extra track study*

1249 Table 40 recaps the different ROE mask used to get the different extra track
 1250 variables. After the full selection, a few additional tracks are still observed. The
 1251 composition of these tracks is studied using MC truth information, and the results
 1252 are shown in Figure 146. Most of the remaining extra tracks are fake tracks or
 1253 secondary particles.

Variable	ROE mask	Description
nROETracksLoose	$d_r < 2$ and $ d_z < 4$ and θ in CDC Set to 0 at the reconstruction level. acceptance	
nROETracksGood	$d_r < 0.5$ and $ d_z < 2$ and θ in CDC acceptance and $n_{\text{CDCHits}} > 20$	Equal to 0 after the BDT selection.
nRemainingTracks	No cuts	Can be equal to 0 or 1 after the BDT selection.

TABLE 40. ROE mask used to get the different extra track variables.

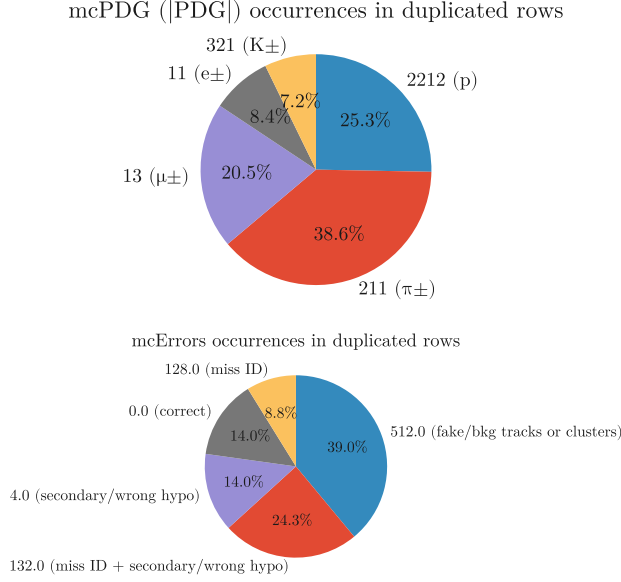


FIG. 146. Composition of the extra tracks observed after the full selection in MC.

E.b. Self-crossfeed study

The self-crossfeed (SCF) component is defined as the signal events where at least one of the reconstructed particles is not matched to the corresponding generated particle. The composition of the SCF events is studied using MC truth information, and the proportions are shown in Table 41. The overlap between the channels using truth-matching and generator information is shown in Tables 42 and 43 respectively. The distributions of the fit variables for the SCF events are shown in Figure 147. Most of the SCF events are due to:

- missID: the reconstructed particle is misidentified. The systematic uncertainty related to the particle identification is already included in the fit.
- $\pi - \rho$ overlap: the reconstructed particle is a π or ρ from a τ decay, but the generated particle is a ρ or π . In the case of $B \rightarrow \rho^+ \rho^-$ decays [34], the post-fit test shows around 10% SCF component due to π^0 miss-reconstruction.
- The majority of the "other" category is due to $\tau \rightarrow \pi 2\pi^0 \nu_\tau$ or $\tau \rightarrow 3\pi \nu_\tau$ decays, which are well measured.

We assign a 10% systematic uncertainty on the shape to the SCF component in the fit as a `histosys`, shown in Figure 148.

τ decay mode	Proportion of SCF events
μ channel	$13.83 \pm 0.14\%$
e channel	$20.79 \pm 0.17\%$
hadronic channels	$44.52 \pm 0.19\%$

TABLE 41. Proportions of SCF events for the different τ decay modes.

Decay mode	μ	e	π	ρ	other
μ	90.1	0.1	9.0	0.0	0.8
e	0.0	99.7	0.2	0.0	0.1
π	1.7	0.1	97.8	0.0	0.4
ρ	0.5	0.0	25.7	73.3	0.4

TABLE 42. Overlap between the channels using truth-matching (`mcPDG`) information (in %).

Decay mode	μ	e	π	ρ	other
μ	88.2	0.8	3.8	5.0	2.1
e	0.9	97.4	0.3	0.8	0.5
π	1.9	0.3	57.7	33.7	6.4
ρ	0.7	0.2	15.3	66.3	17.5

TABLE 43. Overlap between the channels using generator-level information with `GenMCTagTool` (in %).

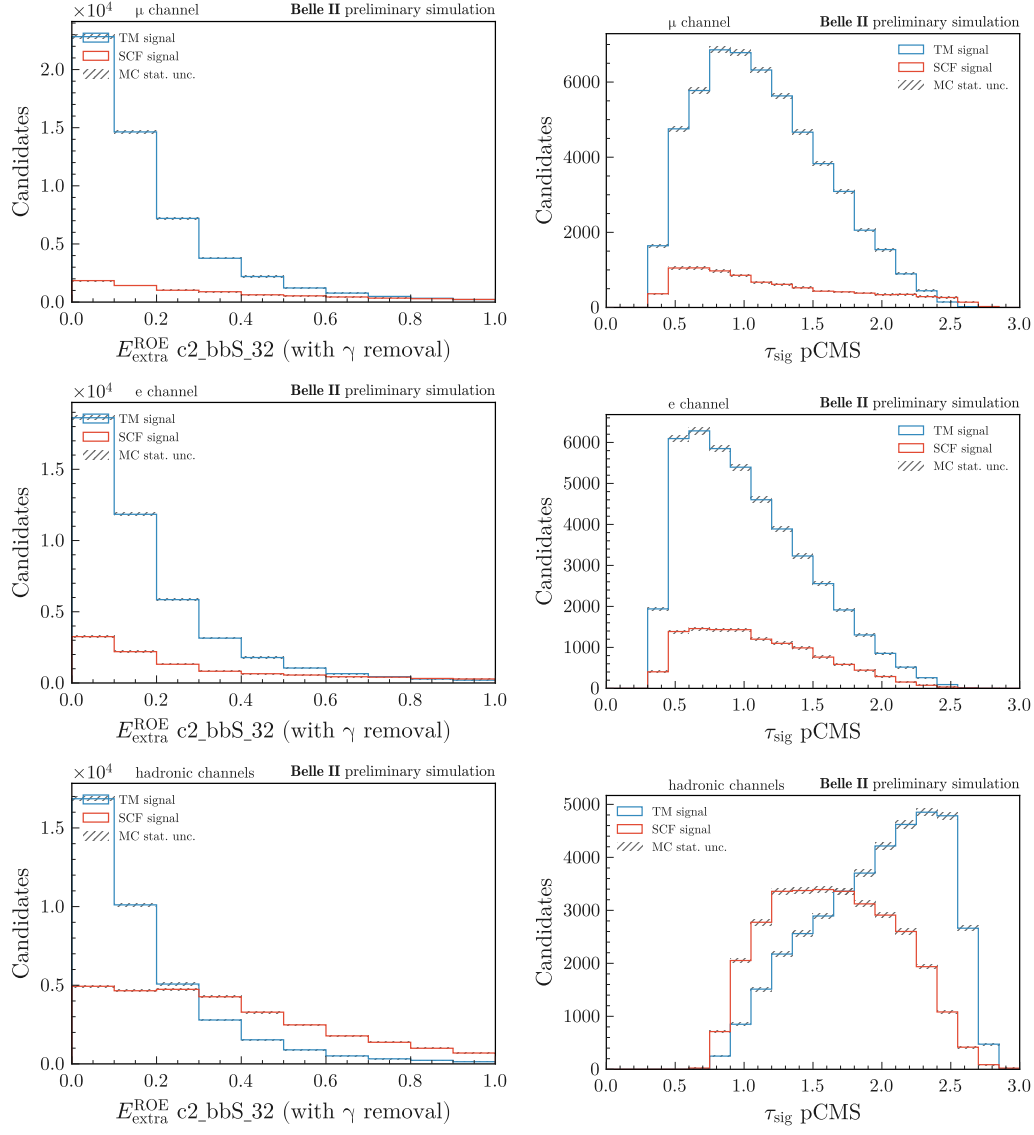


FIG. 147. $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ distributions for the different τ decay modes.

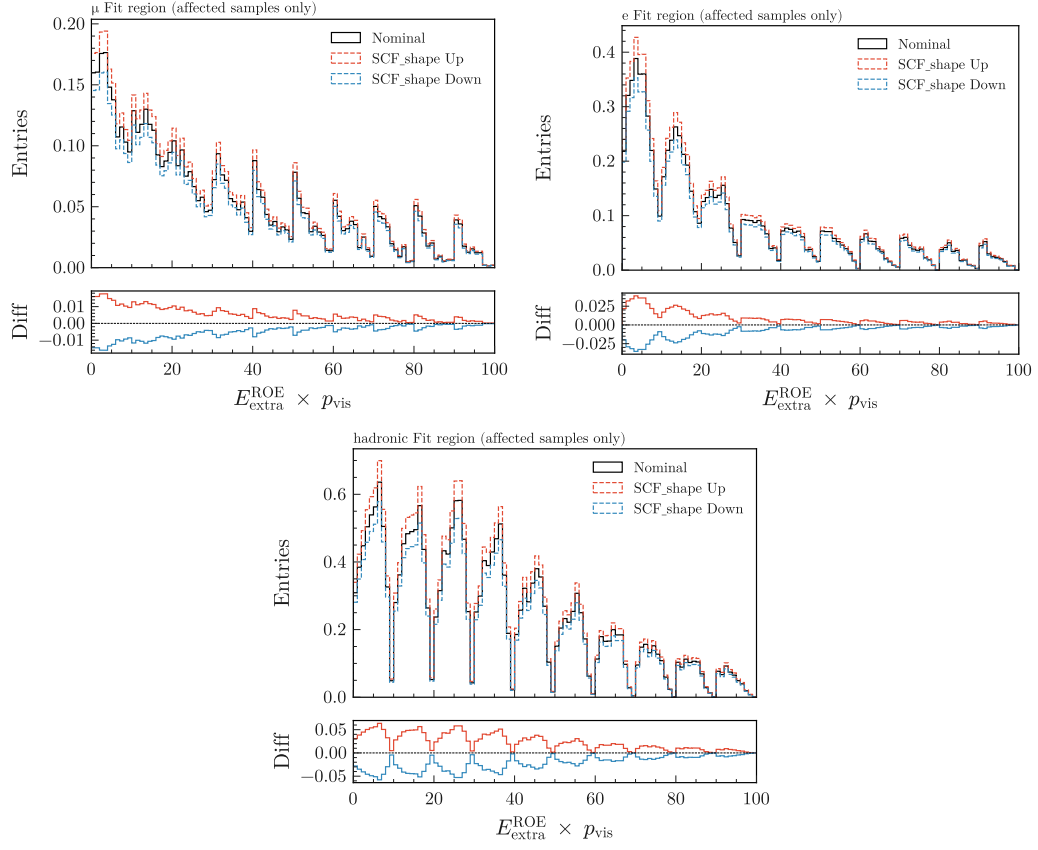


FIG. 148. Shape variation of the SCF component used as a `histosys` in the fit.

1271 **F. Fitting, additional plots**

1272 This section shows additional plots related to the fit procedure.

1273 *F.a. MC templates*

1274 Here are shown unstacked templates for the three channels.

1275

1276 μ channel:

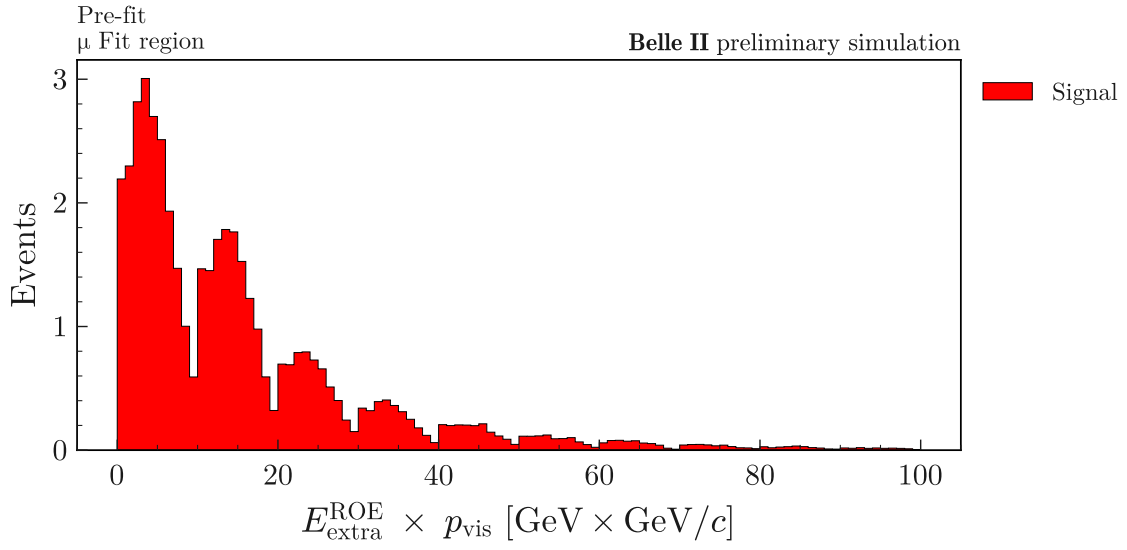


FIG. 149. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the signal sample in the μ channel before the fit.

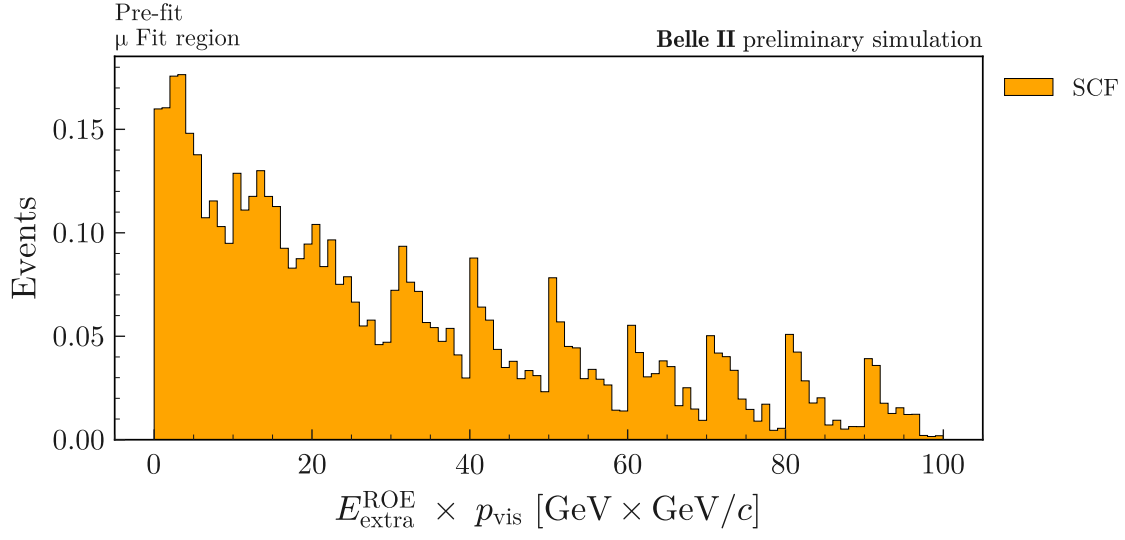


FIG. 150. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the SCF sample in the μ channel before the fit.

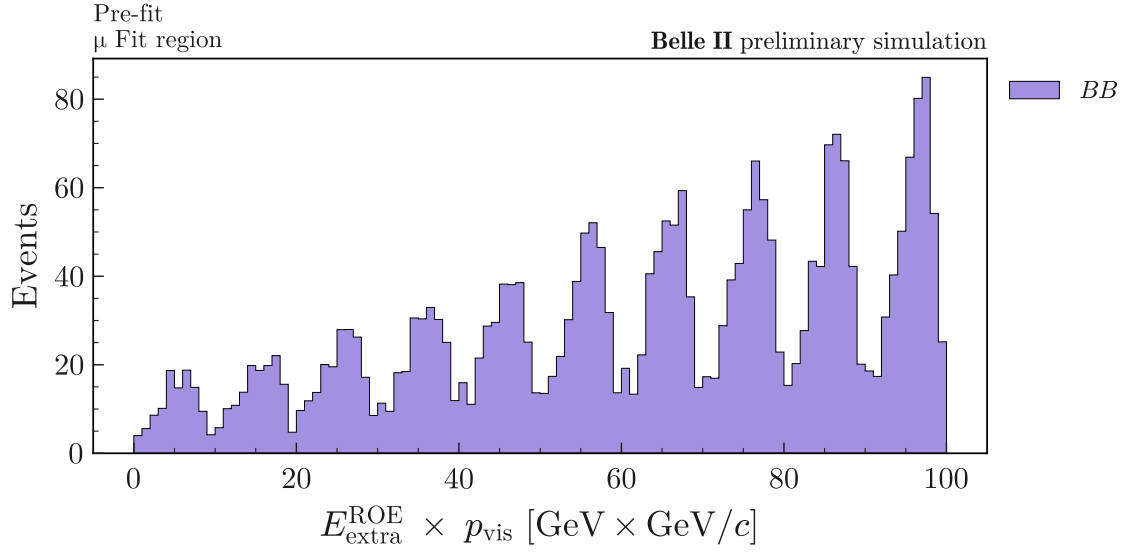


FIG. 151. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the $B\bar{B}$ background in the μ channel before the fit.

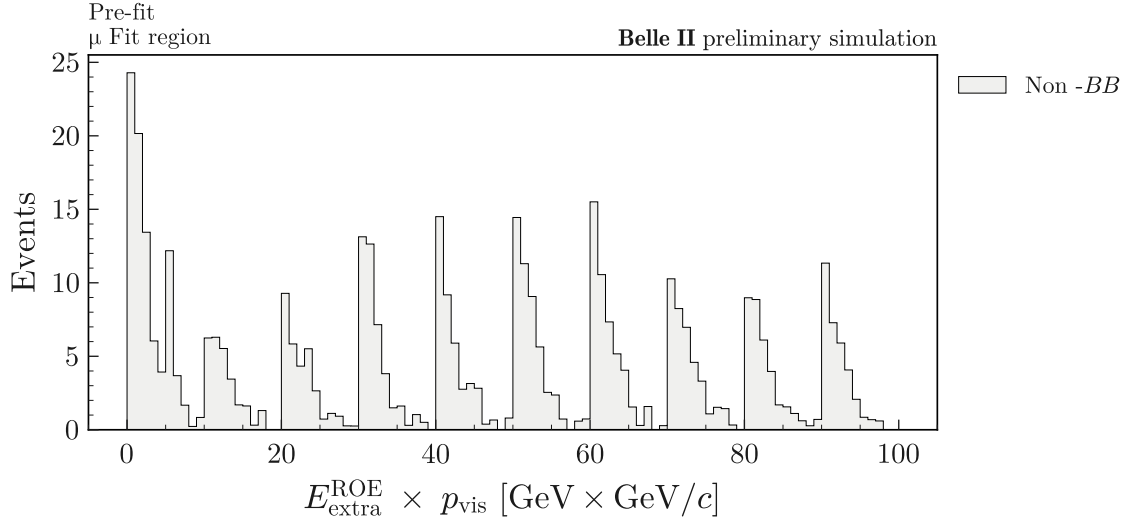


FIG. 152. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the non- $B\bar{B}$ background in the μ channel before the fit.

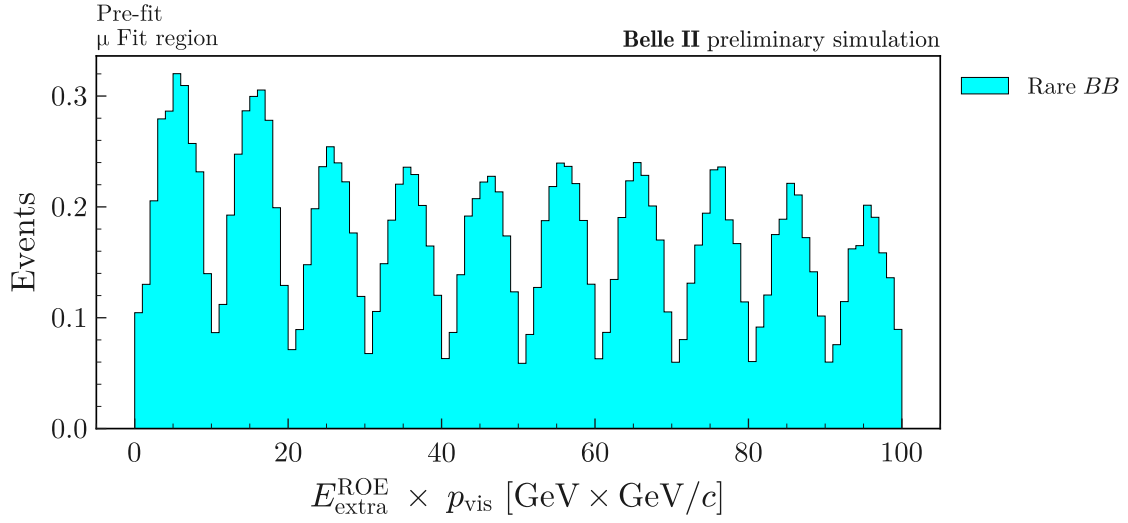


FIG. 153. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the rare $B\bar{B}$ background in the μ channel before the fit.

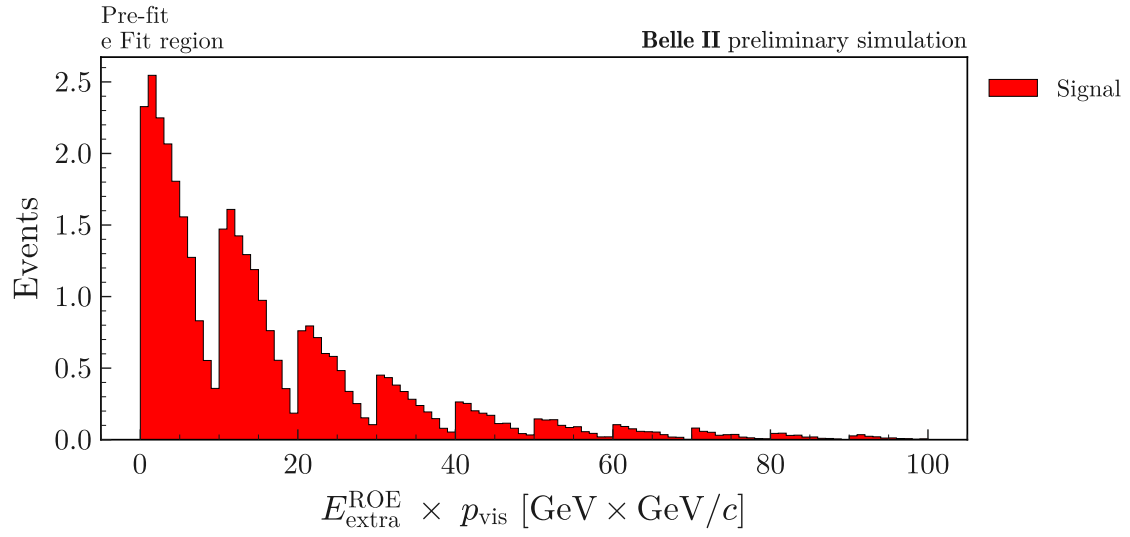
e channel:

FIG. 154. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the signal sample in the *e* channel before the fit.

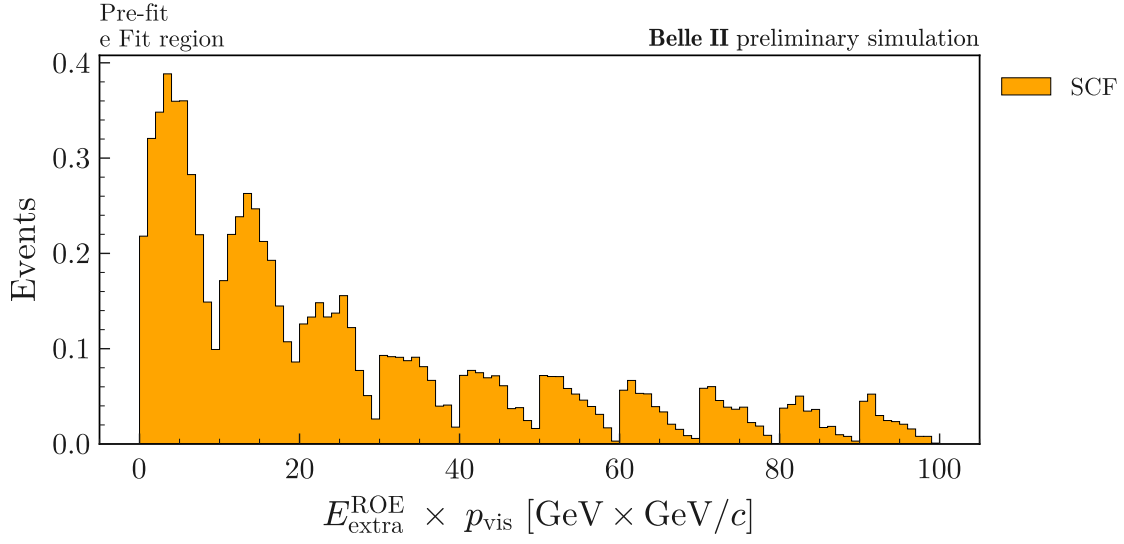


FIG. 155. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the SCF sample in the e channel before the fit.

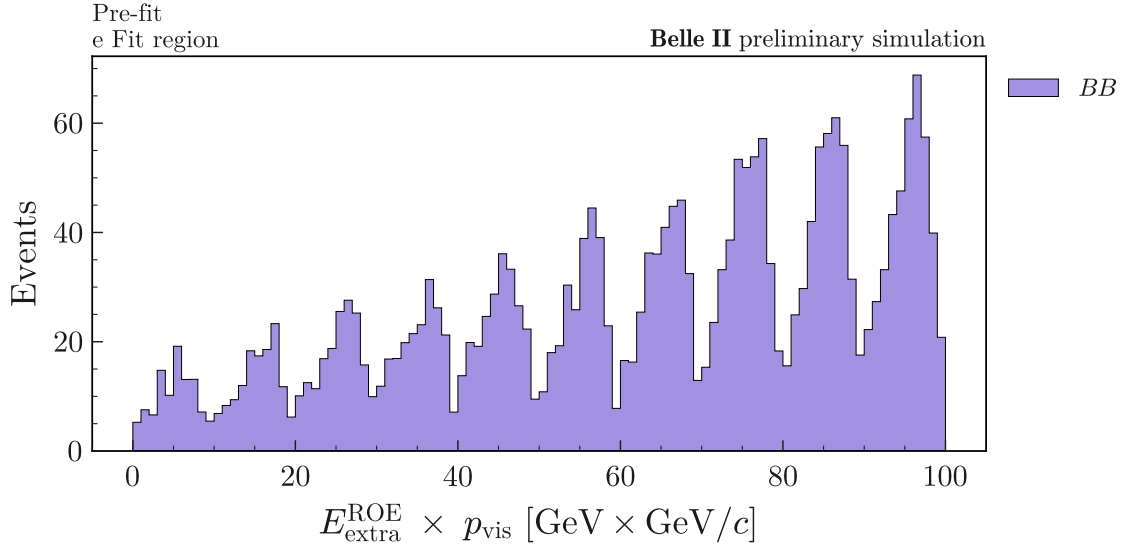


FIG. 156. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the $B\bar{B}$ background in the e channel before the fit.

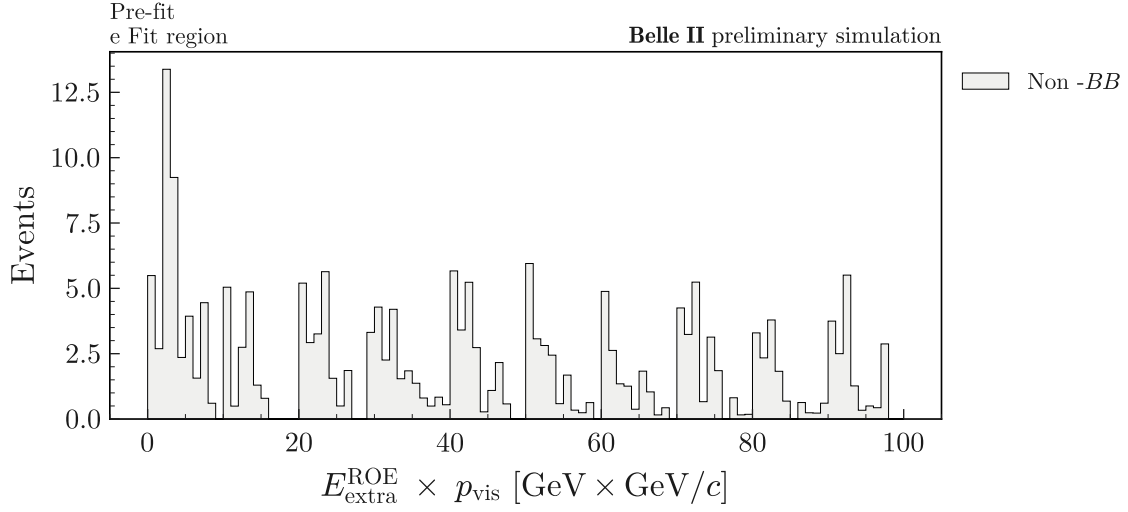


FIG. 157. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the non- $B\bar{B}$ background in the e channel before the fit.

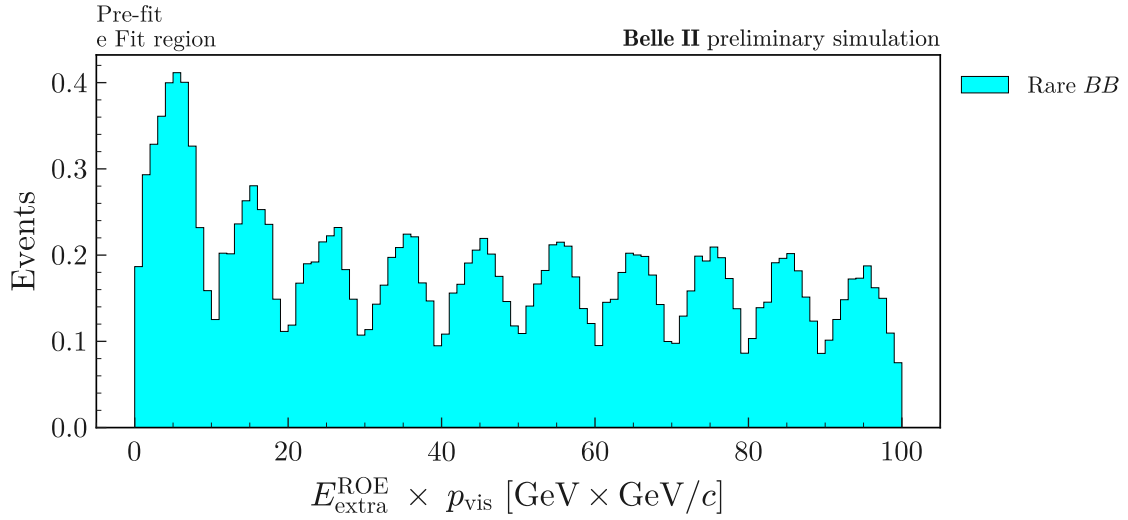


FIG. 158. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the rare $B\bar{B}$ background in the e channel before the fit.

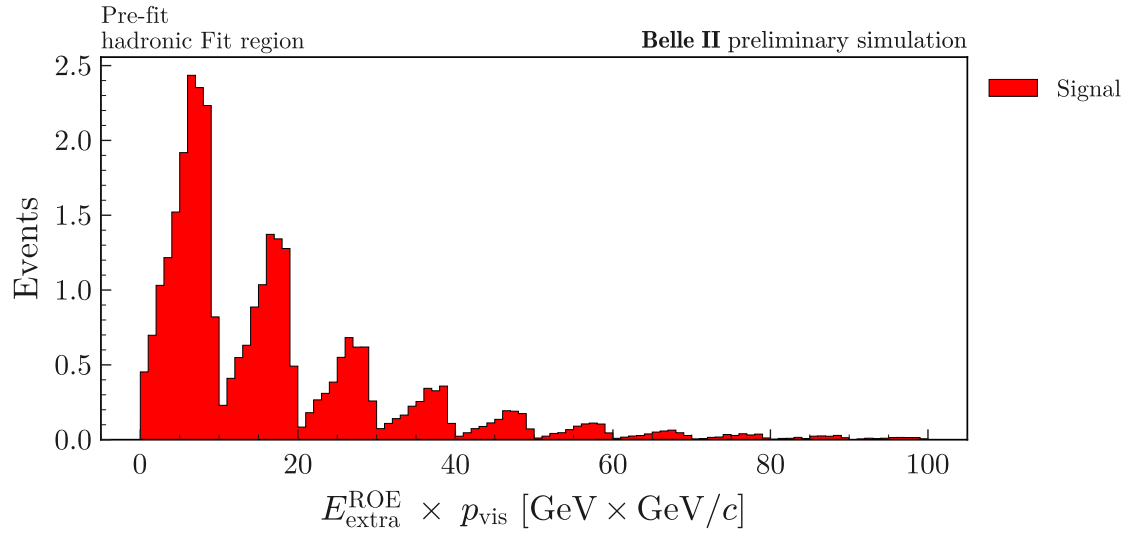
Hadronic channel:

FIG. 159. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the signal sample in the hadronic channel before the fit.

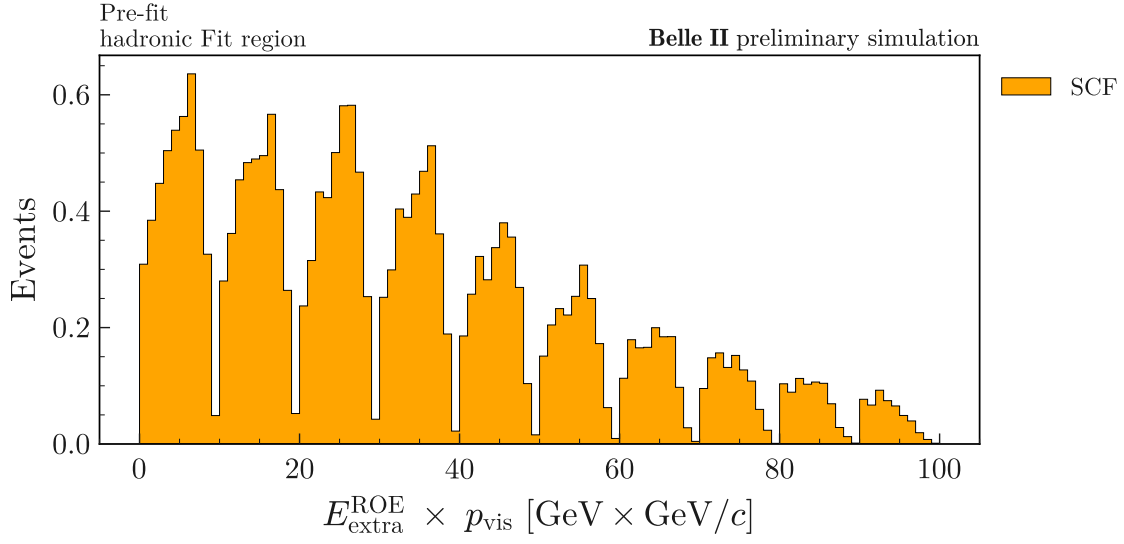


FIG. 160. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the SCF sample in the hadronic channel before the fit.

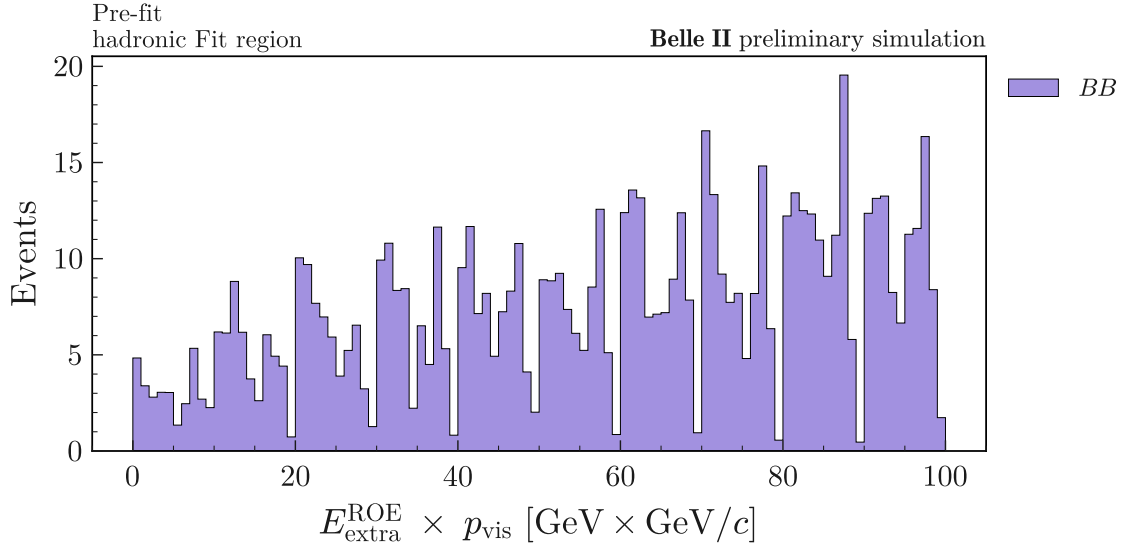


FIG. 161. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the $B\bar{B}$ background in the hadronic channel before the fit.

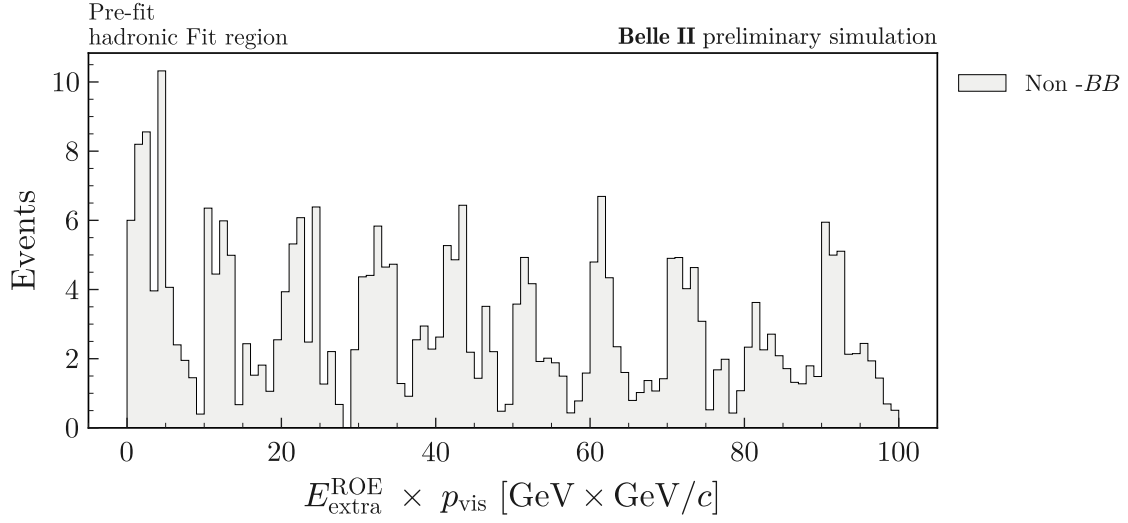


FIG. 162. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the non- $B\bar{B}$ background in the hadronic channel before the fit.

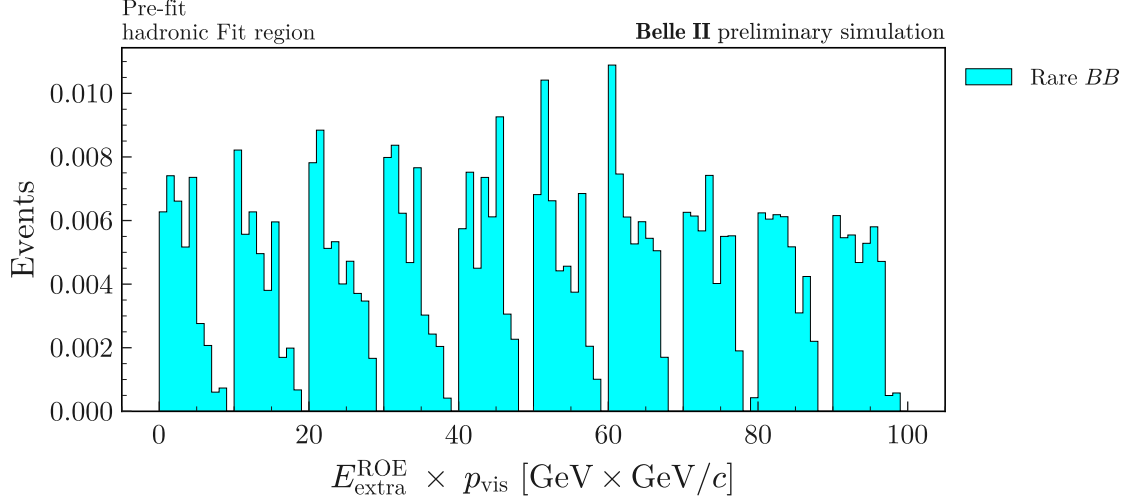


FIG. 163. The $E_{\text{extra}}^{\text{ROE}}$ and $\tau_{p_{\text{CMS}}}$ flattened distribution for the rare $B\bar{B}$ background in the hadronic channel before the fit.

1282

F.c. Pulls of individual NPs

1283

Here are shown the pulls of all the NPs included in the fit on Asimov dataset.

1284

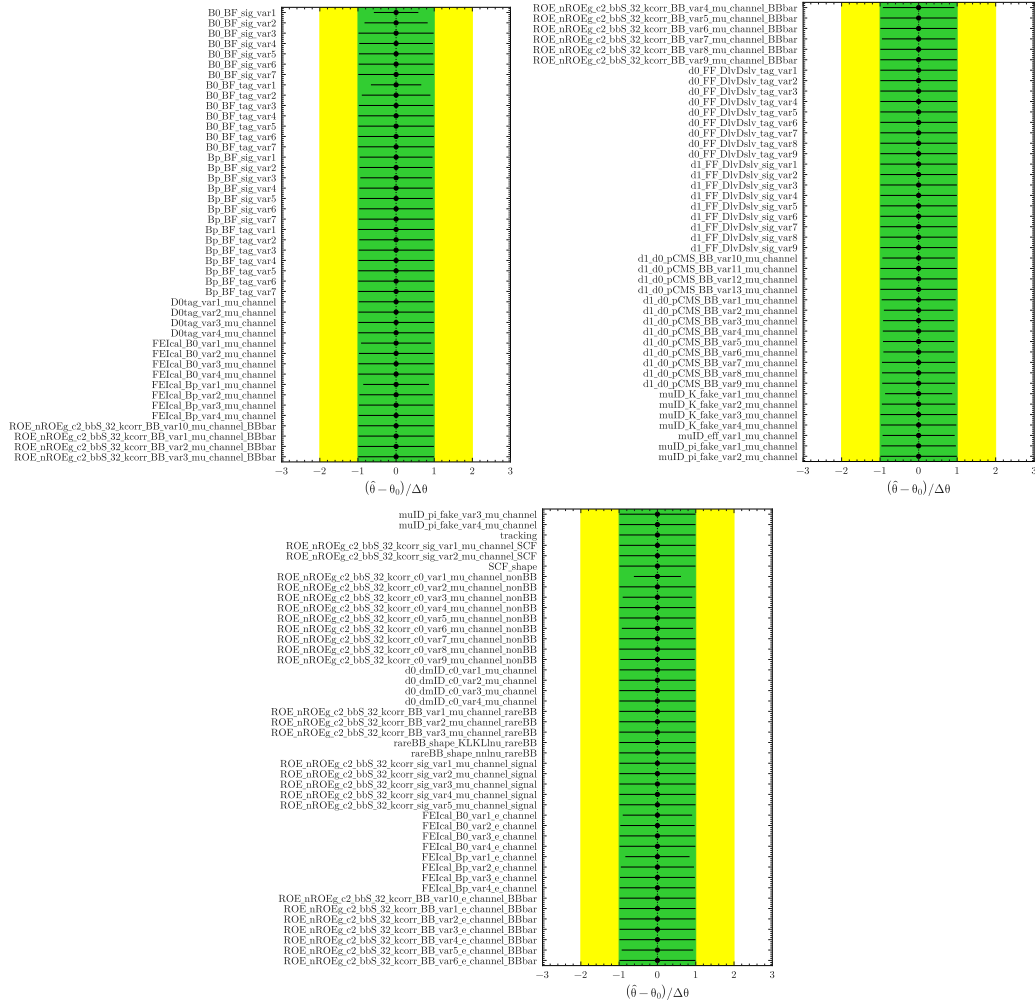


FIG. 165. Pulls of all the NPs included in the fit on Asimov dataset.

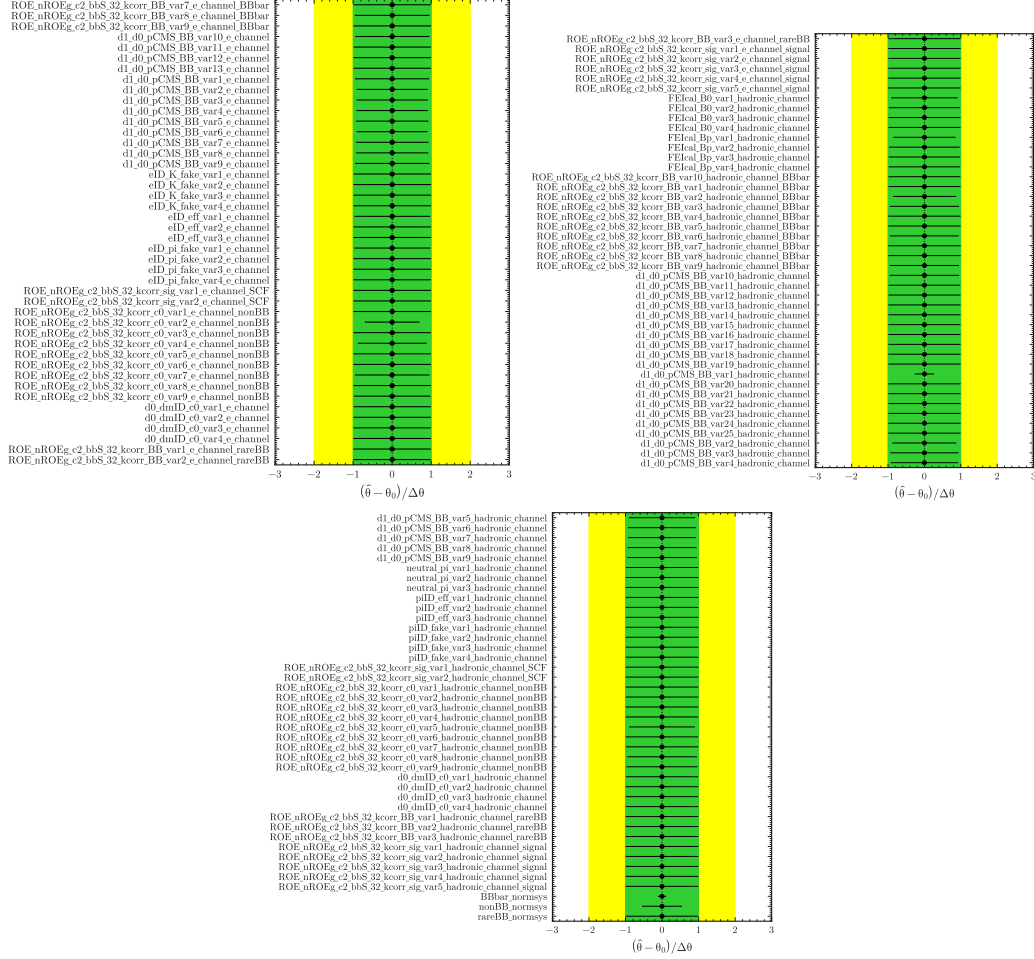


FIG. 166. Pulls of all the NPs included in the fit on Asimov dataset (cont.).

1285 *F.d. Impact of individual NPs*

1286 Here are shown the impact of the most impactful NPs included in the fit on
 1287 Asimov dataset.

1288

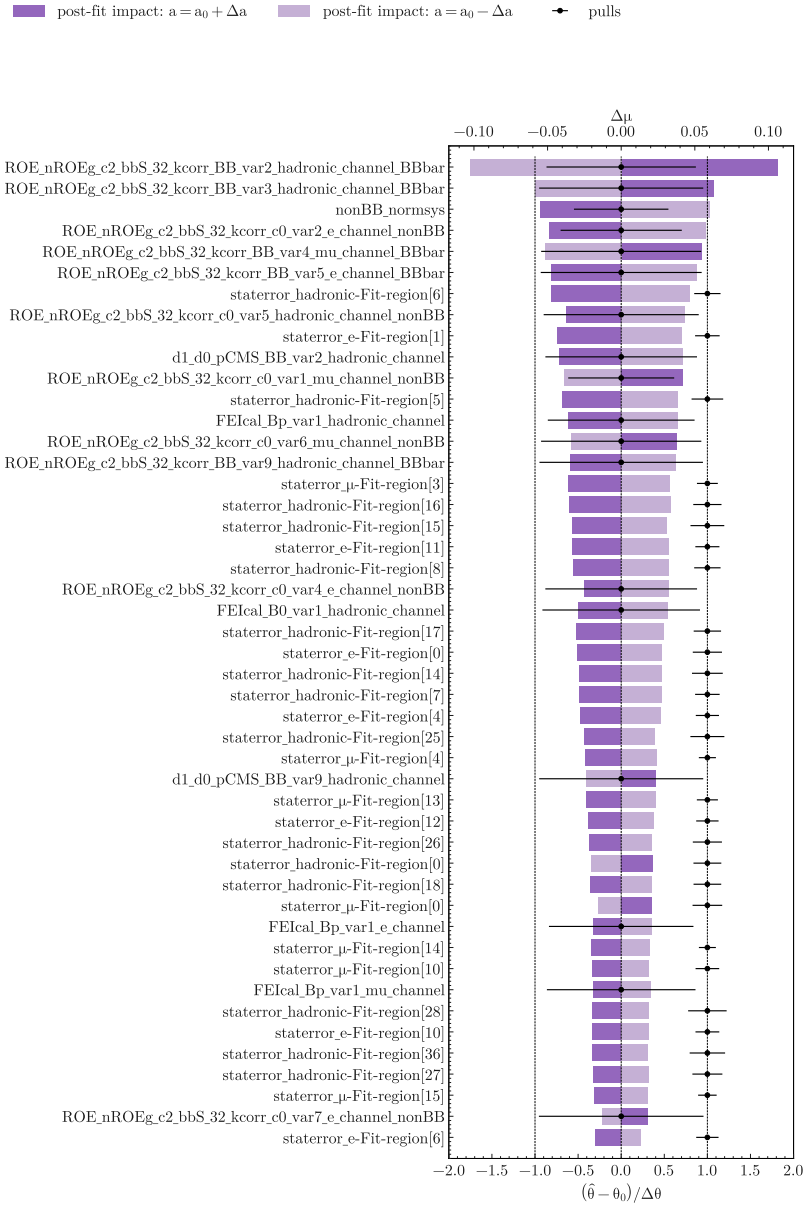


FIG. 167. Impact of all the NPs included in the fit on Asimov dataset.

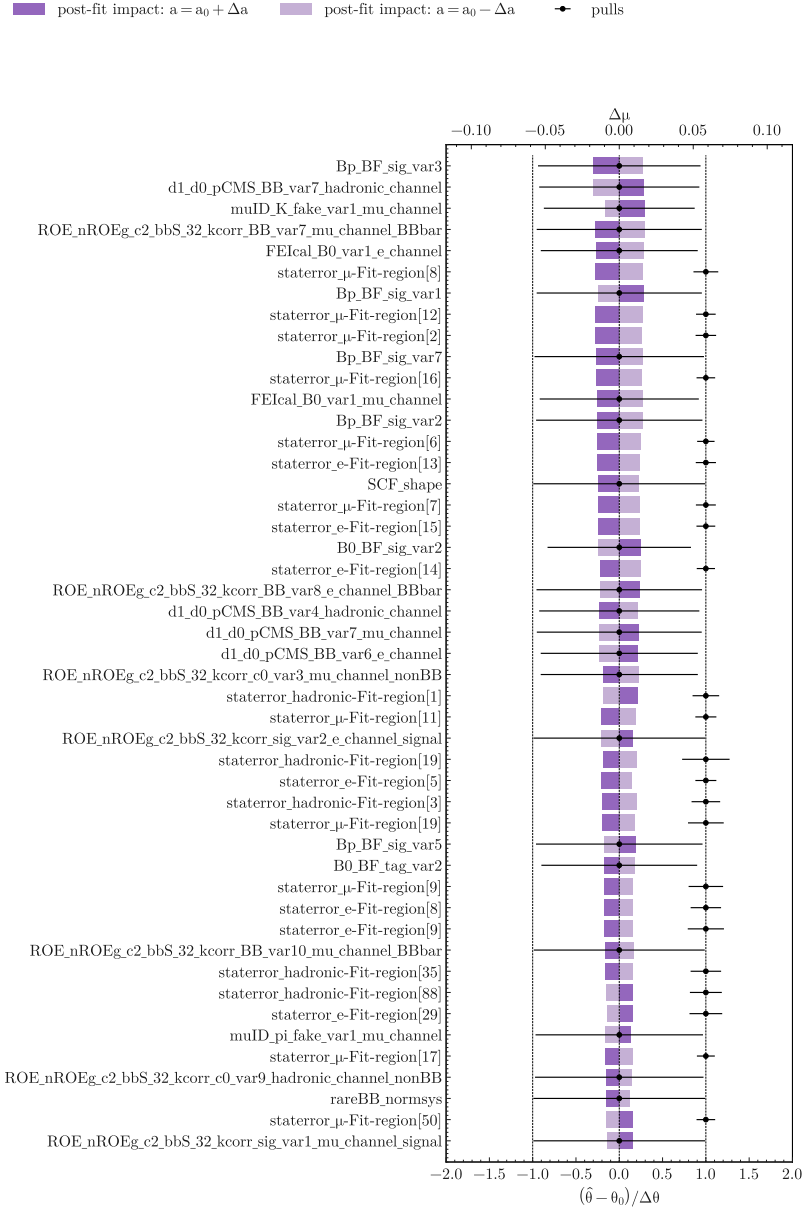


FIG. 168. Impact of all the NPs included in the fit on Asimov dataset (cont.).

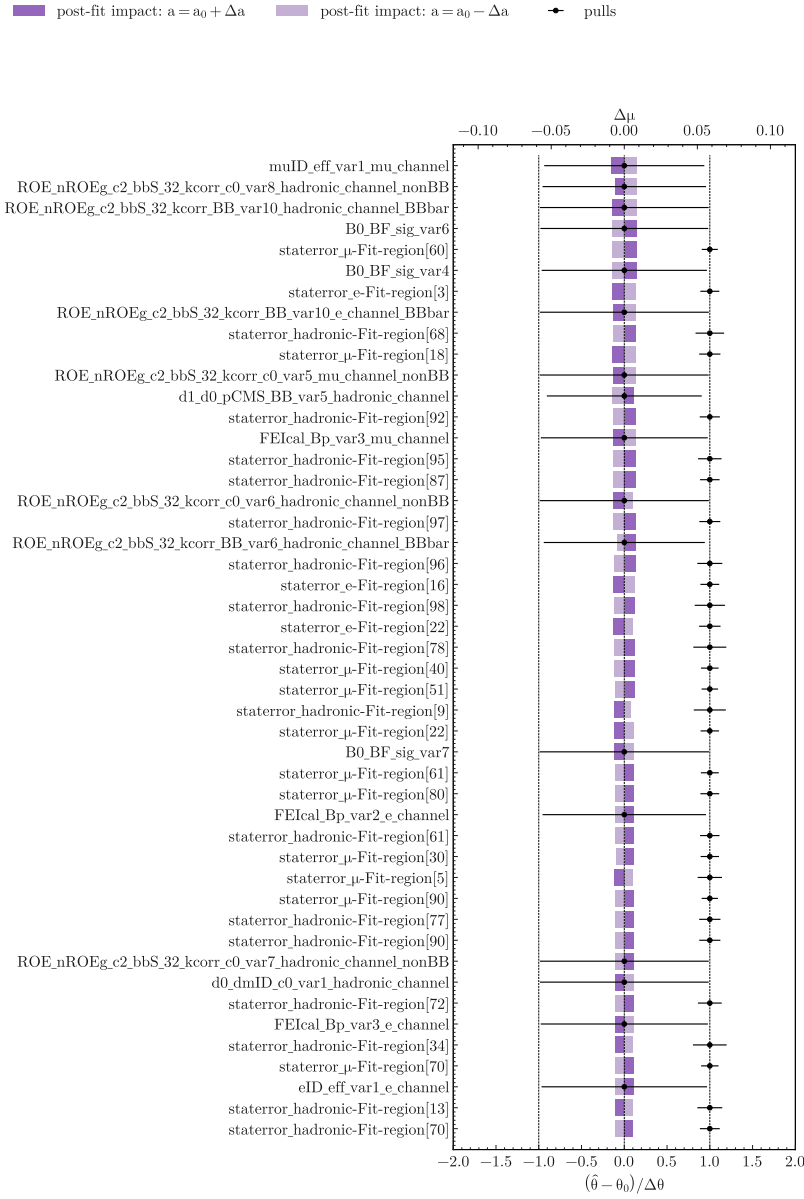


FIG. 169. Impact of all the NPs included in the fit on Asimov dataset (cont.).

1289

G. Unblinding, additional material

1290

Additional plots/checks on the unblinded LS1 dataset.

1291

G.a. Agreement with other measurements

1292

1293

Figure 170 shows the significance between the past measurements, as well as with the null-hypothesis and the PDG average [4], assuming independent measurements.

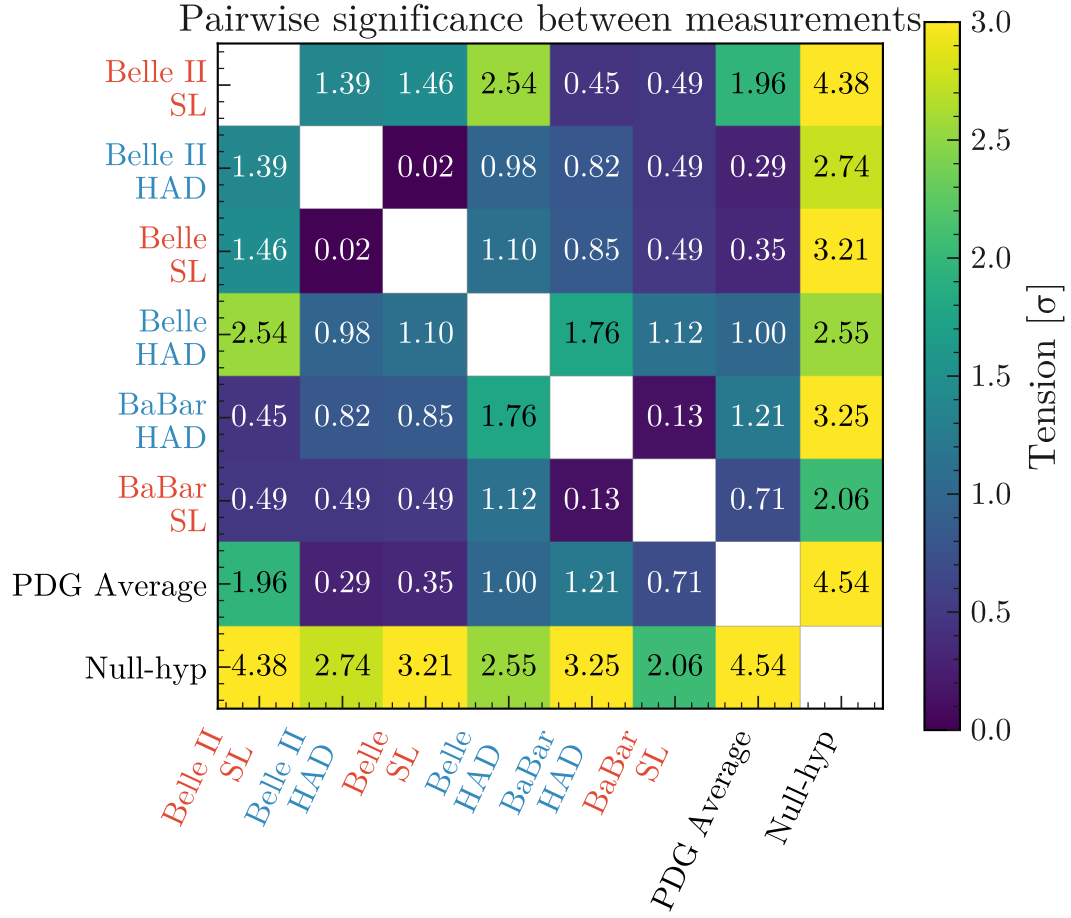


FIG. 170. Comparison of the measured branching fraction with previous measurements.

1294

G.b. Pulls of individual NPs

1295

Here are shown the pulls of all the NPs included in the fit on LS1 dataset.

1296

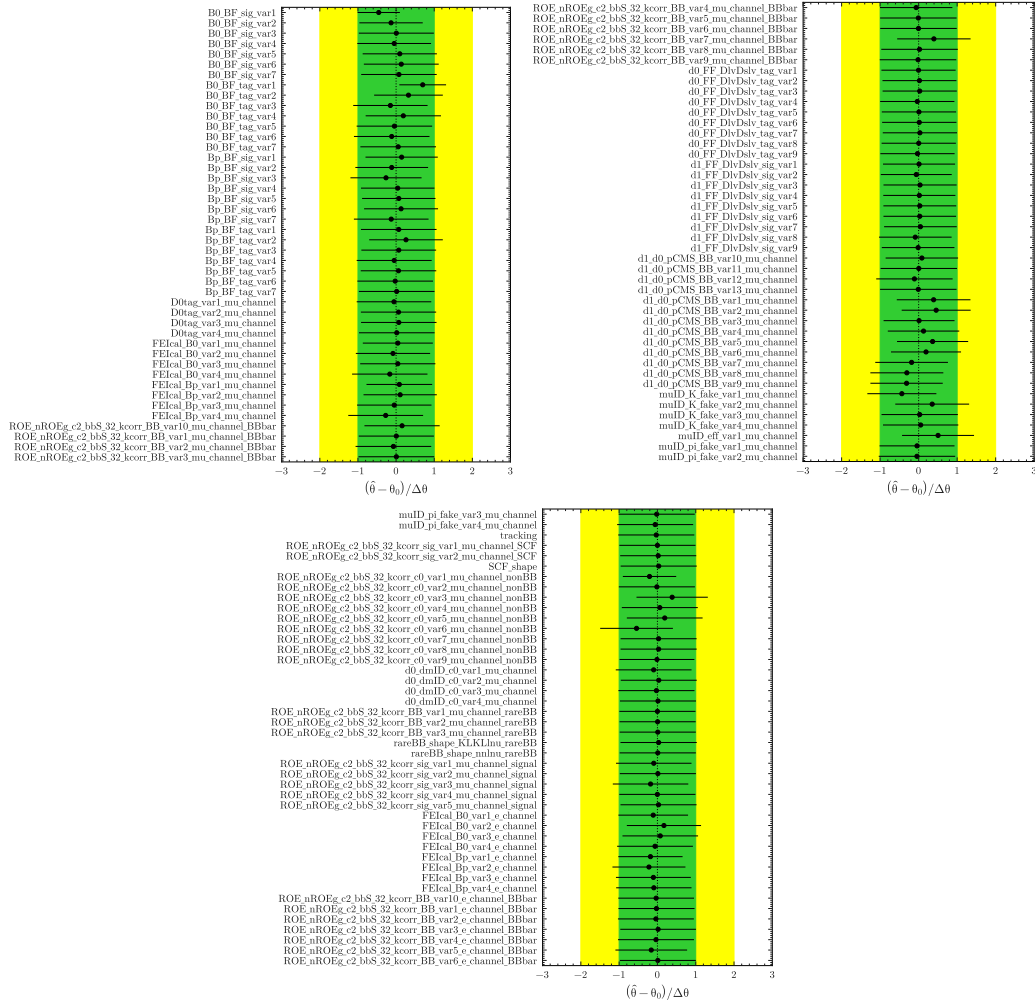


FIG. 171. Pulls of all the NPs included in the fit on LS1 dataset.

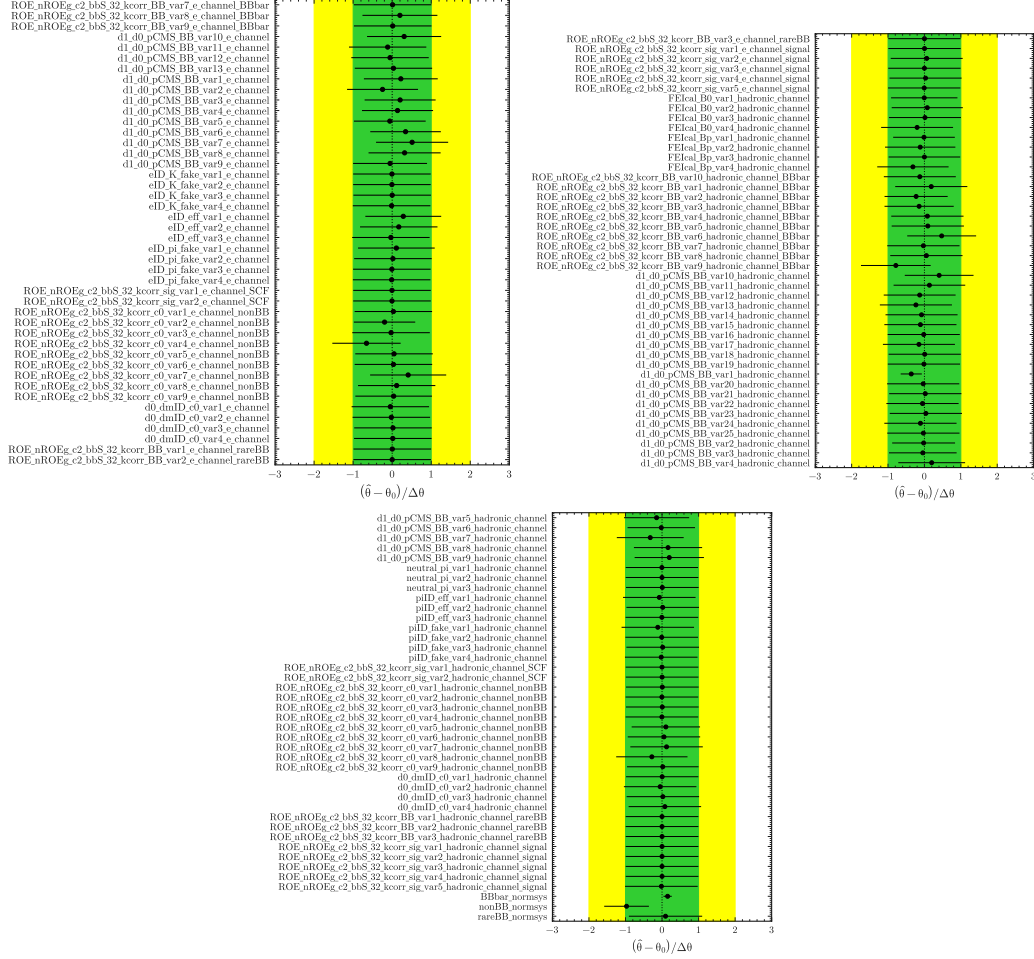


FIG. 172. Pulls of all the NPs included in the fit on LS1 dataset (cont.).

1297 *G.c. Post-fit plots*

1298 Here are show additional post-fit data/MC comparison plots.

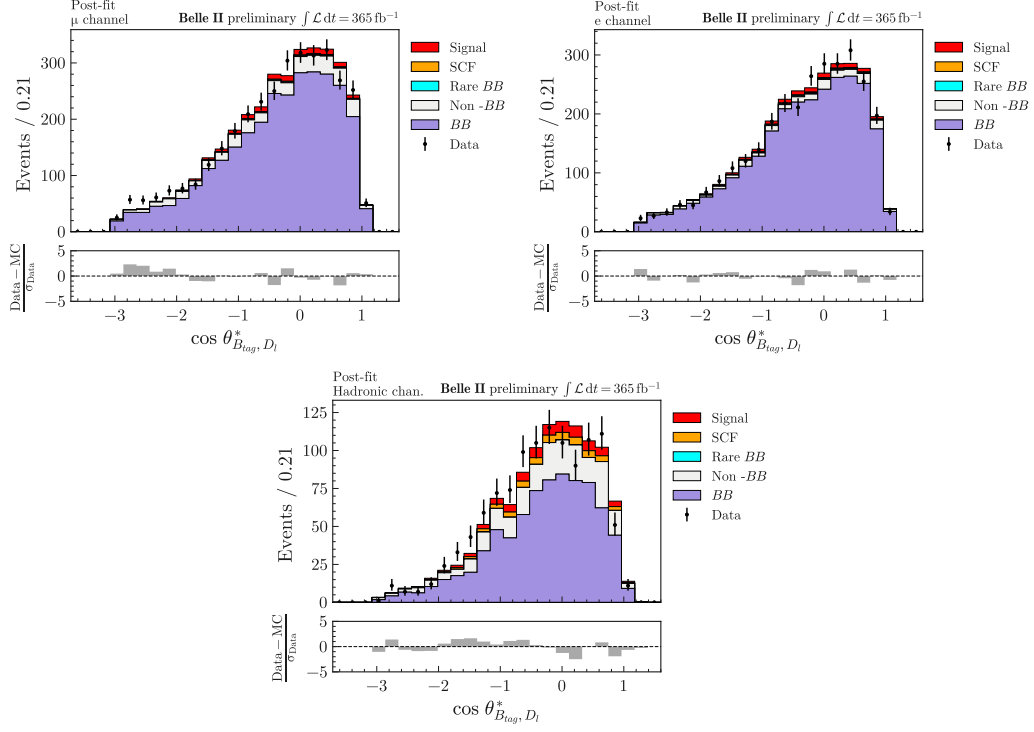


FIG. 173. Post-fit distributions for the 3 channels

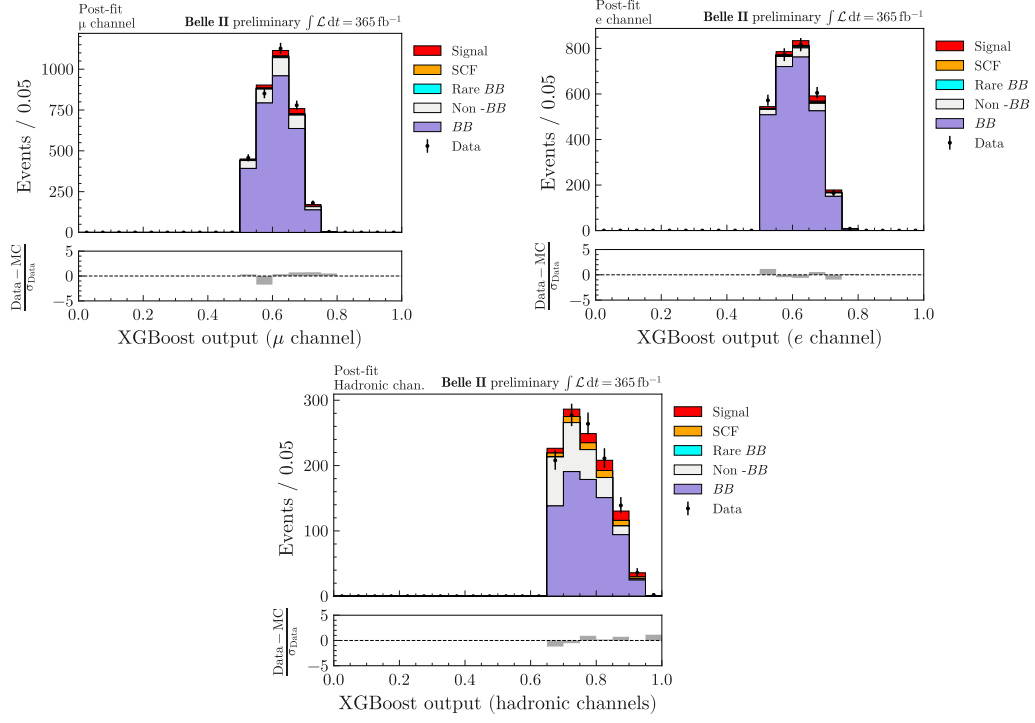


FIG. 174. Post-fit distributions for the 3 channels

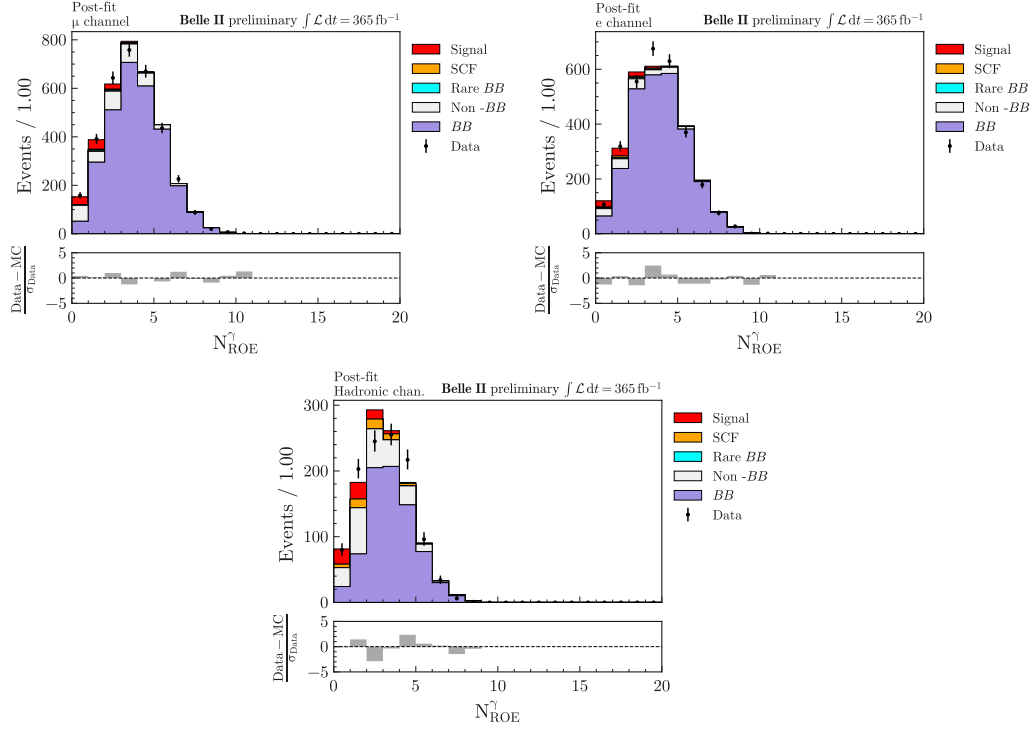


FIG. 175. Post-fit distributions for the 3 channels

1299

G.d. Post-fit plots: SR cut

1300

1301

Here are shown post-fit data/MC comparison plots with a cut on the signal region $E_{\text{extra}}^{\text{ROE}} < 0.3 \text{ GeV}$, corresponding to the first 3 bins of $E_{\text{extra}}^{\text{ROE}}$.

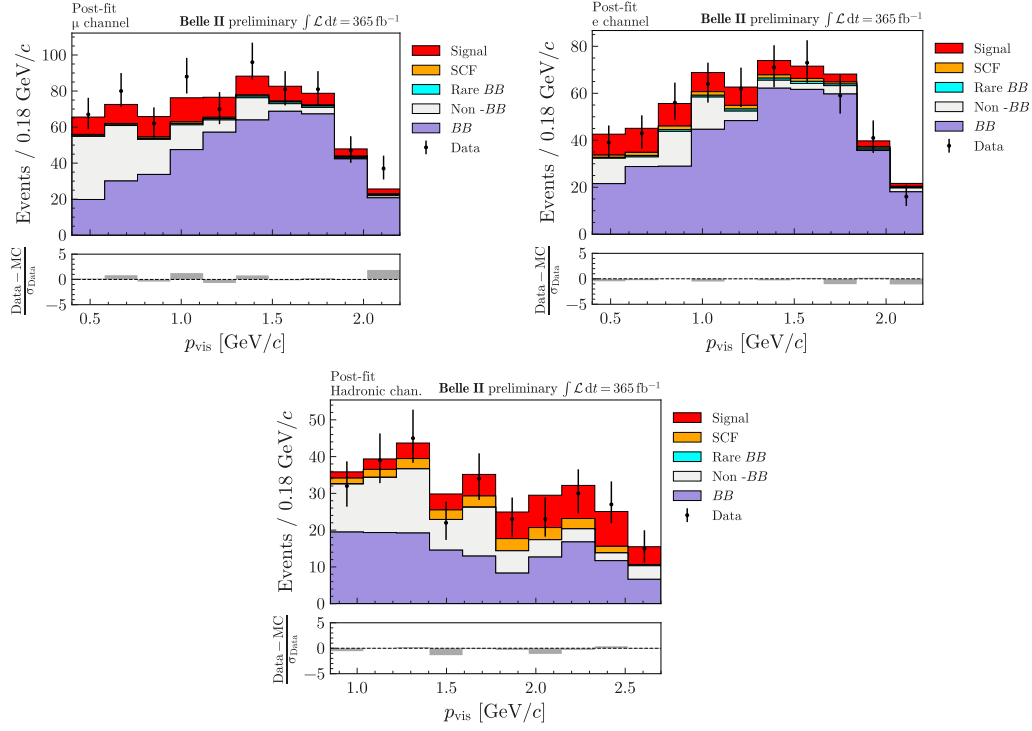


FIG. 176. Post-fit distributions for the 3 channels in the SR.

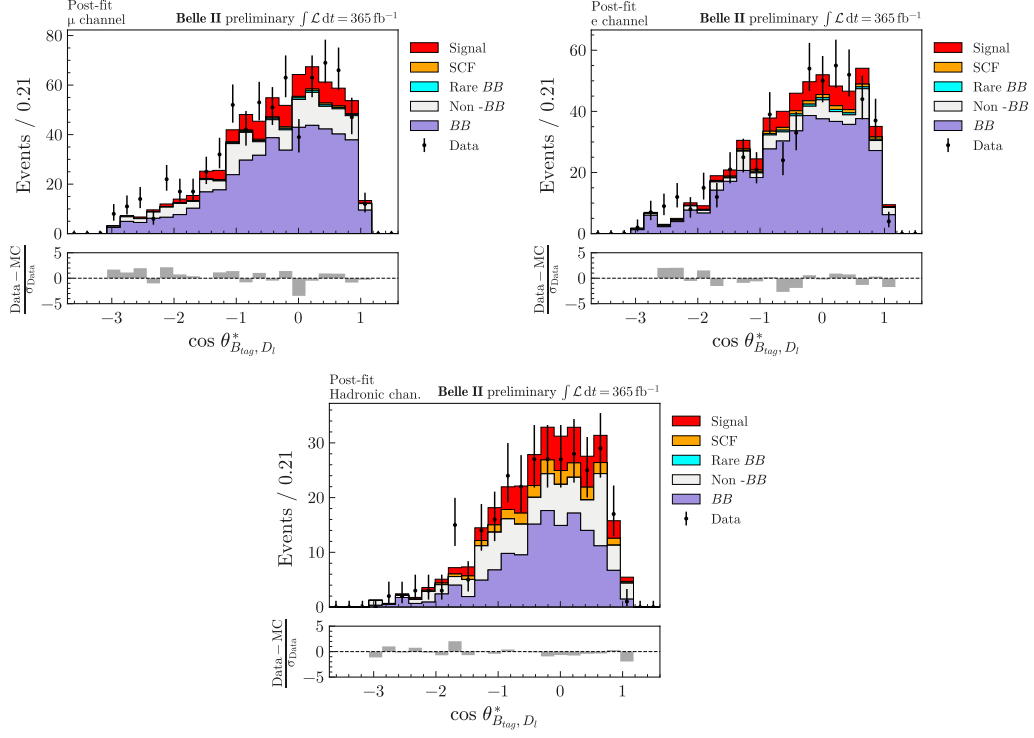


FIG. 177. Post-fit distributions for the 3 channels in the SR.

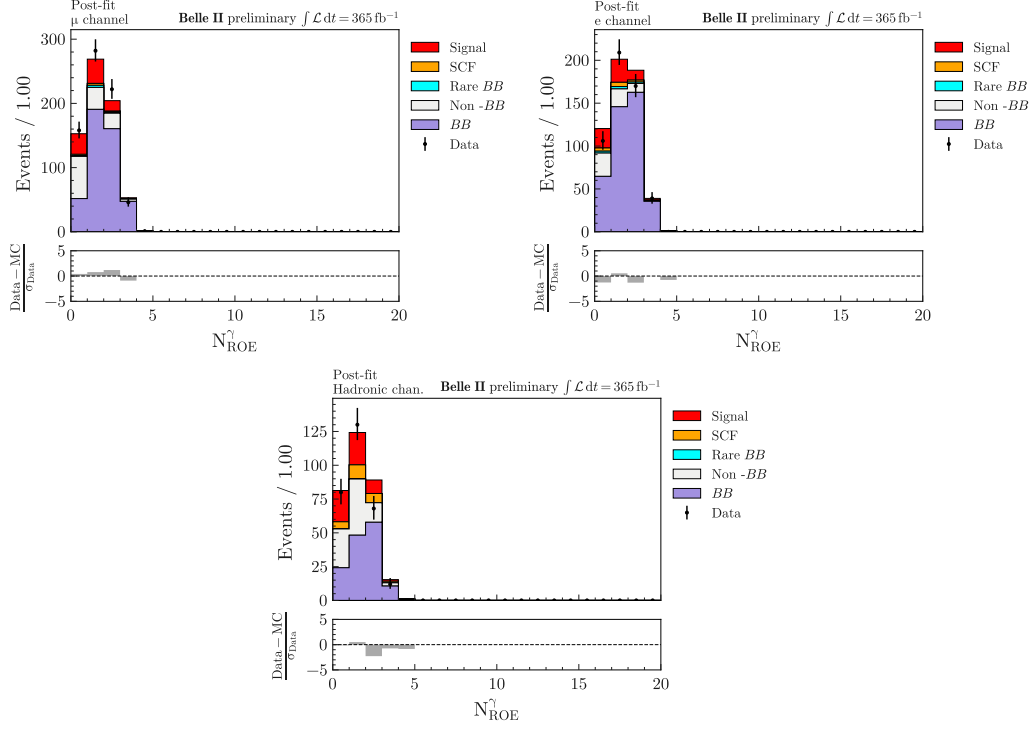


FIG. 178. Post-fit distributions for the 3 channels in the SR.

1302 *G.e. Post-fit plots: p_{vis} cut*

1303 Post-fit data/MC comparison plots with a cut on the momentum of the τ daughter
 1304 $p_{vis} < 1.3 \text{ GeV}/c$ and $p_{vis} < 1.8 \text{ GeV}/c$ for leptonic and hadronic modes respectively,
 1305 shown in Figures 179 to 181. The inverted cut plots are shown in Figures 182 to 184

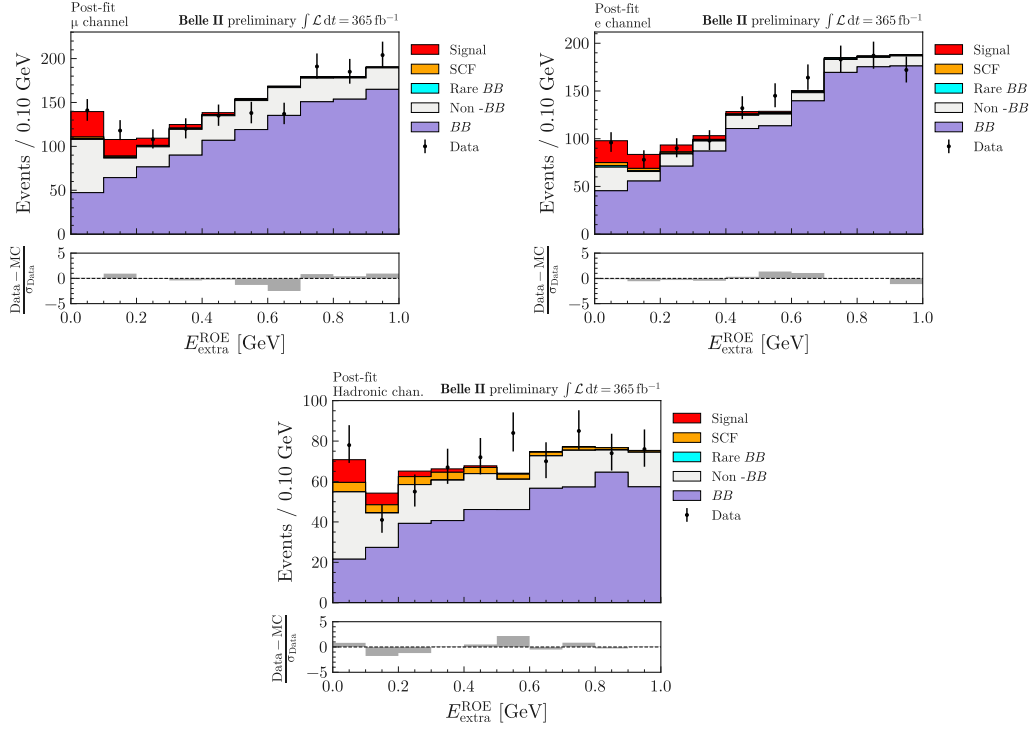


FIG. 179. Post-fit distributions for the 3 channels for low momentum τ daughters

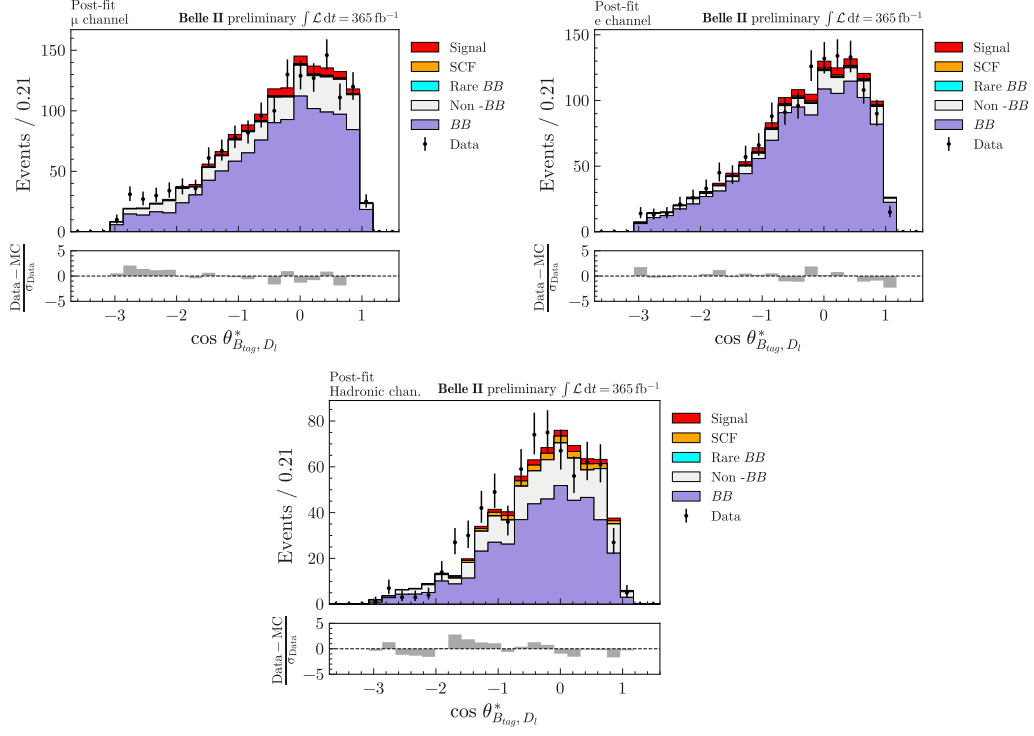


FIG. 180. Post-fit distributions for the 3 channels for low momentum τ daughters

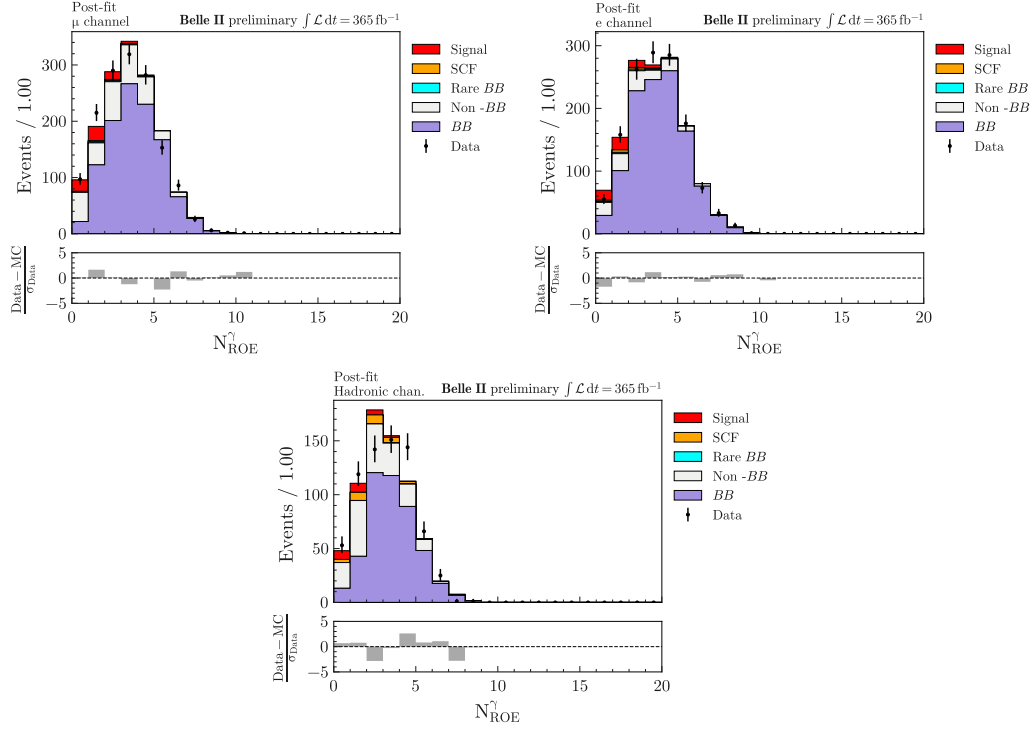


FIG. 181. Post-fit distributions for the 3 channels for low momentum τ daughters

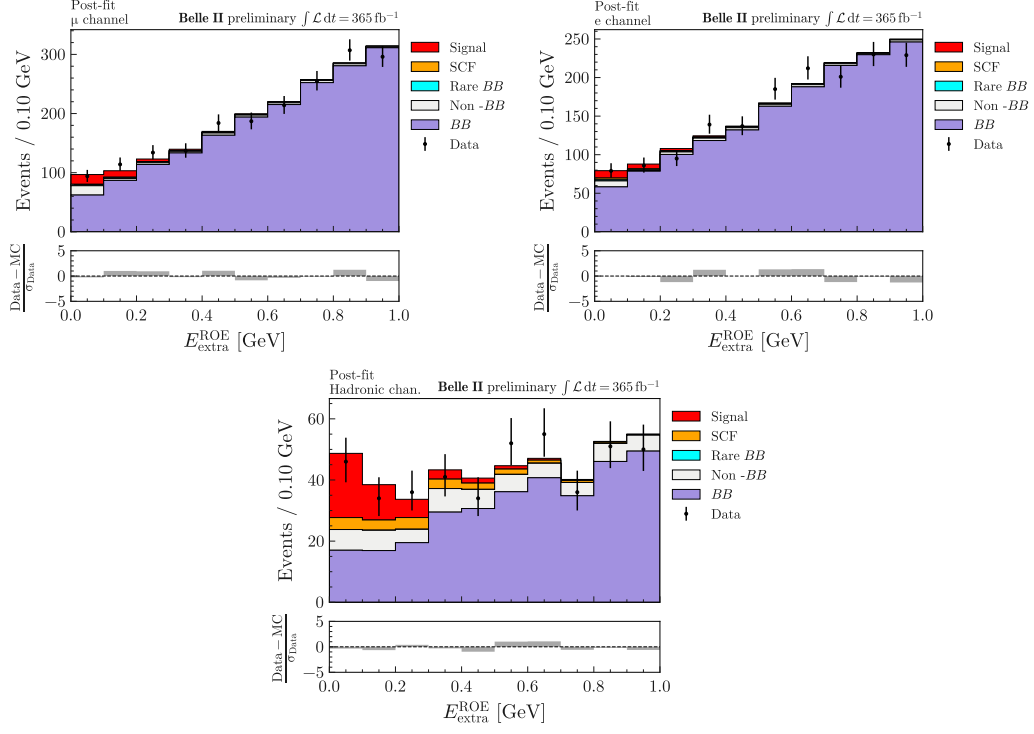


FIG. 182. Post-fit distributions for the 3 channels for high momentum τ daughters

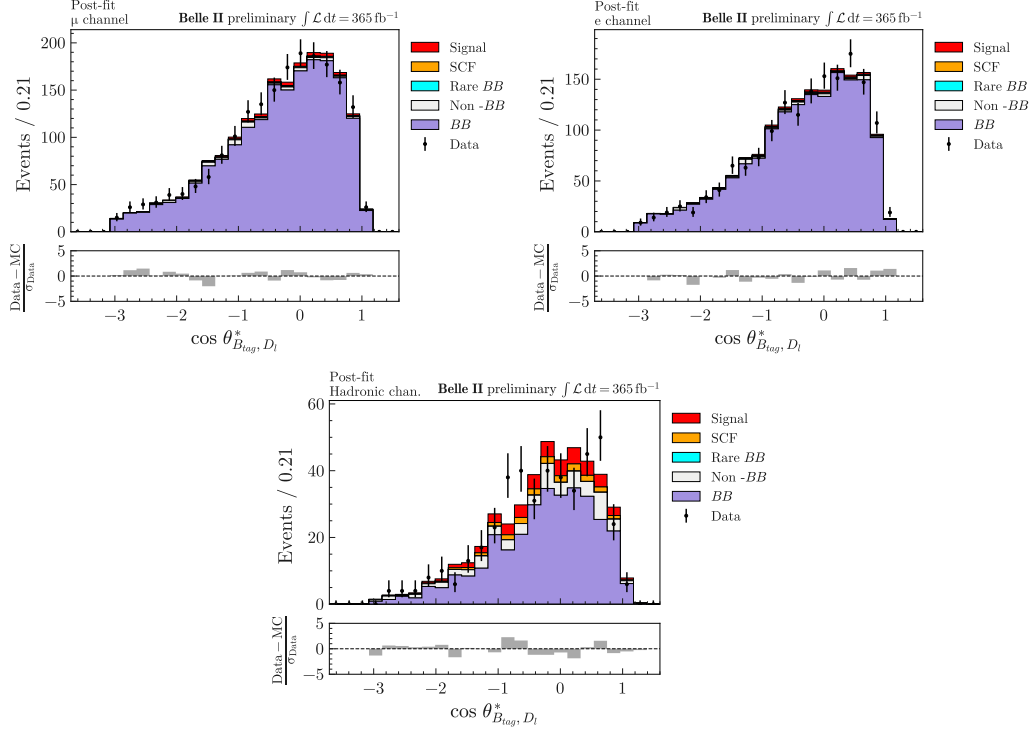


FIG. 183. Post-fit distributions for the 3 channels for high momentum τ daughters

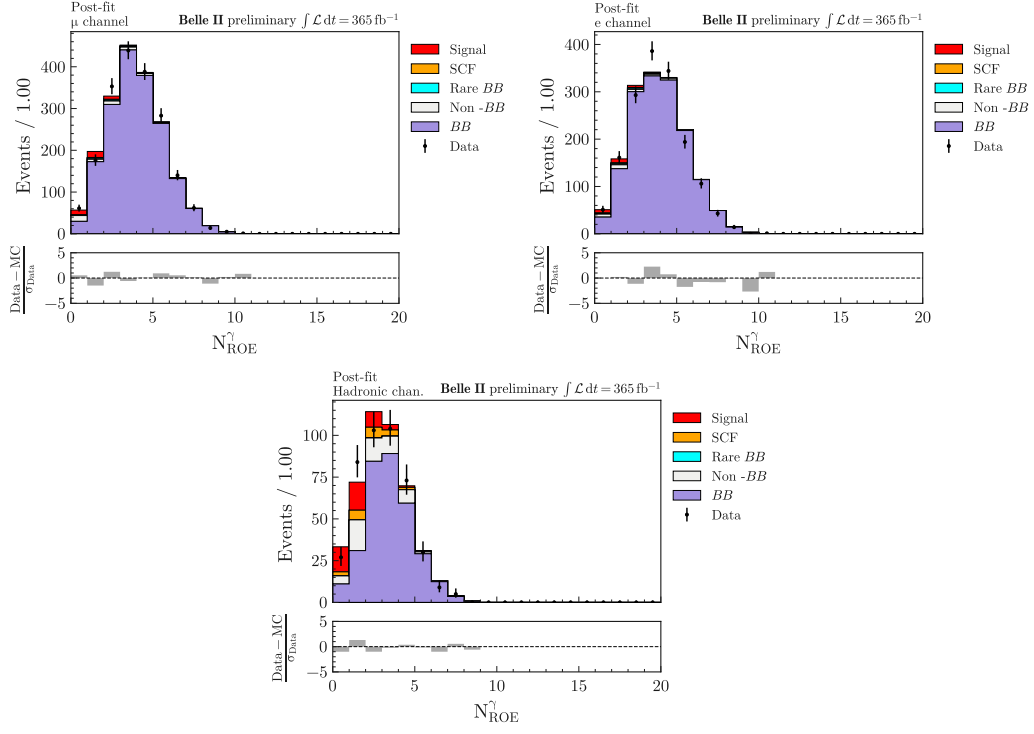


FIG. 184. Post-fit distributions for the 3 channels for high momentum τ daughters

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1383 [LE2-NOTE-PH-2025-031/7/BELLE2-NOTE-PH-2025-031.pdf](https://docs.belle2.org/files/4547/BELLE2-NOTE-PH-2025-031/7/BELLE2-NOTE-PH-2025-031.pdf).
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